LUMINOS Q.namix R

Operator Manual – VB10 and higher

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Legend

i	Indicates a hint Is used to provide information on how to avoid operating errors or information emphasizing important details
»	Indicates the solution of a problem Is used to provide troubleshooting information or answers to frequently asked questions
■	Indicates a list item
✓	Indicates a prerequisite Is used for a condition that has to be fulfilled before starting a particular operation
◆	Indicates a one-step operation
1 2 3	Indicates steps within operating sequences
Italic	Is used for references and for table or figure titles
➔	Is used to identify a link to related information as well as previous or next steps
Bold	Is used for screen outputs of the system
Menu > Menu Item	Is used for the navigation to a certain submenu entry
`Courier`	Is used to identify inputs you need to provide
Orange	Is used to emphasize particularly important sections of the text
⚠ CAUTION	CAUTION Used with the safety alert symbol, indicates a hazardous situation which, if not avoided, could result in minor or moderate injury or material damage. CAUTION consists of the following elements: • Information about the nature of a hazardous situation • Consequences of not avoiding a hazardous situation • Methods of avoiding a hazardous situation
⚠ WARNING	WARNING Indicates a hazardous situation which, if not avoided, could result in death or serious injury. WARNING consists of the following elements: • Information about the nature of a hazardous situation • Consequences of not avoiding a hazardous situation • Methods of avoiding a hazardous situation

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1 Introduction

To operate the LUMINOS Q.namix R system accurately and safely, the operating personnel must have the necessary expertise as well as knowledge of the entire Operator Manual. It must be read carefully prior to starting up the LUMINOS Q.namix R system.

1.1 Intended purpose

Intended Purpose Statement under the European Medical Device Regulation 2017/745:

Radiographic and fluoroscopic X-ray system intended for various procedures that visualizes a variety of anatomical structures.

1.1.1 Intended use

LUMINOS Q.namix R is a device intended to visualize anatomical structures by converting an X-ray pattern into a visible image. It is a multifunctional, general R/F system, suitable for routine radiography and fluoroscopy examinations, including gastrointestinal- and urogenital examinations and specialist areas like arthrography, angiography and pediatrics.

1.1.2 Indication for use

LUMINOS Q.namix R is a device intended to visualize anatomical structures by converting an X-ray pattern into a visible image. It is a multifunctional, general R/F system, suitable for routine radiography and fluoroscopy examinations, including gastrointestinal- and urogenital examinations and specialist areas like arthrography, angiography and pediatrics.

LUMINOS Q.namix R is not intended to be used for mammography examinations.

Warning: This x-ray unit may be dangerous to patient and operator unless safe exposure factors, operating instructions and maintenance schedules are observed.

1.1.3 Patient group

The LUMINOS Q.namix R system is designed for adult and pediatric (removable scattered radiation grid) patients.

The patient table is not approved for use with patients with a weight (incl. accessories) > 330 kg.

For the maximum patient weight of optional tables etc. please refer to Register **Accessories and Auxiliary Devices.**

1.1.4 Contraindications

There are no known contraindications for use of the system.

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1.1.5 Clinical benefits

LUMINOS Q.namix R is a device intended to visualize anatomical structures by converting an X-ray pattern into a visible image for diagnostic purposes as well as fluoroscopic procedures.

1.1.6 Potential side effects

The side effects of the device are derived from the risk analysis. Based on the clinical evaluation, there are no known specific side effects associated with the use of LUMINOS Q.namix R system.

However, due to the nature of X-ray procedures, the patient is generally exposed to radiation. Associated adverse health effects exist and are known. The X-ray equipment described in this Operator Manual is used only by qualified personnel who are aware of the possible side effects. Therefore, the responsible radiologist or urologist must weigh risk and benefit and also evaluate the use of alternative diagnostic technologies. In addition, contraindications due to e.g. contrast agents and other counter indications to specific fluoroscopy procedures have to be considered for the benefit-risk analysis.

Stochastic radiation injuries

The X-rays generated during the use of the device are dangerous for the operator as well as for other persons in the vicinity and may cause stochastic effects.

However, these random radiation injuries may have other causes besides medical examinations (e.g. cosmic radiation, terrestrial radiation, internal radiation from radon, smoking, air travel) and cannot be attributed to X-rays alone.

There is no defined threshold for this stochastic radiation damage, i.e. even low levels of radiation can cause damage in the worst case. These may also occur much later (several years) after exposure.

Deterministic radiation injuries

Deterministic radiation injuries are acute non-random damages that can be caused by ionizing radiation. The likelihood of damage occurring increases with the dose of radiation, i.e. the duration and strength of exposure. The damage usually occurs within a few days or weeks after exposure and is recognizable, for example, by a reddening of the skin.

1.1.7 Residual risks

Residual risks have been assessed and all relevant information is provided to the user via this Operator Manual.

Please note that risks inherently connected to the use of this device (e.g. radiation or moving parts) cannot be avoided due to the nature of this device.

1.1.8 Physical funtionality

A LUMINOS Q.namix R system may be used for projection of radiographic and fluoroscopic examinations of both adults and children (removable scattered radiation grid).

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- On the patient table, X-rays can be taken on the recumbent or sitting patient from head to foot. Exposures in the region of the skull, of the spine, of thorax, lungs and abdomen as well as of the extremities are possible.
- On the wall stand, X-rays can be taken on the standing or sitting patient (chest, and so on).
- In addition, exposures onto MAX detectors and X.wi-D detectors in the form of free exposures are possible.

1.1.9 User profile

This Operator Manual is intended for qualified personnel with the necessary knowledge and expertise to operate X-ray equipment in accordance with the relevant country specific regulations.

1.1.10 Training

Application training is supplied with the equipment according to the hand-over contract. It is mandatory to follow such application training supplied by Siemens Healthineers Representative before any use of the system.

The follow-up training, which is necessary due to change of personnel, is in the responsibility of the operator of the system. Any additional training can be requested from Siemens Healthineers.

1.1.11 Conditions of use

The system is suitable for medical facilities such as hospitals, doctor’s offices, etc. The climatic conditions specified in Register **Technical Data** (→ Page 338 *Environment*) must be complied with.

Minimum requirements concerning hardware

The LUMINOS Q.namix R system is being delivered, installed, and connected to the IT environment as a complete and functioning system by our service organization.

IT network characteristics

LUMINOS Q.namix R may only be run in an environment approved or authorized by the manufacturer. The system may not be operated in public networks without firewalls or isolation.

Refer to the ”MAX Systems Security White Paper and MDS2 Form”.

1.1.12 Maintenance, cleaning, and disinfection

Refer to Register **System Safety**.

1.1.13 Service

The service must only be performed by qualified technical personnel.

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1.1.14 Technical performance characteristics

Refer to Register **Technical Description**.

1.1.15 Frequently used operating functions

Refer to Register **System Operation**.

1.1.16 Operating functions regarding safety

Refer to Register **System Safety**.

1.2 Usability

The systems must only be used by persons with the necessary specialist knowledge after training, for example physicians, radiologists, cardiologists, and medical specialists.

- Patient population: newborn to geriatric.
- Operator profile: The usage of the system described in this Operator Manual requires specific technical and medical knowledge and skills regarding, at a minimum, radiation protection, safety procedures, and patient safety.

  People using, moving, working with the system must have acquired such knowledge and skills during their curriculum.
- Language understanding: Users must understand the language of the Operator Manual before touching the system.
- Equipment training: Application training is supplied with the equipment according to the hand-over contract. It is mandatory to follow such application training supplied by Siemens Healthineers before any use of the system.

  The follow-up training, which is necessary due to change of personnel, is in the responsibility of the operator of the system. Any additional training can be requested from Siemens Healthineers.
- Operator Manual and precautions: Read and understand all the instructions in the Operator Manual before trying to use the system and request additional training from Siemens Healthineers if necessary.

  Keep the Operator Manual with the equipment at all times and periodically review the procedures and safety precautions.

  Failure to follow the Operator Manual and safety precautions could result in serious injury to the patient, others or yourself.
- Patient safety: Give assistance for getting the patient on and off the table. Be sure all patient health lines (IV, oxygen, and so on) are positioned so they will not be caught when moving the equipment. Never leave the patient unattended while in the system room.

  An unattended patient could fall from the table, start a movement control, or encounter other problems which could be hazardous.
- Radiation safety: Always use correct technique factors for each procedure to minimize X-ray exposure and to produce the best diagnostic results.

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- Establish emergency procedures: It is not always possible to determine when some components, such as X-ray tubes, are nearing the end of their operating lives. These components could stop operating during a patient examination. Have an emergency plan in case of non-availability of the system, for example, to use a different system.

  Establish procedures for handling the patient in case of the loss of fluoroscopic imaging or other system functions during an examination.

1.3 Information about this Operator Manual

This Operator Manual is valid for the following product:

- LUMINOS Q.namix R (VB10)

1.3.1 Segmentation of the Operator Manual

To improve readability, your complete Operator Manual has been broken down into several Operator Manuals with thematically distinct content:

- Operator Manual - LUMINOS Q.namix R system
- Operator Manual - imaging system
- User Administration Guide (for privileged users - administrators)

These documents are referenced in this Operator Manual.

1.3.2 Operator Manuals on the internet

The electronic Operator Manuals for the system are available on the internet:
↗*https://doclib.healthcare.siemens.com*.

1.3.3 Names and parameters

All names and data on patients and equipment that are used as examples in this Operator Manual are entirely fictional.

Any resemblance to names of real persons and institutions is entirely coincidental.

All parameters and images shown in this Operator Manual are examples. Only the parameters displayed on your system are relevant.

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i The LUMINOS Q.namix R may be shown with expanded equipment or with optional accessories in some figures.

1.3.4 Illustrations

All illustrations of the equipment and of the user interface in this Operator Manual, are **examples** only.

Differences in detail may occur in your system due to the installed options, configuration and constant development of the system.

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Reproduction of images can cause loss of detail. Pictures in this Operator Manual do not therefore provide any indication of image quality.

All names of patients in images or illustrations, are purely fictional. Any similarities with existing persons are entirely coincidental.

1.3.5 Value statements

All technical data are typical values unless specific tolerances are stated.

Values in pictures of the software user interface, have no clinical meaning.

Only set the values preset in the exam sets provided or the values recom-mended by experienced application specialists.

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2 System Safety

2.1 General safety information

The organization responsible for the safety of the medical device according to IEC 60601 ff. is the user or the operator/operating company of the device.

2.1.1 Trained users

CAUTION

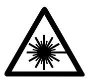
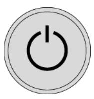
Operation of the system by untrained users

Risk of incorrect diagnosis or treatment due to misinterpretation of image information

- The system must only be used by persons with the necessary specialist knowledge, for example, physicians, trained radiologists, or trained technologists, after an appropriate application training.

2.1.2 Pictograms

The following are pictograms and their meanings as they may apply to your system (IEC standard).

	Alternating current
	Equipotential bonding
	Follow the instructions in the Operator Manual
	Caution: Laser
	Pushing prohibited
	ON
	OFF

	Attention: Hot surface (at the X-ray tube)
	Degree of protection type B

2.1.3 Laws and regulations

The statutory regulations of the respective country must be followed.

If legally binding regulations govern the installation and/or the operation of the product, it is the responsibility of the installer or the operator to observe these regulations.

Modifying the medical product is not permitted.

Incident reporting

Any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority in which the user and/or patient is established.

Country-specific regulations

The legally established country-specific regulations must be followed in all countries. Deviating from this Operator Manual, values may be set according to country-specific regulations.

Regulations in the EU

Due to the Medical Device Directive 93/42/EEC and EU regulation 2017/745 (MDR), LUMINOS Q.namix R complies with the standards of the EN 60 601-1 series.

Additional national regulations

To ensure safety for operators, patients, and third parties, we recommend compliance with the regulations listed here to the extent that local national laws allow.

Germany

Strahlenschutzverordnung (German Radiation Protection Ordinance) - Legally required tests:

Before start-up:

- Acceptance test
- Expert inspection

During operation: According to the time intervals specified

- Constancy test

MedBetreibV: Ordinance for system owners of medical equipment

- Safety check for technical construction at least every 2 years

USA and Canada

Corresponding to the Medical Device Directive in the EU, LUMINOS Q.namix R complies in the U.S.A. and Canada with the applicable standards.

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Federal law stipulates that this system may only be sold to a physician or by order of a physician.

This system is intended for use in radiographic and radioscopic examinations and under the guidance of a trained health care professional.

2.1.4 Explosion protection

LUMINOS Q.namix R is **not** designed for operation in explosion-endangered areas.

WARNING

The system is not designed for operation in explosion-endangered areas. It does not fulfill the requirements for anesthetic-proof (AP/APG) classification.

Danger of explosion!

- Do **not** use the system where explosive atmospheres can arise, for example, where anesthetic gases are being used.

Only products with the AP mark (AP = anesthetics test) may be used in explosion-endangered areas.

2.1.5 Fire safety

In the event of a fire

1. Shutdown the entire system immediately, that is, disconnect the system from the main power supply.
2. Press the emergency SHUTDOWN button or actuate the main or disconnecting switch.
3. Use a CO_2 fire extinguisher.

Do **not** use water!

WARNING

Fire inside or in the vicinity of the system

Risk of injury of patient and personnel and damage to property. Risk of inhalation of fumes from burning plastics.

- Guide patient away from the system.
- Switch off system.
- Know your escape routes.
- Know where the fire extinguishers are and how to use of them.

4. Contact Customer Service before performing any refurbishing work and before starting the system up again.

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2.1.6 Information about the software

Network security

CAUTION

Unauthorized access to the system

Hazards up to and including loss of patient data and non-operational system

- The hospital is responsible for network security at the site.
- Set up firewalls and user account password protections.
- Do not allow users to change configuration files.
- Update virus protection software as required.

Please note that the following are changes to the network that may introduce new risks:

- Network configuration change;
- Connection of additional systems or equipment to the network;
- Disconnection of systems or equipment from the network;
- Update or upgrade of equipment connected to the network.

Please note the following information relating to cybersecurity:

- Appropriate physical access controls shall be implemented;
- Cables shall be protected from interception, interference, and damage;
- Maintenance and inspection of devices shall only be carried out by authorized personnel;
- External personnel who perform maintenance or service work shall be accompanied at all times;
- Devices shall be secured with physical locking cables or securely locked away;
- The system shall lock down automatically or terminate the session after exceeding a reasonable defined idle time limit;
- Applications shall log off the user or terminate the session after exceeding a reasonable defined idle time limit;
- Malicious software shall not be created, downloaded, or distributed on purpose;
- User accounts shall only be assigned to and used by individual persons;
- For each account a different password shall be used;
- IP addresses assigned to systems shall be logged;
- Device log data shall be continuously analyzed;
- Device log data shall be stored for a reasonable time period;
- Devices shall be placed in network zones which fulfill and are appropriate for their information security requirements;

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- The system must be used by persons with necessary specialist knowledge after training supplied by Siemens Healthineers Representative;
- The hospital is responsible for authorized access to the system;
- We recommend installing the system in a separate examination room with a door controller;
- The system can be connected to the internal network. The hospital is responsible for network security at the site;
- The Device Guard built into Windows 10 is used to prevent unauthenticated software from being executed in the system;
- The firewall built into Windows 10 is used to block network connections through illegal ports;
- The security service is used to encrypt/decrypt important files;
- The corespot service is used to enhance the security of the operating system;
- Normal users shall use a unique ID and password to log into the system;
- The system allows access in a special mode without authentication in order to enable examination in the event of an emergency. For this special mode no password is set;
- The system has basic requirements on the length/complexity/expiry time of the password;
- The system predefines several user roles, and users can only access the functions defined in their roles;
- The system has been tested before its launch, making sure that there are no software security vulnerabilities;
- The major 3rd party software used by the system has been registered to the Siemens Healthineers SVM service to monitor security vulnerabilities that may be found in future;
- Siemens Healthineers will analyze vulnerability notifications from SVM in a timely manner and take appropriate measures according to their severity;
- Siemens Healthineers will provide timely patches or mitigations for serious software security vulnerabilities;
- Siemens Healthineers releases a security patch for the system regularly (90 days) to repair general security vulnerabilities.

Copyright

The system and user software used in this product is protected by copyright.

DICOM conformity

The digital imaging system is DICOM-conform. The “DICOM Conformance Statement” is available on request from Siemens Healthineers or via the Siemens Healthineers Internet homepage.

Data security

Personal data are subject to data protection.

- Please observe the relevant legal provisions.

2.1.7 Checks, tests

Before the system is used for examinations, the user must ensure that all safety-relevant devices function correctly and that the system is ready for operation.

Important functional tests and checks must be made at certain time intervals.

(➔ Page 120 *Daily tests*)

2.1.8 Error messages and status indicators

The system displays warnings and error messages as well as information about current system status on the screens.

For further details, refer to the Operator Manual of the imaging system.

2.1.9 Gender inclusivity

At Siemens Healthineers, we want to address all genders equally in our Instructions for Use. It is not our intention to exclude anyone. We continue to update documentation with this in mind.

2.2 Personal safety measures

2.2.1 Warning signs

Special danger points are marked on the unit with a warning sign:

Danger of crushing

These warning signs indicate possible danger of crushing for the patient and examiner.

Cardiopulmonary resuscitation

This warning sign shows the position of the patient table in cardiopulmonary resuscitation (CPR) with pressure compression up to 500N (50 kg).

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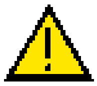
Risk of collision

However, under certain circumstances, additionally installed components, for example lead protective screens, lamps, auxiliary equipment, and so on, can cause collision with the system.



To protect these components and also to protect the patient, such components are provided with the warning sign shown on the left.

We can provide you with these warning signs.

Customer Service will also attach them to additionally installed components on request.

General danger

Caution, always observe the Operator Manual!

This warning notice indicates that special instructions can be found in the Operator Manual. Please read the Operator Manual.

2.2.2 Information about unit movements

Because of the possible speed of movement of the system, it must be operated with particular care.

Unit movements should be started only if neither a patient nor a third party may be endangered and if no object may hinder unit movements.

However, if an object (for example seat) was hindering a movement direction (for example, during a down movement) or the height of the detector unit or the X-ray tube was manually readjusted, the alignment control must be readjusted.

If this alignment is not performed, no acquisitions are possible!

WARNING

System movements cause collision

Injury of persons or system damage

- Perform system movements only when it is certain that neither the operator, the patient, third parties nor pieces of equipment can be endangered by these movements.
- Make sure you are standing outside the collision zone.
- Always watch the patient when performing system movements.

CAUTION

Touching the telescoping column during movement

Risk of crushing

- Do not place your hands on moving or rotating parts.
- Use the provided hand grips to move ceiling-suspended parts.
- Whenever this is not possible, pay special attention to all crush zones on the column especially when it is moving.

WARNING

Movement without intentional activation of control elements.

Risk of collision, risk of injury to patient or operator, risk of damage to the system.

- If movement does not stop, press the nearest emergency STOP button.

WARNING

Devices (such as mobile phones, WiFi components, RFID readers, etc.) emit electromagnetic fields close to the console

Injury of patient or operator due to unintended movement

- Remove the source of interference (for instance, a mobile phone or an electric device) from the area of control console.
- Restart the system.
- Pay attention to error messages during restart.

WARNING

Broken drive chain in wall stand

Risks of injury when moving front unit up or down

- Do **not** activate vertical movements when a broken drive chain is indicated on the user interface.

2.2.3 Abnormal movements

If any part of the system moves although movement was not released, there might be a fault.

Example: The ceiling-suspended X-ray tube moves down by itself.

- Shut down your system and call Customer Service.

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2.2.4 Dead man's grip

All unit movements are controlled with a **dead man's grip** (DMG), that is, movements are performed only while the operating element is being actuated. In the event of danger, the movement can be stopped immediately by releasing the operating element.

2.2.5 Collision protection

The LUMINOS Q.namix R system has a collision control device comprising all motor driven moving components like tube support, detector support, tabletop and wall stand.

Attention: When clinical protocols are used in which the tube was programmed in the collision zone of the table, collisions of the tube and the table are possible.

Never leave the system unattended during system movements.

2.2.6 Red emergency STOP buttons

WARNING

Parts of the body may be injured during motor-driven system movements

Injury of patient or user

- Ensure that the patient is positioned correctly.
- Pay attention to the potential collision and danger points.
- In an emergency, immediately press one of the red emergency-stop buttons located on the system and at the console.

Triggering STOP

If a malfunction of a unit movement causes an emergency situation, danger to the patient, to operating personnel, or to the unit, do the following:

- Press one of the red emergency STOP buttons immediately.

  All system drives are shut down and movements are stopped immediately. Movement can only be continued when STOP is canceled.

  Radiation, fluoroscopy and acquisition are interrupted. Fluoroscopy can be released again if you release the pedal and press it again, even without canceling STOP.

  Injections are interrupted, but can be continued without canceling STOP.

Canceling STOP

Only when the cause of the danger has been unequivocally identified and remedied, the emergency STOP button should be disengaged.

- To do so, turn clockwise and pull the red emergency STOP button.

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In case of a system failure, press the emergency STOP button and unlock it again.

The system is reinitialized.

Where are the emergency STOP buttons?

You find the emergency STOP buttons in the following locations:

On the unit

On the remote control console

On the wall stand (The emergency STOP buttons are located at both sides of the wall stand.)

2.2.7 Emergency SHUTDOWN button (installed on-site)

Use this shutdown method **only in an extreme emergency** because it is an **uncontrolled** process!

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Consequences:

- All unit movements are interrupted
- Radiation is turned off
- The current system program is interrupted
- Current operating sequences are interrupted and cleared
- All current acquisition data are cleared, unless they have been saved to non-volatile memory.

Data could be lost, for example unsaved images, exporting and filming jobs, and so on. The tube cooling system is also disconnected from the power supply, and the tube can overheat.

If a flat detector is used, it is also disconnected from the power supply. After the power has been switched on again, an additional waiting time is required to ensure optimum image quality.

An emergency power supply, if one is connected, will not be activated when emergency SHUTDOWN is pressed.

The UPS of the imaging system is also not activated.

Shutdown in an emergency

Only exclusively, if there is danger for patients, operators, third parties or the unit:

CAUTION

Shutdown using the emergency SHUTDOWN button (if installed)

Risk of data loss or system damage

◆ Use the emergency SHUTDOWN button only in case of emergency or when the system cannot be switched off normally.

◆ Press the on-site emergency SHUTDOWN button.

The entire system is disconnected from the power supply.

Switching on again

Only if the cause of the danger has been **unequivocally** identified and remedied, the emergency SHUTDOWN button can be released and the system operated again.

1. In all other cases, for example system malfunction, contact Customer Service.
2. If the stop function does not respond normally, immediately activate an emergency shutdown to turn off the entire system.

   If this happens, you may not continue to use the system. Contact the Customer Service.

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2.2.8 Power failure

If the power network at the site is very unstable, it may be advisable to install an uninterruptible power supply to prevent data loss.

Contact Siemens Healthineers for more information.

With UPS for imaging system (Uninterruptible Power Supply)

1. In case of a short power failure, please try to switch on the system after some seconds.
2. If it was possible to switch on the system, please wait until the system is ready.
3. Then continue working or rescue the patient.
4. If it was not possible to switch on the system or if the patient has collapsed, remove the patient immediately.

For more details on the UPS, refer to (➔ Page 113 *UPS (for the imaging system)*).

2.2.9 Implanted devices

WARNING

Examination or treatment of patients with implanted devices such as pacemakers or neuro-stimulators

External influences (e.g. during pulsed fluoroscopy) can cause malfunctions in the implanted device.

◆ Find out whether there are known external influences upon the implanted device by the X-ray system (e.g. by checking the technical data or the operator manual of the implanted device).

2.2.10 Maximum load

CAUTION

Too much weight on system components

Risk of damage or patient injury

◆ Observe the maximum allowable table load including the weight of the accessories.

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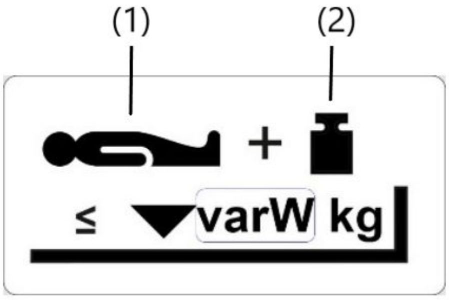

Max. load on table: Patient + additional load (for example, accessory)

(1) Patient
(2) additional load

The actual patient weight with which you can load your tabletop is stated on the real label on the tabletop and in the technical data.

It is important that the load is distributed uniformly over the tabletop. Otherwise there is a risk of material deformation and system malfunction. The patient weight includes any parts permanently or loosely connected with the patient, such as equipment, prostheses, implants, plaster casts.

Example of incorrect use with uneven weight distribution:

A patient with a maximum weight sitting at the end of a fully extended tabletop.

Maximum load on accessory rails

In addition, 30 kg can be attached to any accessory rail or to all accessory rails in total. This value includes **all** accessories, for example injector and radiation protection devices!

In the interest of safety of operation for the patient and unit, the permitted patient weight and accessory weight must not be exceeded.

Maximum load on each component or accessory part

Example

The component, for example tabletop or accessory, must not be loaded with more weight than indicated.

Approved loading of the tabletop

All technical specifications represent typical values, unless specific tolerances are provided.

Patient	Performance figures and restrictions
up to 180 kg	• Full speed

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Patient	Performance figures and restrictions
180 - 210 kg	• All movements possible • All speeds reduced up to 20% Restriction of movement range: • Table longitudinal ± 40 cm • Table transverse: between center and ± 7.5 cm (operator side) Restriction for patient position: • Patient prone or supine position, centered on the table
210 - 260 kg	• Table-tilt -15°/+90°; reduced speed • Lift range not limited; reduced speed • Tabletop: Centered in both directions; no movement allowed Restriction for patient position: • Patient prone or supine position, centered on the table
260 - 330 kg	• Table tilt -15°/+45°; reduced speed; • Tabletop: Centered in both directions; no movement allowed • Table height adjustments are possible with reduced speed. Restriction for patient position: • Patient prone or supine position, centered on the table
330 kg	Maximum table load (patient including accessories)

Approved loading of the foot rest

This warning sign indicates that the foot rest can be loaded with a maximum weight of 230 kg.

All technical specifications represent typical values, unless specific tolerances are provided.

Patient	Performance figures and restrictions
Up to 150 kg	• Table tilt: -90° to +90° • Patient has to be secured with shoulder supports and foot holder
≤ 182 kg	• Table tilt: -45° to +90° • Patient has to be secured with shoulder supports and foot holder
≤ 230 kg	• Table tilt: -15° to +90° • Patient has to be secured with shoulder supports or foot holder

2.2.11 Safety switch

(1) Safety switch between tabletop and stand

On activation of the safety switch (shutdown device), all system movements stop and are blocked.

Movements are possible again only after the obstruction causing the problem is removed (for example, arm/body part).

If it is not possible to remove the obstruction causing the problem, call the Siemens Healthineers Uptime Services.

2.2.12 Equipment damage

WARNING

Collision causes damage to system

Injury of persons, radiation not as desired

◆ Report severe collisions to Siemens Healthineers service so that the system can be inspected.

Avoiding equipment damage

1 Before starting system movements, especially lowering and tilting the table, make sure that the movement range is free of obstructions.

2 Move monitor support systems, operating consoles, gurneys, beds, and instrument tables out of the tilting range of the table.

3 Remove chairs, steps, stands, waste containers, and similar objects from the movement area. (No collision monitoring.)

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4 Do **not** place any objects or consumable material on the cover of the table support, on the detector cover and on the longitudinal guides of the stand carriage.

Considerable forces which can damage these objects in the area of movement of the systems occur during movements of the tabletop.

5 Do **not** place any loose objects anywhere on the table.

These objects could fall down when the table is tilted, causing injury or damage.

6 Do **not** stand at any place on the covers of the table support.

The covers can be deformed.

Components located underneath them can be damaged and thus lead to operating disturbances.

7 Do **not** place any objects on the operating areas of the control consoles and the tableside control.

Do **not** place folders next to the operating area of the control consoles and next to the tableside control.

Folders falling down might cause unwanted system movements.

---

8 In vertical table positions do **not** use the following system parts as a seat or support:

- Stand column
- Tube assembly support arm
- Tube assembly cover
- Primary collimator
- Compression device.

This unallowed loading can lead to material breakage and damage to bearings.

9 Do **not** place contrast medium cups or open containers with liquid or pasty contents on the unit, on the remote console or on the control cabinets.

Contrast medium can spill, leak, or overflow into system parts and lead to operational disturbances of the unit or to misinterpretation of exposures.

10 When storing contrast medium in the cup holder on the compression carriage, use only cups with a maximum volume of 0.25 liters. Use only cups made of unbreakable material, that is under no circumstances glass or porcelain.

11 Remove contrast medium traces immediately!

2.2.13 Danger of crushing

The patient and operating personnel must use only the handles which are intended for the proper handling of the equipment or positioning of the patient. Where it is not possible, you must pay attention to possible risks of injury by crushing in the vicinity of moving parts.

1 Pay special attention to the risks of crushing fingers and hands between moving parts and their guide openings.

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2 Before performing unit movements, make certain that the patients do not grasp the frame of the tabletop.

2.2.14 Danger zones / danger points

The positions marked in the following illustrations show danger zones in which patients or operating personnel could suffer injury by crushing or colliding:

- Due to the mobility of the stands, there is a danger of injury in certain areas during unit movements.
- Make sure that there are neither persons nor objects in the swivel or rotation range of the stands.
- Make sure that there are neither persons nor objects in the lifting and rotation range of the patient table.
- Make sure that the knees of sitting staff members are not in an area under a stand or the patient table.
- Avoid standing or sitting immediately adjacent to the system.
- If it is necessary to position the patient with legs or knees under the cross-beam at the head or foot end of the table, avoid system movements.
- Make sure that, during system movements, no one is in the area between unit base and table.
- Be careful when working with the attached foot rest.
  - There is a risk of collision with the extended cone when the tabletop and/or the longitudinal carriage are moving.
- Be aware of the danger zone between patient table and detector tray.

WARNING

Unintended overtravel when moving or lowering the table

Crushing of patient or operator seated next to the table (for example, a wheelchair patient), injury of patient lying on the table

◆ Avoid movements when somebody is within the dangerous movement zone.

Table movements do not stop immediately when the control elements are released.

The table will move further than expected.

▶ Danger points

Risk of injury by crushing or collision

Unit in horizontal position

If the patient is in the danger zone, always make sure that the operating personnel is in the room and within reach of an emergency stop.

If the operating personnel leaves the room or moves out of reach of an emergency stop, the patient must be moved out of the danger zone.

User is working directly at device during lift or tilting movement. At the point where the distance to the floor is lower than 12 cm, the user can be injured, for example, on the foot.

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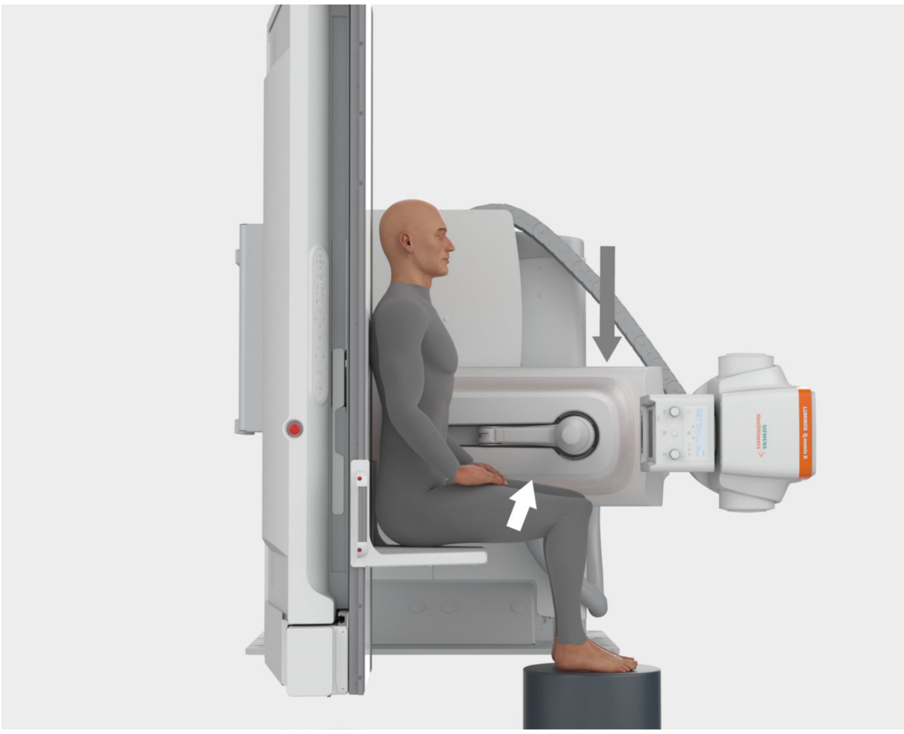
Additional danger zone



Examinations with patient sitting on the foot rest

⚠ WARNING

Downward movement of tube assembly during examinations with patient sitting on the foot rest.

Danger of crushing the patient (knee, hand) with the compression unit

- Pay special attention to the danger zone (1) between the compression unit and the sitting patient.

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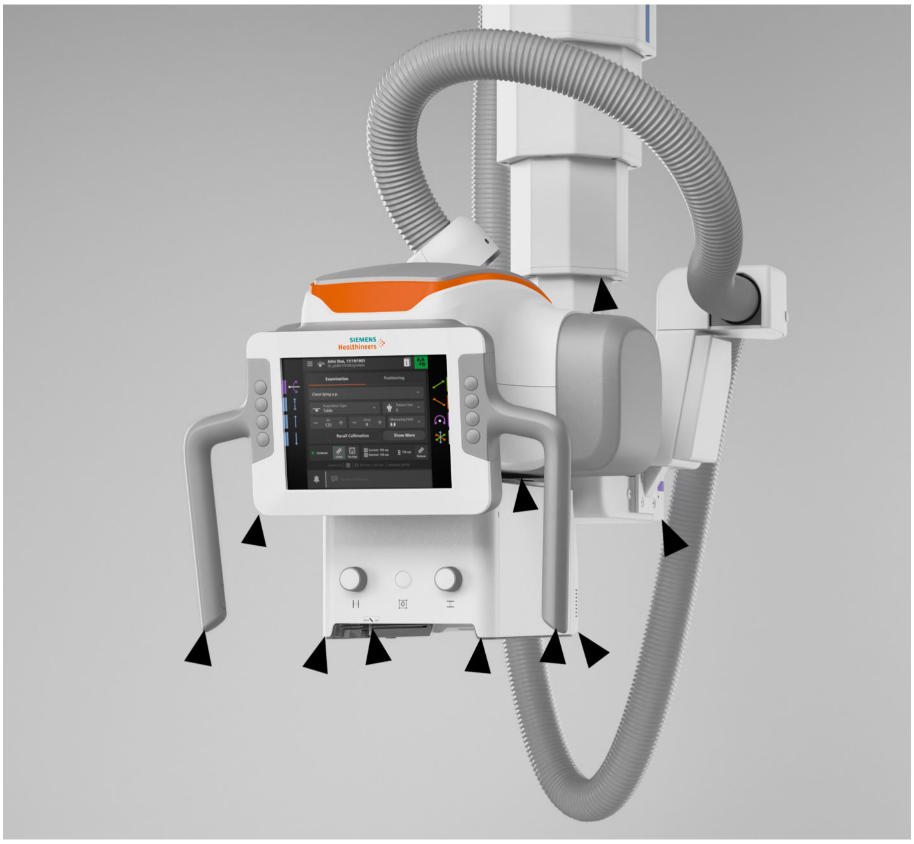
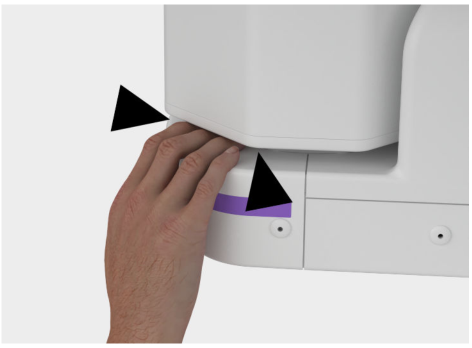
At the ceiling-mounted tube support



Ensure that there are no objects within the hazardous zone of the ceiling-suspended X-ray tube.

Otherwise, collision hazards between the ceiling-suspended X-ray tube and the system may lead to property damage.

Take special care not to crush your fingers between the 3D column and the tube assembly:



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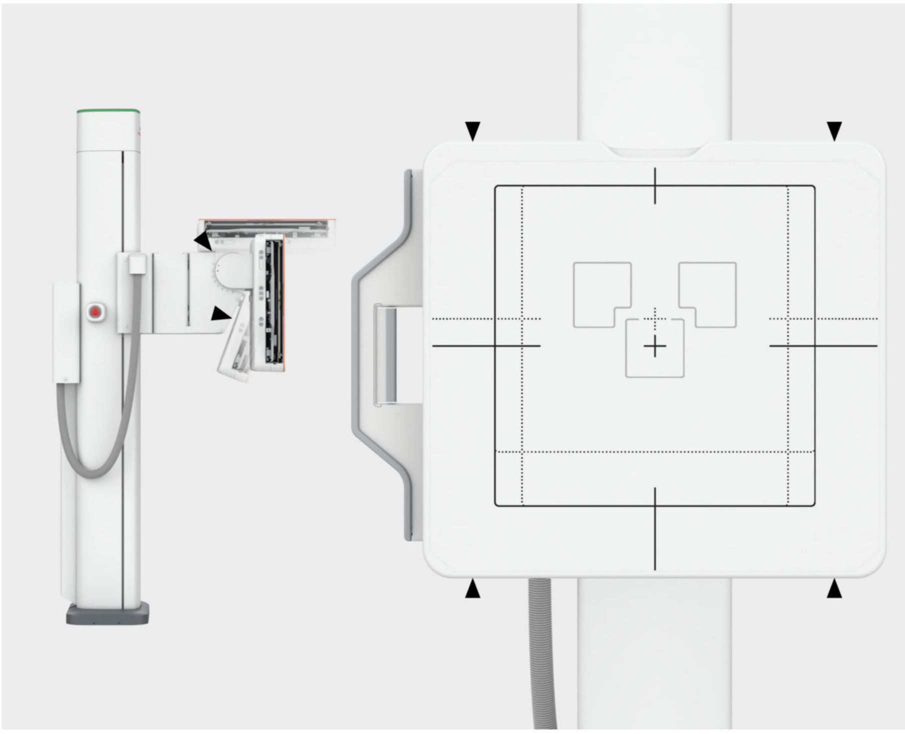
CAUTION

Touching the telescoping column during movement

Risk of crushing

- Do not place your hands on moving or rotating parts.
- Use the provided hand grips to move ceiling-suspended parts.
- Whenever this is not possible, pay special attention to all crush zones on the column especially when it is moving.

At the wall stand



Pay attention to squeezing between floor and front unit.

Downward movement stops at 12-cm distance to floor.

Movement can be reactivated with slower speed.

CAUTION

Lowering the detector of the wall stand onto a seated patient

Crushing of patient seated with legs under the detector (for example, a wheelchair patient)

- Do not move the detector while a patient is seated near the wall stand.
- Move the patient away from the wall stand first.

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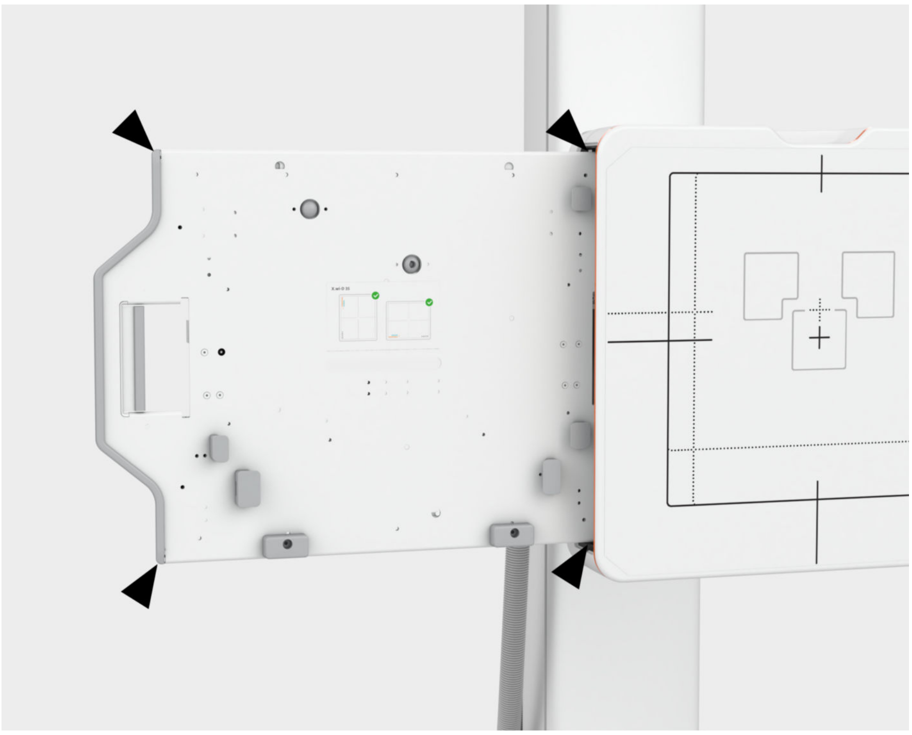
At the wallstand with pulled out detector/cassette tray



2.2.15 Grip locations

When handling the system correctly and when positioning the patient, operators, and patients must use only the grip locations provided for this purpose.

The following grip locations are provided:

- One hand grip rail (front or back)
- Two hand grips (front and back)

Grip attachment

◆ Make sure that the hand grip rail and the hand grips are always attached.

If these grip locations cannot be used:

◆ Pay special attention to the mentioned possibilities of crushing between moving parts and their guide openings.

◆ Make sure during the examination that the patient does not hold on to the edges of the patient table under any circumstances.

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2.2.16 Safety during patient examination

Patient positioning

- All safety-related equipment must be installed and functioning. In particular the hand grip strip, protection strip, hand grip, foot rest, foot holder, compression belt, and shoulder supports.
- The hands, arms, legs, head, and hair of the patient must not extend unsecured beyond the edge of the tabletop.
- Observe the patient while moving the tabletop and during system movements and take care that any catheter is correctly located.
- In examinations with the table tilted up vertically, the foot rest serves for an adjustable step or seat.
- Make sure that the foot rest is locked together with the tabletop on both sides.
- Check the firm location of the foot rest.

WARNING

Hand grips loosen or are not used

Injury of patient due to falling

- ◆ Check that the patient hand grips on the table are in the right position.
- ◆ Ensure that hand grips are tightened prior to use.

CAUTION

Hair, clothes and tubes or cables caught in table.

Risk of patient injury

- ◆ Make sure that hair, clothes and tubes or cables cannot be caught in the table before initiating system movements.
- ◆ Press emergency stop immediately if something gets caught.

Patient positioning with unit in vertical position

During examinations with the unit in the vertical position, there is a risk of crushing to the patient if the X-ray system (stand with tube unit and receptor unit with flat detector) is moved longitudinally.

1. Position the X-ray system in the approximate acquisition position.
2. Move the patient to the acquisition position.
3. Set the X-ray system to object height. Always watch the patient during this movement.

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Motor-driven remote compression

- Take special care for the applied compression forces, especially in the case of sick, elderly, and frail patients (for example, infants).
- There is an increased risk of crushing and injury for the patient.
- Furthermore, there are considerable mechanical shearing forces with risk of damage between the compression cone and attached accessories by collision during the motor-driven tabletop movement.

Patient rescue

In case of, for example power failure, motorized system movements are not available. However, you can move parts of the system manually.

CAUTION

Patient is caught between compression cone and patient table during a power failure

Risk that the patient cannot be removed from the patient table.

◆ To rescue or release the patient from the system, proceed as described in the following (description of patient rescue).

During the following steps, make sure that the patient is not crushed!

Personnel

◆ Use enough personnel for the rescue.

Accessories

◆ Remove any accessories obstructing the rescue.

If the patient is compressed with the cone

1. Press down the hinge and hold the cone at the same time.

   The cone will spring back from the patient.

2. Take care that the hinge does not press into the side of the patient after it releases.

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If shoulder supports or foot holder are loosened in Trendelenburg position, then take care that the operating personnel secure the patient sufficiently.

Get the aid of sufficient personnel for the rescue of the patient.

Immobilization

WARNING

Table tilted in Trendelenburg (head-down) position

Patient slides off table head-first

- Use shoulder supports to prevent the patient from sliding off the table.
- If the table must be tilted more than 15° Trendelenburg (head-down), use the foot holder in addition to the shoulder supports.

- Loosen any fixtures.

Mattress

In difficult cases, you may have to rescue the patient along with the mattress.

1 Lift the mattress at the foot end to unfasten the hook and loop fastener or velcro fastener.

2 Put some paper or cloth between tabletop and mattress so that the velcro does not fasten again.

3 Unfasten safety straps, if applicable.

4 Pull the mattress with the patient from the tabletop at head end or foot end of the table (depending on room situation and accessibility).

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Correct image orientation - patient orientation

It is the responsibility of the operator to ensure correct image orientation on the monitor or film.

CAUTION

Image is incorrectly flipped due to incorrect patient position information or right/left labeling.

Risk of incorrect diagnosis or treatment because of left/right or up/down confusion

- The examiner is responsible for using the image orientation functions and interpreting the images correctly.
- Make sure that the **R**/**L** labels have been placed correctly.
- Compare the patient position data with the anatomic view when diagnosing images to exclude any errors.
- If necessary, use lead letters or similar devices during acquisition.

Visual contact to patient

The operator of the X-ray system must ensure that there is visual and acoustic contact with the patient. Thus, the operator remains informed about the condition of the patient at all times.

Room lighting

According to DIN 6868- 157 (valid in Germany), the lighting in rooms in which diagnoses are made on image display devices (monitors) must fulfill the following requirements:

- The lighting must be adjustable and glare-free.
- The setting of the illuminance must be reproducible (for example, dimmer with scale).
- No mirroring or reflections of windows, lights, view boxes, and so on, must occur in the operating position of the monitors.

Prerequisites for diagnosis and treatment planning

The imaging system software of LUMINOS Q.namix has been designed and tested for use in diagnosis and treatment planning based on digital radiographic X-ray images and series. To ensure that the imaging system produces images suitable for these purposes, the monitor must meet certain criteria for image quality.

Checks

Image quality can deteriorate over time because of aging and normal wear or damage of the monitor and other components.

- The image quality must be checked at regular intervals (**once per month**) after installation.
- Thus, you make sure that the system is still suitable for use in diagnosis and treatment planning.

For further information, refer to (➔ Page 59 *Maintenance plan*).

The operator must ensure that qualified personnel are chosen and that the criteria for image quality specified in the installation and maintenance instructions are fulfilled.

Test images

Test images for calibration of the monitor and to test the quality output of the laser camera, are stored in the system.

For further information, refer to the Operator Manual of the imaging system.

Displays

Please observe the following information:

- The operating indicator must light up.
- Please keep the ventilation slots of the monitors unobstructed at all times.

Do **not** touch the surface of the screen using sharp, pointed objects.

Do **not** open the display under any circumstances.

Monitors are suitable for medical online diagnosis only, if special measures to assure image quality are adopted (especially determining the brightness and contrast values).

In case of doubt, film images on a hardcopy camera, if you want to make a diagnosis.

Siemens Healthineers undertakes no liability for diagnoses, which are performed on monitors of other vendors.

Use of hardcopy cameras

Only hardcopy cameras approved by Siemens Healthineers may be used.

Approvals by Siemens Healthineers refer to DICOM image type XA.

Siemens Healthineers does not accept any liability for diagnoses made on the basis of images from non-approved hardcopy cameras.

2.2.17 Radiation protection

The X-ray equipment for Radiography LUMINOS Q.namix R with radiation protection complies with IEC 60601-1-3 :2008, IEC 60601-2-54:2009 and IEC 60601-2-54:2022.

Mode of operation: Continuous

Important information

The automatic format collimation system for acquisition and fluoroscopy modes, pulsed fluoroscopy, CAREPROFILE and CAREPOSITION as well as the automatic dose rate control including CAREFILTER help reduce the radiation dose to the patient and examiner considerably.

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(➔ Page 50 *Reducing radiation with C.A.R.E.*)

For radiation dose indication during normal use of the equipment, the system is equipped with a measurement system based on an ionisation chamber calibrated in air kerma area product.

Measures for reducing radiation exposure

⚠ CAUTION

Leakage, extrafocal, stray, scatter and residual radiation

Undesired radiation exposure

- ◆ When releasing X-ray, stand as far away as safely possible from the radiation beam path.
- ◆ Use radiation protection as necessary.

Please pay attention to the fact that the air kerma is increased by the following settings:

- Higher kV
- Higher mAs
- Higher ms
- Smaller SID
- Change of fluoroscopy protocol **Basic** → **Standard** → **High**
- Higher frame rate
- Change of zoom stage **F15** → **F43**

Please observe the following:

Radiation may be released only by authorized operators or persons with proper authorization to apply ionized radiation.

Fluoroscopy

1. Release fluoroscopy for as short a time as possible, use the LIH function!
2. Use the dose-saving fluoroscopy with CAREVISION!
3. Collimate radiation-free in the LIH image.

   You can also set the filter diaphragms in the LIH image without radiation: CAREPROFILE
4. Position the patient radiation-free in the LIH image: CAREPOSITION
5. If possible, ensure the best possible protection of the patient during acquisitions in the vicinity of the reproductive organs (use gonadal shields, lead-lined rubber covers).
6. Keep the radiation field as small as possible without reducing the active measuring field.
7. Remove all radiopaque parts from the radiographic field, if possible.
8. Set the X-ray tube voltage as high as possible (not forgetting the image quality).
9. Set the X-ray tube to skin distance as large as is reasonable for each examination.

A larger focal spot to skin distance leads to a lower air kerma.

The maximum air kerma of 2 Gy must not be exceeded.

Recommendation: Change the projection and the air kerma entrance area.

Radiation protection for the patient

1. If possible, ensure the best possible protection of the patient during acquisitions in the vicinity of the reproductive organs (use gonadal shields, lead-lined rubber covers).
2. Keep the radiation field as small as possible without reducing the active measuring field.
3. Remove all radiopaque parts from the radiographic field, if possible.
4. Set the X-ray tube voltage as high as possible (not forgetting the image quality).
5. Set the X-ray tube to skin distance as large as is reasonable for each examination.

   A larger focal spot distance leads to a lower air kerma.

Radiation protection for the examiner

CAUTION

Failure to observe radiation protection regulations

Unnecessary radiation exposure to personnel

This system produces X-rays.

- Wear protective clothing such as lead aprons.
- Monitor radiation exposure using personal dosimeters.

1. If possible, release the acquisition series in the control room.
2. Stay in the examination area as short as possible.
3. If the exposure is to be released in the examination room, use an additional radiation shield (protection for upper and lower body) or radiation-proof window.

   These measures contribute greatly to your personal radiation protection.
4. If the protective shields are not used, wear radiation protection clothing with a 0.25 mm lead layer or similar clothing.
5. Keep the maximum distance from the source of radiation.
6. Check your personal dose by wearing a radiation monitoring badge or a pen dosimeter.

Switching off in an emergency

If a problem, for example a defect, occurs during the examination and radiation can no longer be interrupted by letting go of the radiation release button:

- Press the nearest emergency STOP button.

Avoiding unwanted radiation

- Before starting system movements, make sure that the foot switch for fluoroscopy and radiography in the examination room is not in the travel range of the receptor system.

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CAUTION

Tube is not directed toward the detector during acquisition.

Unwanted radiation exposure

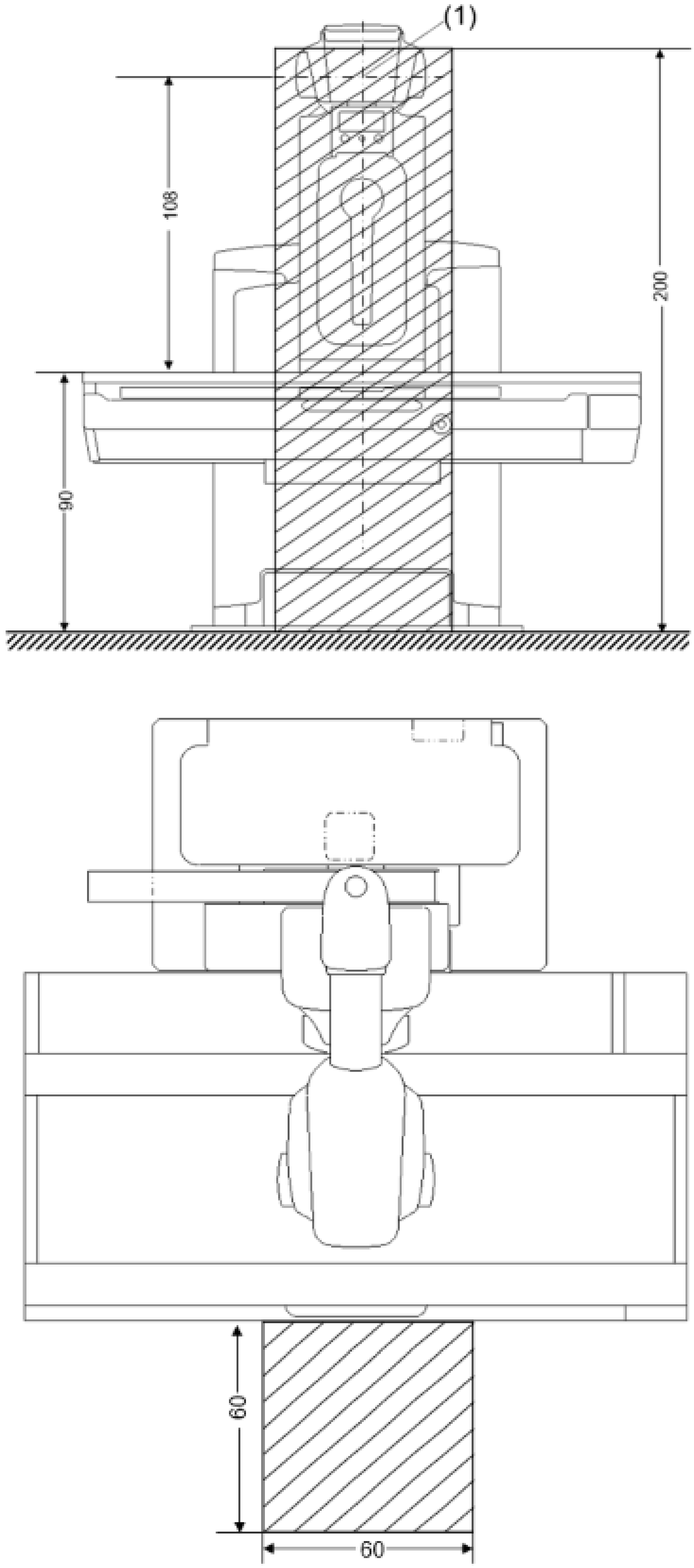
- Use the free exposure mode with care.
- Check that the tube is active and use the light localizer for positioning before releasing X-ray.

Radiation protection zones

Position and dimensions of the main operating area

Print No. XPD3-530G.620.01.04.02

Horizontal patient table



(1) Focus point

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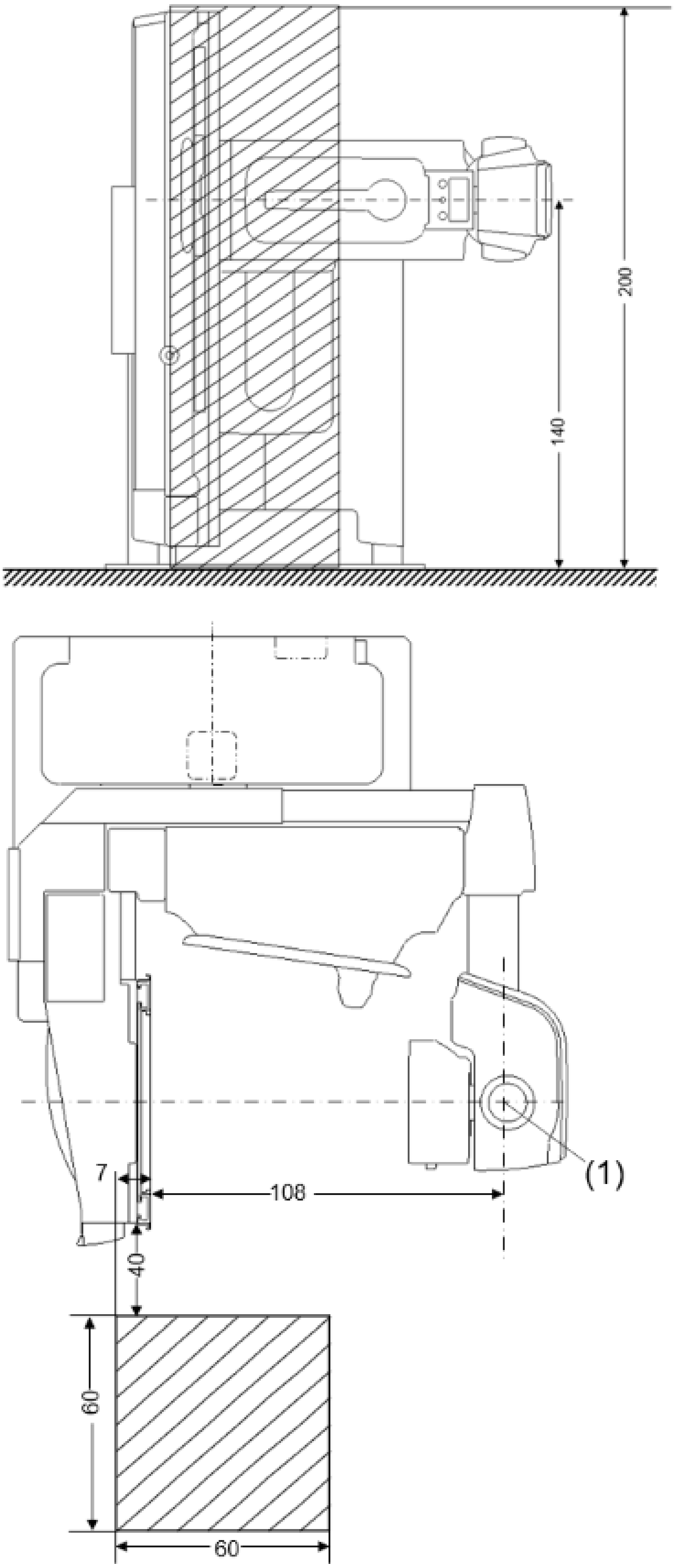
Main operating area

Dimensions in cm

Print No. XPD3-530G.620.01.04.02

Vertical patient table



Print No. XPD3-530G.620.01.04.02

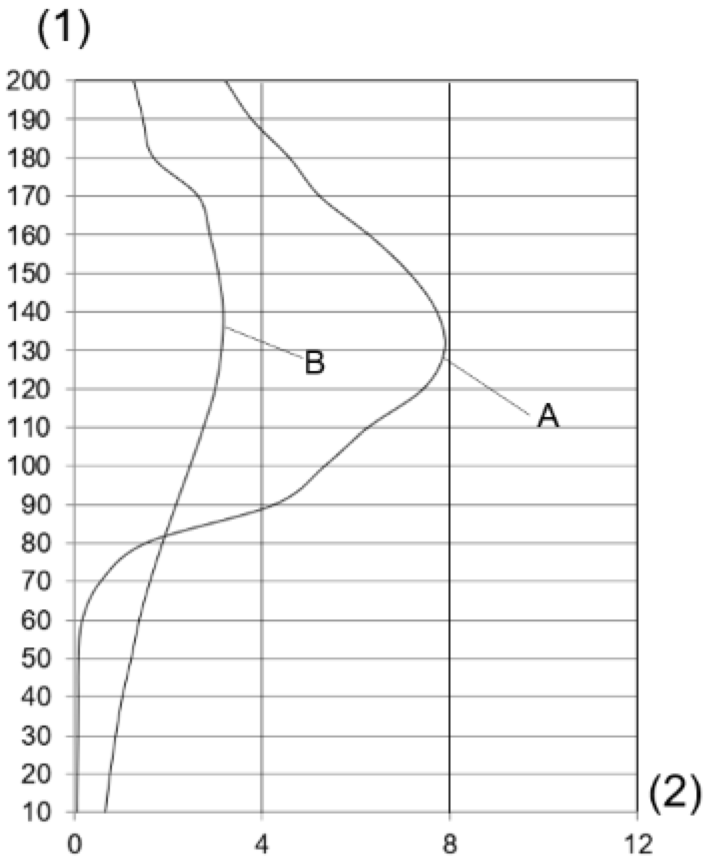
(1) Focus point

Main operating area

Dimensions in cm

Stray radiation in the main operating area

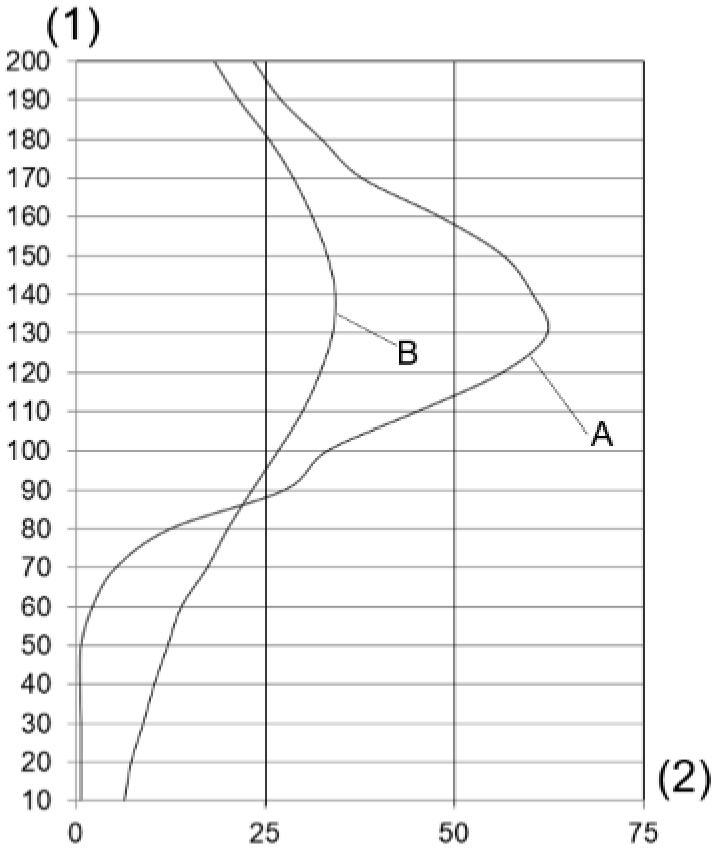
According to IEC 60601-1-3:2021 and IEC 60601-2-54:2022



Characteristic A and B: horizontal/vertical 150 kV, 2.33 mA, Control Fluoro (Service mode)

(1) cm above floor
(2) µGy/h (Air Kerma Rate)

Print No. XPD3-530G.620.01.04.02



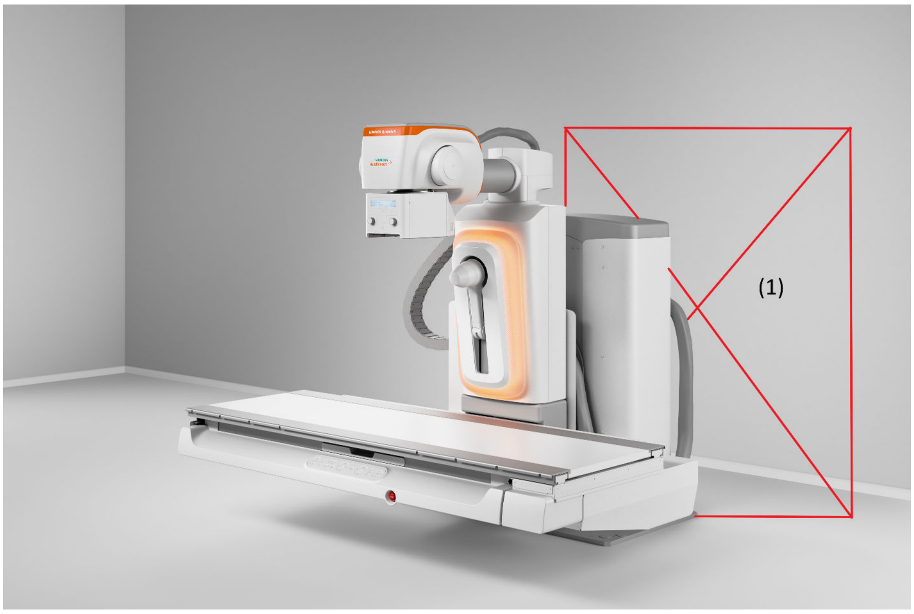
Characteristic A and B: horizontal/vertical Fluoro-Curve: kV_F_P70_16fps_A_P70_Se_16fps, 63,5 kV, 4 mA, 0,2 Cu, 16 p/s, standard dose, patient size M

(1) cm above floor
(2) µGy/h (Air Kerma Rate)

Print No. XPD3-530G.620.01.04.02

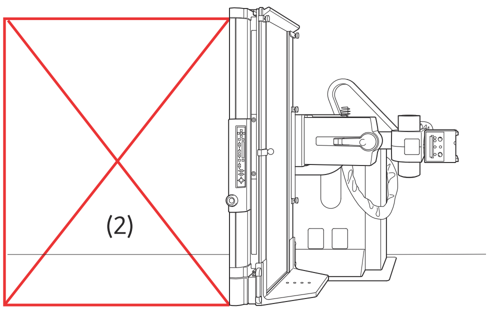
Main operating areas

The following locations are permitted for the operator and the patient:



Patient table in 0° position

- The operator may stand anywhere in the room.
- The patient must not stand behind the unit base (1).



Patient table in 90° position

- The operator may stand anywhere in the room.
- The patient must not stand behind the patient table (2).

2.2.18 Reducing radiation with C.A.R.E.

The C.A.R.E. package (Combined Applications to Reduce Exposure) comprises:

- CAREMATIC
- CAREFILTER
- CAREVISION
- CAREPROFILE

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- CAREPOSITION
- CAREMAX
- CAREWATCH

CAREMATIC

Automatic X-ray control system for fully automatic calculation and optimization of the exposure data based on fluoroscopic values:

CAREFILTER

The CAREFILTER function comprises various copper filters. They filter out the low-energy components of the X-ray spectrum that are not necessary to make the image. This causes hardening of the beam, reducing not only the air kerma for the patient but also the scattered radiation for the examiner.

Automatic dose rate control calculates the water equivalent of the patient from the current kV/mA values and the pulse width. The additional copper filter is automatically moved in or out of the beam path as a function of this value during acquisition if the image quality becomes unacceptable because of a very high patient density.

The CAREFILTER function is automated and cannot be operated manually.

CAREVISION

With CAREVISION, you have at your disposal a selection of fluoroscopy modes with different pulse rates that you can use to reduce the patient dose considerably.

CAREPROFILE

With CAREPROFILE, the positions of the multileaf collimator and the filter diaphragms are displayed graphically in the last fluoroscopic or acquisition image. In this way, you can change the collimation without additional radiation release.

CAREPOSITION

CAREPOSITION allows you to reposition the patient with the aid of the last fluoroscopic image (LIH) without additional fluoroscopy.

CAREMAX

- Electronic unit with KermaX, a measurement chamber integrated into the collimator housing for measuring the dose area product and/or standardized patient entrance dose
- Displayed on the generator display and image system display (CAREWATCH)
- Different displays can be configured for fluoroscopy and for fluoro pause:
  - During fluoroscopy: Air kerma level
  - During fluoro pause: Accumulated air kerma, or dose area product or percentage of a configurable dose limit value (amount of fluoro and radiography)

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The dose measuring chamber must be calibrated at regular intervals. This calibration is done under the terms of a service contract. If you do not have a service contract, Customer Service or the manufacturer can calibrate the dose measuring chamber.

If the dose-area product exceeds a certain limit value, radiation injury (initially reddening of skin) can occur on the patient.

This information allows the examiner to avoid radiation injury, for example by changing the angulation.

Display data

During digital fluoroscopy and when a fluoro LIH image or fluoro loop is displayed, the following system parameters are displayed in the control area of the live monitor and on the Touch User Interface:

- Cu filter for fluoroscopy
- Cu filter for exposure
- Dose-area product in μGycm² (only on live monitor)

  (With dose-area product meter only)

  The unit for display of Dose-area product is configurable by Customer Service.
- Dose or dose rate during fluoroscopy (only on live monitor)
- Error text
- Door open display
- Emergency power supply
- X-ray disable display

Reset of the dose-area product

The dose-area product is reset to 0 when a new patient is selected

CAREWATCH

The dose-area product and the patient dose (air kerma) are measured and displayed on the system console and recorded in the examination data.

The measuring device must be calibrated at regular intervals. This is done within the scope of a maintenance contract. Should no maintenance contract have been concluded, the measuring device can be calibrated by the Customer Service or the manufacturer.

- During fluoroscopy or acquisition

  The **current** values of the patient dose rate

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- Without radiation

  The **accumulated** values of the patient dose or the relative patient dose in percent (%) related to a configurable limiting value (usually 2 Gy)

  (configurable. Contact your Siemens Healthineers representative, if you prefer a different display.)

  As well as:

- Permanently

  The accumulated **dose-area product**

If the dose-area product exceeds a certain limit value, radiation injury (initially reddening of skin) can occur on the patient.

This information allows the examiner to avoid radiation injury, for example by changing the angulation.

During fluoroscopy and when a LIH image or fluoro loop is displayed, the following system parameters are displayed on the touchscreens.

2.2.19 Information about dose measurement

To provide you with perfect image quality for your daily work without additional operating effort, we install the most frequently used clinical protocol steps (preferred clinical protocol steps) in the factory.

Measurement configuration

- PMMA phantom (25 cm x 25 cm x 20 cm) centered in X-ray beam placed on tabletop
- Measurement at patient entrance reference point
- Grid r13 N92 F115 is used
- Focus skin distance is 115cm

The measurements were performed according to IEC 60601-2-43:2022.

	Basic	**Standard**	**High**
Dose level	21	40	55
Max. air kerma	45 mGy/min	65 mGy/min	80 mGy/min
Pulse rate	6 p/s	8 p/s	
kV plateau	102 kV	96 kV	85 kV
Max. pulse width	10 ms		15 ms
Dose rate adaptation	0%	25%	
Cu filter	0.2 mm Cu		

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Mode	Resulting exposure parameters Air kerma rate		
Zoom: F43 Field Size: 42.6 cm x 42.6 cm	0.8 mGy/min	1.6 mGy/min	2.5 mGy/min
Zoom: F30 Field Size: 30.3 cm x 30.3 cm	1.1 mGy/min	2.3 mGy/min	3.7 mGy/min
Zoom: F23 Field Size: 24.7 cm x 24.7 cm	1.4 mGy/min	3.2 mGy/min	5.4 mGy/min
Zoom: F15 Field Size: 15.2 cm x 15.2 cm	2.2 mGy/min	6.3 mGy/min	10.5 mGy/min

Table 1: Resulting parameters & air kerma rate with different zoom & dose levels

Pulse rate	0.5 p/s	1 p/s	2 p/s	3 p/s	4 p/s	6 p/s	8 p/s	12 p/s	16 p/s	24 p/s	32 p/s
Mode	**Resulting air kerma rate**										
Dose level: 40 Dose rate Adaptation: 0% Max Air Kerma: 65 mGy/min kV Plateau: 96 kV Max Pulse width: 10 ms Field Size: 42.6 cm x 42.6 cm Cu Filter: 0.2 mm Cu	Tube voltage: 95 kV Tube Current: 15 mA Pulse width: 2 ms AKR: 0.1 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.2 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.4 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.6 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.8 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 1.2 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 1.6 mGy/min	Tube voltage: 95 kV Tube Current: 14 mA Pulse width: 2 ms AKR: 2.5 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 3.2 mGy/min	Tube voltage: 95 kV Tube Current: 14 mA Pulse width: 2 ms AKR: 4.9 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2.1 ms AKR: 6.6 mGy/min

Table 2: Resulting air kerma rate with varying pulse rates when dose rate adaptation is 0 %.

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Deterministic effects

At the Air Kerma rate of 13,86 mGy/min at the reference point, a reference dose of 2 Gy is reached after 8658 s fluoroscopy.

Deterministic effects are probable at this dose (2 Gy).

Clinical protocol steps

To optimize operation, protocols (clinical protocol steps) for the most common standard examinations are delivered with the system. In general, these protocols can be taken as examples to derive individual protocols for more specific examinations.

For editing protocols, refer to the **Clinical Protocol Editor** in the Operator Manual of the imaging system.

- Normal dose CP step: **Standard**
- Low dose CP step: **Basic**
- IQ optimized CP step: **High**

Fluoro protocol steps

	Basic	Standard	High
Dose level	21	40	55
Max. air kerma	45 mGy/min	65 mGy/min	80 mGy/min
Pulse rate	6 p/s	8 p/s	
kV plateau	102 kV	96 kV	85 kV
Max. pulse width	10 ms		15 ms
Dose rate adaptation	0%	25%	
Cu filter	0.2 mm Cu		
Mode	**Resulting exposure parameters** **Air kerma rate**		
Zoom: F43 Field Size: 42.6 cm x 42.6 cm	0.8 mGy/min	1.6 mGy/min	2.5 mGy/min
Zoom: F30 Field Size: 30.3 cm x 30.3 cm	1.1 mGy/min	2.3 mGy/min	3.7 mGy/min
Zoom: F23 Field Size: 24.7 cm x 24.7 cm	1.4 mGy/min	3.2 mGy/min	5.4 mGy/min
Zoom: F15 Field Size: 15.2 cm x 15.2 cm	2.2 mGy/min	6.3 mGy/min	10.5 mGy/min

Table 1: Resulting parameters & air kerma rate with different zoom & dose levels

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Mode	**Resulting air kerma rate**										
Dose level: 40 Dose rate Adaptation: 0% Max Air Kerma: 65 mGy/min kV Plateau: 96 kV Max Pulse width: 10 ms Field Size: 42.6 cm x 42.6 cm Cu Filter: 0.2 mm Cu	Tube voltage: 95 kV Tube Current: 15 mA Pulse width: 2 ms AKR: 0.1 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.2 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.4 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.6 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 0.8 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 1.2 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 1.6 mGy/min	Tube voltage: 95 kV Tube Current: 14 mA Pulse width: 2 ms AKR: 2.5 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2 ms AKR: 3.2 mGy/min	Tube voltage: 95 kV Tube Current: 14 mA Pulse width: 2 ms AKR: 4.9 mGy/min	Tube voltage: 95 kV Tube Current: 13 mA Pulse width: 2.1 ms AKR: 6.6 mGy/min

Table 2: Resulting air kerma rate with varying pulse rates when dose rate adaptation is 0 %.

RAD protocol steps

	80 kW	highest dose	Example RAD1	Example RAD2
Measuring fields			l, r	m
Techniques	3-point	3-point	1-point	1-point
Collimation [cm²]	25 x 25	25 x 25	25 x 25	25 x 25
SID [cm]	115	115	115	115
Grid	yes	yes	yes	yes
Copper [mm]	0	0	0	0
U [kV]	79	150	109	70
Q [mAs]	40	200	3.5	46.1
t [ms]	40	500	10.8	56.8
Focus	big	big	small	big
Power	100%	100%	80%	80%
DAP (System) [µGy x m²]	100.70	1363.03	15.33	93.57

2.3 Equipment safety measures

2.3.1 Protection against electric shock

WARNING

Use of defective electrical devices/components

Risk of electrical hazard

◆ Please do not use defective or damaged electical devices or components near the system, especially to the table top, where the patient is positioned.

Attention: To avoid the risk of electric shock, a protective conductor must be implemented when connecting this device to mains power.

Power supply

For all products that are operated within an X-ray system, the power supply must be made through a contactor or other multipole circuit breaker installed on-site.

The room installation must comply with DIN VDE 0100-710 or the corresponding national regulations.

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Injector connector

When the injector is removed from the table, the adapter must also be removed from the connector on the table for safety reasons.

Power outlet

The country-specific multipurpose power outlet on the patient table is **not** switched off and on with the system.

Only devices which accord to IEC 60601-1 must be connected.

Covers

If socket covers (especially those of the operating modules) are damaged, they must be replaced.

In the event of defects, for example, if a covering cap has broken off, call Customer Service.

Protection class

The system belongs to protection class I with a type B applied part according to IEC 60601-1.

Protection against ingress of liquids:

- Foot switches: IPx8 = Protection against long periods of immersion under pressure
- Tableside control panel: IPx3 = Protected against spraying water
- Control Room Module: IP22 = Protection against contact with any large area by hand and against solid foreign bodies with Ø > 12 mm; Protection against diagonal water drips (up to a 15° angel)
- MAX detectors: IP67
- X.wi-D detectors: IP67
- Rest of the system: IPx0 = No special protection

Protection class IPx3 is only valid as long as the tableside control panel is originally built into the table. When it is removed or replaced, protection class IPx3 may not be met anymore.

Equipotential bonding

Systems for which equipotential bonding is recommended must only be operated in medical facilities where supplemental equipotential bonding has been installed and tested according to the specifications in DIN 57107/VDE 0107/6.81 section 5 (Federal Republic of Germany) or the relevant local and federal regulations.

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Attention: To avoid the risk of electrical shock, a protective conductor must be implemented.

Opening the units

Only authorized service personnel are permitted to open the units.

2.3.2 Electromagnetic compatibility (EMC)

(➔ Page 341 *Electromagnetic compatibility (EMC)*)

CAUTION

Radio interference from the system affects nearby equipment

Risk of malfunction of nearby equipment.

- Reorient or relocate the equipment or system or shield its location.
- Test the operation of the equipment in its new location.

CAUTION

Devices (such as mobile phones, WiFi components, RFID readers etc.) emit electromagnetic fields close to the system

Dysfunctional system, repetition of examination due to poor image quality

- Remove the source of interference (for instance, a mobile phone or an electric device) from the area.
- Restart the system.
- Pay attention to error messages during restart.

2.3.3 Maintenance plan

WARNING

Wear or material fatigue in the system or accessories

Risk of injury or system damage

- Follow the maintenance guidelines to maintain safety and proper function of the system.
- Check accessories for wear before use.

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CAUTION

Image artifacts due to unperformed maintenance

Incorrect diagnosis basis

- Check that the image quality criteria described in the installation and maintenance manual are met before using the system.
- Maintenance work should be performed by trained technical personnel only.
- Purchasing a maintenance contract will keep the system in optimum condition.

CAUTION

Incorrect exposure or defect in the automatic exposure control

Excessive radiation exposure for the patient

- Have the system tested regularly to a national standard such as §16 RöV (German X-ray ordinance).

This maintenance plan includes the required tests according to IEC 62353

Periodic maintenance

Periodic maintenance includes:

- Safety inspection
- Preventive maintenance
- Quality and function tests
- Replacement of safety-relevant parts subject to wear

This work may be performed only by qualified and authorized service engineers. Qualified in this context means that the engineers have been trained accordingly or have acquired the necessary experience in practice. Authorized means that the engineers have been authorized by the operator of the system to perform maintenance work.

Prior to the first system startup, we recommend that you name a staff member who will be responsible for ensuring that routine checks and preventive inspection and maintenance work are performed. This employee has to archive all certificates in the folder for the "Technical Documentation".

In addition to our repair service, Siemens Healthineers also offers you a complete range of services for preventive inspection and maintenance of your system. These services can be called on as required or agreed in a flexibly drafted service contract.

If you have not received a quotation from our UPTIME Services service organization, please contact the Siemens Healthineers representative responsible for your facility.

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Safety inspection

The following checks are intended to ensure the safety of the system. Where appropriate, preventive measures must be adopted or repairs performed. For the most part, the points to be checked are prescribed by laws and standards.

Maintenance interval: 24 months for the entire system

Listing of work steps to be performed

Object or function	Reason	What is checked
General checks		
Overall system	Safety of patients and personnel	Visual check of the system (cover panels) for damage and sharp edges
Cables and how cables are laid	Protection of patient and personnel against electric shocks	Visual check of cables and how cables are laid for safety deficiencies
Accessories	Safety of patients	Check for safety deficiencies
System radiation protection devices	Protection of patient and personnel against radiation injuries	Check unit radiation safety devices as well as any additional safety devices that are configured, e.g. lower body radiation shield, upper body radiation shield, ceiling-mounted radiation shield, for installation and for damage.
Operator documents	Protection against incorrect operation and thus protection of patients, personnel and unit	Check for the presence of operator documents
Warning notices	Protection against incorrect operation and thus protection of patients, personnel and unit	Check for presence of the required warning labels visible to the user for operation of the system.
Insertable fuses	Protection of patients, personnel and unit	Check all insertable fuses that are accessible without tools to determine whether manufacturer data (nominal value and switch-off characteristics) are met.
User interfaces	Protection against incorrect operation and thus protection of patients, personnel and unit	Check for legibility and operating symbols.
Electrical checks		
Electrical safety	Protection of patient and personnel against electric shocks	• Measurement of ground wire resistance • Device leakage current measurement • Patient leakage current measurement of the complete system per national regulations

Object or function	Reason	What is checked
Mechanical checks		
Wall and floor mounting	Protection of patients, personnel and unit	Mounting (visual check)
Pull and support cables	Protection of patients, personnel and unit	Check for wear and fraying
Chain drives	Ensure power transmission and safety functions	Check for wear, tension, ease of movement and function of the guide rollers
System movements (manual)	Safety of the system latching mechanisms	Check of brake holding and end stops
Movable components	Protection of patients, personnel and unit	Cable lead-in, movement behavior and, if applicable, brakes
Accessories	Safety of patients, personnel	Check for functionality and for good condition
User interfaces	Protection against incorrect operation and thus protection of patients, personnel and unit	Check for damage
Safety-relevant parts subject to wear	Protection of patients, personnel and unit	Check for damage
Function check		
Emergency stop devices	Preventing the first malfunction caused by application errors and collapsing patients	Switch-off of system functions after activating the emergency stop device
Control devices and warning indicators	Inform the operator about relevant system conditions and overload situations	The functions of the following displays: • Radiation • X-ray tube load • Blocking • Unit movements at the collision limit
System movements (motorized)	Protection of patients, personnel and unit	Safety shutdown of the movements
Collision protection	Prevent damage to unit components	Automatic shutdown of system movements in the collision zone (for example ceiling, wall, floor)
Function test	Prevent damage to unit components	Final function test of all components

Preventive maintenance

The purpose of preventive maintenance is to reduce unforeseen failures to a minimum. Thus the prerequisites for the system to meet the assured characteristics in the long term are created.

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The effects of different operating conditions (full or partial load operation, temperature, size of dust particles, humidity, gases, vapors) are checked and the condition of parts subject to wear is determined by recording and analyzing characteristic values. Preventive measures must be adopted or repairs must be undertaken as appropriate.

The specified maintenance intervals correspond to the minimum requirements. Compliance with stricter national legislation may be necessary

Maintenance interval: 24 months for the entire system

Listing of work steps to be performed

Object or function	Reason	What is checked
Overall system	Preventive measures to avoid: • Safety risks • Overheating • Wear and tear • Image artifacts	• Check of operating data • Check of cables, cable routing and cable connections for damage • Cleaning contrast medium, blood, disinfectants from areas not accessible to the customer • Inspection and cleaning of coolant and air circulation passageways • Inspection and cleaning of optical transmission paths • Removal of foreign objects, for example positioning aids and injection needles • Touch-up of paint to prevent corrosion and infection • Check of measurement values with tolerance ranges • Check of the movement forces • Check of the drive characteristics, acceleration and deceleration movements • Measures to ensure that all components move smoothly • Check and analysis of points of friction • Reading and analyzing the error log files • Repairing minor damage

Quality and function tests

Quality and function tests are used to check whether the system meets the assured characteristics. Image quality tests determine differences from the original status (for example resolution, contrast range, minimum contrast, image signal and, if applicable, check of Digital Subtraction Angiography).

If there are differences, preventive measures must be taken or repairs performed, whenever is appropriate.

Maintenance interval: 24 months for the entire system

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Listing of work steps to be performed

Object or function	Reason	What is checked
Overall system	Optimum operation on the basis of the specifications listed in the data sheet	Interaction of all system components according to the assured characteristics
Power line connection	Unrestricted power input from the line power source for the purpose of maximum system load. (The power input can be limited by oxidation and corrosion, which can result in exposure fluctuations and system shutdowns).	Internal line impedance
Vacuum components: • X-ray tube assembly	Assuring the system specifications (vacuum components are subject to aging)	Image quality
Beam geometry, centering, beam collimation	Observance of the specifications and legal regulations* to minimize radiation exposure of patient and personnel	Centering the central beam onto the center of the image receptor. Coincidence of radiation field size and image receptor size or light field size and radiation field size.
Radiation dose	Compliance with specifications and legal requirements* to minimize radiation exposure of patient and personnel	Check of dose rate/cutoff dose** during • Fluoroscopy (DL) • Acquisition in all operating modes
Detail recognition	Compliance with specifications and legal requirements* Ensuring image quality	Resolution** during • Fluoroscopy (DL) • Acquisition in all operating modes
Image contrast	Compliance with specifications and legal requirements* Ensuring image quality	Minimum contrast and dynamic range** during • Fluoroscopy (DL) • Acquisition in all operating modes
Image display	Compliance with specifications and legal requirements* Ensuring image quality	Brightness, focus and the geometry of the configured monitors
Image artifacts	Compliance with specifications and legal requirements* Ensuring image quality	Image display for intolerable image artifacts in all existing application techniques
Dose measurement device**	Maintenance of specifications	Accuracy of the display
Image documentation systems	Compliance with specifications and legal requirements* Ensuring image quality	Gray scale reproduction, geometric imaging characteristics, resolution, optical density, artifacts

* DHHS and country-specific regulations must be observed

** Depending on the system configuration

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Safety-relevant parts subject to wear

Safety-relevant parts which are subject to wear must be replaced in periodic intervals.

3D V ceiling stand

Object or function	Reason	What is replaced?	Interval
3D	Wear can cause springs to break and the support cable to split or fray.	Spring-loaded mechanism with steel cables	8 years

2.3.4 Combination with other products/components

To ensure the necessary safety, only products or components expressly approved by Siemens Healthineers, may be used in combination with the system.

Interfaces

Accessories connected to the analog or digital interfaces must be certified according to the respective IEC standards.

Example: IEC 60950-1 for Information Technology Equipment and IEC 60601-1 for Medical Equipment.

The person connecting additional equipment to the medical device is considered to be the system configurer and is therefore responsible for ensuring that the current system configuration complies with the relevant standards (for example system standard IEC/EN 60601-ff and/or other applicable standards).

In case of doubt, please consult your local contact person.

Furthermore all configurations shall comply with the valid version of the system standard IEC 60601-1.

Everybody who connects additional equipment to the signal input part configures a medical system, and is therefore responsible that the system complies with the requirements of the valid version of the system standard IEC 60601-1.

If in doubt, consult the technical service department or your local representative.

Display ceiling suspension (DCS)

Example

⚠ WARNING

Display not properly mounted in suspension system

Electrical shock, injury of persons

- ◆ Only certified electricians should make changes to the display suspension system.
- ◆ The customer is responsible for the proper mounting of non-Siemens-provided displays.

  This includes proper grounding, cabling and fixation of the displays using the provided equipment.
- ◆ Make sure that external cables do not create an additional risk of injury.

  Siemens Healthineers is not responsible for injuries resulting from improper installation of foreign display monitors.

2.3.5 Installation recommendations

- Interlocks must not be present on the doors of the room containing the interventional X-ray equipment. No other measures, whether or not employed for radiation protection, should be able to cause the interruption of irradiation or any other disturbance of a procedure in progress, unless the operator has the means to prevent such action from occurring during the procedure.
- All emergency stop controls in the system must be protected against accidental actuation.

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- Sufficient space must be available around the patient support for the unimpeded conduct of CPR.
- One or more warning lights must be present in order to indicate the loading state to persons at all positions in the room containing the interventional X-ray equipment.
- Appropriate warning lights to indicate the loading state must be present adjacent to doors opening into the procedure room when warning lights within the procedure room are not visible.

2.3.6 Installation, repair, or modifications

Modifications of or additions to the product must be made in accordance with the legal regulations and generally accepted engineering standards.

Siemens Healthineers does not accept responsibility for the safety features and for the reliability and performance of the equipment as the manufacturer, assembler, installer, or importer, in the following cases:

- Installation, equipment expansions, readjustments, modifications or repairs are not carried out by persons authorized by Siemens Healthineers to do so.
- Components affecting safe operation of the product are not replaced by original spare parts in the event of a malfunction.
- The electrical installation of the room does not meet the requirements of the VDE regulation or the corresponding national regulations.
- The product is not used in accordance with the Operator Manual.

Technical documents

On request Siemens Healthineers can provide you with technical documents for the product for a charge.

This does not imply authorization to undertake repairs.

Siemens Healthineers recommends that you obtain a report indicating the nature and the extent of the work performed from the persons carrying out such work. The report should include all changes in rated parameters or operating ranges and the date, the name of the company and a signature.

Siemens Healthineers does not take responsibility for repairs performed without express written approval.

Installation

All information and installation requirements contained in the service documentation require strict adherence.

If you should require additional information, please contact the qualified people specified by the manufacturer.

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2.3.7 Product service life

Our medical engineering products are designed for operation under average conditions in accordance with the Operator Manual for a useful life of 10 (ten) years. If the products are operated for longer than this time, additional checks and possibly repairs beyond the usual maintenance procedures may be necessary to ensure the functional integrity and operating safety of these products. Please talk to your Siemens Healthineers Representative about these measures early enough!

2.3.8 Disposal

On disposing of the complete system or parts thereof, currently valid environmental legislation must be observed.

Examples of environmentally relevant components are:

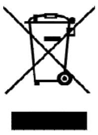
- Accumulators and batteries
- Transformers
- Capacitors
- Flat panel monitors
- Phantoms

For details contact your local Customer Service representative or your Siemens Healthineers regional office.

NOTICE: System components which are hazardous to persons or the environment must be disposed of with care and in compliance with legally binding ordinances.

For all countries of the EU:

In the member states of the European Union (EU), Siemens Healthineers will take back and will dispose of the packing material of your system.



Products bearing this symbol are subject to EC directive 2002/96/EC on waste electrical and electronic equipment (WEEE), amended by 2003/108/EC. This directive determines the framework for the return and recycling of used equipment as applicable throughout the EU.

For details about return and disposal of the product or its components or accessories please contact your local Customer Service or your Siemens Healthineers regional office.

2.4 Cleaning and Disinfection

2.4.1 Specific safety information

This subsection lists safety information for your accessories or system components.

✓ Switch off the system and disconnect it from the main power prior to cleaning or disinfecting.

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CAUTION

Use of unsuitable cleaning and disinfection agents can accelerate the aging process of plastics or leave residues

Damage of equipment, allergic reaction, injury of persons

- Wipe off the system with a damp, lint free wipe.
- Moisten the wipe only with water.
- Use compatible and non-inflammable cleaning and disinfection agents (see list of compatible active ingredient classes).

CAUTION

Image artifacts due to cleaning and disinfection agent residue

Repetition of examination, damage to system

- Take care that cleaning agents do not penetrate into the system.

CAUTION

Inadequate cleaning and disinfection

Risk of infection

- Clean and disinfect all contaminated surfaces and all parts which may come (or have come) into contact with patient after each examination.
- Use compatible cleaning agents and disinfectants.
- Protect the portable detector with a single-use plastic bag.

Ventilation slots

Keep the ventilation slots of all components unobstructed.

CAUTION

Objects on top of the generator cabinet block ventilation

Generator becomes inoperable because it is too warm.

- Do not store objects on top of the generator cabinet.

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Generator cabinet

CAUTION

Cleaning liquids drip into the generator cabinet.

Electrical parts are damaged.

- Make sure that no liquids are used at the top of the generator cabinet.
- Clean the generator cabinet covers very carefully, making sure that no liquid seeps into the cabinet.

Detector protection

Avoid contact of detector with iodine-containing or other strongly dyeing substances. We recommend using a protective cover during examination.

2.4.2 Cleaning and disinfection

All surfaces (system and accessories) that may have come into contact with the patient, the personnel, or any body fluids must be cleaned and disinfected immediately after each examination. In addition, if the system remains unused for a longer period of time (for example, overnight), you must clean and disinfect these surfaces again, before the next examination.

For manual cleaning and disinfection, you should use a non-protein fixing disinfectant based on peroxide compounds with proven cleaning efficacy that is certified by the relevant national authorities for example, EPA (Environmental Protection Agency in the USA) or VAH (Verbund für Angewandte Hygiene e. V. in Germany).

The cleaning and disinfection instructions for the LUMINOS Q.namix R system surfaces have been validated using the following materials:

- Disinfectant wipes ECOLAB Incidin OxyWipe S
- Disinfectant cleaners ECOLAB Incidin OxyFoam S, ECOLAB Incidin Active and BODE Chemie Dismozon Plus
- Lint-free wipes
- Potable water at room temperature

Active ingredient classes

The active ingredient classes listed in the compatibility matrix in this section provide an overview of the chemicals that are compatible or incompatible with the surfaces of the unit to be cleaned.

The results of the compatibility tests were generated using representative products and specific test methods. Although process-chemical manufacturers are using the same base-active ingredient for many of the commercially available disinfectants, these products often differ in adjuvants or additives that frequently are unnamed or confidential.

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In addition, product instructions for use recommend a variety of parameters, such as concentration, procedure, contact time, and others. Consequently, incompatibilities cannot be completely ruled out when using these generic classes of active ingredients.

Active ingredient class	Compatible	Incompatible
pH-neutral cleaning products	☑	
Quaternary compounds	☑	
Guanidine derivatives	☑	
Pyridine derivatives	☑	
Alkylamines	☑	
Organic acids	☑	
Peroxide compounds	☑	
Aldehyde	☑	
Surgical spirit		X
Alcohol solutions		X
Phenol derivatives		X
Hypochlorites		X
Accelerated hydrogen peroxides		X

Compatibility matrix of active ingredient classes

The list of active ingredients does not claim to be complete. Double check the safety data sheet (SDS) of the manufacturer that the disinfectant does not contain such compounds.

Accelerated hydrogen peroxides are not compatible (for example, SciCan Optim 1 or Diversey Oxivir 1 Wipes AHP)!

Safety-data sheets provided by the manufacturer of cleaning and disinfection products provide information on the active ingredient, the pH and relevant safety information.

Disinfectants may contain substances that are hazardous to health. The concentration of such substances in the air must not exceed the legally defined limit.

To ensure safe use, always follow the instructions provided by the manufacturer of the cleaning and disinfection product.

Do not use a cleaning and disinfection product in spray form:

- Sprays can enter the equipment and damage electrical components.

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Some cleaning agents or disinfectants may cause discoloration of some surfaces. This does not impact proper functioning.

Remove dust on parts mounted above the patient regularly.

Never use abrasive cleaners or cleaning agents with solvents, since they may damage housing surfaces.

Preparing the system components for cleaning and disinfection

Before you start cleaning and disinfection, you need to prepare the system components, protect yourself and prepare the equipment required for cleaning and disinfecting.

To avoid personal injury and damage to the equipment, carefully follow the cleaning and disinfection instructions provided.

1. Switch off the power of all devices that require cleaning and disinfection or switch them to standby mode.
2. Check the components for damage. In case of material damage to any parts of the system, notify Siemens Healthineers Customer Service.
3. If any parts of the equipment are detachable, detach them before cleaning and disinfecting.
4. Clean and disinfect detached parts separately.
5. Movable parts need to be positioned so that all surfaces can be reached.
6. Ensure that all surfaces, crevices, and notches are accessible.

Preparing the cleaning and disinfection equipment

1. Wear personal protective equipment. Put on chemical-resistant gloves and a water-repellent apron. Wear a face protection mask, or alternatively, protective goggles and a mask.
2. Prepare the cleaning and disinfection solution. Follow the instructions provided by the manufacturer of the cleaning or disinfection product. You can also use ready-to-use cleaning and disinfectant wipes.

Patient contacting parts can be cleaned and disinfected without switching off the system.

3. Use lint-free wipes and potable water at room temperature for removing any residues.
4. Always adhere to the instructions provided by the disinfectant manufacturer for handling and exposure time required for disinfection.

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General cleaning and disinfecting

All surfaces that may have come into contact with the patient, the personnel, or any body fluids must be cleaned and disinfected immediately after each examination. In addition, if the system remains unused for a longer period of time (for example, overnight), you must clean and disinfect these surfaces again, before the next examination.

Use commercially available cleaning and disinfectant solutions. Follow the instructions provided by the manufacturer of the cleaning and disinfectant product. Refer to (➔ Page 70 *Active ingredient classes*).

In order to avoid possible damage to the system, the following information must be considered:

1. Do not mix different products together.
2. Do not pour or spray any fluids onto surfaces - always use a damp wipe for cleaning and disinfection.
3. Do not immerse objects in water or cleaning fluids, do not rinse.
4. Ensure that no liquids seep into the openings of the system, for example, air openings or gaps between covers.
5. Do not use hard or sharp objects (for example, knives or tweezers) to remove residue.
6. Do not insert any objects into areas that are hard to reach.
7. Do not touch or wipe electrical contacts or outlets. Cover electrical contacts before cleaning, if possible.
8. Ensure that all surfaces, crevices, and grooves of the system are accessible during cleaning and disinfection.
9. Work from top to bottom and from clean to dirty areas.
10. Only use wipes if they leave a continuous liquid film on the surface.
11. After removing blood or other body fluids, immediately discard the used wipe.
12. Always dispose of used wipes according to the instructions of your facility.

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Cleaning and disinfection areas

Overview of system surfaces

Overview of unit surfaces

Print No. XPD3-530G.620.01.04.02

User console control room

Bucky Wall Stand for Max Static

Print No. XPD3-530G.620.01.04.02

Bucky Wall Stand for mobile detectors (e.g. X.wi-D 43)

3D Stand with Tube, TUI and Collimator

Print No. XPD3-530G.620.01.04.02

Manually cleaning and disinfecting

The following instructions for cleaning and disinfecting are applicable to system components.

Make the appropriate preparations for cleaning and disinfection.

(➔ Page 70 *Cleaning and disinfection*)

Cleaning

1. Thoroughly wipe all surfaces with sufficiently saturated cleaning wipes.
2. Ensure that all surfaces, crevices, and notches are accessible during the cleaning process and are completely moistened.
3. Check all parts and surfaces for cleanliness. If soiling is still visible, repeat the cleaning steps above.
4. To remove cleaner residue, moisten wipes with sufficient water and wipe down the cleaned surfaces.

Disinfecting

1. Thoroughly wipe all surfaces with sufficiently saturated disinfection wipes.
2. Ensure that all surfaces, crevices, and notches are accessible during the disinfection process and are completely moistened.
3. Ensure that all surfaces remain exposed to the disinfectant for the exposure time required by the disinfectant manufacturer.
4. Remove disinfectant residue with damp wipes moistened with sufficient water.
5. Let all surfaces dry completely before use.

Aditional information

Cleaning the collimator

⚠ CAUTION

Use of harsh cleaning agents, liquids or sprays

When cleaning the collimator, liquids can seep into the openings of the system and cause electrical shock, short circuits, or corrosion of electrical parts.

Risk of electrical hazard or damage to the system.

- Use recommended substances for cleaning and disinfection only, but no sprays.
- Do not let cleaning liquids seep into the openings of the system (e.g. air openings, gaps between covers).
- Observe the following cleaning and disinfection instructions.

Cleaning the DAP chamber

High voltage is present in the DAP ionization chamber. It is therefore very sensitive.

Use a soft cloth to avoid scratching the chamber.

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Cleaning the MAX and X.wi-D detectors

CAUTION

The battery of the portable detector is not resistant to fluids.

Electrical shock, damage to the detector

- Remove the battery from the portable detector before cleaning the detector.
- Check that the detector is waterproof before immersing it (symbol on the detector).

  The battery can be wiped off but it is not waterproof.

The charge contact plate on the back of the detector must be completely dry before the detector is returned to the docking station, in the Bucky tray of the table or the Bucky wall stand. Otherwise, the detector will not charge sufficiently, and the charge contact surfaces may corrode.

It is recommended to clean the back of the detector first so that it has more time to dry while the front is being cleaned.

Accessory parts

Please note that for some accessory parts, there are special instructions on cleaning in the corresponding chapters. Refer to Register Accessories and Auxiliary Devices in the Operator Manual.

Unless special instructions are given there, the manual cleaning and disinfection instructions are valid.

Sterilization

The system does not require sterilization.

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3 System Overview

Example of system configuration

(1) Basic system
(2) Overhead support with X-ray tube and multileaf collimator (second plane)
(➔ Page 96 *Ceiling-suspended X-ray tube with multileaf collimator*)
(3) Wall stand (second plane)
(➔ Page 103 *Wall stand*)
(4) Trolley with system remote control console and footswitch for fluoroscopy and exposure
(5) System remote control console
(6) Touch monitor of the imaging system
(7) Display ceiling support (DCS)

3.1 System description

The LUMINOS Q.namix R system with MODEL NO. 11574002 bears the CE 0123 marking for the entire system. The label is attached to the back of the table frame.

3.1.1 Features

- Swiveling overtable X-ray tube assembly, with oblique projection and gently starting and braking system movements.
- Motor-driven height adjustment of the tabletop and three source-image distance positions settable.
- Table tilt +90° to -90° with two speeds.
- Fluoroscopy with high frame rate at 32 p/s
- Motor-driven longitudinally and transversely moving tabletop.
- System remote control system with system movement control, generator parameters, and Imaging System functions.
- Integrated system movement control on the table.

The basic system is equipped with a solid-state flat panel detector for universal use. It is suitable both as intensively used universal workstation and as a highly loaded special workstation.

3.2 System configurations

3.2.1 Configuration (1)

Basic configuration without second plane consisting of:

- LUMINOS Q.namix R examination unit
- Flat detector
- OPTITOP X-ray tube assembly
- X-ray generator:

  POLYDOROS RFX 65

  POLYDOROS RFX 80 (option)
- Primary collimator
- 24" or 32"(option) display
- Display trolley or ceiling suspension for one display
- Digital imaging system with DICOM functions
- Footswitch for fluoroscopy and exposure
- Interface for high-pressure contrast medium injector
- Measuring device for dose-area product
- Mobile tableside console
- Compression device

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3.2.2 Configuration (2)

Basic configuration plus fully synchronized 3D overhead support.

3.2.3 Configuration (3)

Basic configuration plus fully synchronized 3D overhead support and wall stand for mobile detectors like MAX wi-D or X.wi-D 43/35.

3.2.4 Configuration (4)

Basic configuration plus fully synchronized 3D overhead support and wall stand with fixed detector (MAX static).

3.2.5 Configuration (5)

Basic configuration plus fully synchronized 3D overhead support and the mobile detectors MAX wi-D, MAX minior X.wi-D 24/35/43.

3.2.6 Examination techniques

You can perform examinations with the following techniques:

- Fluoroscopy with flat detector
- Digital fluororadiography DFR
  - Single acquisitions
  - Series acquisitions
- Digital radiography DR
  - Single acquisitions
- Free exposures

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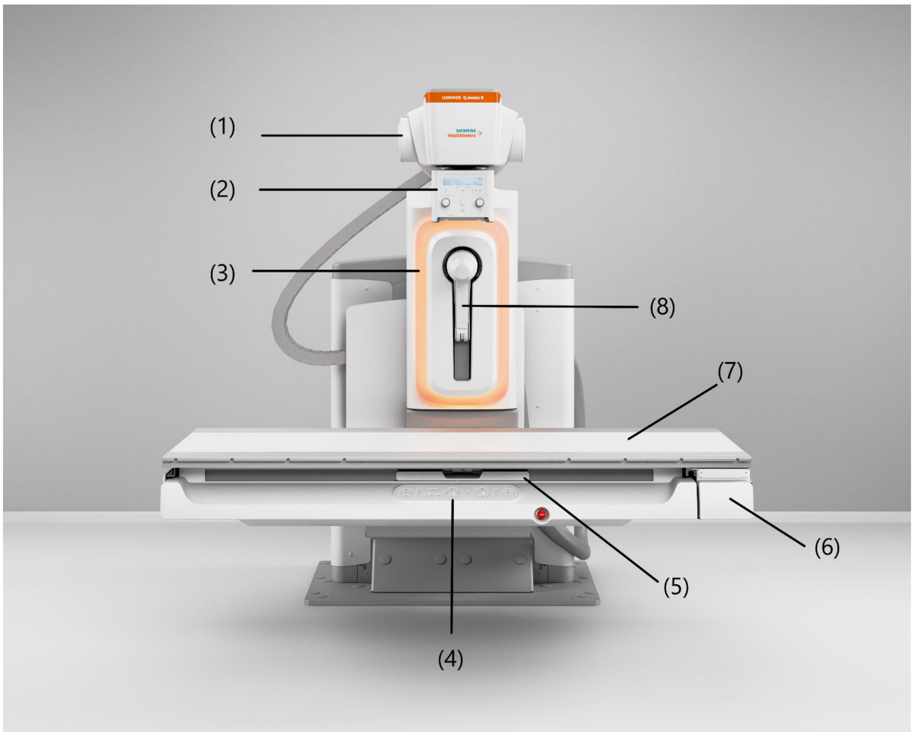
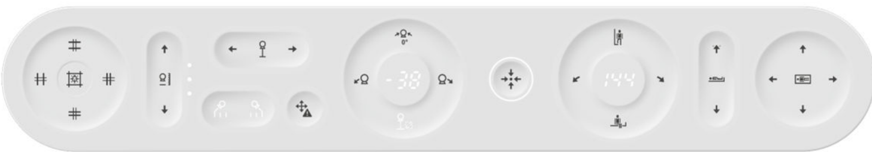
3.3 Unit overview

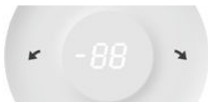


(1) Air-cooled X-ray tube assembly, partly enclosed, can be swiveled
(2) Collimator with automatic format collimation and numerical format indication, with integrated motor-driven additional Cu filters
(3) Mood light
(4) Tableside control panel
(5) Detector tray
(6) Table frame, height-adjustable by motor drive, can be tilted + 90° to - 90°
(7) Tabletop with flat accessory rails, motor-driven longitudinal and transverse travel
(8) Detachable compression device with continuously adjustable compression force

3.3.1 Tableside control panel



System settings

		Coordinate switch for opening or closing rectangular primary collimator Middle: Switching light localizer on or off
		Moving the X-ray system longitudinally
		Tube angulation
		Move tube angulation to 0°
		Angulating / rotating tube to the left side
		Angulating / rotating tube to the right side
		Display: • Oblique projection angle (maximum ±45°) or • Display of a flashing "E" for an error message
		Enabling or disabling tube detector coupling
		Automatic system positioning for upright patient transfer
		Tilting the table Display showing • Table tilt angle or • Flashing "E" for an error message
		Automatic system positioning for horizontal patient transfer
		Increasing or decreasing SID. Three programmable options possible: • min SID 115 cm, max SID 180 cm (ignore SID of 150 cm) • min SID 115 cm, max SID 150 cm (ignore SID of 180 cm) • all 3 SID's available: min SID 115 cm, softstop at 150 SID, max SID 180 cm

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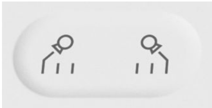
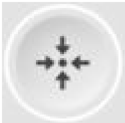
	Ortho start / end
	Park override
	**SmartMove** button with LED for starting movement control: Moves the system into the position which is defined in the clinical protocol step.
	Raising or lowering the table
	Switch for moving the tabletop longitudinally or transversely Combination of longitudinal and transversal movement is possible.

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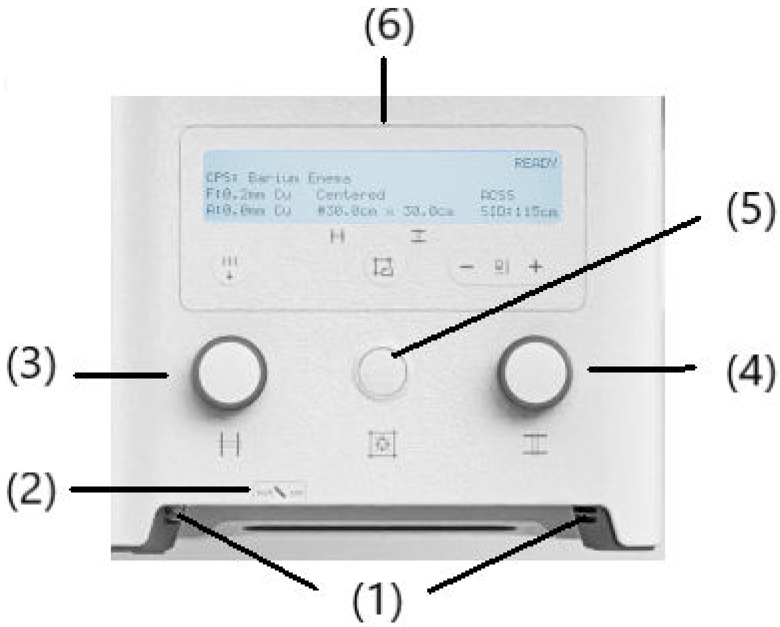
3.3.2 Multileaf collimator



(1) Accessory rails
(2) Tape measure for SID setting (cm and inch)
(3) Manual vertical collimation
(4) Manual horizontal collimation
(5) Button for full-field light localizer and laser light localizer, switches off automatically after 10 - 90 s (configurable)
(6) Display field

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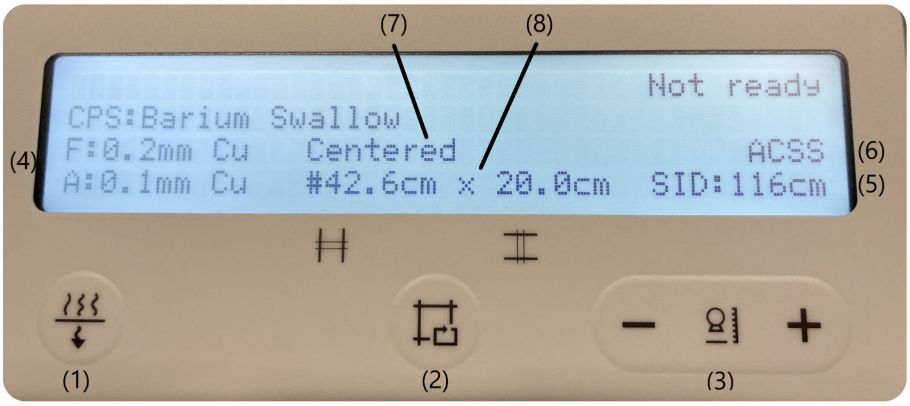
Display field



(1) Motorized prefilter selection button
(➔ Page 169 *Selecting the Cu filter*)
(2) Selection of the collimated radiation field of the last exposure
(3) Buttons for entering the SID for free setting
(4) Display for HVL filter in beam
F = Fluoroscopy
A = Acquisition
(5) Display for selected SID (cm or inch - configurable by Siemens Healthineers Customer Service)
(6) Operating mode:
ACSS (automatic format collimation) or
Manual (no automatic format collimation)
(7) Information if tube is centered
(8) Display for width and height of collimated radiation field (cm or inch - configurable by Siemens Healthineers Customer Service)

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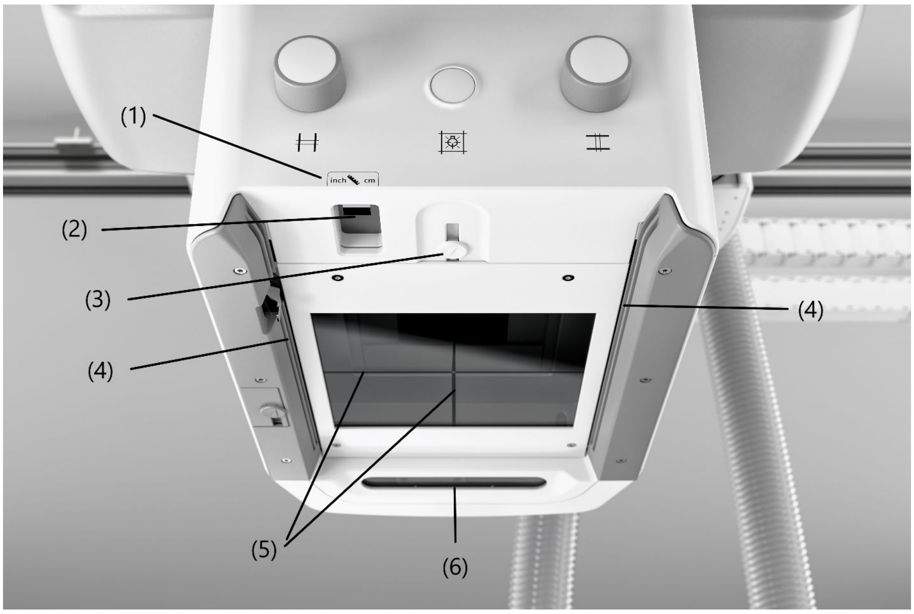
Control elements at the underside



(1) Measuring tape for SID/SOD measurement
(2) Laser-line light localizer
(3) Slider to cover the laser outlet opening
(4) Accessory rails
(5) Crosshairs in the light localizer window
(6) Camera

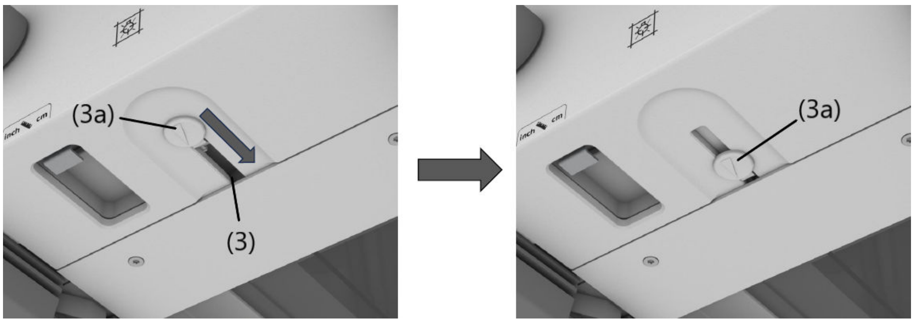
Measuring tape for SID/SOD measurement (1)

The measuring tape has two scales:

- **inch** (left scale)
- **cm** (right scale)

Laser-line light localizer (2) and sliding cover (3)

The laser-line light localizer projects the axis mark for longitudinal centering, which is aligned with the centering mark on the receptor.



If necessary, the laser radiation exit of the laser-line light localizer (3) can be closed with the slider (3a).

Crosshairs (5)

The crosshairs project the longitudinal and transverse axis of the radiation field onto the cassette or directly onto the patient.

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Accessories and auxiliary devices

Accessories and auxiliary devices can be attached to the accessory rail.

(➔ Page 213 *Accessories and Auxiliary Devices*)

The maximum permissible weight of accessories and auxiliary devices attached to the collimator must not exceed 7 kg.

3.4 System components

3.4.1 Touch display

- The display is switched on and off with the system.

  Do not switch it off manually.

- The display has already been configured for optimal performance.

  Do not change any settings.

- If no input signal is present, e.g. during startup, **No Signal** will be displayed.
- If no image or a blurred image, vertical lines or other defects, are displayed, notify the Customer Service.

You can configure the language of the user interface, the display scheme (bright or dark), and the display of patient data in the **myExam Cockpit > Client Settings**.

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3.4.2 System remote control console

(1) Touchscreen user interface (TUI)
(➔ Page 92 *The touchscreen user interface at the system remote control console (remote UI)*)

(2) System On/Off button
(➔ Page 110 *On/off*)

(3) SmartMove button

(4) Joysticks and buttons
(➔ Page 90 *Hardware buttons panel user interface*)

(5) Emergency STOP button
(➔ Page 22 *Red emergency STOP buttons*)

(6) Button with LED to block or unblock radiation

ℹ The user interface is identical for the mobile tableside control console.

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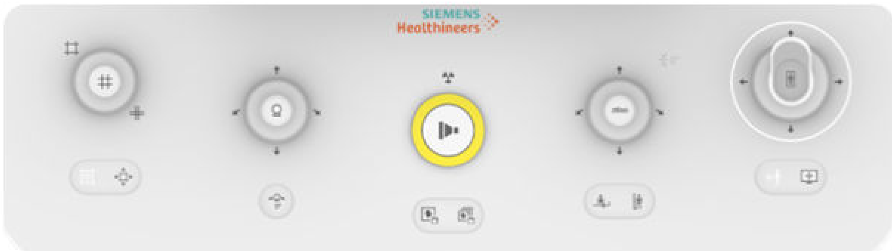
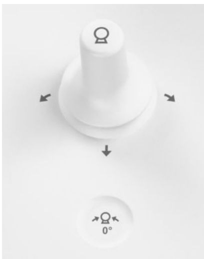
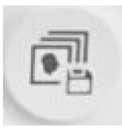
Hardware buttons panel user interface



	Collimator functions: • Joystick module: – Horizontal deflection opens or closes width blades (left: opens; right: closes) – Vertical deflection opens or closes height blades (up: opens; down: closes) – Diagonal deflection up left opens width and height blades – Diagonal deflection down right closes width and height blades • Button on the bottom left for switching the laser light localizer on or off • Button on the bottom right for opening the collimator completely
	• Joystick module for: – Oblique beam angulation (deflection to the left or right) – Vertical movement of the tube (deflection up or down) Around the joystick is a blue LED ring for the activation state of the dead man's grip (DMG). When the LED ring is on, the DMG is activ. • Button for moving the tube into 0° position
	Exposure release button with pre- and main contact and radiation indicator
	Store image
	Store loop

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	Joystick module for: • Table tilt: Deflection to the left or to the right • Table vertical movement: Deflection up or down
	Selects automatic system positioning for horizontal patient transfer.
	Selects automatic system positioning for upright patient transfer.
	• Multifunctional joystick for – Switch between column and table motion (button on top of shaft) – Horizontal deflection to the left or right moves table transversely. – Vertical deflection up or down moves the X-ray system longitudinally. – Vertical deflection up or down with the thumb button depressed moves the table longitudinally. • Buttons for toggling joystick function between patient orientation (left button) and monitor orientation (right button)

We recommend that the button to block or unblock exposure release is used at all times, except when a procedure is in progress. This is to prevent the possibility of radiation being emitted through the inadvertent actuation of an exposure release button.

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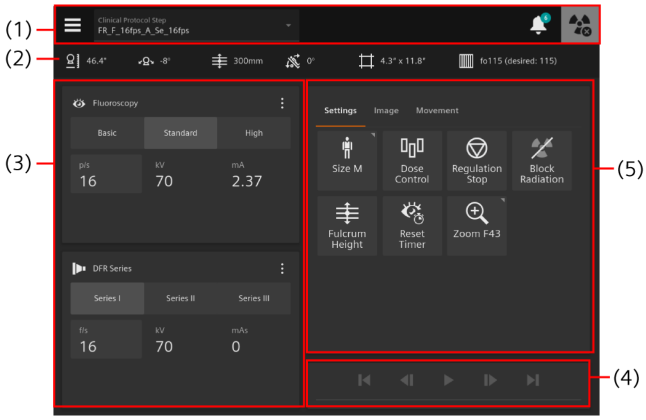
The touchscreen user interface at the system remote control console (remote UI)



Example

(1) Menu bar
(➔ Page 92 *The menu bar on the remote UI*)

(2) Display of important system parameters
(➔ Page 93 *Displays on the remote UI*)

(3) Fluoroscopy and acquisition parameters
(➔ Page 93 *Fluoroscopy and acquisition parameters on the remote UI*)

(4) Player for scene review

(5) Most important functions
(➔ Page 94 *Further functions on the remote UI*)

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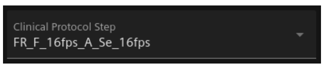
i You can configure the language of the user interface, the display scheme (bright or dark), and the display of patient data in the **myExam Cockpit > Client Settings**.

The menu bar on the remote UI

	Opens the menu. (➔ Page 93 *The menu on the remote UI*)
	Selection list for scheduled clinical protocol step.
	Indicates system messages. The number of active messages is indicated in the turquoise dot. Opens the **Notification Center**.
	System is ready for acquisition.
	System is not ready for acquisition.

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The menu on the remote UI

	Display of the logged-in user account
	Lock Screen: Locks the touchscreen (depends on configuration).
	Cleaning screen: Activates the touch screen lock for cleaning.
OFF	**Room light**: Switches on and off the room light.
ON	

Displays on the remote UI

	Indication of source-image distance (SID)
	Indication of tube angulation
	Indication of fulcrum height
	Indication of table angulation
	Indication of collimation size
	Indication of grid status and grid focus f_0

Fluoroscopy and acquisition parameters on the remote UI

	Fluoroscopy parameters	
	Acquisition parameters	
	Opens the menu for additional parameters.	
		Closes the menu for additional parameters.

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The fluoro/acquisition clinical protocol steps consist of two triplets of protocols.

- **Fluoroscopy** - **Basic**: Fluoroscopy protocol with low dose
- **Fluoroscopy** - **Standard**: Fluoroscopy protocol with medium dose
- **Fluoroscopy** - **High**: Fluoroscopy protocol with higher dose
- **Acquisition** - **I**: Acquisition protocol 1 (single image or series)
- **Acquisition** - **II**: Acquisition protocol 2 (single image or series)
- **Acquisition** - **III**: Acquisition protocol 3 (single image or series)

Further functions on the remote UI

Further functions for controlling the system and imaging system are available on tab cards.

(➔ Page 94 *Settings tab*)

(➔ Page 95 *Image tab*)

(➔ Page 95 *Movement tab*)

Settings tab

	**Size**: Selects patient size.
 	**Dose Control**: Selection of automatic dose control or selection of measuring fields. The icon indicates the selected function. (Default depends on the selected clinical protocol step.)
	**Regulation Stop**: Stops the dose regulation in order to continue with the current fluoroscopy parameters.
	**Block Radiation**: Blocks radiation release.
	**Reset Timer**: Resets the fluoroscopy timer.
	**Zoom**: Display and selection of zoom stage/input field

We recommend usage of the **Block Radiation** function at all times, except when a procedure is in progress. It is in order to prevent the possibility of radiation being emitted through the inadvertent actuation of an radiation release button.

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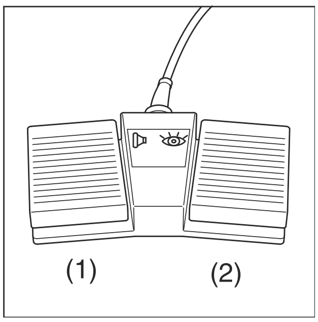
Image tab

	Store Frame: Stores the currently displayed image.
	Set as Reference: Sets/copies the displayed image to the reference image segment or monitor (depending on installation).
	Mark: Marks the currently displayed image for automatic transfers.
	Flip vertically: Preselection of a flipping the next acquired image vertically about the horizontal axis.
	Flip horizontally: Preselection of a mirroring the next acquired image horizontally about the vertical axis.
	L/R Label: Preselection of a laterality label

Movement tab

	Block Movement: Blocks the system against unintended unit movements.
	Inverse transver.: Inverts transversal movements.
	Inverse longitud.: Inverts longitudinal movements.
	Compres. Cone: Activates the compression cone.
	Detector Align.: Activates the alignment of the detector (tracking).
	Override Softstop: Overrides the stop in horizontal position when tilting the table.

3.4.3 Footswitch for fluoroscopy and exposure



(1) Switch for exposure (without precontact)
(2) Switch for fluoroscopy

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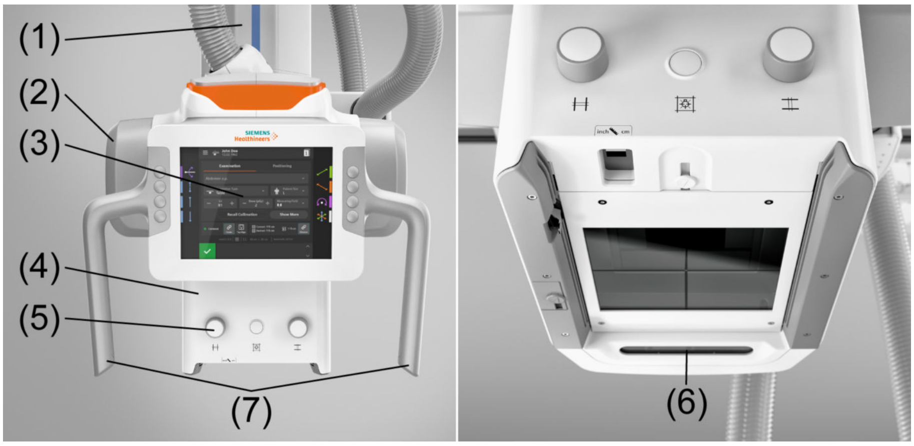
⚠ CAUTION

Operator or patient trips over cables or foot switches on the floor

Risk of injury

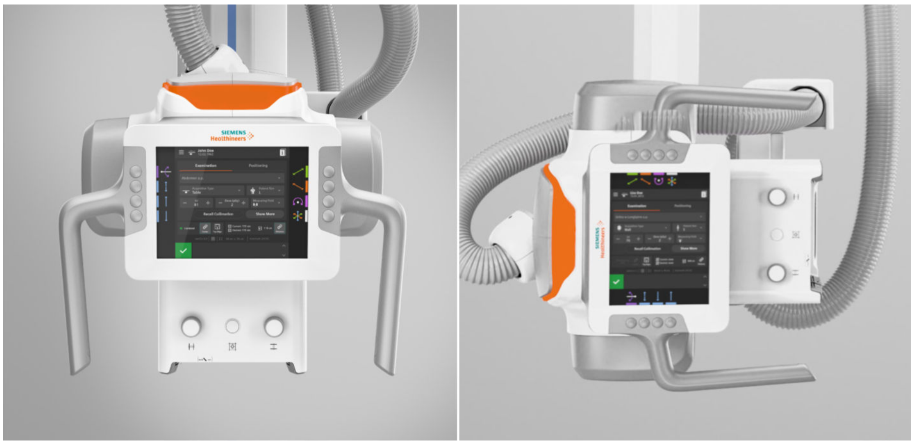
◆ Position the cables and foot switches on the floor so that nobody trips.

3.4.4 Ceiling-suspended X-ray tube with multileaf collimator



(1) Lifting column
(2) X-ray tube unit
(3) Touchscreen user interface (tube UI)
(➔ Page 96 *User interface at the tube (tube UI)*)
(4) Multileaf collimator
(➔ Page 101 *Multileaf collimator*)
(5) Collimator control panel
(6) 3D camera
(7) Handles with buttons
(➔ Page 97 *Buttons for movements and indications on the tube UI*)

User interface at the tube (tube UI)



Landscape orientation (left) and portrait orientation (right)

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The tube UI is a color touchscreen for control of multiple functions. The touchscreen corrects its display orientation from portrait to landscape, when the X-ray tube is tilted by 90°.

The display on the tube UI informs you about the position of the support (SID, tube angle), the generator settings, patient name, clinical protocols, and so on.

(→ Page 98 *The touchscreen user interface at the tube (tube UI)*)

(→ Page 97 *Buttons for movements and indications on the tube UI*)

(→ Page 99 *The menu on the tube UI*)

(→ Page 99 *General displays and settings on the tube UI*)

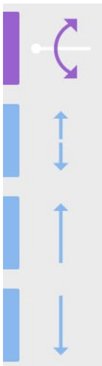
(→ Page 100 *Positioning tab*)

(→ Page 101 *Positioning tab - Measuring Tape*)

Buttons for movements and indications on the tube UI

Next to the handles on the tube unit, there are four buttons on each side of the tube UI.

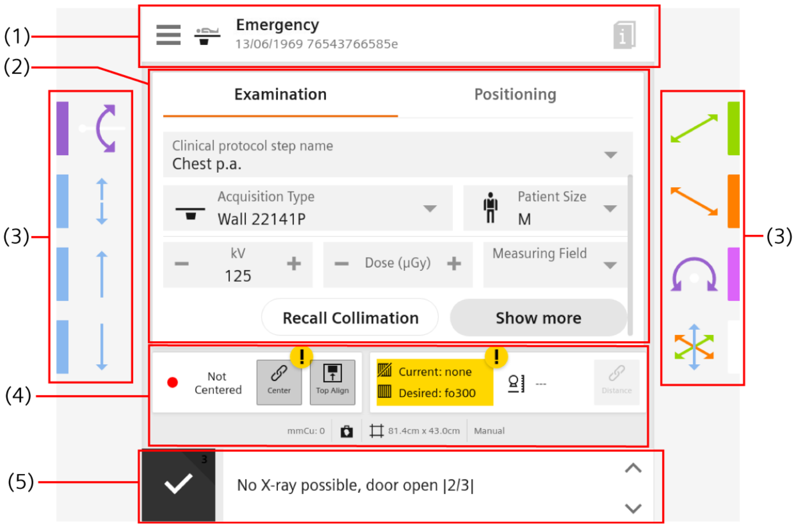
The functions of the buttons are indicated by symbols and colors on the tube UI.

Vertical alignment of the tube unit		Horizontal alignment of the tube unit
		
		

Button symbol	Function of the button
	Tube rotation about the horizontal axis
	Floating movement up and down
	Motorized movement up
	Motorized movement down
	Longitudinal movement

Button symbol	Function of the button
	Longitudinal movement is in lock position.
	Transverse movement
	Transverse movement is in lock position.
	Tube rotation about the vertical axis
	Floating tube movement (longitudinal, transverse, vertical)

The touchscreen user interface at the tube (tube UI)



Example

(1) Menu and general displays
(➔ Page 99 *The menu on the tube UI*)
(➔ Page 99 *General displays and settings on the tube UI*)
(2) Tab cards for parameter display and function selection
(3) Movement button indicators
(➔ Page 97 *Buttons for movements and indications on the tube UI*)
(4) Display of and/or selection of alignment, grid status, SID, copper filter, and collimation
(➔ Page 99 *General displays and settings on the tube UI*)
(5) System messages

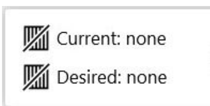
You can configure the language of the user interface, the display scheme (bright or dark), and the display of patient data in the **myExam Cockpit > Client Settings**.

The menu on the tube UI

	Display of the logged-in user account
	**Block movement**: Blocks the system against unintended unit movements.
	**Block radiation**: Blocks radiation release.
 OFF  ON	**Rotation 0° detent**: Activates or deactivates the automatic rotation of the tube to the 0° detent in the configured range The range can be configured by Customer Service from ±1° to ±5° Default is ±3° The detent will be active if the detector is not inserted or inserted in the corresponding format. The detent will be inactive or deactivated if the detector is inserted in the opposite format as configured.

We recommend usage of the **Block Radiation** function at all times, except when a procedure is in progress. It is in order to prevent the possibility of radiation being emitted through the inadvertent actuation of an radiation release button.

General displays and settings on the tube UI

	Opens the menu. (➔ Page 99 *The menu on the tube UI*)
	Access **Positioning Guide** information
	Selected workflow, patient name, and configured patient data
	Centering information
	Enable/disable center tracking
	Enable/disable top alignment
	Grid status and grid focus f_0

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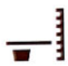
	Grid status and grid focus f_0 do not match the setting in the clinical protocol step.
	Indication of source-image distance (SID)
	Enable/disable SID tracking
	Indication of additional information: copper filter, detector type, detector orientation, collimation size, rotated collimator, and if automatic collimation size sensing is active (**ACSS**) or not (**Manual**)
	System is ready for acquisition.
	System is not ready for acquisition.
	Tap an arrow at the message bar to see all previous/next message.

Positioning tab

The **System Position** blind on the **Positioning** tab displays the rotation angles of the tube unit, detector unit, wall stand (detector tilt and height), and patient table height.

	Tube rotation horizontally
	Tube rotation vertically
	Detector rotation horizontally
	Detector rotation vertically
	Detector tilt on wall stand
	Table height

Tap the arrow down icon to display the **Measuring Tape** settings.

(➔ Page 101 *Positioning tab - Measuring Tape*)

Positioning tab - Measuring Tape

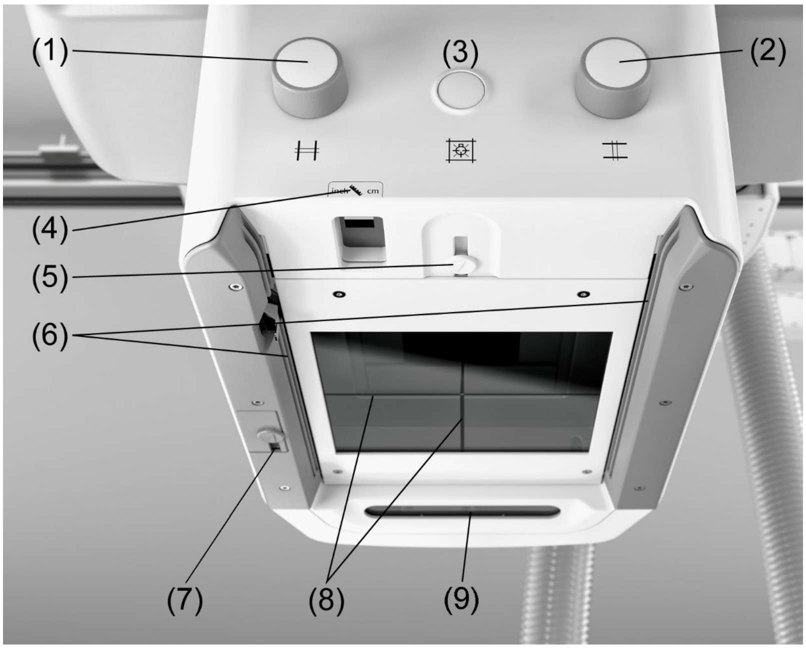
The **Measuring Tape** blind on the **Positioning** tab displays the settings for tape measurement.

	Select SID measurement (Source-Image Distance)
	Select SOD measurement (Source-Object Distance)
	Magnification display: Calculated magnification result from SID and SOD

Tap the arrow up icon to display the **System Position** settings.

(➔ Page 100 *Positioning tab*)

Multileaf collimator



(1) Manual vertical collimation
(2) Manual horizontal collimation
(3) Button for full-field light localizer and laser line light localizer
(4) Measuring tape for SID/SOD measurement
(5) Laser line light localizer with sliding cover
(6) Accessory rails
(7) 2nd Laser line light localizer with sliding cover
(8) Crosshairs in the light localizer window
(9) 3D camera

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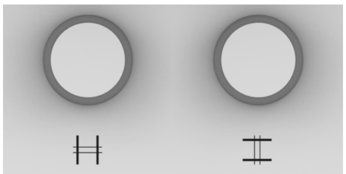
Manual collimation (1 and 2)



Collimation of the desired acquisition field.

- Left knob to adjust the format width.
- Right knob to adjust the format height.

The format which results in the exposure plane corresponding to the predetermined SID appears in the indication for the radiation field size in the display of the multileaf collimator.

Check illumination of the format and collimate as close as possible to the object.

Collimation close to the object reduces the scattered radiation and thus improves image quality. The radiation exposure to the patient is also reduced.

To protect the patient from unnecessary radiation, always check the collimator position with the light localizer.

During collimation, the selected measuring field may be automatically changed to another measuring field.

The currently active measuring fields are visible on the display.

Pay attention that the automatic change-over is correct.

Button for full-field light localizer and laser line light localizer (3)

Double clicking the light localizer button will extend the time until automatic turn off of the light localizer. As a feedback that the longer time is activated, the light shortly flashes.

Measuring tape for SID/SOD measurement (4)

The measuring tape has two scales:

- **Inch** (left scale)
- **cm** (right scale)

Laser line light localizer (5)

The laser line light localizer projects the axis mark for longitudinal centering, which is aligned with the centering mark on the receptor.

The laser line light localizer is switched on and off jointly with the full field light localizer.

To protect the eyes of the patient or any other person, the LASER radiation outlet (5) of the LASER line light localizer can be closed with the sliding cover (5a).

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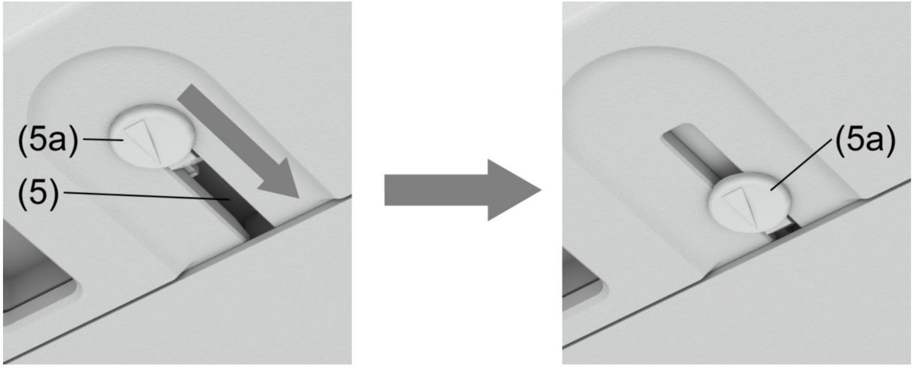

Accessory rails (6) The accessory rails may be loaded with maximum 70N (7kg).

Locking spring The locking spring is located on the left guide rail at the underside of the collimator.

The locking spring locks the compensating filters, templates and so on, inserted in the accessory rails of the collimator, thus securing them against falling out.

To remove the accessories, press the locking spring to the left until the compensating filter, template and so on, can be removed from the collimator.

3.4.5 Wall stand

Description

- The wall stand contains a detector unit (wireless detector or MAX static) with grid.
- The detector unit is fully counterbalanced and features continuous vertical adjustment.
- The detector unit can be continuously tilted from the vertical into the horizontal position (+90°) and up to -20° with a detent at 0° and 90°.
- Automatic synchronization (detector height adjustment)

  X-ray tube follows synchronized and keeps centering on middle of detector or its upper edge (top alignment).
- The wall stand has a centering function
  - Fully automatic centering of X-ray tube on middle of detector or its upper edge (top alignment)
- The wall stand is mounted to the floor.

Application

The wall stand can be used as an examination unit for universal medical applications performed on standing, sitting or recumbent patients and involving exposures of the abdominal, pelvic, skull and vertebral regions as well as radiography of the extremities.

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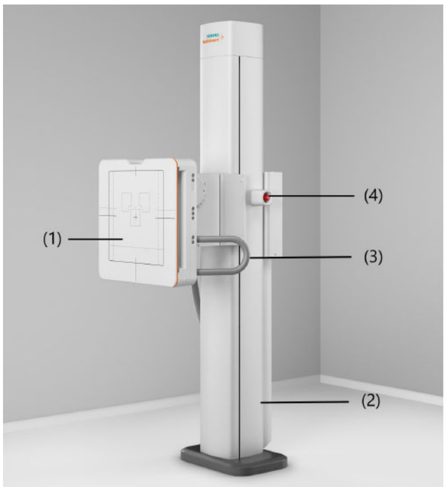
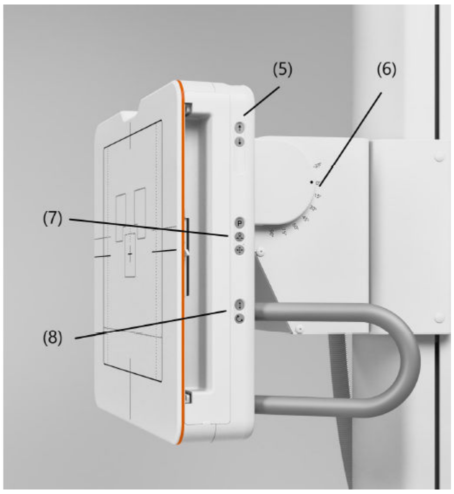
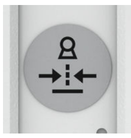
Operating elements





(1) Front panel with imprint for detector position and measuring fields
(2) Stand column
(3) Patient hand grips (on both sides)
(4) Emergency STOP push button (on both sides)
(5) On both sides: Buttons for motorized vertical movement
(6) Degree of tilting angle
(7) On both sides: Park position button and centering button
(8) On both sides: Button for manual vertical movement (above) and button for unlocking the tilt mechanism (below)

Details to legend point (7):

Key	Description
	**Park position** The ceiling-suspended X-ray tube moves to a pre-programmed position (lift and tilt axis).
	**Centering button** (tube to detector)
	**SmartMove button** The system moves the ceiling-suspended X-ray tube and the BWS into a pre-programmed CP position (for example, height).

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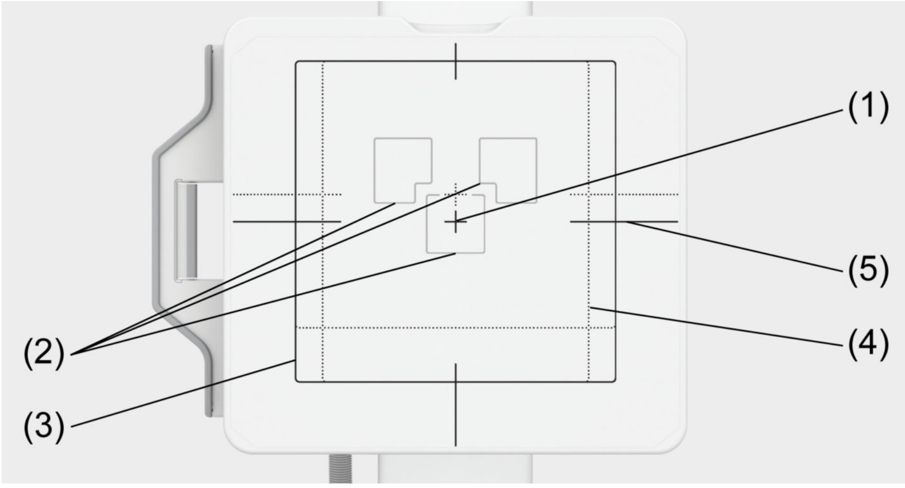
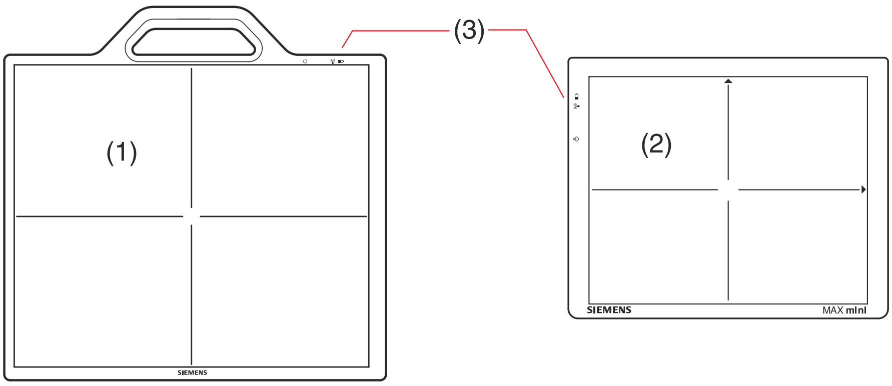
Front panel markings



(1) Center crosshair
(2) Position of measuring fields
(3) Detector field for portrait format
(4) Detector field for landscape format
(5) Center marking for portrait format

3.4.6 Mobile detectors

Wireless detectors (MAX wi-D or MAX mini)



(1) Detector MAX wi-D
(2) Detector MAX mini
(3) Detector indicators

The wireless detector is part of a digital image acquisition chain in an overall radiological system. It features a portable equipment designed for mobile applications.

The communication is managed through a WiFi interface.

MAX wi-D The MAX wi-D has a handle within the housing for easy manipulation, LED indicators for internal status, and connector for service issues. The back side (with respect to the active sensitive array) includes electrical contacts to recharge the replaceable battery.

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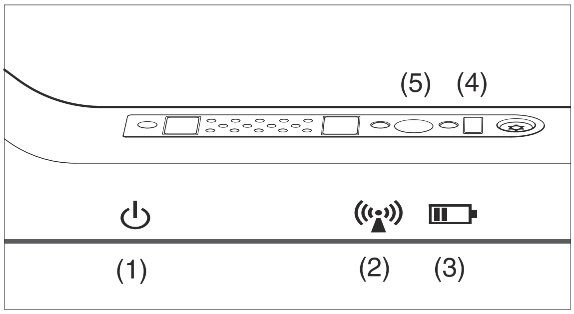
Detector indicators



(1) Detector status LED
(2) Wi-Fi status LED
(3) Battery status LED
(4) Infrared sensor, needed for attachment of a detector (MAXswap)
(5) Power button
Short push (wake up)
Long push (switch off/on)

⚠ CAUTION

Wrong detector size connected or wrong detector used

Repetition of examination

- Check that detector identification on the detector label matches the detector identification shown on the user interface.
- Check that the green ready light on the correct detector is lighted before releasing radiation.

It may be possible that even when the Wi-Fi Status LED and the detector status LED of the detector are green, the release of radiation is blocked by the system.

Check the system messages.

LEDs LEDs are provided on the detector (close to handle), to give basic information on the detector behavior.

If the detector is restarted during examination mode, e.g. by replacing the battery, the status LED will stay orange even if the detector is ready.

Please observe the system status on the imaging system or on the tube UI instead. If the status is green (OK), the system is ready for acquisition.

LED color / status	Detector status LED (1)	Wi-Fi status LED (2)	Battery status LED (3)
LED off	OFF state	Wi-Fi disabled	No battery
LED green	Detector selected for examination (A clinical protocol for this detector is selected.)	Wi-Fi available (detector connected to an access point)	Battery OK

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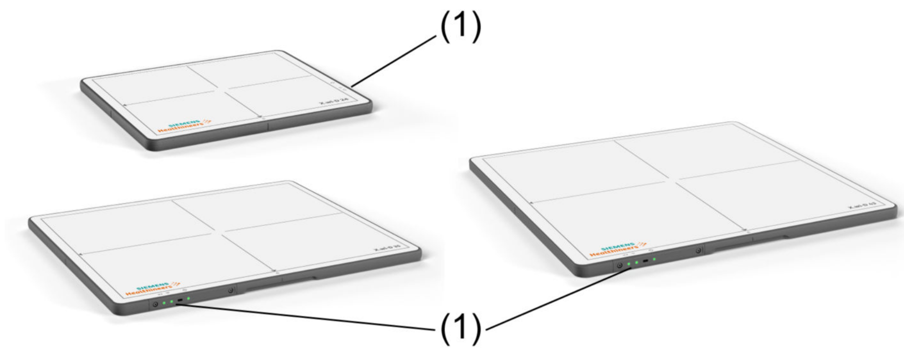
LED color / status	Detector status LED (1)	Wi-Fi status LED (2)	Battery status LED (3)
LED green blinking slow	n.a.	Wi-Fi available (detector connected to an access point)	Battery charging
LED orange	Offline, or download state, or online (but not selected for examination)	Wi-Fi not ready (detector not connected to an access point)	Battery getting low
LED orange - blinking slow	Error state or switching ON	n.a.	n.a.
LED orange - **blinking fast**	n.a.	n.a.	Battery low
LED orange - **blinking fast , then red flashlight on the 3 LEDs, then OFF**	Switching OFF (going to OFF mode) when requested by message or by time-out	n.a.	n.a.

Wireless detectors (X.wi-D 24, X.wi-D 35 or X.wi-D 43)



(1) Detector indicators

The wireless detector is part of a digital image acquisition chain in an overall radiological system. It features a portable equipment designed for mobile applications.

The communication is managed through a WiFi interface.

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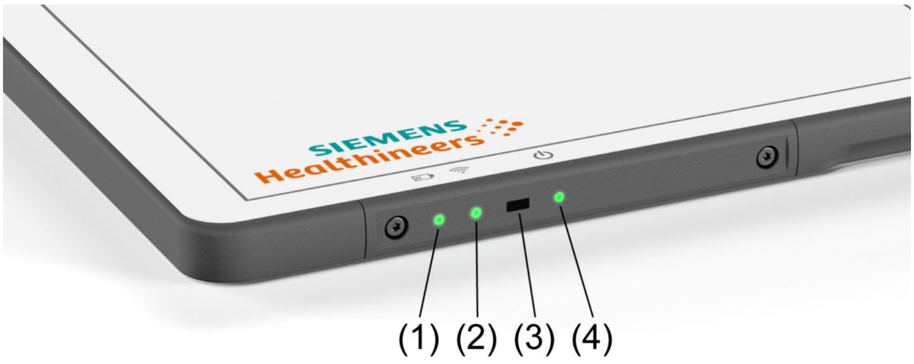
Detector indicators



(1) Battery status LED
(2) Wi-Fi status LED
(3) Infrared sensor, needed for attachment of a detector
(4) Detector status LED

CAUTION

Wrong detector size connected or wrong detector used

Repetition of examination

- Check that detector identification on the detector label matches the detector identification shown on the user interface.
- Check that the green ready light on the correct detector is lighted before releasing radiation.

It may be possible that even when the Wi-Fi Status LED and the detector status LED of the detector are green, the release of radiation is blocked by the system.

Check the system messages.

LEDs LEDs are provided on the detector, to give basic information on the detector behavior.

If the detector is restarted during examination mode, e.g. by replacing the battery, the status LED will stay orange even if the detector is ready.

Please observe the system status on the imaging system or on the tube UI instead. If the status is green (OK), the system is ready for acquisition.

LED color / status	Battery status LED (1)	Wi-Fi status LED (2)	Detector status LED (4)
LED off	No battery or detector is turned off or booting	Wifi disabled by software or detector is turned off or booting	OFF state
LED green	External power supply not present and Battery charge level between 100% and 10%	Wi-Fi available (detector connected to an Access Point)	Connected to the system and operational
LED green blinking slow	External power supply present	n.a.	Energy low consumption

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LED color / status	Battery status LED (1)	Wi-Fi status LED (2)	Detector status LED (4)
LED orange	External power supply not present and Battery charge level between 10% and 5%	Wifi not ready (detector not connected to an Access Point)	Not connected to the system
LED orange - blinking slow	n.a.	n.a.	ERROR state
LED orange - **blinking fast**	External power supply not present and battery capacity below 5%	n.a.	Switching OFF

3.5 Acoustic signals

In the following cases, acoustic signals will sound:

- System ready (only once after startup/restart)
- Movement blocked
- During motorized movement
- End of CP step movement = position reached
- Limit signal
- X-ray ON
- End of X-ray
- Skin dose threshold alert
- X-ray tube too hot
- User information display
- Error message requiring user confirmation

The sounds can be configured by Customer Service.

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4 System Operation

4.1 On/off

Before using the system:

If the power network at the site is very unstable, it may be advisable to install an uninterruptible power supply to prevent data loss.

Contact Siemens Healthineers for more information.

CAUTION

Image artifacts due to unperformed maintenance

Incorrect diagnosis basis

- Check that the image quality criteria described in the installation and maintenance manual are met before using the system.
- Maintenance work should be performed by trained technical personnel only.
- Purchasing a maintenance contract will keep the system in optimum condition.

4.1.1 Switching on a switched-off system

✓ No indicators are lit on the system ON/OFF console.

1 Press the system “ON” button.

The ON indicator is lit.

The system is powered up.

The system performs a self-test.

The imaging system is started.

2 Wait until the application program appears on the touch display.

Depending on configuration, the home screen or the login dialog is displayed.

Depending on country-specific regulations, a quality control procedure is necessary according to quality assurance protocol of your country.

3 In this case: Start the quality control procedure.

Follow the instructions in order to perform quality control.

The system is now ready for use.

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The detector needs a warm-up time between 60 and 120 minutes to achieve good image quality, depending on environmental temperature and switch-off time.

During warm-up time acquisitions are possible but reduced image quality may occur.

(This is the case if the system has been turned off completely via the on-site main switch and has not been in standby mode. If it has been turned off on the system ON/OFF console, no warm-up time is necessary.)

If the battery of the detector is completely empty:

◆ Charge the detector battery and wait at least 60 minutes.

In case of a fault:

◆ Shut down the system and turn it on again.

If the fault remains:

◆ Call Customer Service.

Power failure or emergency SHUTDOWN switch has been activated.

The whole system has been cut-off from the power supply.

The last image might be lost.

◆ After power recovery or removing the reason for activating the emergency SHUTDOWN switch, switch on the system.
◆ Check the images of the last examined patient.

If a fault is indicated when the system starts up again:

◆ Call Customer Service.

4.1.2 Switching off the system

CAUTION

Data not saved before shutdown

Loss of data

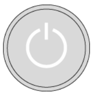
◆ Save data before switching user, shutting down or restarting the system.

◆ Shut down the computer before switching off the system.

1 Terminate the current examination or review.

2 On the Home screen or Lock screen, tap the **Shutdown** icon.

– or –



Press the system “OFF” button.

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If a patient is open for examination or postprocessing, a confirmation dialog will appear.

3 Tap **No**, check/close the active workflow, and invoke the shutdown again.

– or –

Tap **Yes** to close the active workflow and shut down the system.

If there are active background jobs, e.g. transfer jobs, a confirmation dialog will appear.

4 Let the background jobs finish.

The imaging system monitors the status of the transfer jobs. When the last job is completed, the system will be shut down. This may take some time.

– or –

Tap **Force Shutdown** to stop the active background jobs and shut down the system anyway.

The transfer jobs are aborted. The transfer jobs are kept in the list and can be restarted when the system is started up again. The system is shut down immediately.

The application program is terminated.

The operating system of the computer shuts down when finished (dark display).

The entire system including all connected devices is switched off.

Power off

Do not switch off the system using the on-site main switch **before** the operating system of the computer has been shut down.

◆ Wait until the imaging system has been shut down (display gets dark while being still switched-on).

Only then, you may switch off using the on-site main switch.

Emergency SHUTDOWN switch activation

◆ **If and only if** patients, operators, third parties, or the unit are in danger, press (activate) the on-site emergency SHUTDOWN switch.

4.1.3 Restarting the system

CAUTION

Long continuous computer runtime can lead to system errors.

Dysfunctional system, repetition of examination

◆ Restart the system once a day.

1 Terminate the current examination or review.

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2 Shut down the system.

(➔ Page 111 *Switching off the system*)

3 Wait until the imaging system has shut down (dark monitor).

4 Switch on the system by pressing the system "ON" button.

(➔ Page 110 *Switching on a switched-off system*)

5 Wait until the system is ready again.

Depending on configuration, the home screen or the login dialog is displayed.

4.1.4 UPS (for the imaging system)

UPS = Uninterruptible Power Supply

A UPS is a battery-backed system which provides emergency power in the event of a mains power supply failure.

To prevent data/image loss, your system may have a UPS which shuts down the **imaging system** in a controlled way in the event of a power failure.

WARNING

Inappropriate device connected to UPS

Risk of electrical hazard or damage to the system

- Do **not** connect any additional device to the UPS power outlets.

Functioning in the event of power failure

In the event of power failure, all indicators and displays are dark. Only the imaging system as well as the main monitor in the control room is still being powered by the UPS to end all active tasks, for example archiving of images.

1 Stop the examination.

2 Do not press any keys.

After a configurable time (typically 2 minutes), the imaging system shuts down to prevent the backup battery to be exhausted.

3 Wait for power to return or start emergency power.

4 Restart the system.

Description of UPS

The UPS has a five-button graphical LCD. It provides useful information about the UPS itself, load status, events, measurements and settings.

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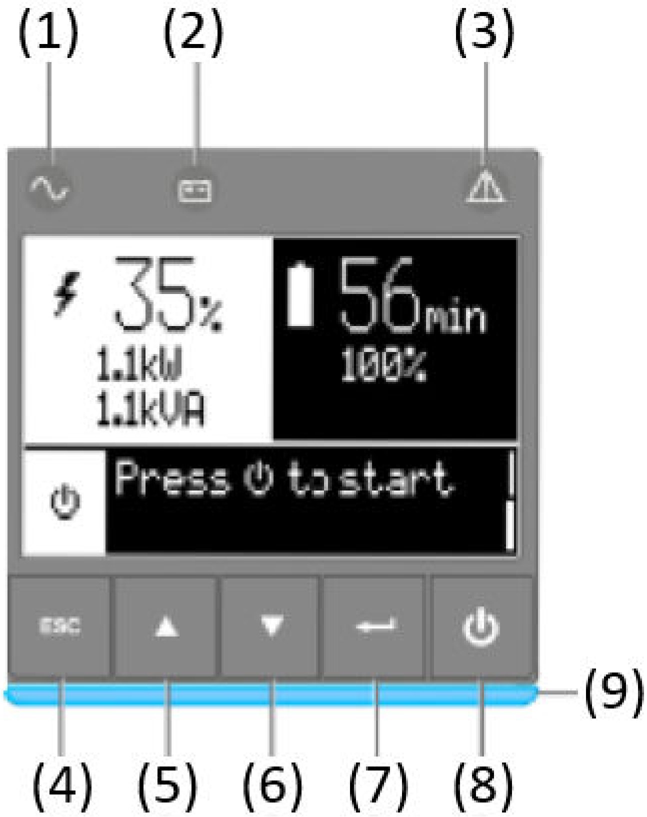
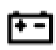


(1) Power ON indicator (green)
    The UPS is operating normally
(2) Battery ON indicator (yellow)
    The UPS is on battery mode
(3) Alarm indicator (red)
    The UPS has an active alarm or fault
(4) Escape
(5) Up
(6) Down
(7) Enter
(8) ON/OFF button
(9) Led bar (horizontally for rack / vertically for tower)

LED indicator

The LED bar (9) has been implemented to provide a quick visual reference of UPS status "at-a-glance".

The following table shows the indicator status and description :

Indicator	Status	Description
 green	On	The UPS is "ON" and the load is protected.
	On	The UPS is in battery mode and the load is protected.

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Indicator	Status	Description
Orange	Flashing	The battery voltage is below the warning level.
 Red	On	The UPS has an active alarm or fault. See troubleshooting page for additional information.
LED bar	Static blue	The UPS is "ON" and the load is protected.
	Flashing blue	The UPS is on battery or the battery service age warning is reached.
	Static red	The UPS has an active alarm or fault.
	Flashing red	The UPS output has stopped due to a fault.

UPS LCD description

(1) Load status and measurements
(2) Equipment status icon
(3) Status / Message
(4) Battery status

As default, or after 5 minutes of inactivity, the LCD displays the screen saver.

The backlight LCD automatically dims after 5 minutes of inactivity. Press any button to restore the screen.

If other indicator appears, see troubleshooting page for additional information.

The following table describes the status information provided by the UPS:

Operation status	Possible cause	Action
Standby mode 	The UPS is OFF, waiting for start-up command from user	System is not powered until button is pressed during start up and the green "normal mode" LED indicator is illuminated.

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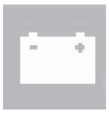
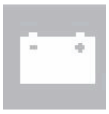
Operation status	Possible cause	Action
Normal mode 	The UPS is operating normally.	The UPS is powering and protecting the system.
On battery  1 beep every 10 seconds	A utility failure has occured and the UPS is in battery mode.	The UPS is powering the system with the battery power. Prepare your system for shutdown.
End of backup time  1 beep every 3 seconds	The UPS is in battery mode and the battery is running low.	This warning is approximate and the actual time to shutdown may vary significantly. Depending on the UPS load, the ”Battery low” warning may occur before the battery reaches 20% capacity remaining.
In AVR mode  No beep	The UPS is operating normally, but the system voltage is outside normal mode thresholds.	The UPS is powering the system through an Automatic Voltage Regulation device. The system is still protected normally.

UPS operation

Start-up and normal operation

Check that the indications on the name plate located on the back of the UPS meets to the AC power source and the true electrical consumption of the total load.

Battery charge

The UPS charges the battery as soon as it is connected to the AC outlet, whether the ON/OFF button is pressed or not. It is recommended that the UPS be permanently connected to the AC power supply to ensure the best possible autonomy.

On the first startup of the UPS, you will need to configure the output voltage and time of the UPS.

To start the UPS:

1 Verify that the UPS power cord is plugged in.

The UPS front panel display illuminates and shows Eaton logo.

2 Verify that the UPS status screen shows  .

3 Press the button on the UPS front panel for at least 2 seconds.

4 Check the UPS front panel display for active alarms or notices. Resolve any active alarms before continuing.

(➔ Page 118 *Troubleshooting*)

5 Verify that the indicator illuminates solid, indicating that the UPS is operating normally and any loads are powered and protected.

The UPS should be in normal mode.

If the indicator is on, do not proceed until all alarms are clear.

- Check the UPS status from the front panel to view the active alarms.
- Correct the alarms and restart if necessary.

Starting the UPS on battery

✓ Before using this feature, the UPS must have been powered by utility power with output enabled at least once.

1 When the UPS is disconnected from the AC power source, press the button on the UPS front panel.

The UPS transfers from Standby mode to Battery mode.

The indicator illuminates solid.

The UPS supplies power to your system.

2 Check the UPS front panel display for active alarms or notices besides the “Battery mode” notice and notices that indicate missing utility power.

3 Resolve any active alarms before continuing.

(➔ Page 118 *Troubleshooting*)

4 Check the UPS status from the front panel to view the active alarms.

5 Correct the alarms and restart if necessary.

UPS shutdown

To shut down the UPS:

- Press the button on the UPS front panel for 2 seconds.

  A confirmation message appears.

  When confirmed, the UPS starts beeping and shows the status “**UPS shutting OFF...**”

  The UPS then transfers to Standby mode and the indicator turns off.

Operation on battery power

Transfer to battery power

- The connected devices continue to be supplied by the UPS when AC input power is no longer available.

  The necessary energy is provided by the battery.

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- The and indicators illuminate solid.
- The audio alarm beeps every 10 seconds.

The connected devices are supplied by the battery.

Low-battery warning

- The and indicators illuminate solid.
- The audio alarm beeps every 3 seconds.

The remaining battery power is low.

Shut down all applications on the connected system because automatic UPS shutdown is imminent.

End of battery backup time

- LCD displays “**End of backup time**”.
- All LEDs go OFF.
- The audio alarm stops.

Return of AC input power

Following an outage, the UPS restarts automatically when AC input power returns and the load is supplied again.

Troubleshooting

The Eaton 5P Gen2 is designed for reliable, autonomous operation while providing you with notifications and alerts whenever a potential operational or performance issue occurs.

Usually the alarms shown by the control panel do not mean that the output power is affected. Instead, they are preventive alarms intended to alert the user.

- Events are silent status information that are recorded into the Event log. Example = "AC freq in range".
- Alarms are recorded into the Event log and displayed on the LCD status screen with the logo blinking. Some alarms may be announced by a beep every 3 seconds. Example = "Battery low".
- Faults are announced by a continuous beep and red LED, recorded into the Fault log and displayed on the LCD with a specific message box. Example = Out. short circuit.

Use the following troubleshooting chart to determine the UPS alarm condition.

Typical alarms and faults

To check the Event log or Fault log:

1. Press any button on the front panel display to activate the menu options.
2. Press the down button to select Event log or Fault log
3. Scroll through the listed events or faults.

The following table describes typical conditions:

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Conditions	Possible causes	Action
Battery mode LED is on. 1 beep every 10 seconds	A utility failure has occurred and the UPS is in battery mode.	The UPS is powering the equipment with battery power. Prepare your equipment for shutdown.
Battery low LED is on. 1 beep every 3 seconds	The UPS is in Battery mode and the battery is running low.	This warning is approximate, and the actual time to shutdown may vary significantly. Depending on the UPS load and number of Extended Battery Modules (EBMs), the "Battery Low" warning may occur before the batteries reach 20% capacity.
No battery LED is on. Beep continuous	The batteries are disconnected.	Verify that all batteries are properly connected. If the condition persists, contact your service representative.
Battery fault LED is on. Beep continuous	The battery test is failed due to bad or disconnected batteries.	Verify that all batteries are properly connected. If the condition persists, contact your service representative.
The UPS does not provide the expected backup time.	The batteries need charging or service.	Apply utility power for 48 hours to charge the batteries. If the condition persists, contact your service representative.
Power overload LED is on.	Power requirements exceed the UPS capacity (greater than 100% of nominal).	Remove some of the equipment from the UPS. The UPS continues to operate, but may shut down if the load increases. The alarm resets when the condition becomes inactive.
UPS overtemperature LED is on. 1 beep every 3 seconds	The UPS internal temperature is too high or a fan has failed. At the warning level, the UPS generates the alarm but remains in the current operating state. If the temperature rises another 10°C, the UPS shuts down.	Clear vents and remove any heat sources. Allow the UPS to cool. Ensure the airflow around the UPS is not restricted. Restart the UPS. If the condition continues to persist, contact your service representative.
The UPS does not start	The input source is not connected correctly.	Check the input and battery connections.

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Conditions	Possible causes	Action
	The Remote Power Off (RPO) switch is active or the RPO connector is missing.	If the UPS Status menu displays the "Remote Power Off" notice, deactivate the RPO input.

Silencing the alarm

Press the ESC (Escape) button on the front panel display to silence the alarm. Check the alarm condition and perform the applicable action to resolve the condition. If the alarm status changes, the alarm beeps again, overriding the previous alarm silencing.

4.1.5 Standby power supply (inside the hospital)

No radiation is possible in standby operation.

The message “**Standby power supply active**” is displayed.

A standby power supply (inside the hospital) switches on if the line voltage fails. However, this does not always happen without interruption and can take some time.

The system is switched off and must be switched on anew.

4.2 Functional and safety check

4.2.1 Daily tests

After system switch-on

- Perform a visual inspection of all displays and signal lamps on the operating units.

  No errors must be indicated.

Damage check

- Check system and accessories daily for damaged or loose parts.

  If problems are found, the parts must be repaired before resuming patient examinations.

Radiation indicators

- Release radiation.

  All radiation ON indicators which may be present must light up.

Collimation

- Check collimation during fluoroscopy.

At least one edge of the collimator leaves is visible in the fluoro image.

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Prior to examination

1. Remove unnecessary objects and equipment from the area of action of the system.
2. Remove unnecessary accessories from the accessory rails of the table and the primary collimator.
3. Attach the devices required for patient positioning and immobilization securely to the system.
4. Attach all safety-related accessory parts correctly (for example, footboard, grip protection strip, hand grip, hand grip strip) and make sure that they are properly secured.
5. Clean any contrast agent residues from the patient table, protective plate, and spotfilm device cover.
6. Recommendation: As a test, take a full-format exposure using the largest possible format.
7. Run a functional test of the emergency STOP buttons by performing an arbitrary system movement and pressing the emergency STOP button during the movement.

   The movement concerned must stop immediately.

Check functionality of emergency STOP daily.

In case of any damage of the emergency STOP button, the system must not be operated any further.

Please contact Siemens Healthineers service immediately.

---

8. Then unlock the button turning it clockwise.
9. Run a functional test of the proximity switches (collision protection) by performing an arbitrary system movement and pressing a proximity switch during the movement.

   The movement concerned must stop immediately.
10. Release the proximity switch.

    All system movements are possible again.

Overhead support

WARNING

Abnormal sinking of the overhead column

Injury of persons

- Perform maintenance at regular intervals (at least every 24 months) to maintain the safety and proper functioning of the product.

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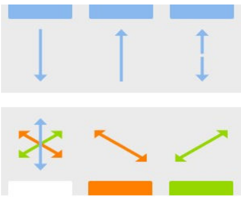
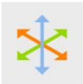


1 Check manual longitudinal, transverse and vertical movements of the overhead support.



2 Check 'floating' overhead support movement.

3 Check tube unit rotation.

4 Check motor driven overhead support height adjustment.

5 Observe the indications in the display.

See also (➔ Page 135 *Positioning the overhead support with tube unit*).

Wall stand

Verify up and down movement of the detector/cassette unit: (only in combination with ceiling-suspended X-ray tube)

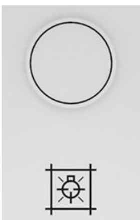
- Press the up and down buttons at the detector unit.

  See also Detector unit movements

Checking the light localizer and centering/collimation

Check the center position of radiation unit and Bucky wall stand with the light localizer and check the collimation.

Light on



- Press this button at the multileaf collimator.

  The light beam for collimation appears (time limited).

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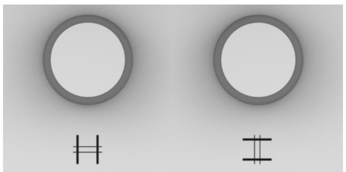
Centering

The accuracy of the stop positions of the tube unit can vary +/- 3 degrees due to hard wear. Lateral image information may be cut off.

Always check the alignment of the tube unit to the detector with the light localizer.

- Check the alignment of the tube unit to the detector with the light localizer.

Collimation

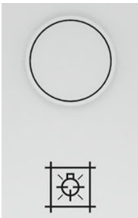


- Open and close diaphragm.

  The light beam changes its height or width.

Checking the automatic collimation (ACSS): When **ACSS** is displayed on the tube UI, the automatic collimation is active.

Light off



- Press the button again if light is on.

  The light beam is turned off.

Function control of automatic collimation

1. Switch on the full-field light localizer.
2. Switch over to another zoom format.
3. The collimator settings should adjust to the new zoom format.
4. If this does not work, please call the Customer Service.

4.2.2 During examination

1. Check the radiation ON indicator.

   It should light up only if one of the fluoroscopy switches has been pressed or for the duration of an X-ray exposure.

   CAUTION

   X-ray exposure unintentionally released or radiation does not stop.

   Undesired radiation exposure

   - In case of unwanted radiation, press the nearest emergency STOP button.

2. Check the patient positioning devices, for example the hand grips.
3. Initiate system movements only under the following conditions:

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There is no danger of injury to the patient or third parties.

There are no objects blocking the path of system movements.

4.2.3 Monthly tests

Functional check of the automatic systems

Run a functional test of the automatic dose rate control.

Automatic exposure control - RAD mode

(If the values obtained in the following tests deviate from those specified, switch the system off and contact Customer Service immediately.)

1. Select the **RAD Standard** clinical protocol step.
2. Close the collimator on the tube assembly to a format of 10 cm x 10 cm.
3. Place a lead rubber apron, folded four times, in the beam path as a cover.
4. Release an exposure and keep the exposure switch pressed down.

   The automatic exposure control automatically switches off when the limit is reached. A message appears: "Acquisition abort on limit excess".
5. Release the exposure switch.
6. Open the collimator on the tube assembly and remove the lead rubber apron.
7. Press the exposure switch.

   The automatic exposure control switches off automatically at < 1 mAs.

Checks concerning radiology

1. IQAP check at delivery
2. Maintenance every 24 months
3. Daily image check by customer, for example, for artifacts

Dose indication check

1. Adjust the SID to 90cm.
2. Set the maximum collimation size.
3. Put an external dose rate measurement device 1cm above the tabletop, place the sensor of the measuring device towards X-ray tube.
4. Use any clinical protocol in fluoroscopy and release radiation until the system and the measuring device deliver stable dose rate.
5. Compare the system displayed dose rate and the dose rate measured from external measuring device. The deviation shall not be bigger than 35%.

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4.3 System settings

WARNING

Movement without intentional activation of control elements.

Risk of collision, risk of injury to patient or operator, risk of damage to the system.

- Do not load the remote console with any objects, accessories, folders or documents.
- If movement does not stop, press the nearest emergency STOP button.

4.3.1 Park position and movement override

To prevent a possible collision, system movements and radiation release on the LUMINOS Q.namix R examination unit are only possible when the ceiling-suspended X-ray tube is in park position.

The park position is indicated on the remote UI.

Moving to park position

A system movement can be enabled by simultaneously pressing the **Park override** key in addition to the corresponding operating element.

1. Press and hold the **Park override** key.
2. Perform the required movements.

Overriding blocked movements

In case of a position sensor inconsistency, all movements are blocked. It is possible to perform movements with slow speed mode by simultaneously pressing the **Park override** key in addition to the corresponding operating element.

WARNING

Collision monitoring is inactive due to selected override function

Collision of patient and device with ceiling-suspended X-ray tube or other parts of the room

- Actuate system movements with override only with direct visual control.

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1 Press and hold the **Park override** key.

2 Perform the required movements.

4.3.2 Operating elements for system positions

Dead man’s function

All operating elements for the system movements are designed as a deadman function. That is, system movements are active only as long as the operating element is actuated.

LEDs

Some operating elements have an LED showing the status of the selected function. The general statuses are the following:

LED lights up	The function is selected or the system is in the the indicated position.
LED does not light up	The function is not selected yet.
LED flashes	The selected function is not yet concluded (see relevant chapter for description).

Start positions

Automatically determined system movements are performed with the SmartMove button depending upon the selected operating mode.

- LED flashes: The approved system movements are not yet concluded.
- LED does not light up: The approved system movements are concluded.

General notes on error messages

The system is constantly monitored internally during operation.

When a fault is detected, the system is blocked. The fault is displayed as an error in the message line of the TUI and stored.

Canceling blockage

1 Select the button on the error message on the imaging system or the TUI.

Normally, the fault message is canceled by this action.

However, if an error message is still displayed, there is an error.

2 Call the Siemens Healthineers Customer Service in this case.

Further operation of the system may be made only under increased attention of the user.

System movements in case of a fault

If there is a fault during some movements, they are stopped and the fault is displayed as an error in the message line of the TUI and stored.

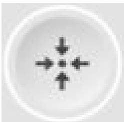
- Tilting the table up or down
- Moving the tabletop or the X-ray system longitudinally
- Moving the table transversely.

If a fault occurs again when the same movement is actuated, then this movement stops again. The fault is displayed as an error in the message line of the TUI and stored.

Canceling the fault

1 Briefly press the exposure release button on the system remote control console up to the first pressure point.

Normally, the fault is canceled by this action.



2 Press the SmartMove button

Error when actuating the same movement again

If the same movement is actuated again and there is a fault once again, then a warning signal sounds and this movement is blocked. A message is displayed in the message line of the remote UI.

All other movements are possible. A warning signal sounds on actuating and during the movement.

WARNING

System movements cause collision

Injury of persons or system damage

◆ Perform system movements only when it is certain that neither the operator, the patient, third parties nor pieces of equipment can be endangered by these movements.

◆ Make sure you are standing outside the collision zone.

◆ Always watch the patient when performing system movements.

1 Rescue the patient.

2 Call Siemens Healthineers Customer Service.

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4.3.3 Preparation and movements of the wall stand

Safety information

Information about stabilization aids

CAUTION

Patient exerts too much weight on the patient hand grip on the wall stand.

Risk of injury to the patient and damage to the wall stand

- Do not let the patient hang on the patient hand grip.
- Do not exert more than 25 kg weight on the patient hand grips.

Information about system movements

WARNING

Broken drive chain in wall stand

Risks of injury when moving front unit up or down

- Do **not** activate vertical movements when a broken drive chain is indicated on the user interface.

When the patient uses the stretch grips during exposures onto the wall stand, take care to ensure that the detector unit is fixed in the 0° position.

Detector unit movements

CAUTION

Patient overloads wall stand detector

Detector damage or injury of the patient.

- Do not allow the patient to sit on the detector unit or to push down on it.

Vertical movement (height adjustment) of the detector unit

The detector unit can be moved vertically using the respective buttons at either side of the detector housing.

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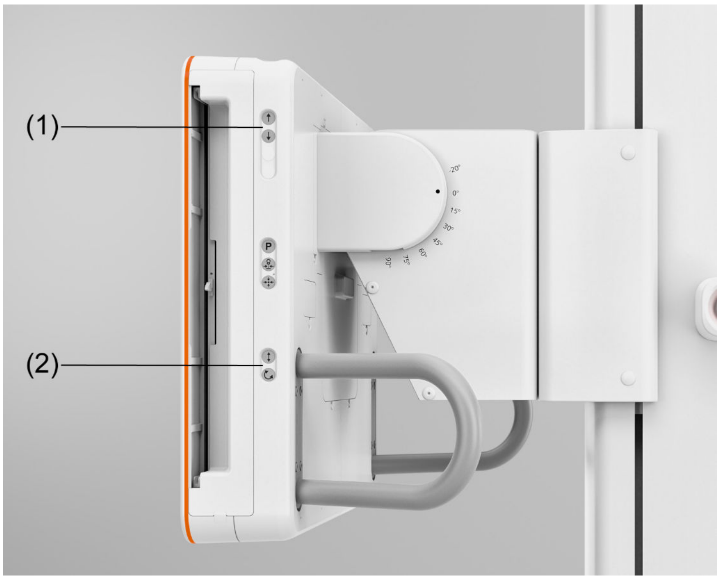
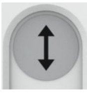


(1) Buttons for height adjustment
(2) Buttons for manual height adjustment and tilting



- Press the button for the manual vertical movement and move the detector unit up or down to the required working height.

  – or –

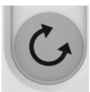
  Press one of the buttons for the motorized vertical movement to move the detector up and down to the required working height.

Downward movement stops at 12 cm distance to floor.

You can continue movement manually.

Tilting the detector unit

The detector unit can be **continuously** tilted to the required position between 0°/vertical and +90°/horizontal or up to -20°.



- Press the button for unlocking the tilt mechanism and tilt the detector tray to the required position.

Inserting the detector

(with mobile detectors only)

1 Move the detector unit to a convenient working height.

2 Bring the detector unit in 0° position.

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Example: Tray for X.wi-D 43

3 Press the brake handle (A) and pull out the detector tray (B) until it reaches the stop position.

Example: X.wi-D 43

4 Insert the detector from above into the holders (C) with the centering cross facing toward the front.

For the correct positions, refer to the labels on the tray.

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5 Press the brake handle.

6 Carefully and slowly slide in the detector tray with the detector.

Be careful not to damage the detector when sliding it into the tray as the tray does not have a mechanism to absorb the shock.

MAX wi-D and X.wi-D 35 can be inserted into the detector tray in portrait or landscape format .

Removing the detector

(with mobile detectors only)

1 Move the detector unit to a convenient working height.

2 Bring the Bucky unit in 0° position.

3 Use the brake handle (A) to pull out the detector tray with the detector.

4 Remove the detector pulling it upwards.

5 Carefully and slowly insert the detector tray again.

Removing and inserting a grid

The responsibility for selecting and inserting the scattered radiation grid to be used for an exposure lies with the examiner.

Removing the grid

1 Move the detector unit to the 0° position.

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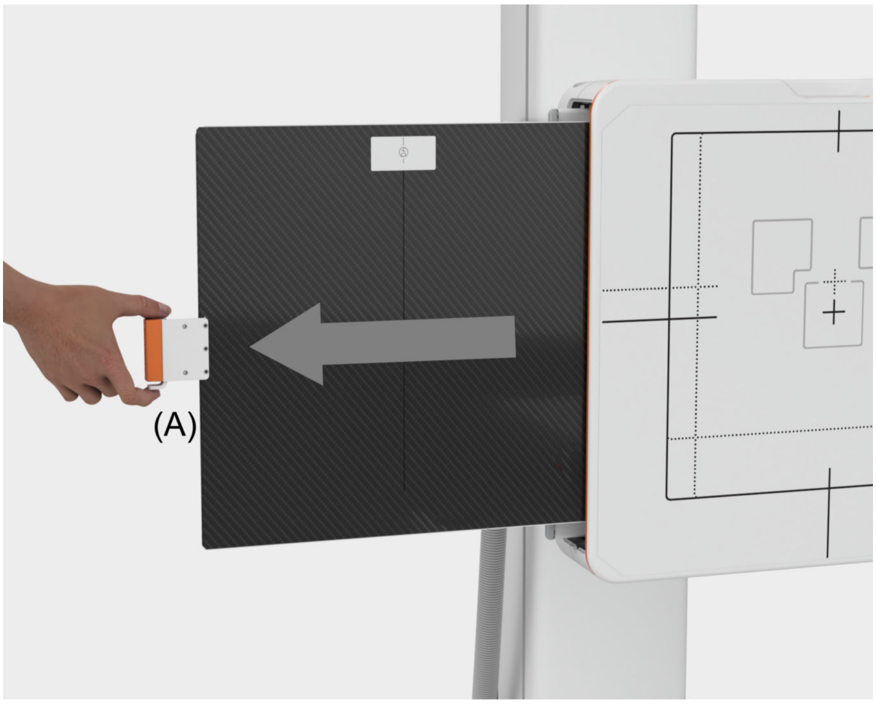


Example: with wireless mobile detector

2 Press the locking lever (A) and, keeping it straight, pull the grid out towards the side.

3 Remove the grid **with both hands**.

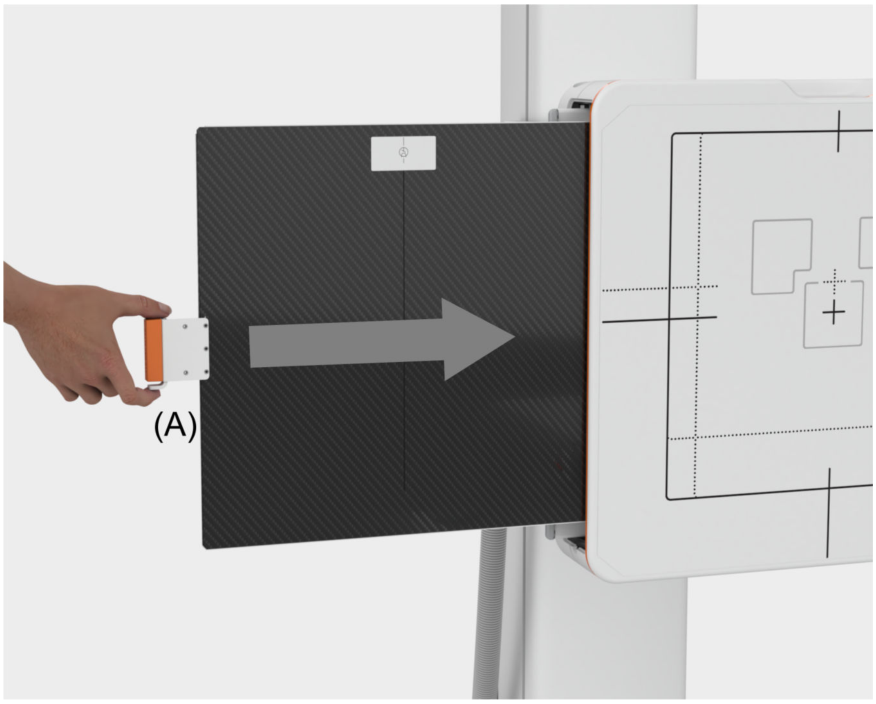
Inserting the grid

When inserting the scattered radiation grid, make sure that the **grid** being inserted is correctly assigned to the intended **source-image distance** (SID) on which the grid is focused.

1 Move the detector unit to the 0° position.

2 Hold the grid vertically, labels facing the tube unit.

The label with the tube symbol on the grid must **point toward** the tube unit.

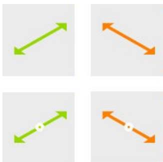


3 Place the grid against the top and bottom guide rail.

4 Holding it vertically and straight, push the grid all the way in.

When the grid has reached the end position, the locking lever engages.

Positioning the tube unit manually



1 Position the tube unit manually using the buttons for longitudinal and transverse movement on the tube UI.

When the tube unit has reached the correct longitudinal and transverse centering detents, the centering displays (white ring) at the movement buttons light up.

2 Press the centering button on the left or right side of the detector device.

– or –

Move the tube unit to the vertical center position with the help of the light localizer.

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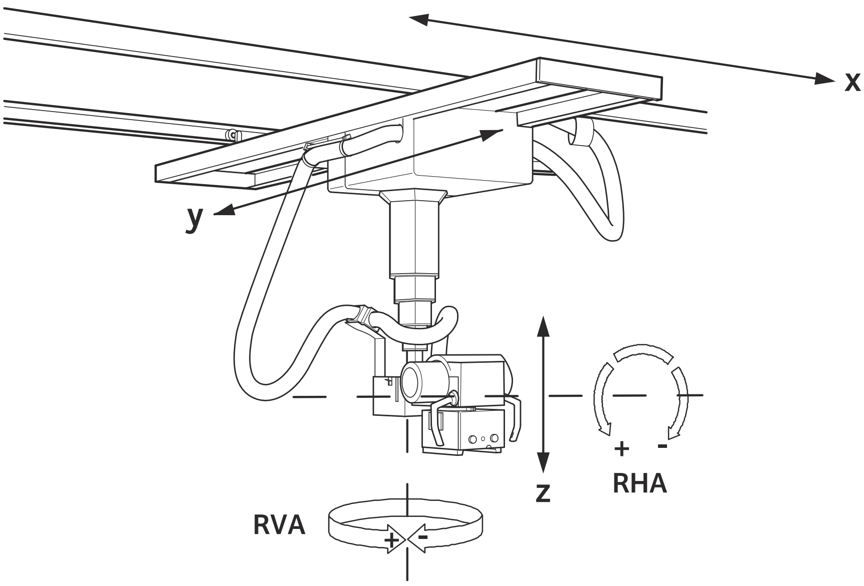
4.3.4 Preparation and movements of the overhead support

Directions for moving and rotating the tube



Directions for moving

The overhead support can be moved in the following directions:

- x direction: longitudinal movement
- y direction: transverse movement
- z direction: height movement

Directions for rotating

The X-ray tube unit (with multileaf collimator) can be rotated manually about two axes:

- about the horizontal axis (RHA = rotation horizontal axis)
- about the vertical axis (RVA = rotation vertical axis)

Safety information for overhead support

WARNING

Damaged cables inside the suspension of the overhead tube.

Overhead tube can fall down.

Risk of patient or user injury

◆ If the overhead tube cannot be moved vertically, do not force the movement.
◆ Call Siemens Healthineers Customer Service.

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CAUTION

Overload of ceiling-mounted devices.

Parts could be damaged or fall down, injury of persons

- Do not allow the patient to hang onto ceiling-mounted devices.
- Do not hang any additional load (such as lead aprons) on the devices.

CAUTION

In extreme operating mode, the X-ray tube assembly can heat up.

Injury of persons, burns

- Avoid contact with the tube housing to avoid burns.

Positioning the overhead support with tube unit

The temperature of the tube surface during normal use reaches 50°C.

In case of high energy radiation the temperature may exceed the 50°C.

CAUTION

In extreme operating mode, the X-ray tube assembly can heat up.

Injury of persons, burns

- Avoid contact with the tube housing to avoid burns.

Adjusting height manually (z direction)

1 Grasp the handles of the overhead support with both hands

2 Press the button to release the brake for vertical movement.

Releasing the button will stop the movement.

3 Pull the overhead support into the required position.

Adjusting height motor driven (z direction)

- Press one of the buttons on the overhead support control panel for up or down movement.

Adjusting longitudinal position manually (x direction)

1 Grasp the handles of the overhead support with both hands

2 Press the button to release the brake for longitudinal movement.

Releasing the button will stop the movement.

3 Pull the overhead support into the required position.

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Stops are provided in the longitudinal and transverse direction of the overhead support, i.e. the support can be moved only by applying increased force or by releasing the button and pressing it again to go through certain positions (e.g. table center in longitudinal and transverse direction or defined SID values in longitudinal direction).

Adjusting transverse position manually (y direction)

1 Grasp the handles of the overhead support with both hands

2 Press the button to release the brake for transverse movement.

Releasing the button will stop the movement.

3 Pull the overhead support into the required position.

Stops are provided in the longitudinal and transverse direction of the overhead support, i.e. the support can be moved only by applying increased force or by releasing the button and pressing it again to go through certain positions (e.g. table center in longitudinal and transverse direction or defined SID values in longitudinal direction).

“Floating” support movement manually

- If you want to move the support in longitudinal, transverse and vertical direction simultaneously, press one of the buttons on the handles or the button on the support control panel for ’floating’ support movement.

Rotating the tube unit manually (RHA)

1 Grasp the handles of the overhead support with both hands.

2 Press the button for the brake for rotating the tube unit about the horizontal (support arm) axis.

3 Turn the tube unit into the required position.

The angle of rotation is displayed in degrees on the **Positioning > System Position** tab card on the tube UI.

The tube unit can be rotated about the horizontal axis continuously, at a maximum by ±137° (fully automatic) or ±140° (semi). There are stop positions at 0° and at ±90°.

The accuracy of the stop positions of the tube unit might decrease to ±3°, due to hard wear for example. Lateral image information might be “cut”.

Always check the alignment of the tube unit with the light localizer.

Rotating the tube unit manually (RVA)

The tube unit can be rotated about the vertical axis from +182° to -154° with stop positions at +180°, +90°, 0° and -90° (see drawing).

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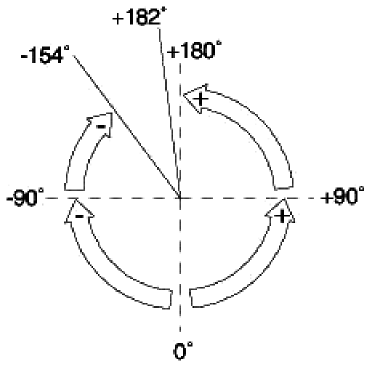

Rotation of the tube unit about the vertical axis (view from above: from the ceiling to the floor)

1 Grasp the handles of the overhead support with both hands.

2 Press the button to release the brake for rotating the tube unit about the vertical (overhead support) axis.

3 Turn the tube unit into the required position.

The angle of rotation is displayed in degrees on the **Positioning > System Position** tab card on the tube UI.

Turning the collimator

CAUTION

Hands between the collimator and the overhead column

Risk of injury

◆ Grasp the collimator on the sides when turning it so that there is no risk of crushing your hands.

CAUTION

Rotation of the collimator

Risk of injury of the hands and the fingers

◆ Always pay attention to your hands to ensure that they are not pinched or crushed between the collimator and surrounding objects.

Tube tracking in table acquisition mode

Prerequisites:

- System is in table acquisition mode and centered.
- Tracking is selected.

With rotation of the tube up to 45° in both directions, the detector inside the table follows the tube rotation; the central beam is locked on the detector.

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Tube tracking in wall acquisition mode

Prerequisites:

- System is in wall acquisition mode and centered.
- Tracking is selected.

When rotating the tube about the horizontal axis outside the 90° position, the tube support follows vertically; the central beam is locked on the detector.

This works in both directions up to 45°.

SID tracking in table acquisition mode

In addition to the tracking control perpendicular to the beam direction (in x direction), a further tracking control (in z direction) can be activated for keeping the SID (source-image distance) constant.

Activation of a tracking control

Touch the **Distance** tracking button on the tube UI.

CAUTION

Parts of the system move unexpectedly during automatic positioning or tracking.

Risk of injury to the patient or user

- Keep the patient within visual contact whenever performing any positioning functions.
- Use the emergency STOP button if motion needs to be stopped quickly.

Deactivation of a tracking control

If the state of the system is changed so that one or several of the conditions for active tracking control are not fulfilled, then the tracking control is automatically switched off.

SID display

The SID (source-image distance) is displayed on the tube UI (in cm or inch according to configuration).

The displayed SID is only accurate when ACSS is active.

Otherwise, the SID must be determined manually set with the tape measure.

4.3.5 Movement of X-ray system

The X-ray system consists of:

- Stand with tube assembly and primary collimator
- Image receptor unit.

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When moving the X-ray system towards the end position, please be aware that the table will start to move in the opposite direction once the X-ray system has reached the end position.

At the system remote control console

- Deflect the joystick up or down.

  The X-ray system moves headwards or footwards.

Use this operating element primarily during fluoroscopy.

The speed of the movement increases the further you deflect the joystick.

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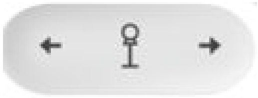
At the tableside control panel



- Press a key.

  The X-ray system moves headwards or footwards.

The movement takes place at constant speed.

4.3.6 Tabletop movements

CAUTION

Patient falls and gets fingers caught in moving parts.

Risk of crushing or falling

- Make sure that patient always uses the hand grips.
- If this is not possible, make sure that the patient's hands cannot get caught in the moving parts of the system.
- Do not allow the patient to grasp the sides of the table.

Longitudinal and transverse movement of tabletop

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Travel range

- In order to avoid collisions with the floor, the system will retract the tabletop automatically as needed if you tilt the table up or down. The X-ray system follows this movement synchronously if the available travel range is sufficient. The object always remains centered in this case.
- If you have tilted the table more than -30° down or +30° up, the system continuously calculates the maximum travel range of the tabletop. This calculated range depends upon the room height and the system position. You can move the tabletop only within this range.

At the system remote control console

1 Deflect the joystick to the left or right.

The tabletop moves transversely.

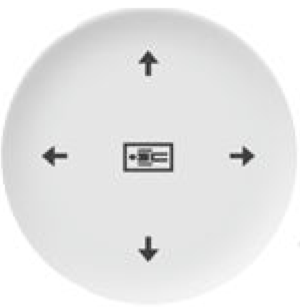
2 Deflect the joystick in y-direction up or down.

The X-ray system moves longitudinally.

3 Deflect the joystick in y-direction up or down with the thumb button pressed.

The table moves longitudinally.

At the tableside control panel



1 Press the left or right key on the coordinate switch.

(Display in horizontal table position)

The tabletop moves longitudinally.

2 Press the upper or lower key on the coordinate switch.

The tabletop moves transversely.

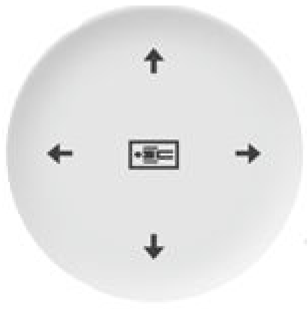
Complete lowering of tabletop in vertical position

An acoustic signal sounds during the tabletop movement.

You hear a warning tone if the tabletop is lowered to less than 12 cm above the floor.

Print No. XPD3-530G.620.01.04.02

At the tableside control panel



✓ Table in vertical position (+ 90° or - 90°).

1 Press the left or right button until the tabletop stops automatically at 12 cm from the floor.

(Display in horizontal table position)

2 Release the button and press it once again.

The tabletop moves to a minimum of 4 cm from the floor.

4.3.7 Movement of tube assembly stand

Oblique projection

The setting range of the projection angle is ± 45° at most.

The swivel can be adjusted in height by motor drive from 10 mm to 300 mm above the tabletop. Thus you can adjust it to the position of the object to allows isocentric viewing on the monitor.

The additional mechanical centering device ensures that the central beam is centered onto the image receptor for each projection angle.

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Setting of projection angle

1 Press the right tube angulation key for cranio-caudal projection.

2 Press the left tube angulation key for caudo-cranial projection.

If the tube assembly stand or the image receptor unit reach the end of their respective travel ranges, the movement will stop.

Performing further oblique projections

1 Press the key for oblique projection again.

The tube assembly stand or receptor unit moves to a position for additional projections.

The object centering is lost.

2 Reposition the object by moving the tabletop longitudinally provided that the range of travel allows such repositioning.

For fluoroscopy, position the isocenter for oblique projection at the object height.

The object centering on the monitor will remain the same without shift if you select oblique projection.

Oblique projection ±0°

Setting of orthogonal projection

- Press the upper tube angulation key.

  (➔ Page 82 *Tableside control panel*)

  If the central beam is orthogonal to the receptor unit, the LED lights up.

  The display for oblique projection indicates 0°.

Reading the projection angle

1 Read the angle in degrees in the status area of the remote control console TUI.

– or –

Read the angle in degrees on the system tableside control console.

2 Press this key at the TUI of the remote control console to disable the tube-detector coupling.

The LED lights up.

3 Press the left key for caudocranial projection.

Press the right key for craniocaudal projection.

If the tube reaches the end of its travel range, the movement will stop.

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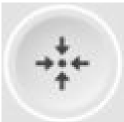
The setting range of the tube rotation is maximally -90°/+180° with stops every 90°.

In case of tilting the tube, a manual adjustment of the collimation may be necessary to avoid irradiation outside of the active area of the detector.

4.3.8 Automatic system positioning



The system can be moved automatically in the position which is defined in the clinical protocol step using the SmartMove button on the system remote control console or the tableside control panel.

4.3.9 Table movements

WARNING

System movements cause collision

Injury of persons or system damage

- Perform system movements only when it is certain that neither the operator, the patient, third parties nor pieces of equipment can be endangered by these movements.
- Make sure you are standing outside the collision zone.
- Always watch the patient when performing system movements.

0° position of table

- Tap **Override Softstop** to override the stop in the horizontal position when tilting the table.
  - Active: The table moves through the 0° position when tilting up or down. This can be useful for myelography examinations.

    Whend the tilting-down or tilting-up operating element is activated longer than 0.5 s, the tabletop lowers to the minimum table height.
  - Inactive: The table stops in the 0° position when tilting up or down

    The movement stops even if the tilting-down or tilting-up operating elements are kept activated in the 0° position.

    You can continue the movement beyond the 0° position by releasing the operating element and activating it again.

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Table tilt

The movement always stops automatically in the 0° position of the tabletop if the “Stop in 0° position” function is switched on (LED is not active).

At a Trendelenburg position >15°, use the provided shoulder supports or the foot restraints available as accessory to secure the patient. The weight of the patient must not exceed 300 kg.

At a Trendelenburg position >30°, you must use the foot restraints available as accessory in addition to the shoulder supports. The weight of the patient must not exceed 210 kg.

At a Trendelenburg position >45°, the weight of the patient must not exceed 150 kg.

WARNING

Table tilted in Trendelenburg (head-down) position

Patient slides off table head-first

- Use shoulder supports to prevent the patient from sliding off the table.
- If the table must be tilted more than 15° Trendelenburg (head-down), use the foot holder in addition to the shoulder supports.

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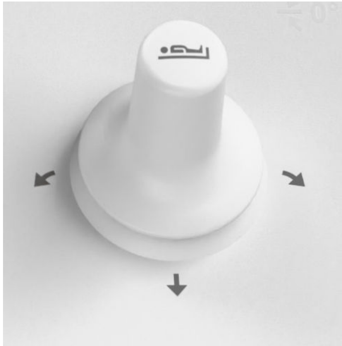
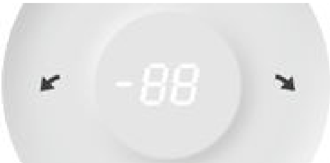
At the system remote control console



- Deflect the joystick to the left or right.

  The tabletop tilts down or up.

At the tableside control panel



- Press one of the keys.

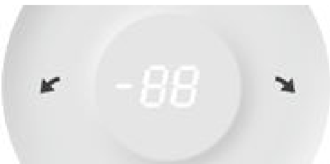
  The tabletop tilts down or up.

Reading off table tilt

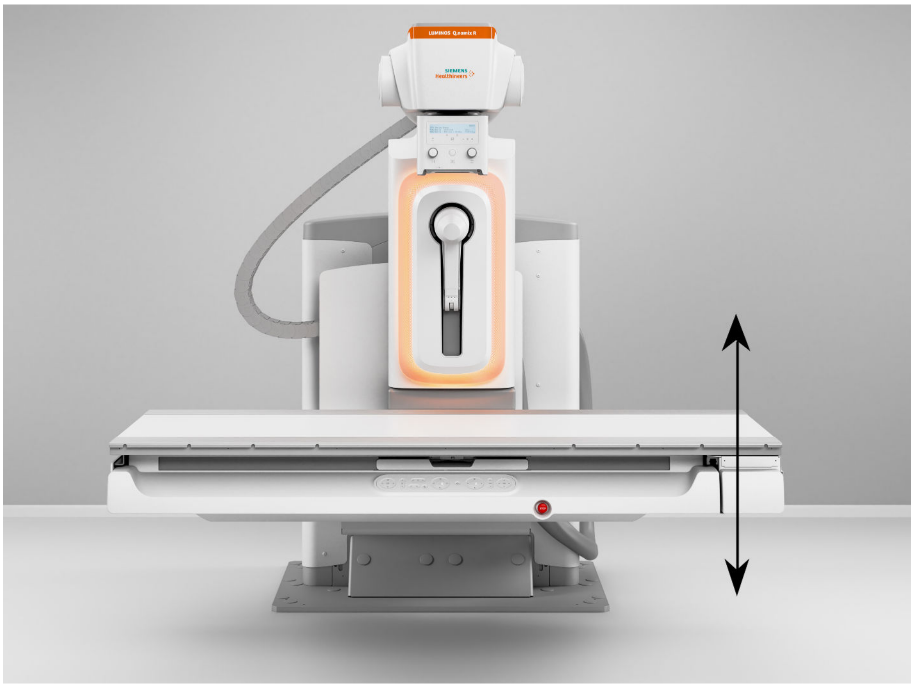
- Read off the tilt in degrees in the status area of the TUI.

  – or –

  Read off the tilt in degrees on the tableside control panel.



Vertical movement of table



Vertical movement of the table is possible only with the table horizontal or slightly inclined (16°/-16°).

WARNING

Lowering the table onto a seated patient

Crushing of patient or operator seated with legs under the table (for example, a wheelchair patient)

- Do not position a patient with legs under the table.
- Do not move the table while a patient is seated near the table. The table will NOT stop automatically.
- Move the patient away from the table first.

- Press one of the keys.

  The tabletop moves up or down.

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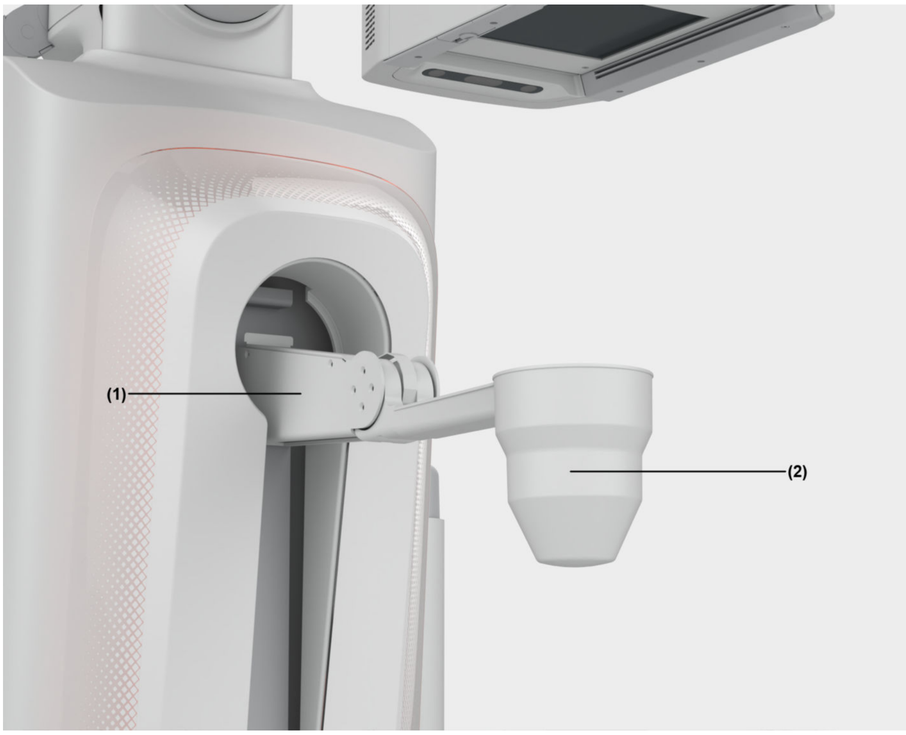
4.3.10 Compression device



(1) Compression device
(2) Compression cone

The LUMINOS Q.namix R can be equipped with a stationary mounted compression device with detachable compression cone.

Using the compression device

- You can apply a maximum compression force of 150 N or 80 N (configuration for Japan only), adjustable in 15 or 8 (configuration for Japan only) steps.
- Movements are no longer possible at a compression force of 50 N and higher unless the override button is selected.

  Override is possible as long as the **Compres. Cone** button is pressed.

  (➔ Page 95 *Movement tab*)
- The cone carriage moves automatically into the park position:
  - When setting an oblique projection angle greater than 20°.

CAUTION

Compressed area not in the field of view during oblique projection

Risk of crushing

◆ Be careful using compression during oblique projection.

1 Set an oblique projection angle of less than 20°.

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2 Touch the icon for compression and decompression (can be configured).

It is the full responsibility of the examiner to pay attention to the applied compression forces, especially in the case of frail, sick, and elderly patients.

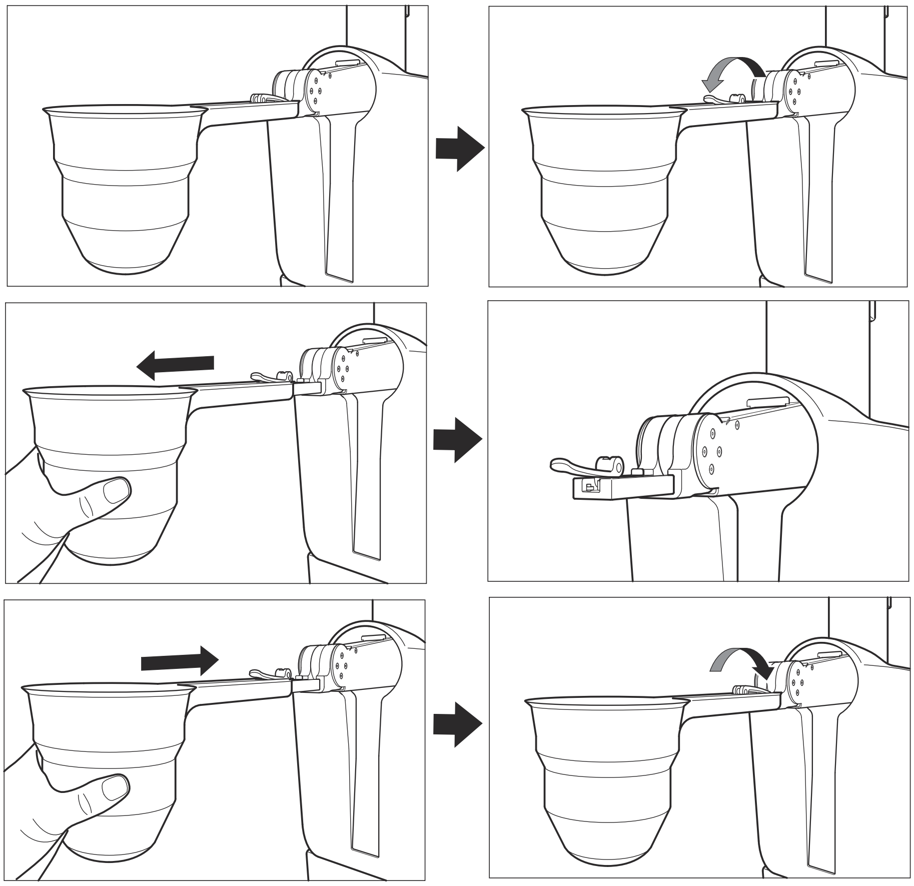
The compression level is indicated on the touchscreen.

At ± 90°, compression varies from displayed value.

For further information, refer to (➔ Page 175 *Positioning the patient on the table*).

For information about patient rescue in case of a power failure, refer to (➔ Page 38 *Patient rescue*).

Detaching the compression cone



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4.3.11 Preparation and handling of the wireless detectors

Safety information for wireless detectors

CAUTION

Improper or careless detector handling

Damage to the detector or detector surface

- Do not drop the detector.
- Only use the detector on a level surface to avoid sagging.
- Exercise absolute care in avoiding pressure or sudden impact on the detector surface.

(➔ Page 154 *Notes on handling the mobile detectors*)

CAUTION

Detector cover broken

Toxic contamination

- Handle portable detector with care!
- If the detector is broken do **not** use it.
- Inspect if any particles are spilled.
- Leaked particles can be toxic - do **not** touch them!
- Use protective gloves to collect and keep them in a sealed container.
- Return spilled particles to Siemens Healthineers.

CAUTION

Detector cover breaks because patient is too heavy

Possible injury of patient or personnel by carbon or glass splinters or electrical shock.

- Use the detector protector for patients heavier than 100 kg (maximum 250 kg).
- Handle the portable detector with care!
- When the detector is broken, do not use it.

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CAUTION

Portable detector is too hot

Burns or heat injuries

- Pay attention to the warning messages pertaining to detector surface temperature.
- The surface of the portable detector can reach 43°C (109°F), which may be too hot for high risk patients, such as infants.

  Protect such patients from exposure to the surface of the detector.
- Remove the detector from contact with the patient when the surface temperature exceeds 41°C (106°F).

CAUTION

Image cannot be made because the battery in the portable detector is low

Portable detector not available for use

- Connect the portable detector to a loading port when it is not in use.
- Consider purchasing the battery loading station and additional batteries if portable detector availability is important.

The X.wi-D detectors have a protection class IP67 incl. battery.

Detector protection

Do not spray the detector directly for cleaning and disinfection purposes.

Avoid contact of detector with iodine-containing or other strongly dyeing substances.

We recommend to use a protective cover during examination.

Do not install the detector close to facilities where water is used.

Do not spill liquid or chemicals onto the detector or, in cases where the patient is injured, allow it to become wet with blood or other body fluids, as doing so may result in fire or electric shock.

In such situation, protect the detector with a single use plastic bag.

Waterproof detector

Newer versions of the MAX detectors and also repaired/upgraded MAX detectors are provided with a higher degree of protection against liquids.

It is only valid **without** battery in the battery compartment!

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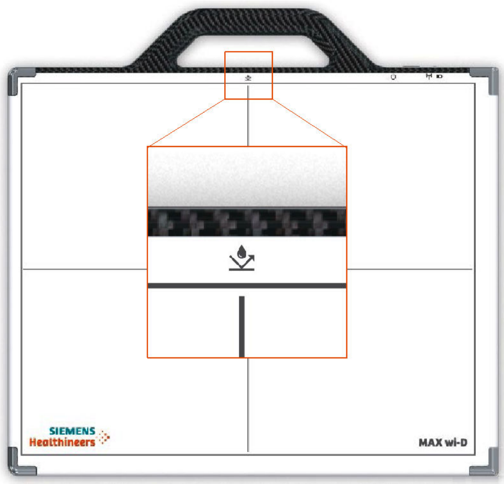
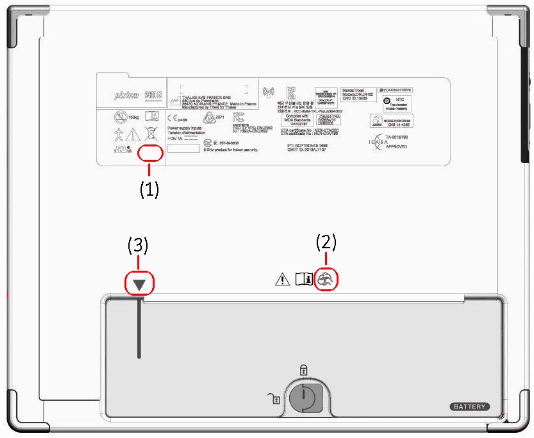
You can recognize these detectors as follows:



Additional symbol on the detector top side

Symbol indicates waterproof detector.



Additional symbols on the detector underside

(1) IP code moved to a different label
(2) Symbol for manual cleaning
(3) Triangle for correct battery position

Symbol indicates that manual cleaning is necessary.

CAUTION

The battery of the portable detector is not resistant to fluids.

Electrical shock, damage to the detector

- Remove the battery from the portable detector before cleaning the detector.
- Check that the detector is waterproof before immersing it (symbol on the detector).

  The battery can be wiped off but it is not waterproof.

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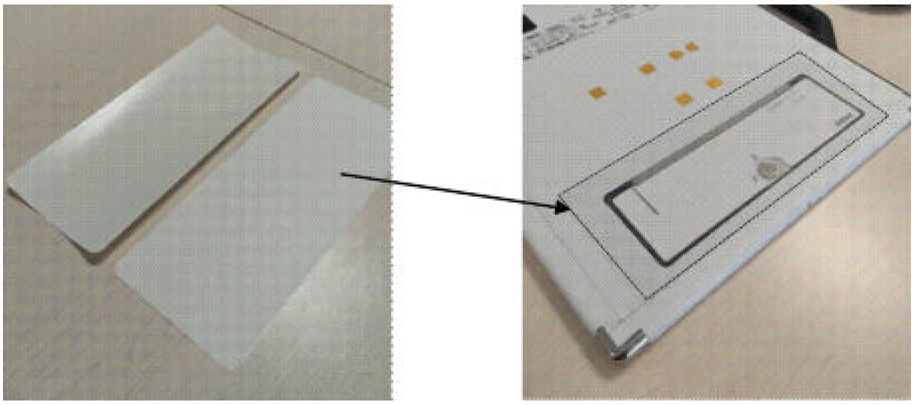
With every MAX wi-D detector, you receive re-usable adhesive protective films to protect the battery in the battery compartment against moisture during cleaning and disinfection.

Just apply the protective film over the battery in the battery compartment as shown in the following figure.



Interference with other devices

Portable and mobile RF communication equipment can affect the detector operation.

The portable wireless detectors may be interfered with by other equipment, including portable and fixed RF communication equipment, even if such equipment meets the applicable emissions requirements.

WARNING

As with many wireless components, the portable detector and the wireless footswitch may have electromagnetic fields that could disturb a life-supporting device such as a pacemaker or ventilator.

Interference with and possible malfunction of a life-supporting device

◆ Keep a safety separation distance of more than 30 cm between the wireless component and the life-supporting device, except for pacemakers. For pacemakers, a safety distance of minimum 3 cm is required.

◆ In case of interference with other equipment, increase the distance between the interfering devices.

Image transfer may be interrupted sporadically.

The operator must ensure that other wireless devices in the 2.4 GHz or 5 GHz ISM band should not be operated in the vicinity of the Radiography X-ray system.

Please observe and verify normal operation of the portable wireless detectors prior to using it.

Status and system messages

In the following situations, the battery will probably be empty or the voltage will be too low:

- The wireless detector has been accidentally left outside one of the charging locations for too long, for example over night.
- The wireless detector is outside the room.

Proceed as indicated in the system messages to solve the problem.

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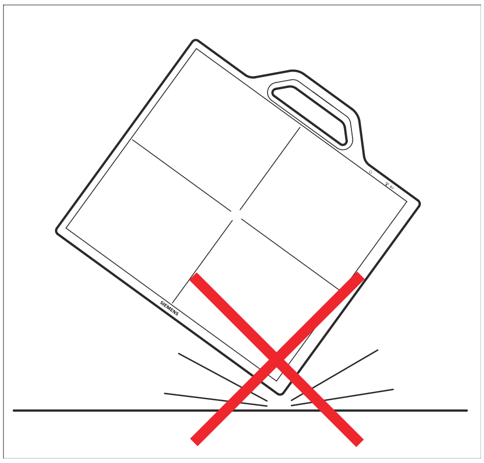
Maintenance

No other maintenance action is required except the calibration needed for image quality, whose frequency is defined by the system program.

There are no parts that need to be replaced due to limited lifetime, except for the battery. In case of malfunction, the detector will be returned to the manufacturer for repair.

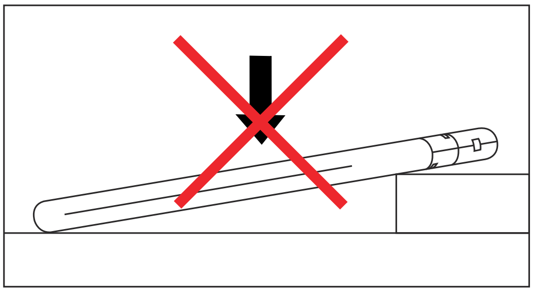
Notes on handling the mobile detectors

These notes are valid for all mobile detectors used with the system. The images shown in the following are samples only.

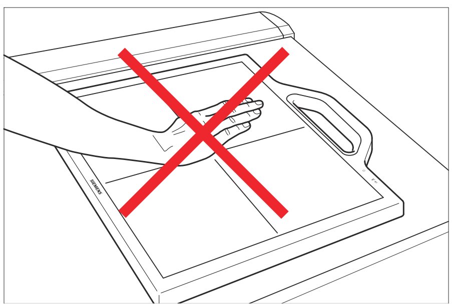


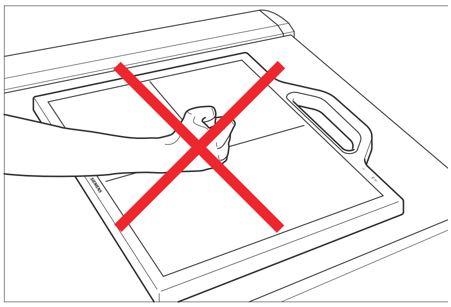
Handle the detector carefully, as it may be damaged if something is hit against it, if is dropped from high altitude, or if it is a strong jolt.

The detector is monitored by integrated sensors, so that data can be read out by a technician.



Be sure to use the detector unit on a flat place so it will not bend. Otherwise, the detector may be damaged.





Do **not** hit the detector surface in any way.

Detector temperature

If one of the following messages appears:

- "**Detector temperature below valid range**" or
- "**Detector temperature above valid range**"

Please check if the warm-up time from a cold start (120 minutes) is respected. If yes, we recommend a recalibration.

Warm-up time from a cold start

When the system and the detector have been switched off completely for a longer time, or when the detector battery became completely empty, image quality will be impaired.

Depending on ambient temperature, acquisition will be possible after 5 minutes with reduced image quality and after 120 minutes until full image quality without restrictions.

Avoid the detector battery getting completely empty.

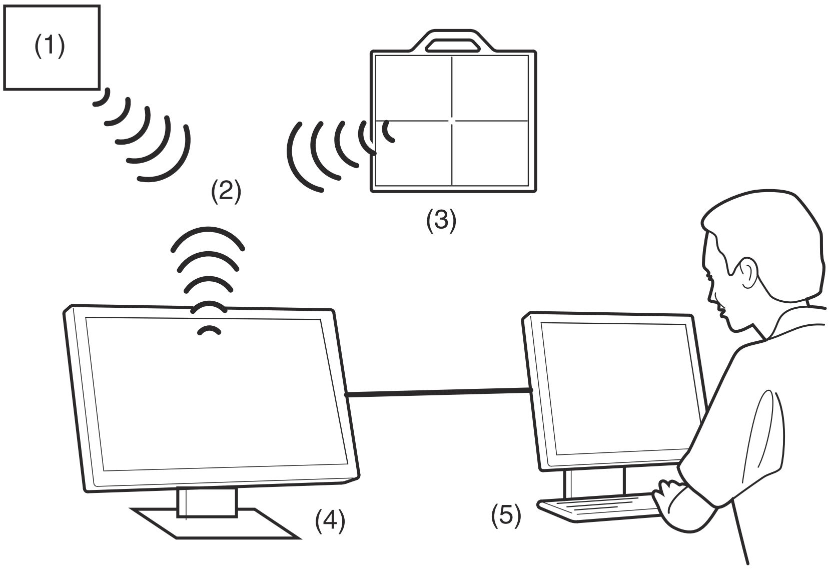
CAUTION

Detector temperature out of range.

X-ray without diagnostic value

◆ Check the detector temperature indication regularly.

Operation and setting of WLAN



(1) Base station (WLAN access point)
(2) Wireless connection
(3) Mobile detector
(4) Imaging system
(5) PACS

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The wireless connection of the mobile detector is established between the mobile detector and the imaging system.

Therefore, the phase of wireless connection and transfer does not utilize the existing wireless network that may be installed in the clinic. However, some precautions should be taken that both can coexist without interferences.

Properties of wireless connection

The system uses its own WLAN access point which is connected to the imaging system and the mobile detector as its sole client.

The communication is based on the WLAN standard 802.11. Within this standard, in most countries, there is the choice between the standards 11b, g, n (operating at 2,5 GHz) and the standards 11a, n (operation between 5 and 6 GHz).

Within these standards (11a, b, g, n), several channels (frequencies) are available. A specific channel (frequency) can be selected by the site for use.

However, the available channels depend on legal regulations and requirements of the country with the installed system (wireless detector). The wireless connection is encrypted. For this the widely used and very secure WPA2 is applied.

The standard (11a, g, n) and the channel can be set by a Siemens Healthineers service technician via the service software installed on the imaging system. For a list of available channels in your country please contact your Siemens Healthineers project manager or sales representative.

The detector utilizes a power dependent on the standard and channel of the typical order of 32-79mW (always below 100mW). As the emission is not purely isotropic, more power is emitted in some directions and less in other directions. The maximum power emitted in a direction (max. EIRP) is 150 mW. This is defined based on the assumption that power would be emitted in all directions.

Functional specifications

MAX detectors

Operating frequency	• 2.412 GHz up to 2.484 GHz • 5.180 GHz up to 5.320 GHz • 5.500 GHz up to 5.700 GHz • 5.725 GHz up to 5.825 GHz
Standard	IEEE 802.11a; IEEE 802.11b; IEEE 802.11g; IEEE 802.11n
Theoretical data rate	• 801.11a: 54 Mbps (OFDM) • 802.11b: 11 Mbps (CCK) • 802.11g: 54 Mbps (OFDM) • 802.11n: 65 Mbps (OFDM) • 802.11n: 260 Mbps (OFDM)
Modulation techniques	OFDM / CCK
Network architecture	Infrastructure mode

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X.wi-D detectors

Operating frequency	• 2.400 GHz up to 2.4835 GHz • 5.150 GHz up to 5.850 GHz
Standard	IEEE 802.11ax/ac/a/b/g/n (2T2R)
Theoretical data rate	• 802.11b: 11 Mbps • 802.11a/g: 54 Mbps • 802.11n: MCS0~15 • 802.11ac: MCS0~9 • 802.11ax: HE0~11
Modulation techniques	• 802.11b: DSSS (DBPSK, DQPSK, CCK) • 802.11g: OFDM (BPSK, QPSK, 16-QAM, 64-QAM) • 802.11n: OFDM (BPSK, QPSK, 16-QAM, 64-QAM) • 802.11a: OFDM (BPSK, QPSK, 16-QAM, 64-QAM) • 802.11ac: OFDM (BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM) • 802.11ax: OFDMA (BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM)
Network architecture	Infrastructure and ad hoc mode
Transmission	MIMO 2x2

EIRP (according to EN 300 328 (2006-10) and EN 301 893 (2011-11))

Frequency band	MAX detectors: MAX wi-D and MAX mini
2.4 GHz	12.28 dBm
5.18 GHz up to 5.24 GHz	10.73 dBm
5.26 GHz up to 5.32 GHz	10.85 dBm
5.50 GHz up to 5.70 GHz	11.06 dBm

Coexistence with other wireless connection

Wireless devices (e.g. your wireless hospital network and the mobile detectors) can influence each other if they are operated at the same frequency.

For the mobile detector, this can lead to a slowdown or disruption of the wireless connection.

To avoid such a situation, it is necessary to enquire about wireless devices that are used in or within the vicinity of the installed system. The contact person for enquiries regarding wireless network and devices inside your hospital is probably your network or IT expert.

In case a WLAN-network is present in your work environment, the channels of the access points can be detected in the room where the system will be installed. The frequencies in use (channels if you have WLAN) should be communicated to the Siemens Healthineers project manager.

Alternatively, select the channel to be utilized by the mobile detectors and inform the service technician/project manager of the channel (frequency) accordingly.

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It is important to ensure that this channel (frequency) will not be used at a later stage by other wireless devices.

Changing the battery (Max detectors)

CAUTION

Incorrect handling of detector battery

Explosion or fire

- Handle the battery pack carefully.
- Do not drop, heat, open or mishandle it.

CAUTION

Damaged detector batteries

Explosion, fire or burns

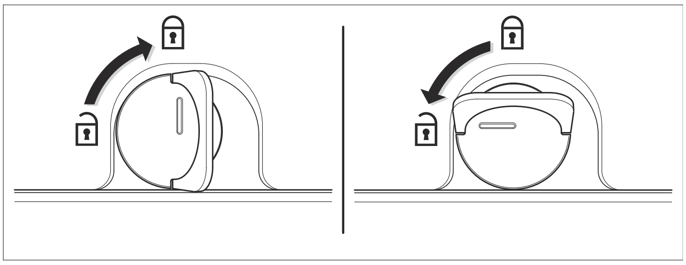
- Check detector batteries for cracks or other damage before using.
- Do not use damaged batteries and return them to Siemens Healthineers.

Do not remove the battery from the detector during image processing or image transfer. An image could be lost.

Before changing the battery, make sure that all data have been transferred.

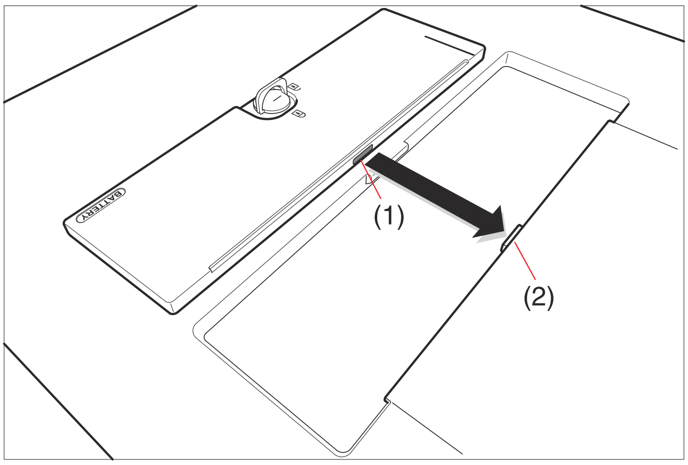
When you take out the battery, any data on the detector will be erased!

✓ Switch off the detector with a long push of the power button.

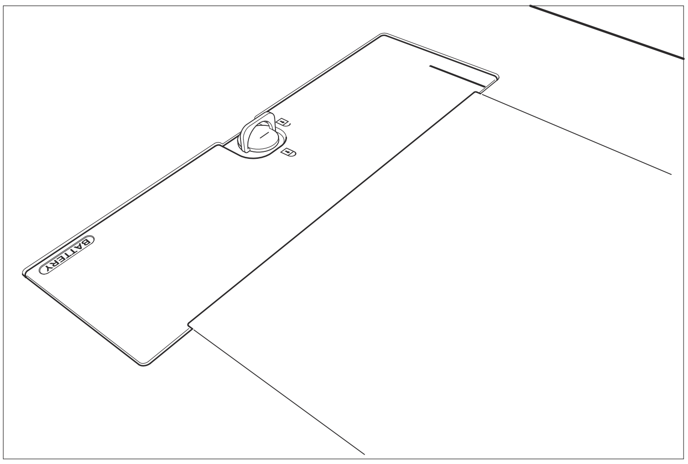


1 Open the locking system of the battery as shown in the figure.

2 Remove the discharged battery.

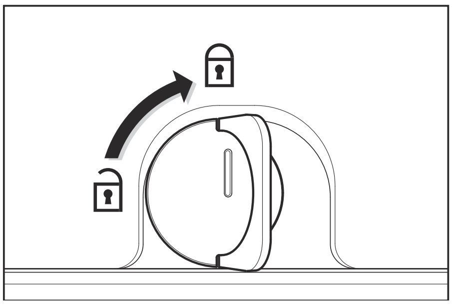


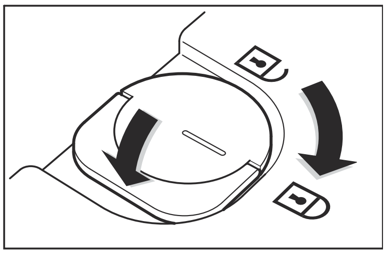
3 Orientate the charged battery in the correct position: battery connector (1) facing the device connector (2).



4 Slide battery in the recess area of the device.

5 Push down the battery in the recess area.





6 Close the locking system of the battery and push down the latch to ensure no protrusion of the locking system.

When the battery is inserted correctly, the detector switches on automatically.

7 Before using the detector, wait until all LEDs are green.

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Changing the battery (X.wi-D)

During battery change, avoid touching the detector connector pins.

CAUTION

Incorrect handling of detector battery

Explosion or fire

- Handle the battery pack carefully.
- Do not drop, heat, open or mishandle it.

CAUTION

Damaged detector batteries

Explosion, fire or burns

- Check detector batteries for cracks or other damage before using.
- Do not use damaged batteries and return them to Siemens Healthineers.

Do not remove the battery from the detector during image processing or image transfer. An image could be lost.

Before changing the battery, make sure that all data have been transferred.

When you take out the battery, any data on the detector will be erased!

Safety distance of the battery to pacemakers of min. 25 mm

The X.wi-D detectors have a protection class IP67 incl. battery.

Attention: Keep distance to pacemakers

To ensure the watertightness according protection class IP67 of the detector and battery, the battery is equipped with a magnet.

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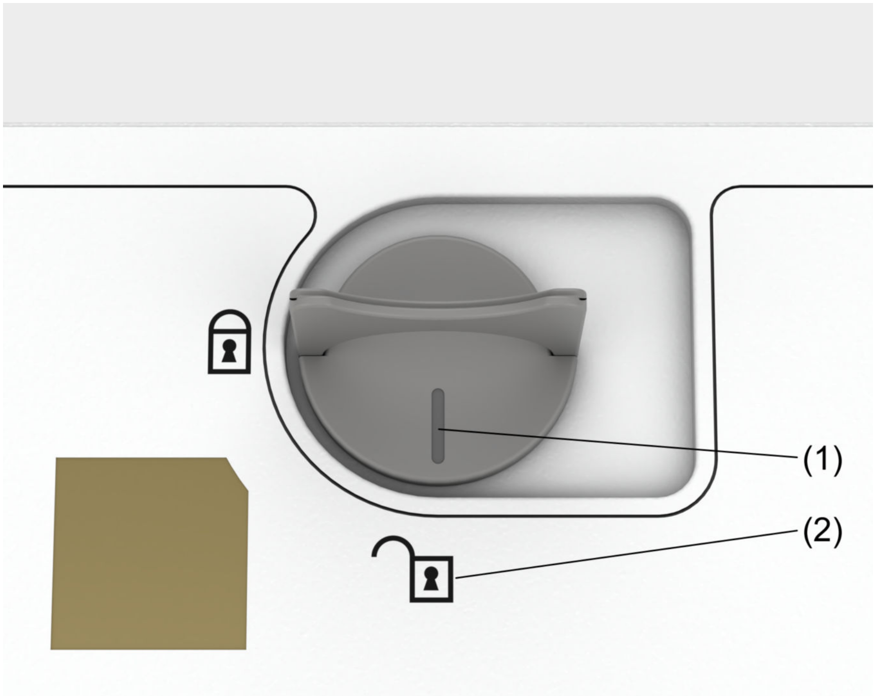
During change of battery, please pay attention to keep a safety distance of min. 25 mm to pacemakers.

There is no hazard for persons which wear a pacemaker if the battery is installed in the detector.

Hot-swap functionality (only for X.wi-D 35 and X.wi-D 43)

The hot-swap functionality allows to maintain the detector on during the battery exchange operation. Nevertheless, the operation must be complete in less than 30 seconds, otherwise the detector will start a switch off procedure and you will have to wait for the detector to automatically switch on again.

Workflow



1 Open the locking system of the battery as shown in the figure. The marking (1) must point to the open lock symbol (2).

2 Remove the discharged battery.

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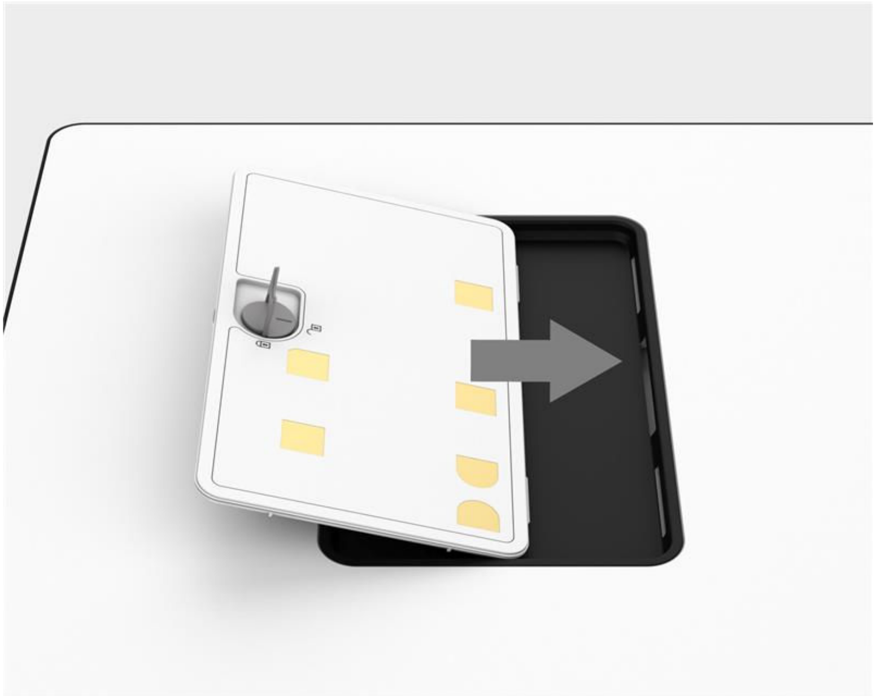
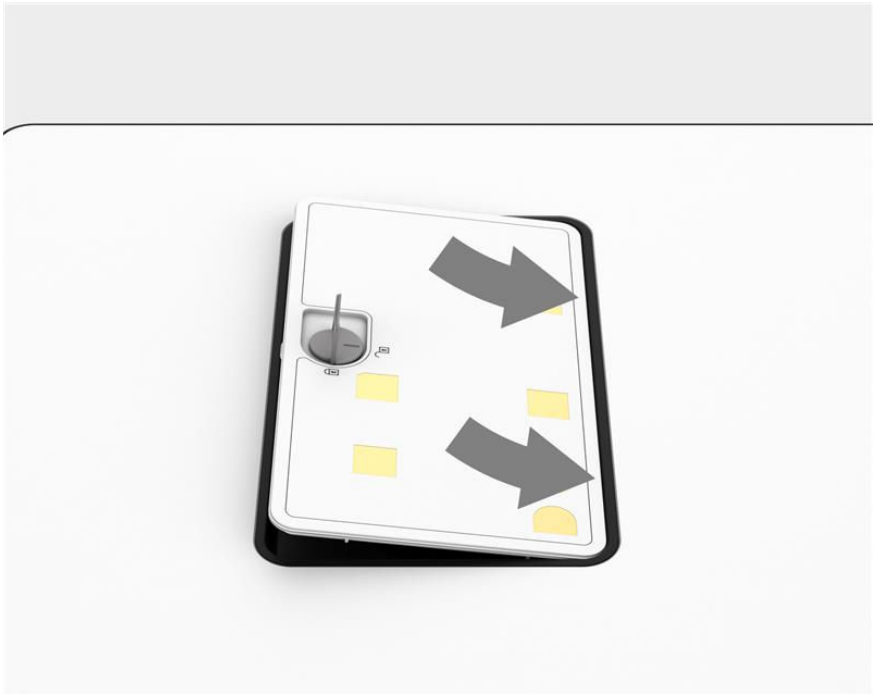
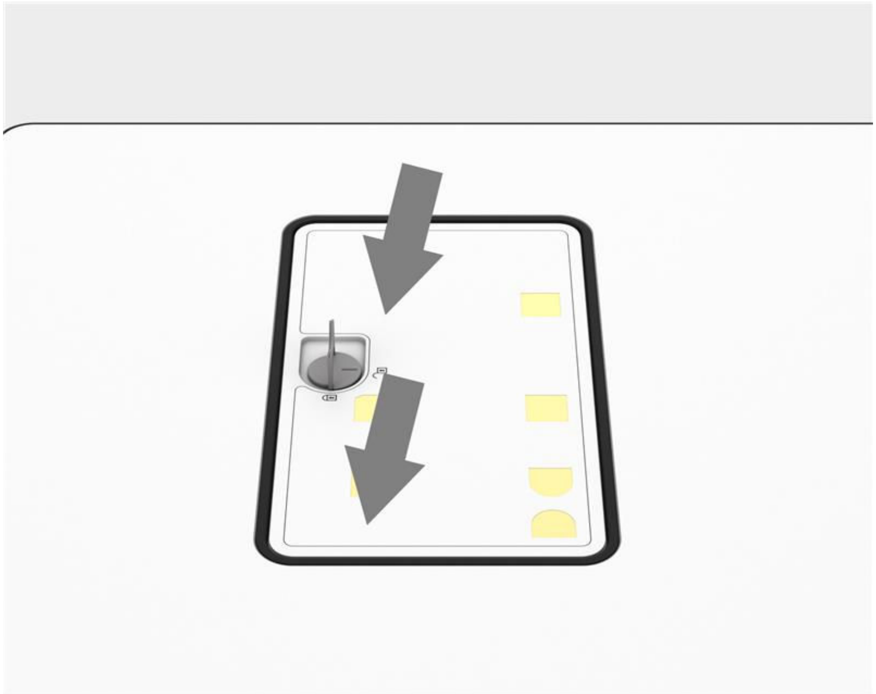


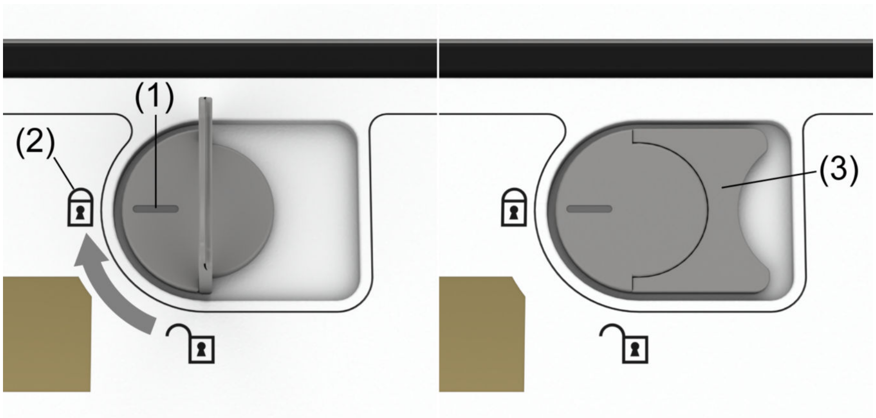
3 Orientate the charged battery in the correct position: battery connectors facing the device connectors.



4 Slide battery in the recess area of the device.



5 Push down the battery in the recess area.



6 Close the locking system of the battery. The marking (1) must point to the closed lock symbol (2).

Push down the latch (3) to ensure no protrusion of the locking system.

7 When the battery is inserted correctly, wait until all LEDs are green before using the detector (refer to (➔ Page 161 *Hot-swap functionality (only for X.wi-D 35 and X.wi-D 43)*)).

Charging of the detector battery

MAX detector battery

The battery of the MAX detectors can be charged:

- with the detector in the closed detector tray of the Bucky wall stand
- with the detector in the wall charger

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- in the battery charger

  (➔ Page 222 *Battery charger for MAX detector batteries*)

If the battery of the detector is completely empty, charge it and wait at least 60 minutes. The detector needs a warm-up time between 60 and 120 minutes to achieve good image quality, depending on environmental temperature and switch-off time.

X.wi-D detector battery

The battery of the X.wi-D detectors can be charged:

- with the detector in the closed detector tray of the Bucky wall stand
- with the detector in the wall charger

  (➔ Page 227 *Wall charger for X.wi-D and MAX detectors (option)*)
- in the battery charger

  (➔ Page 224 *Battery charger for X.wi-D detector batteries*)

4.3.12 Scattered radiation grid

If necessary for pediatric exposures, the grid can be removed from or inserted into the beam path.

The responsibility for selecting and inserting the scattered radiation grid to be used for an exposure lies with the examiner.

You must remove or insert the correct grid manually as required.

CAUTION

Grid is dropped or not handled properly

Risk of invisible damage and impaired image quality

- ◆ Handle the grid with special care.
- ◆ Be especially careful changing the grid when the table is not at 0°.

CAUTION

Image made with incorrect grid status

Radiation not as desired, poor image quality, too much dose

- ◆ The expected grid status (inserted, removed) is indicated on the imaging system.

  Manually remove or insert the grid as required before making the X-ray exposure.

Pay attention to the label which indicates the correct focus side and attach the grid correctly.

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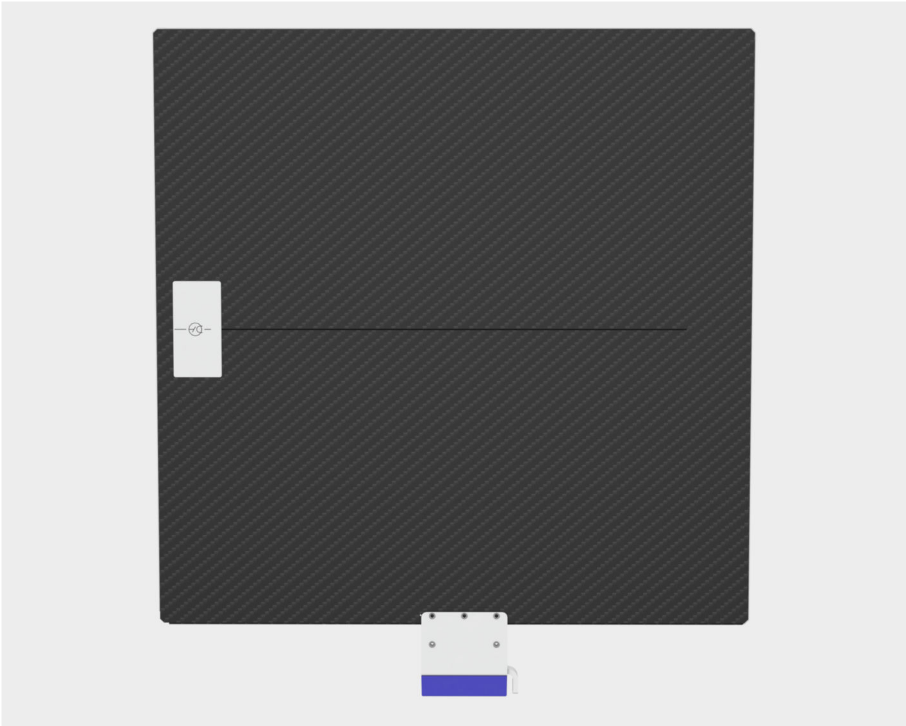

Top of grid (facing the tube unit)

Take care that the SID range of the grid is compatible to the selected SID.

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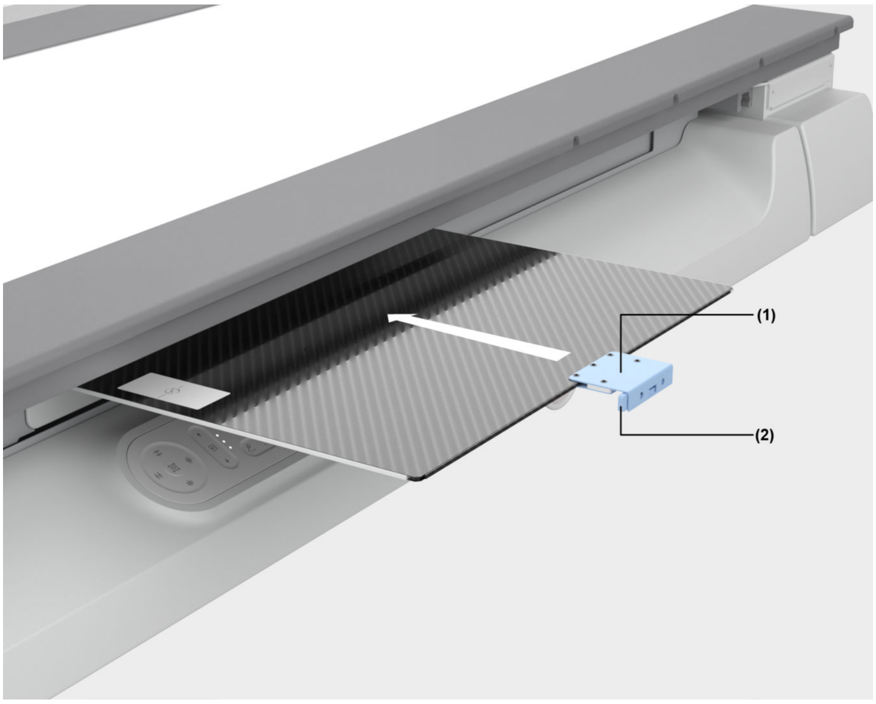
Inserting or removing the grid

Inserting the grid



(1) Grid handle
(2) Release button

- Insert the grid (with the label facing upwards) into the slot at the detector until an audible click can be heard.

Removing the grid

- Press the release button (2) at the left side of the grid handle (1) and remove the grid.

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4.3.13 Setting the source-image distance (SID)

If an SID position other than 115 cm, 150 cm or 180 cm is set, only free exposure technique is possible.

Setting the SID on the multileaf collimator or the system tube unit

- Press the - key or the + key to set the measured SID.

  The set SID can be seen on the display of the collimator.

  (➔ Page 85 *Multileaf collimator*)

With an SID lower than 90 cm, the X-ray field is not identlical/congruent to the lightfield because of physical reasons (anode shade). The X-ray field is smaler than the lightfield. We recommend not to use too small SIDs.

4.3.14 Multileaf collimator

If collimation not parallel to the table edges is required, the multileaf collimator can be rotated by ± 45° about the vertical axis.

Rotating the multileaf collimator

CAUTION

Hands between the collimator and the overhead column

Risk of injury

- Grasp the collimator on the sides when turning it so that there is no risk of crushing your hands.

CAUTION

Rotation of the collimator

Risk of injury of the hands and the fingers

- Always pay attention to your hands to ensure that they are not pinched or crushed between the collimator and surrounding objects.

- Grasp the collimator and turn it by the required angle in the intended direction.

The respective icon on the tube UI is visible, if the collimator is rotated.

Rotating back into the 0° lock-in position

- Grasp the collimator and turn it into the centered 0° lock-in position.

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Using the laser-line light localizer

The laser-line light localizer and the full-field light localizer **cannot** be switched independently of one another.

Laser light!

Do **not** use optical lenses, mirrors and similar instruments if working with laser light. The optical instruments within the laser beam may amplify the laser intensity to dangerous values for eyes and skin.

Switch off the laser light when optical instruments are used.

To protect the eyes of the patient or any other person, the laser radiation exit of the laser line light localizer can be closed with the **sliding cover**.

CAUTION

Looking into the laser light.

Eye injury, burns

- Take care that neither you nor anybody else looks directly into the laser beam.
- Close the sliding cover over the laser light to protect the eyes of the patient and other people.
- Note that the collimator can heat up when the light localizer is used for a longer period of time.

CAUTION

Photobiological effect of ultraviolet radiation

Eye injury

- Do not look into the light beam for longer than 15 seconds.
- Always keep enough distance to the collimator.

CAUTION

If the LED of the light localizer burns for a long time, parts of the light localizer can heat up.

Danger of burns

- Avoid contact with the light localizer to avoid burns.

Switching on the light localizer manually

You can switch the light localizer on or off at any time independent of automatic switching on and off.

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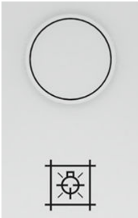

- To switch on the light localizer, press the button at the front of the collimator.

Switching off the light localizer manually

- To switch off the light localizer, press the button again.

  The light localizer can also be switched off automatically by an internal time switch.

Selecting the Cu filter

- Press the prefilter selection button.

  The Cu prefilter changes to the next value each time the button is pressed.

  Possible settings are: 0.0 mm Cu (no prefilter); 0.1 mm Cu, 0.2 mm Cu and 0.3 mm Cu.

If a Cu filter is set in the clinical protocol step, it is shown in the display.

(➔ Page 86 *Display field*)

4.3.15 Collimation

Collimation can be set in the following manners:

- Initial collimation is defined in the clinical protocol step.
- Via ACSS

  (➔ Page 170 *ACSS (Automatic Collimation Size Selection)*)
- With the corresponding button and joystick
- With the light localizer using the knobs on the multileaf collimator at the ceiling-suspended X-ray tube

  (➔ Page 101 *Multileaf collimator*)

CAUTION

Collimation defined with camera view is incorrect

Radiation not as desired

- Check the final collimation with the light marker before applying radiation to the patient.

Only manual collimation is possible for free exposure.

ACSS (Automatic Collimation Size Selection)

LUMINOS Q.namix R supports ACSS (Automatic Collimation Size Selection).

ACSS limits the maximum opening of the collimator to the borders of the selected detector in the patient table and the Bucky wall stand.

In case ACSS is not possible, you have to take care that the X-ray field will not go beyond the border of the detector.

The collimation format can be set in the clinical protocol. If the selected clinical protocol requires a new position of the tube and/or the detector, the collimation is set when the detector and the tube have reached the final position stored in the clinical protocol.

Conditions for ACSS:

- All system components (collimator, generator, tube stand, imaging system, detector) are ready for use.
- A detector is inserted in the tray of the table or wall stand (ACSS is not supported for free mobile detector operation).
- The central beam is perpendicular to the solid state detector with a tolerance of not more than ±3 degrees.
- The collimator is not rotated.

  The respective icon on the tube UI is not visible.

Manual collimation is always active if the above conditions are not fulfilled.

The type of collimation is displayed on the tube UI:

- **ACSS** is displayed: Automatic Collimation Size Selection (ACSS) is active.
- **Manual** is displayed: Manual collimation is active.

Make sure the X-ray beam does not extend the dimensions of the detector.

If **Manual** is active, the user is responsible for the adjustment of the multileaf collimator.

If **Manual** is active, but the image is very flat (low contrast), auto cropping may not work properly. In this case, disable auto cropping at the imaging system, and use the manual shutter after acquisition to crop the image manually.

◆ Manually move the detector and the tube to its final positions.

Even when the SID is changed, the collimator position in the multileaf collimator is adapted with active ACSS so that the collimation of the organ remains the same.

Rectangular collimation

Fluoroscopy and digital radiography (DR): You can view the collimation on the monitor under fluoroscopy.

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The maximum collimator aperture is never greater than the selected format.

The collimator is opened or closed synchronously and symmetrically to the image receptor center.

Within the selected format, the radiation field can be collimated as desired.

- On selection of a lower ZOOM level (larger field) or full format, the rectangular collimation is retained , or the larger field is collimated (can be configured).
- On selection of a higher ZOOM level (smaller field), the collimator always switches to the smaller field.

Setting collimation

At the system remote control console

◆ Deflect the joystick

  - Horizontal deflection opens or closes width blades (left: opens; right: closes)
  - Vertical deflection opens or closes height blades (up: opens; down: closes)
  - Diagonal deflection up left opens width and height blades
  - Diagonal deflection down right closes width and height blades

At the tableside control panel

◆ Press a key.

  - Upper or left key: Width or height collimator blade pair opens.
  - Lower or right key: Width or height collimator blade pair closes.
  - Left or right key: Both collimator blade pairs open or close simultaneously.

Storing/restoring the last collimation

You can restore the collimation which was used for the last exposure.

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Please note that restoring collimation is only possible if the preceding exposure was actually released.

◆ Press this key on the TUI of RCC.

The collimation set from the last taken exposure is saved and will be implemented for the next clinical protocol step.

Collimating without radiation - CAREPROFILE

With the CAREPROFILE function, the position of the collimator blades and the semi-transparent filter positions are displayed graphically in the last image. In this way you can change the collimation without additional fluoroscopy.

The indication comes on during any change of collimation or filter leave positions, or a change in the zoom stage.

The position of the measuring fields is also indicated when a change occurs.

You can configure the duration of the CARE display.

✓ A LIH is displayed.

◆ Perform collimation or position filters.

The collimator position and size or/and the filter leave position is indicated in the LIH image.

After a configurable time, the display disappears.

When triggering fluoroscopy, any existing display is also removed.

With the next fluoroscopy or acquisition, an electronic shutter automatically covers the collimator blades (set to black).

4.3.16 Positioning without radiation - CAREPOSITION

With CAREPOSITION, any move of table or stand is indicated in the last acquired image or LIH.

You can configure the duration of the CARE display.

After selection, the CAREPOSITION function is active and stays active.

You can reposition without applying radiation.

✓ A LIH is displayed.

◆ Tap the **CAREPOSITION** button.

The CAREPOSITION graphics is displayed: the current central beam ray position as a cross and the graphical frame of the image area (rectangle from collimator).

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The CAREPOSITION graphics is updated whenever the position of the system is changed: table longitudinal, table transverse or x-ray system longitudinal.

The CAREPOSITION graphics is removed after a configurable period of time.

4.3.17 Additional Cu filter

The clinical protocol steps determine which additional Cu filter is moved into the beam path on selection of the program.

A temporary change of the automatic additional Cu filter selection with clinical protocol step is possible by manual switchover. The manually selected additional Cu filter is maintained until renewed switchover or selection of an clinical protocol step with configured additional Cu filter.

The currently selected additional Cu filter is displayed on the control units and primary collimator.

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5 Examination

5.1 Patient positioning

Patient and/or equipment transfer

WARNING

Hand grips loosen or are not used

Injury of patient due to falling

- Check that the patient hand grips on the table are in the right position.
- Ensure that hand grips are tightened prior to use.

WARNING

Patient cannot maintain position or becomes unconscious

Injuries to patient from falling

- Check that the patient is physically able to hold the examination position, especially when standing.
- Use the provided positioning aids.
- Choose a different position for weak or very unstable patients.

5.1.1 General information

1. Attach all safety accessories, especially the grip protection strip, hand grip strip, hand grips, foot rest, and shoulder supports.
2. Use only positioning aids expressly approved by Siemens Healthineers.

   Contact your local Siemens Healthineers representative if you have any questions.
3. Materials located in the X-ray beam have a possible adverse effect (for example, attenuated X-ray beam and scattering effects).
4. Refer to (→ Page 213 *Accessories and Auxiliary Devices*) for all accessories available for positioning. Attaching the accessories is also described there.
5. Ensure that these devices and positioning aids are secured correctly and are functioning properly.
6. Make sure that the body parts of the patient, especially arms and legs, do **not** extend over the edge of the tabletop.
7. Make sure that the patient uses the provided grip locations.
8. Let the patient get on or off the table only in the central area.

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CAUTION

Incorrect positioning of the patient.

Risk of injury

- Ensure correct positioning of the patient, especially when using accessories.

5.1.2 Positioning the patient on the table

CAUTION

Patient activates motorized movement

Injury of patient or user

- Make sure that movement controls are out of reach of the patient during patient transfer.

Positioning the patient horizontally (table at 0°)

1 To position the patient comfortably in horizontal position, press this key.

The system moves into a position favorable for patient transfer.

2 Now transfer the patient.

3 Immobilize the patient with the corresponding accessory parts.

4 Move the X-ray system into the required position.

The collision protection system does not recognize accessory parts which project beyond the system contours.

The system and the accessories can be damaged and possibly the patient can also be injured.

Positioning the patient vertically (table upright at 90°)

1 To position the patient comfortably in vertical (upright) position, press this key.

The system moves into a position favorable for patient transfer.

2 Now transfer the patient.

3 Immobilize the patient with the corresponding accessory parts.

4 Move the X-ray system into the required position.

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Patient repositioning

For safe transfer of the patient between patient bed/stretcher and tabletop of the system, and to avoid damage to the system due to collisions of beds/stretchers with the system, please note the following instructions.

Patient transfer from the patient bed/stretcher to the system

1 Before transferring the patient from the bed or stretcher to the system, please move the tabletop of the system laterally away from the device column (Figure 1) until the tabletop protrudes beyond the front of the table (Figure 2). This prevents the bed/stretcher from colliding with the front of the table or the emergency stop button.

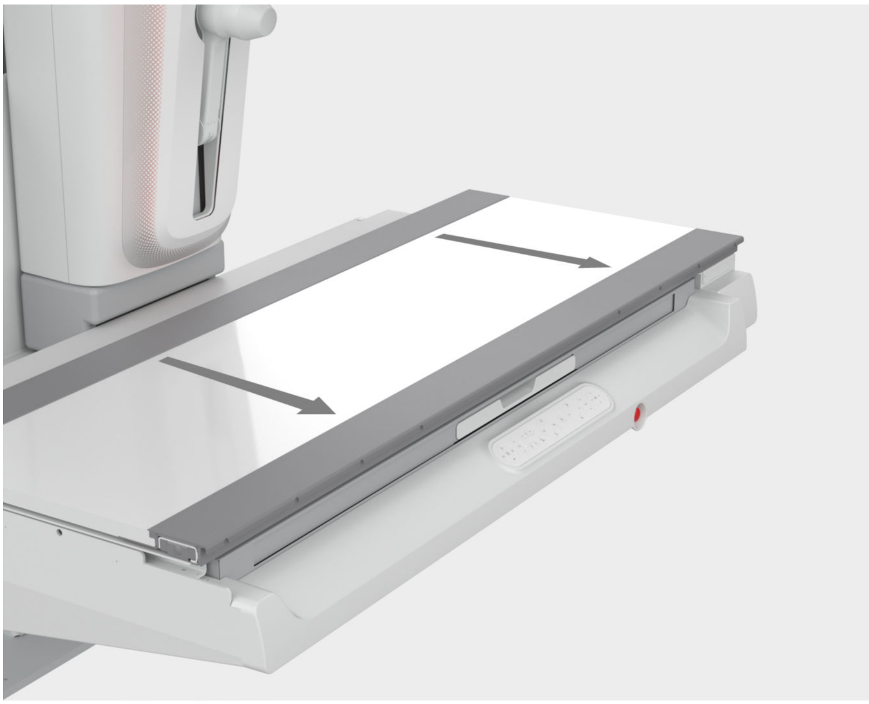

Figure 1: Lateral move

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Figure 2: Lateral move until the tabletop protrudes beyond the front of the table

2 Then move the bed/stretcher parallel to the patient table at a distance of approx. 15 cm.

3 Move the bed/stretcher and the system to a comfortable height for repositioning the patient and adjust the heights in such a way that the bed/stretcher is approx. 3 cm higher than the tabletop of the system.

This makes it easier to reposition the patient.

4 Then move the bed/stretcher close up to the system and lock the brakes of the bed/stretcher.

Make sure not to collide the bed/stretcher with the system and that there is no further movement or height adjustment of the bed/stretcher or the system when in close position.

5 Transfer the patient from the bed/stretcher to the system.

6 Release the brakes of the bed/stretcher and move it carefully away from the system.

7 Only now the height of the system and/or bed/stretcher may be adjusted again.

Patient transfer from the system to the patient bed/stretcher

1 Before transferring the patient from the system to the bed/stretcher, please move the tabletop of the system laterally away from the device column until the tabletop protrudes beyond the front of the table.

This prevents the bed/stretcher from colliding with the front of the table or the emergency stop button.

Figure 1: Lateral move

Figure 2: Lateral move until the tabletop protrudes beyond the front of the table

2 Then move the bed/stretcher parallel to the patient table at a distance of approx. 15cm.

3 Move the bed/stretcher and the system to a comfortable height for repositioning the patient and adjust the heights in such a way that the bed/stretcher is approx. 3 cm lower than the tabletop of the system.

This makes it easier to reposition the patient.

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4 Then move the bed/stretcher close up to the system and lock the brakes of the bed/stretcher.

Make sure not to collide the bed/stretcher with the system and that there is no further movement or height adjustment of the bed or the system when in close position.

5 Transfer the patient from the system to the bed/stretcher.

6 Release the brakes of the bed/stretcher and move it carefully away from the system.

7 Only now the height of the system and/or bed/stretcher may be adjusted again.

Please pay attention that the bed/stretcher is moved parallel to the system and does not collide with the system.

Additional information

Check the functionality of the emergency STOP daily.

5.1.3 Positioning the patient at the wall stand

Preparing positioning

1 Check the unit for cleanliness.

If necessary, cover the detector unit with fresh paper from the roll.

2 Attach the required radiation protection to the patient.

3 Position the tube unit.

(➔ Page 135 *Positioning the overhead support with tube unit*)

4 Observe messages on the display and if necessary on the patient display.

5 Adjust the detector unit on the wall stand approximately to the required operating height by pressing the up/down buttons.

6 Swing the overhead patient hand grip to the required position or to the side if it is not needed.

(➔ Page 289 *Overhead patient hand grip*)

7 If necessary, remove or insert the grid.

Automatic Collimation Size Selection (ACSS) is active when the central beam is vertical to the detector.

8 To achieve this, position the tube using the centering button.

Patient positioning and acquisition

1 Insert the wireless mobile detector in the detector tray.

(With wireless mobile detector only)

2 Guide the patient to the wall stand.

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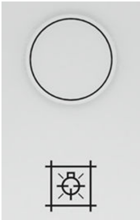
3 Using the handles bring the X-ray tube unit into position, if necessary into the pull-in range of the tracking control.



4 Press the button on the multileaf collimator to switch on the light localizer / line light localizer, if necessary.

CAUTION

Looking into the laser light.

Eye injury, burns

- Take care that neither you nor anybody else looks directly into the laser beam.
- Close the sliding cover over the laser light to protect the eyes of the patient and other people.
- Note that the collimator can heat up when the light localizer is used for a longer period of time.

To protect the eyes of the patient or any other person, the LASER radiation exit of the LASER line light localizer can be closed with the sliding cover.

(➔ Page 101 *Multileaf collimator*)

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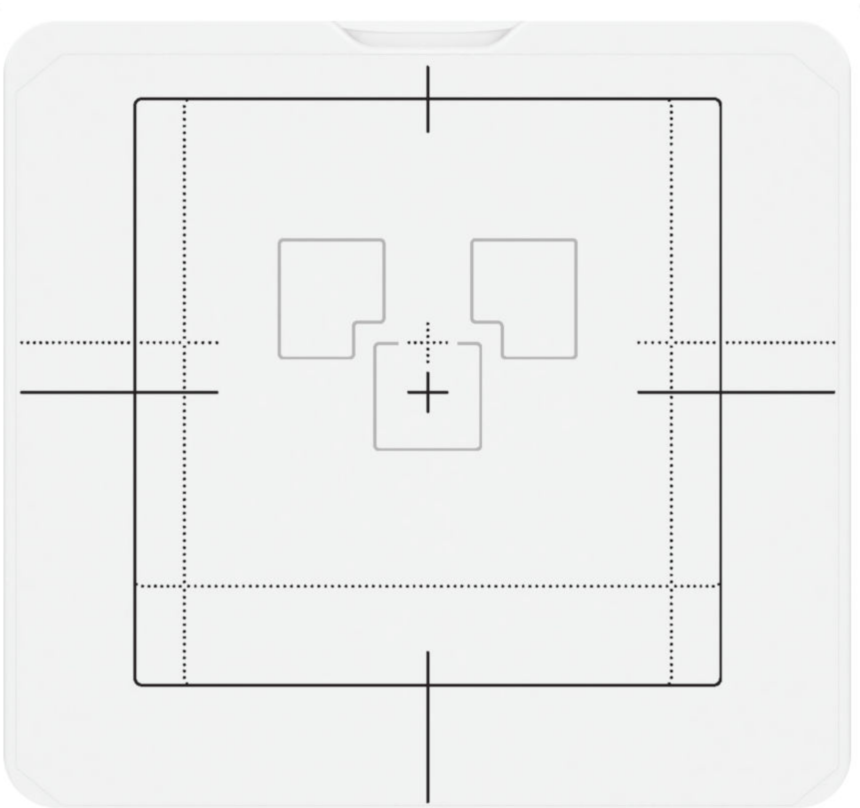
5 Pay attention to the positioning with respect to the Iontomat measuring chambers.



Position of the Iontomat measuring chambers (example: wall stand with mobile detector)

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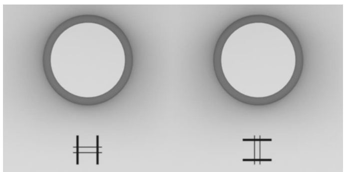
When the left and/or right chamber is/are selected and one or both of them is covered during collimation, the system automatically switches to the middle chamber.

An incorrect selection of measuring chambers by the user or by communication network disturbances leads to unnecessary X-ray exposure.

- Always select the measurement chambers carefully.
- Always check the selection of the measurement chambers before starting the measurement.
- If several measuring chambers were selected simultaneously, check the selection carefully.



6 Collimate the desired acquisition field.

Rotate the left knob to adjust the format width.

Rotate the right knob to adjust the format height.

- **ACSS**: The automatic format preselection controls the collimation by the multileaf collimator for **constant** collimation onto the format determined in the clinical protocol step with variable SID within the SID range.
- **Manual**: You can collimate additionally further close to the object if required with the control knobs for format collimation on the multileaf collimator:

CAUTION

Collimation defined with camera view is incorrect

Radiation not as desired

- Check the final collimation with the light marker before applying radiation to the patient.

7 Check the radiation protection.

8 Perform acquisition.

Refer to the Operator Manual of the imaging system.

5.2 Fluoroscopy

5.2.1 Fluoroscopy operating modes

Primary pulsed fluoroscopy (CAREVISION)

Pulsed fluoroscopy is possible with the following pulses per second: 0.5, 1, 2, 3, 4, 6, 8, 12, 16, 24, 32.

With lower pulses per second it is possible to reduce dose with a lower temporal resolution.

Selection of operating mode

All parameters necessary for fluoroscopy are stored in clinical protocol steps.

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◆ Select a corresponding clinical protocol step at the imaging system or on the remote UI.

Refer to the Operator Manual of the imaging system for more details.

Changing the selection of the operating mode

◆ It is possible to change the selection of the operating modes in connection with CAREVISION.

The current operating mode is displayed in the operating menu of the live monitor.

---

i You can also switch from one operating mode to another during fluoroscopy.

5.2.2 CAREVISION

Selecting the frame rate

◆ Press the **f/s** button on the remote UI and select the required frame rate.

Pulsed fluoroscopy with a pulse rate of 0.5, 1, 2, 3, 4, 6, 8, 12, 16, 24, 32

A pulse rate of 0.5 p/s is the lowest setting and 32 p/s is the highest setting.

---

i You can also switch over from one operating mode to another during fluoroscopy.

5.2.3 Automatic format collimation in fluoroscopy

Automatic format collimation is also switched on automatically when fluoroscopy is released.

That is, the fluoroscopic field is never larger than the selected zoom format.

You can limit the fluoroscopic field within the area to the required size with the corresponding operating elements.

5.2.4 Releasing fluoroscopy

✓ The correct patient and procedure is selected on the imaging system.

◆ Release fluoroscopy with the foot switch pedal.

Radiation is released as long as the foot switch for fluoroscopy is pressed. When the foot switch is released, the last image is retained and displayed on the monitor (LIH).

---

i For a radioscopy of more than 0.5 s, X-ray is terminated within 0.1 s after releasing the foot switch pedal or hand switch button.

For a radioscopy of 0.5 s or less, X-ray is terminated within 0.5 s after releasing the foot switch pedal or hand switch button.

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When the red emergency STOP button is actuated, fluoroscopy can be switched back on consciously after releasing the fluoroscopy switch and pressing it again.

No processing functions apart from storing the image are possible during fluoroscopy.

The radiation ON indicators must light up.

5.2.5 Fluoroscopic data

The fluoroscopic data of voltage in kV, current in mA, and time in min are displayed on the TUI.

The dose area product is displayed on the monitor.

Display of kV and mA values:

- During fluoroscopy: Real-time display of the actual voltage kV and mA values
- After fluoroscopy: Voltage kV and current in mA mean value of the examination

Display of fluoroscopic time:

- Continuous display of the time in min accumulated in the examination.

Display of dose area product:

- Continuous display of the dose area product accumulated in the examination.

Reset of fluoroscopic data:

- The displays for the fluoroscopic data and the dose area product are reset to zero when a new patient is selected.

Fluoroscopy elapsed time:

- Display of the elapsed fluoroscopy time for the current patient.

5.2.6 Turning off fluoroscopy warning signal

The fluoroscopy warning signal sounds when the elapsed fluoroscopy time has reached a configurable limit (maximum 5 minutes or a certain dose). The warning signal can be reset by selecting this key on the remote UI or on touch display.

(➔ Page 94 *Settings tab*)

The signal will sound again when the time limit has expired again.

Refer to the Operator Manual of the imaging system.

5.3 Digital radiography

5.3.1 Misuse of radiography

If the radiography mode is misused on purpose for real-time imaging, the image display delay may be longer than in fluoroscopy.

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5.3.2 Power supply

If the mains power supply does not meet the standard, for example if the internal resistance is too high, it must be protected against overload by restricting the power.

In this case, the selection of the mA and mAs values may be limited.

5.3.3 Automatic format collimation in digital radiography

The exposure field is never larger than the detector format.

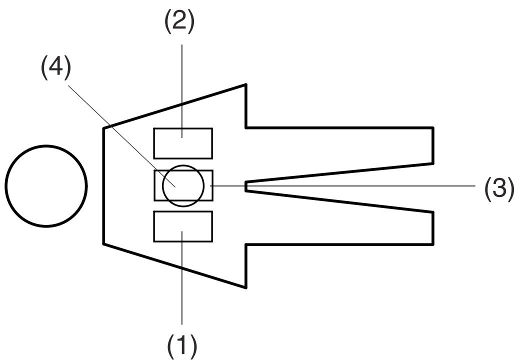
Within this area, you can limit the exposure field to the required size with the corresponding operating elements.

5.3.4 Exposure measurement in digital radiography

Arrangement of measuring fields

At the table

The following figure shows the position of the measuring fields in the monitor in relation to the a.p. patient position. That is head end at the left, looking onto the patient with the table horizontal.



(1) Left measuring field
(2) Right measuring field
(3) Middle measuring field
(4) Central circular measuring field

An additional "measuring field" is the Histogram Based Dose Regulation which cannot be shown in the above figure.

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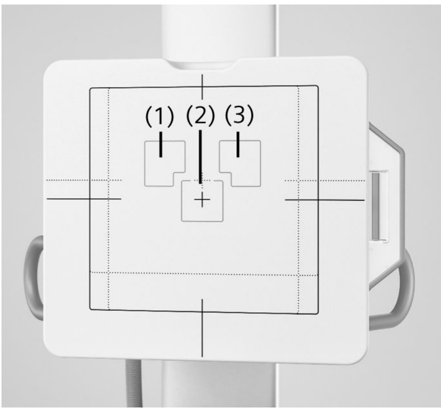
At the wallstand



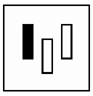
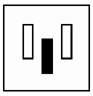
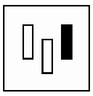
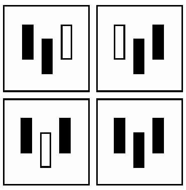
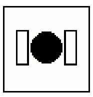
(1) Left measuring field
(2) Middle measuring field
(3) Right measuring field

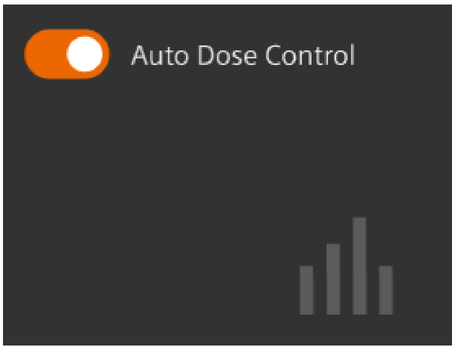
Selecting measuring fields

The measuring fields are stored in the clinical protocol steps or can be set on the TUI.

Multiple selection of the measuring fields is possible.

Exception: Central circular measuring field at the table.

	Left measuring field
	Middle measuring field
	Right measuring field
	Selectable measuring field combinations
	Central circular measuring field (at the table only)

	Auto Dose Control (Histogram Based Dose Regulation) "On" (only on the table)

◆ Press one of the keys.

If a selected left and/or right measuring field becomes invalid temporarily, for example due to collimation, then the central circular measuring field is selected and displayed automatically (only at the table).

The originally selected measuring field will be active and displayed again when the collimator is opened again.

When collimating under fluoroscopy (at the table only), take care that the selected left or right measuring field lies completely in the radiation field. Otherwise there will be overexposure because of the inappropriate acquisition of the measured exposure values.

The position of the measuring field in relation to the object can be checked on the LIH image by means of brief fluoroscopy (at the table only). A diagram with the position and size of the relevant measuring field is displayed briefly on the monitor.

5.3.5 Acquisition

Clinical protocol steps

To optimize operation, protocols (clinical protocol steps) for the most common standard examinations are delivered with the system. In general, these protocols can be taken as examples to derive individual protocols for more specific examinations.

For editing protocols, refer to the **Clinical Protocol Editor** in the Operator Manual of the imaging system.

- Low dose CP step: Basic
- Normal dose CP step: Standard
- IQ optimized CP step: High

Single image - series

The display is in the operating area of the monitor.

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The zoom format cannot be changed during an exposure series. Only the collimation is corrected on changing over to a larger zoom level.

Temporary changes of clinical protocol steps

For the following parameters stored in the clinical protocol step, it is possible to change them temporarily with the corresponding operating elements:

- Switching over from single image or series
- Dominants
- Cu prefilter
- X-ray parameters
- Collimation
- Focus size
- Fluoroscopy program Basic, Standard, High

The changed parameters apply only until you select another or the same clinical protocol step or switch off the system.

Exposure release

Please note that following acoustic signals:

- acoustic signal for fluoroscopy
- acoustic signal for series acquisition
- acoustic signal for X-ray by the basic unit

  can only be set by Siemens Healthineers Customer Service)

At the table

◆ Release the exposure with the exposure release button at the system remote control console or the multifunctional joystick or via the footswitch in the control room, or table-side.

The radiation ON indicators must light up briefly during the exposure.

At the wall stand

◆ Release the exposure with the exposure release button at the control console in the control room.

The radiation ON indicators must light up briefly during the exposure.

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Letting go of the exposure release button

⚠ CAUTION

Exposure button released too soon during long exposures

Repetition of examination or contrast agent dose

- Keep the exposure button pressed until the acquisition has finished, recognizable by the exposure indicator light and beep.

Radiation interruption

If you let go of the exposure release button during radiation, the exposure is interrupted immediately in single exposure. In series, the last exposure is still exposed correctly.

Exposure end

In single exposure only one image is exposed, even if the exposure release button continues to be pressed.

In series, letting the exposure release button go, a full image memory or reaching the programmed maximum scene time ends the exposure series.

5.4 Exposures onto detector in wall stand

5.4.1 System configurations

Exposures on the mobile detector or the fixed detector in the wall stand are only possible with one of the following system configurations:

- Configuration (3)

  Basic configuration plus fully synchronized 3D overhead support and wall stand for mobile detectors like MAX wi-D or X.wi-D 43/35.

  In this configuration, exposures onto the mobile detectors in the wall stand are only possible with the fully synchronized 3D overhead support.

- Configuration (4)

  Basic configuration plus fully synchronized 3D overhead support and wall stand with fixed detector (MAX static).

5.4.2 Exposure workflow

Preparations

✓ Workplace “Detector at the wall stand”

1 Check the unit for cleanliness.

  If necessary, cover the detector unit with fresh paper from the roll.

2 Attach the required radiation protection to the patient.

3 If necessary, remove or insert the grid.

4 Insert the mobile detector in the detector tray.

5 Adjust the detector unit on the wall stand approximately to the required operating height.

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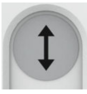

6 Press the button for the manual vertical movement and move the detector unit up or down to the required working height.

– or –

Press one of the buttons for the motorized vertical movement to move the detector up and down to the required working height.

7 Swing the overhead patient hand grip to the required position or to the side if it is not needed.

(➔ Page 289 *Overhead patient hand grip*)

Positioning the tube assembly at the 3D overhead support

1 Tilt the tube assembly to 90°

2 Position the tube assembly using the corresponding buttons at the 3D overhead support.

(➔ Page 135 *Positioning the overhead support with tube unit*)

3 Observe messages on the display and if necessary on the patient display.

Patient positioning and acquisition

1 Position the patient at the wall stand.

2 Attach the required radiation protection to the patient.

3 If necessary, switch on the light localizer / line light localizer on the multileaf collimator pressing this button.

4 Pay attention to the positioning with respect to the measuring fields.

When the left and/or right chamber is/are selected and one or both of them is covered during collimation, the system automatically switches to the middle chamber.

An incorrect selection of measuring chambers by the user or by communication network disturbances leads to unnecessary X-ray exposure.

- Always select the measurement chambers carefully.
- Always check the selection of the measurement chambers before starting the measurement.
- If several measuring chambers were selected simultaneously, check the selection carefully.

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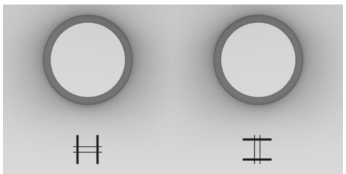


5 Collimate the desired acquisition field.

Rotate the left knob to adjust the format width.

Rotate the right knob to adjust the format height.

- **ACSS**: The automatic format preselection controls the collimation by the multileaf collimator for **constant** collimation onto the format determined in the clinical protocol step with variable SID within the SID range.
- **Manual**: You can collimate additionally further close to the object if required with the control knobs for format collimation on the multileaf collimator:

CAUTION

Collimation defined with camera view is incorrect

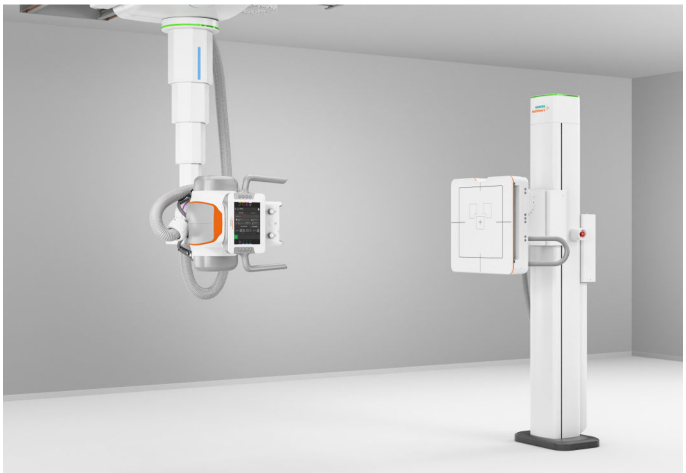
Radiation not as desired

◆ Check the final collimation with the light marker before applying radiation to the patient.

6 Perform acquisition.

Refer to the Operator Manual of the imaging system.

Positioning for horizontal projection



Example: Tube unit position for exposures on the wall stand with horizontal projection with the ceiling-suspended X-ray tube

Example for a thorax aquisition

⚠ CAUTION

Patient exerts too much weight on the patient hand grip on the wall stand.

Risk of injury to the patient and damage to the wall stand

◆ Do not let the patient hang on the patient hand grip.

◆ Do not exert more than 25 kg weight on the patient hand grips.

⚠ CAUTION

Patient overloads wall stand detector

Detector damage or injury of the patient.

◆ Do not allow the patient to sit on the detector unit or to push down on it.

1 Select a suitable clinical protocol step on the imaging system or on the tube UI.

2 Position the tube unit manually as required for the exposure.

Please take care especially for Bucky exposures on the wall stand to exact fixation of the ceiling-suspended X-ray tube in one of the predetermined SID stop positions.

You avoid image quality defects due to grid shadows if the set SID agrees with the focusing of the grid.

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3 Center the tube unit to the detector.

(➔ Page 135 *Positioning the overhead support with tube unit*)

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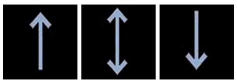
The tube unit automatically follows the detector in height adjustment within a certain range (tracking control).

Positioning for vertical projection

✓ Configuration (3) and (4) only

1 Select a suitable clinical protocol step on the imaging system or on the tube UI.

2 Position the tube unit manually as required for the exposure using the movement buttons tube UI and/or the buttons on the handles.



3 Set the SID by using the buttons on the tube UI.

– or –

Set the SID by using the buttons on the detector unit.

4 Make an exposure.

Attention: No SID tracking is possible for this mode.

Positioning for oblique projection

WARNING

Distortion due to exposure geometry

Incorrect basis for diagnosis

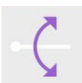
◆ Pay special attention in the diagnosis to distortions of the image geometry when using oblique projections.

✓ With configuration (3) and (4) only.

1 Tilt the detector unit into the horizontal position.

Tilting the detector unit

The mechanism locks automatically into place.



2 Rotate tube unit about the horizontal axis (RHA) and then about the vertical axis (RVA) into the required direction.

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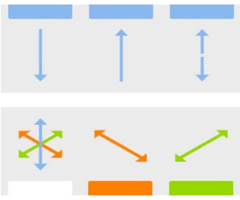
3 Release the brakes with the pushbuttons on the control panel or with the buttons on the handles and position the tube unit.

The **angle in degrees** is displayed in the display of the tube unit control panel.



4 Press the buttons for the longitudinal, transverse brake and the vertical brake or for the ‘floating’ movement and bring the tube unit into the required position.

In the case of oblique projection, the angle in degrees is displayed in the display of the overhead support control panel instead of the SID.

You must therefore absolutely **measure the SID** with the tape measure.

The automatic format preselection is not effective in the oblique projection. (**Manual**)

You must **collimate manually**.

5 Make an exposure.

Pay especial care in the diagnosis to distortions of the image geometry due to the oblique projection!

5.5 Free exposures

WARNING

Unintended activation of control elements for movements during bed exposures when table is in vertical position

Collision of tube with patient, operator or equipment

- Select free exposure mode, then rotate tube, tilt stand and move to correct SID. Only then can the patient be positioned under the tube.
- Do not leave the patient under the tube during absence of the operator.

CAUTION

Tube is not directed toward the detector during acquisition.

Unwanted radiation exposure

- Use the free exposure mode with care.
- Check that the tube is active and use the light localizer for positioning before releasing X-ray.

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5.5.1 MAXalign

MAXalign makes free exposures easier and more accurate by displaying the current detector angle on the touchscreen.

It is not necessary to guess the tube angle. You just align the tube according to the indicated detector angle.

Procedure

1 Position the mobile detector (tilted up to 90°).

The detector angle is shown on the touchscreen.

2 Rotate the tube until the angle corresponds to the shown detector angle.

⚠ CAUTION

Wrong angles displayed, tube not aligned to detector

Radiation not usable

◆ After the system has been positioned, do a final check to make sure that the tube, anatomy and detector are properly aligned before releasing radiation.

5.5.2 Collimation during exposure

The automatic format collimation of the system is not effective if you have selected "free mode", or if you have turned the tube assembly into a position outside 0°.

- You can set the exposure format freely.
- The limits of the maximum radiation field are determined by the mechanism of the primary collimator.
- The collimator does not react to a change of the SID.
- The digital display field of the primary collimator shows "**Manual**".

With the tube assembly swivelled out, the system automatically switches over to "Free mode". The automatic format collimation system is disabled.

5.5.3 Releasing an exposure

1 To release an exposure, actuate the exposure release button.

2 Before every exposure release check:

The format limitation with the aid of the light localizer on the collimator

Set SID with the tape measure on the collimator

The central ray on detector center with the aid of the laser line light localizer on the collimator.

The radiation ON indicators must light up.

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5.6 SmartOrtho

If configured, you can acquire a series of images from different patient heights at the Bucky wall stand or on the patient table.

SmartOrtho provides Ortho acquisition with automatic movement, whereby the system will automatically move in the correct position for each image of the Ortho sequence after the user selected start and end position.

(➔ Page 195 *SmartOrtho on patient table*)

(➔ Page 195 *SmartOrtho onto wall stand*)

(➔ Page 196 *Clinical protocol and clinical protocol step for SmartOrtho*)

(➔ Page 196 *Preparing SmartOrtho*)

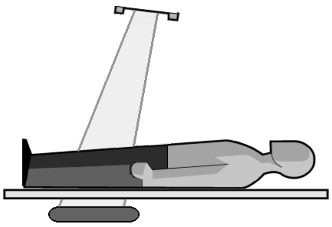
(➔ Page 202 *Releasing the Ortho exposures*)

(➔ Page 203 *Adjusting/repeating an Ortho sequence*)

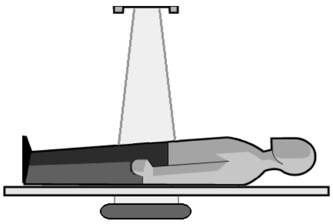
5.6.1 SmartOrtho on patient table

The SmartOrtho acquisition is done by tilting the X-ray tube unit and moving the detector unit. The X-ray tube unit is centered over the table. When an Ortho sequence is started, the detector moves automatically to the different positions (up to 3). The tube follows the detector, by tilting itself accordingly.

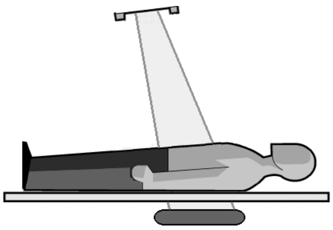
Example: SmartOrtho a.p. lying (region with 3 positions)



Acquisition position 1



Acquisition position 2



Acquisition position 3

Ortho acquisitions on the table can be performed with a SID up to 1.8 m.

5.6.2 SmartOrtho onto wall stand

The SmartOrtho acquisition is done by tilting the X-ray tube unit and moving the detector unit. The X-ray tube unit is positioned in the middle of Ortho cover range. When an Ortho sequence is started, the detector moves automatically to the different positions (up to 4). The tube follows the detector, by tilting itself accordingly.

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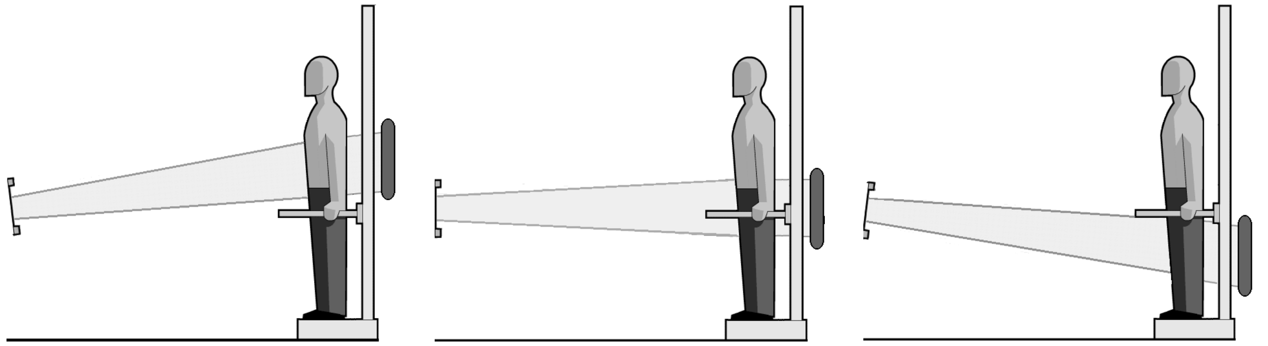
Example: SmartOrtho a.p. standing (region with 3 positions)



Acquisition position 1 | Acquisition position 2 | Acquisition position 3

For automatic Ortho imaging onto the wall stand, the SmartOrtho stand is needed.

Ortho acquisitions onto the wall stand can be performed a SID up to 3 m.

5.6.3 Clinical protocol and clinical protocol step for SmartOrtho

Clinical protocols and clinical protocol steps for SmartOrtho allow up to four acquisitions, belonging to one Ortho series. For each step individual settings for the generator are available.

The position of the units must be saved for the SmartOrtho clinical protocol steps. It must be ensured that the unit axes are located orthogonally to each other.

CAUTION

Wrong clinical procedure used for ortho

Delayed diagnosis or repeated examination because results cannot be composed

◆ Use the correct clinical procedure for ortho.

◆ Make a test run.

5.6.4 Preparing SmartOrtho

Pay attention to the messages displayed on the imaging system display and on the tube UI.

If the series acquisition is interrupted, the whole sequence must be started again from the beginning!

(➔ Page 196 *Clinical protocol and clinical protocol step for SmartOrtho*)

✓ The correct patient is registered.

✓ The correct clinical protocol step for Ortho has been selected.

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✓ The correct detector is connected and selected.

✓ The system and the patient are positioned correctly.

✓ The imaging system is in examination mode.

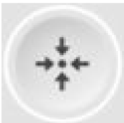
✓ The correct acquisition parameters are set.

✓ The collimation is set as required.

✓ The collimator must be in the 0° lock-in-position.

✓ For Ortho acquisitions onto the wall stand, the detector unit must be in the 0° position.

✓ The Ortho parameters are displayed on the on the tube UI.



1 Press the SmartMove button to move the ceiling-suspended X-ray tube and the BWS or the table automatically into a pre-programmed CP position (for example, height).

Once in position, the integrated camera (option) will switch on.

The SmartMove button does not work for clinical protocols with tilted table positions (e.g. Ortho with table tilted to +90°). In these cases the table has to be moved to the required position by using the buttons on the table control console or the joysticks on the remote control console

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2 Position the patient lying supine on the patient table or standing with its back directed to the SmartOrtho stand.

(➔ Page 197 *Positioning the patient on the patient table*)

(➔ Page 198 *Positioning the patient on the SmartOrtho stand*)

(➔ Page 198 *Changing/entering the TOD*)

(➔ Page 199 *Adjusting the ROI with start and end position (ceiling-suspended X-ray tube)*)

(➔ Page 200 *Adjusting the ROI with camera in the control room (Smart Virtual Ortho)*)

Positioning the patient on the patient table

CAUTION

Patient hand is under the table during ortho

Risk of crushing between table and detector tray

- Check the position of the patient's hands.
- Make sure the patient is using the hand grips.

1 Position the lying patient on the table.

2 Position the calibration element (sphere, ruler) in diagnostically relevant area.

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Positioning the patient on the SmartOrtho stand

WARNING

Patient cannot maintain position for ortho

Injury to patient, repetition of examination

- Always use the ortho support for ortho examinations.
- Check that the hand grips are at the right position and are tightened prior to use.
- Make sure that the patient is stable enough to maintain position and is using the hand grips.

1 Carefully position the SmartOrtho stand as close as possible to the detector tray.

Check that the detector tray does not grind on the semi-transparent stripes of the complete relevant area.

2 Position the patient standing on the SmartOrtho stand in a way that the longitudinal axis of the object runs parallel to the image receptor.

3 Position the calibration element (sphere, ruler) in diagnostic relevant area.

Changing/entering the TOD

For later measurements and correct composition of Ortho images it is required to enter the correct TOD = "Table"-Object Distance, i.e. SmartOrtho stand to object plane distance.

✓ After selecting a clinical protocol step for Ortho, the TOD defined in the clinical protocol step is displayed on the tube UI.

✓ The X-ray tube unit is positioned for Ortho.

1 Measure the SOD (source-object distance) using the tape measure.

– or –

Measure the TOD using a ruler.

2 Select the correct TOD on the tube UI in the examination room or on the touch display in the control room.

(The units depend on the settings.)

Accuracy: 1 cm (0.4 inch)

Adjusting the ROI with start and end position (patient table)

✓ (➔ Page 196 *Preparing SmartOrtho*)

Use the laser line light localizer to set the image area.

1 Ask the patient to close eyes and turn on the light localizer.

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For acquisitions on patient table, it is possible to change the SID for easier adjustment.

Align the laser line on the upper edge of the region of interest to the start position.

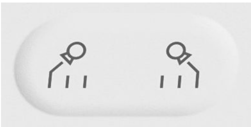
2 Press the Ortho button for the start position.

– or –

Press this icon on the remote UI.

The position is stored.

Align the laser line on the lower edge of the region of interest to the end position.



3 Press the Ortho button for the end position.

– or –

Press this icon on the remote UI.

The position is stored.

The start and end positions should not be outside the travel limits of the detector. If so, a message is displayed and new measurements are required.

The system automatically calculates any intermediate acquisition positions required for coverage of the acquisition range.

4 Turn off the laser light.

5 Collimate the region of interest in width.

(Collimation in height will be adjusted automatically.)

6 Check acquisition parameters, collimation, and measuring fields.

When collimated narrowly in width, the system switches automatically to the center measuring field.

Adjusting the ROI with start and end position (ceiling-suspended X-ray tube)

✓ (➔ Page 196 *Preparing SmartOrtho*)

Use the laser line light localizer to set the image area.

1 Ask the patient to close eyes and turn on the light localizer.

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Align the laser line on the upper edge of the region of interest by tilting the X-ray tube unit to the upper start position.

2 Press this icon on the tube UI.

The position is stored.

Align the laser line on the lower edge of the region of interest by tilting the X-ray tube unit to the lower end position.

3 Press this icon on the tube UI.

The position is stored.

Both icons are marked by check marks in case both positions are valid.

A red cross is shown in case the defined positions are not valid.

The upper and lower limits should not be outside the travel limits of the detector. If so, a message is displayed and new measurements are required.

The system automatically calculates any intermediate acquisition positions required for coverage of the acquisition range.

If more than four acquisition positions would be required to cover the acquisition range, a message is displayed. The target area has to be adjusted and the start and end positions have to be set again.

4 Turn off the laser light.

5 Collimate the region of interest in width.

(Collimation in height will be adjusted automatically.)

6 Check acquisition parameters, collimation, and measuring fields.

When collimated narrowly in width, the system switches automatically to the center measuring field.

Adjusting the ROI with camera in the control room (Smart Virtual Ortho)

Using the integrated camera, you can smoothen the preparation for orthopedic procedures, especially long leg and full spine, and thus enhance the conditions patients undergo.

✓ Smart Virtual Ortho is properly licensed.

✓ (➔ Page 196 *Preparing SmartOrtho*)

✓ The camera is configured for Ortho.

✓ The X-ray tube unit is positioned for Ortho.

✓ The X-ray tube unit is in 0° position.

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1 Select the upper and lower edge of the region of interest on the touchscreen by adjusting the upper and lower line.

The virtual collimation may slightly be displaced to the true collimation range during acquisition, especially for long spine acquisitions. It is because the virtual collimator of the Ortho range is drawn as a projection from the current tube position instead of from the acquisition tube position.

In addition, the camera is not installed at the same position as the X-ray source, deviations in image geometry may occur.

The dotted line of the virtual collimator shows the correct projection of the Ortho range onto the patient.

The top and bottom lines indicate the average height.

2 Collimate the region of interest in width on the touchscreen by adjusting the right or left line.

3 Adapt acquisition parameters for each planned Ortho image, if needed.

For each image, touch the number of the image to be acquired and change the acquisition parameter as needed.

4 In case the Ortho acquisition range (ROI) has been changed after adapting the exposure values, check acquisition parameters and measuring fields.

If the desired ROI cannot be fully covered in the camera image, you might move the tube vertical to a higher position.

Or if no automatic movement is triggered, you may switch the acquisition type back and forth.

Or if the virtual collimator/ Ortho sub-steps are not shown, you may reselect the Ortho clinical protocol step.

◆ Afterwards, reselect the upper and lower edge of the region of interest on the touchscreen.

5.6.5 Auto Collimation Long leg / Full Spine for Smart Virtual Ortho

Auto Collimation Long leg / Full Spine provides collimation support for long leg and spine examinations. Auto Collimation detects the legs/body with the myExam 3D camera and suggests a collimation on the tube UI.

By using the virtual collimation functionality or the manual collimation via the knobs the suggested collimation can be adapted.

✓ Smart Virtual Ortho and Auto Collimation Long leg / Full Spine are properly licensed.

✓ The camera is configured for Auto Collimation Long leg / Full Spine.

✓ A long leg / spine protocol step is selected.

✓ The collimator must be in the 0° lock-in-position.

✓ Detector unit must be in the 0° position.

✓ The X-ray tube unit is positioned in a valid Ortho position for the long leg / spine protocol step.

1 Position the patient.

Collimation is being suggested automatically and shown with a white frame.

In the protocol step it is defined (**Composing algorithm**) if spine, one leg, or two legs will be detected.

2 You can enable or disable collimation support by tapping the **Auto Collimation** icon on the touch display in the control room.

3 Check collimation in the examination room or in the control room and adjust, if necessary.

(➔ Page 202 *Releasing the Ortho exposures*)

5.6.6 Releasing the Ortho exposures

✓ The correct clinical protocol step is selected.

✓ The units and the patient are correctly positioned.

✓ The Ortho acquisition range (ROI) has been defined.

✓ The TOD and the acquisition parameters are correctly set.

✓ The collimation is set as required.

✓ The system is ready for acquisition.

It is highly recommended to collimate properly in order to prevent direct radiation on the flat detector.

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⚠ CAUTION

Motion between ortho images

Incorrect diagnosis basis

- Check results of composing for motion artifacts due to mismatched ortho images.
- Compare the positions of the collimator blades in each image.

  If the blades are correctly matched, but the anatomy shows a mismatch, then the patient moved between images.
- Use single images to check anatomy. Consider also manual composing.

1 Press an exposure release push button half (pre-contact) and hold it in that position.

The system moves to the first acquisition position.

2 Instruct the patient to keep still.

3 Press the exposure release push button fully down and hold it pressed until all the acquisitions are finished.

Depending on the configuration:

The first position is acquired and displayed.

The system moves to next acquisition position.

The next position is acquired and displayed.

and so on

– or –

All the positions are acquired and displayed.

After all exposures are done, the acquired images are automatically stitched together.

Every Ortho image within an Ortho series is numbered and stored.

- The first acquired image gets the image number 1.
- The last acquired image gets the highest number.

5.6.7 Adjusting/repeating an Ortho sequence

In some cases, it may be necessary to adjust, restart, or repeat an Ortho sequence.

For example, if:

- The camera is too close to the table so that the correct ROI cannot be set with the camera in the control room.
- The Ortho position has been reached but the tube has been moved afterwards.

  The Ortho region is wrongly indicated and the actual SID value is incorrect.

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- The tube is not in the correct start position.

  No automatic movement takes place.

In case you cannot perform an Ortho exposure as expected, proceed as follows:

◆ Switch to a clinical protocol step with a different **Acquisition Type** (e.g. wall stand in case of table) and back.

  The required movements will be released.

5.7 Interventional procedures

WARNING

Medical devices are not infallible and unexpected errors may occur

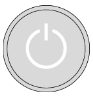
System is not available in an emergency or may stop working during an examination

◆ Have an emergency plan in case of non-availability of the system, for instance, to use a different system.
◆ Have an emergency plan for continuing or ending different types of examinations if the system stops working.
◆ Consider whether an uninterruptible power supply is necessary at your site.

5.7.1 System recovery

System recovery in case of a failure during examination in interventional procedures

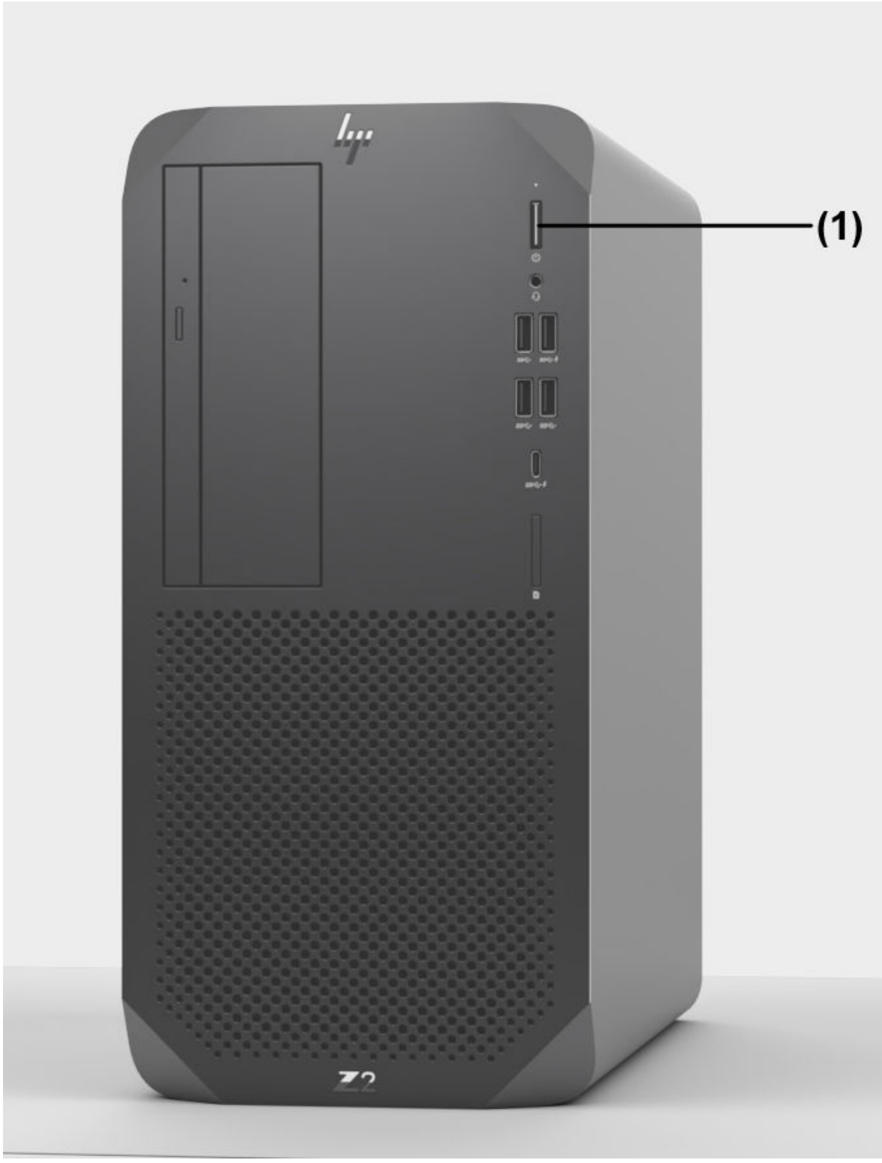
All functions of the system can be recovered within 10 min by the following procedure:



1 Switch OFF the system with the OFF button.

Wait a maximum of 60 seconds until the entire system and all further connected devices are switched off.

» The imaging system did not shut-down automatically.



◆ Press the power button (1) of the imaging system PC for 10 seconds to shut-down the imaging system separately.

2 When the imaging system has also shut-down, press the ON button.

Wait a maximum time of 240 seconds until all functions are recovered and fluoroscopy can be continued.

5.7.2 Contrast medium

Some contrast media are very aggressive and can react with some materials of the system (for example tabletop).

To prevent this, please double-check the used contrast media with your Siemens Healthineers representative.

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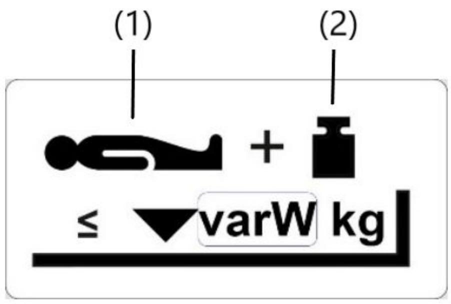
Avoid contact between contrast media and tabletop.

Certain contrast media can harm the surface of the tabletop.

For more information and details, refer to your Siemens Healthineers contact person.

5.7.3 Important labels



Max. load on table: Patient + additional load (for example, accessory)

(1) Patient
(2) additional load

The permitted maximum load on your tabletop is stated on the label on the tabletop and in the technical data.

This warning sign shows the position of the patient table for cardio-pulmonary resuscitation (CPR) with a pressure compression of up to 500 N (50 kg).

5.7.4 Prerequisites for interventional procedures

- CAREMAX (Mandatory in the EU, USA, and Canada)
- For interventional examinations, the system must be horizontal.
- The tabletop must be centered.
- The system must be in a position in which CPR can be performed. (➔ Page 208 *Cardiopulmonary Resuscitation (CPR)*)
- The interventional kit (part no. 10357938) is to be used as protection against ingress of liquids.

  (➔ Page 228 *Interventional kit*)

5.7.5 Fluoroscopy guided interventional procedures

The Luminos Q.namix R/T fulfills the “Particular Requirements for the Safetyof X-ray Equiment for Interventional Procedures” according to IEC 60601-2-43:2010 + AMD1:2017 + AMD2:2019 / IEC 60601-2-43: 2022

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Typical interventions:

- Ballon enteroscopy
- Lumbar puncture
- Pain management
- Port system implantation
- Percutaneous transluminal angioplasty (PTA)
- Basic stent implantation for example:
  - esophagus stent
  - bile duct stent
- Central venous catheter (CVC)

This system is designed for cardiopulmonary resurection (CRP) with pressure compression of up to 500 N (50 kg) for patients up to 250 kg.

5.7.6 Radiation protection for interventional procedures

Notice that the dose rate increases with increasing pulse rate and increasing zoom factor.

High air kerma values

The system is intended for procedures that, with proper use of the system, involve high air kerma (mainly due to the long examination times), which presents a risk of radiation injury.

Clinical protocols

The dose is determined in the exam sets (fluoro and acquisition protocols).

The available clinical protocols depend on the system type and the profile set.

The system comes preinstalled with a number of clinical protocols.

For further information on the preinstalled clinical protocols ask your applications specialist.

- Select only suitable clinical protocols for the examination.
- If possible, select a dose-saving fluoro/acquisition protocol.

  (Fluoro protocols with a lower dose are further to the left).

Clinical protocols for interventional procedures

Contact your Siemens Healthineers service engineer in this regard.

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Disabling radiation

Door contacts

Recommended: In case of interventional applications, a current procedure should neither be interruptible by door contacts nor by other measures.

Remedy: Install door contacts (or have them installed) in such a way that an interruption can be disabled.

External buttons

To prevent the unintentional release of radiation, for example during cleaning or patient positioning, radiation can be disabled by the **Block Radiation** button on the **Settings** tab of the remote UI or in the menu of the tube UI without impairing other system functions.

If the button is activated, an according message is displayed.

5.7.7 Free system movements

Avoid unnecessary interruptions of system movements:

- Before starting system movements, in particular lowering and tilting the unit, ensure that there are no obstacles in the movement area.
- Make sure that the patient does not actuate the switch rail between the tabletop and the stand.

5.7.8 Cardiopulmonary Resuscitation (CPR)

The LUMINOS Q.namix R system is designed for cardiopulmonary resuscitation (CPR=Cardio-Pulmonary Resuscitation).

CAUTION

System cannot be placed in position for CPR

Resuscitation (CPR) cannot be performed.

◆ For interventional procedures, the system must be horizontal.

Cardiopulmonary resuscitation can only be performed with the following settings:

- Horizontal tabletop
- The pressure point (patient thorax) must be between the head-end and foot-end tabletop support.

Preparations for CPR

There should be sufficient free space for CPR.

Only begin CPR when the tabletop is positioned over table support or table base or table column.

◆ Remove any objects, such as chairs, from the examination area.

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Moving the system to a position for CPR

1 Press the key until all movements, except table lowering, have been completed.

The system moves to the patient transfer position with horizontal tabletop.

The system moves to the configured table height.

The system moves into the orthogonal beam path.

The system moves to the foot-end position of the tube assembly stand respectively the image receptor unit.

The table is centered in the longitudinal direction.

The table moves toward the front in the transverse direction.

2 Keep pressing the key to lower the table until it reaches a comfortable height.

5.7.9 Fluoroscopy in interventional procedures

Fluoroscopy mode

Using the pulse rate button on the remote UI, you can select the low dose values with pulsed fluoroscopy (CAREVISION).

For further information, refer to (➔ Page 181 *Fluoroscopy*) and to the Operator Manual of the imaging system.

Releasing fluoroscopy

◆ Use the foot switch for fluoroscopy and acquisition in the examination room.

If the Radiography mode is misused on purpose by the operator for realtime imaging, the image display delay may be longer than in Radioscopy!

5.7.10 Cleaning and disinfection for interventional radiology

Prevention

Use sterile covers.

The following system parts can come into contact with body fluids, contrast medium and so on:

System part	Special cleaning measures
• Patient table • Tabletop	If necessary, remove the tabletop
Control modules	To prevent system movements, press an emergency STOP button.
Flat detector	Protect by sterile covers.

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System part	Special cleaning measures
• Hand switch • Foot switch	To prevent radiation release, activate **Block Radiation** in the **Exam** task card on the touchscreen control.
Injector	See Operator Manual of the injector

5.7.11 Concept of the interventional patient entrance reference point

The interventional patient entrance reference point for the measurement of the air kerma strength was defined in the central ray 30 cm above the tabletop as the location of the patient entrance reference dose for a medium patient thickness.

5.7.12 Isodose curves

According to IEC 60601-2-43:2022

The measurements were performed at the LUMINOS Q.namix R with a fixed table height of 92 cm.

The isodose values behind the tilting/lifting base (160°-200°) are reduced considerably.

Measuring conditions

- PMMA cube phantom with a side length of 25 cm
- X-ray field 10 cm x 10 cm at phantom entrance
- CPR setting, tabletop in center position
- Source - phantom surface 79 cm
- Small focus, SID 115 cm, without Cu prefilter
- Vertical, downward projection.

Tabletop horizontal

- Without mobile radiation protection
- Scattered radiation of isodose curves in µGy/(Gy cm²) at 150 kV, 2.33 mA
- The conversion factor to the unit (µGy/µGy m²) according to IEC 60601-2-43:2022 is 1/100

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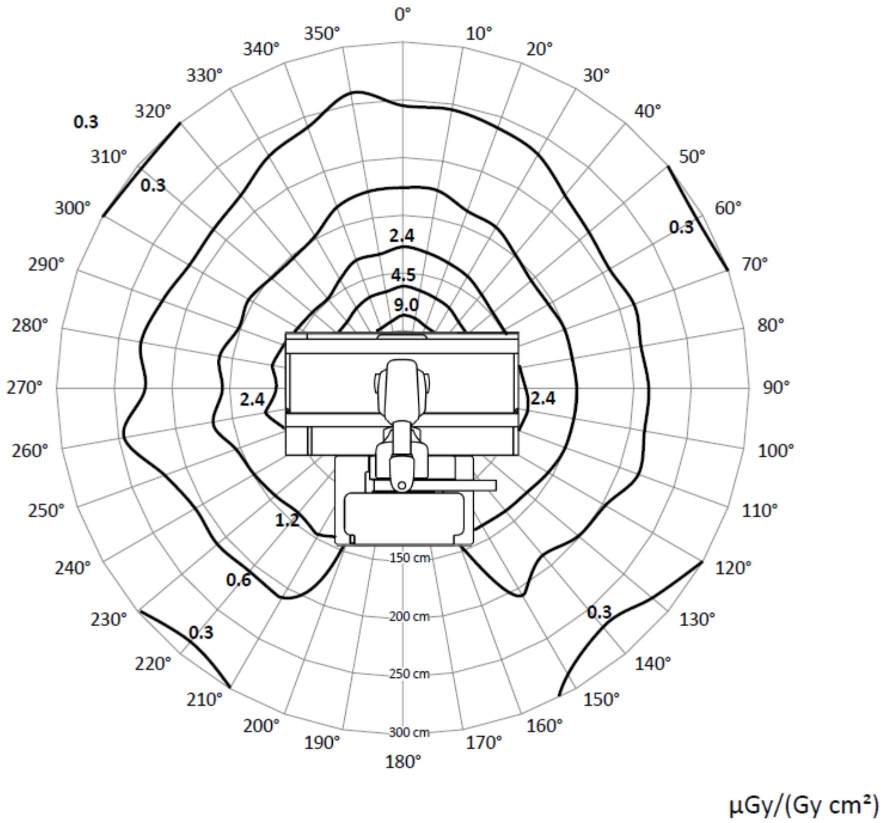
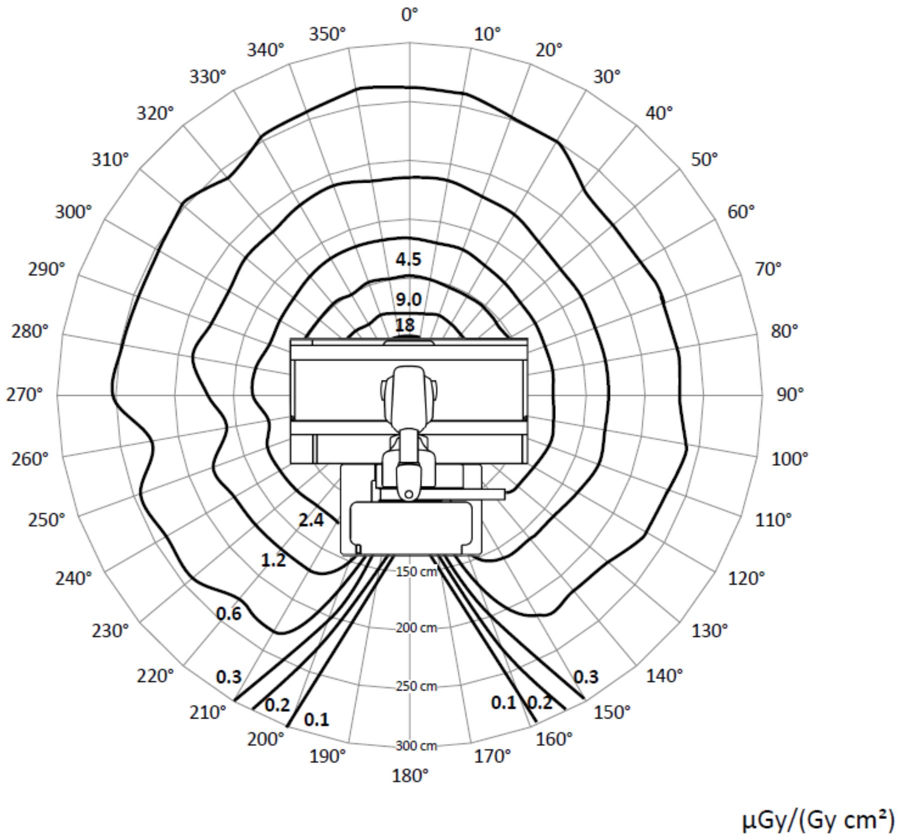


Isodose 100 cm above the floor



Isodose 150 cm above the floor

6 Accessories and Auxiliary Devices

6.1 Preliminary Remarks

This part of the Operator Manual describes all the system accessories provided by Siemens Healthineers for the LUMINOS Q.namix R system.

6.1.1 Proper use of the accessories

Proper use of the accessories presupposes that the user has read and understood this part of the Operator Manual in detail.

In addition, the user must be thoroughly familiar with the rest of the Operator Manual.

Finally, detailed knowledge of the applicable safety regulations for the medical equipment being used is indispensable. These regulations are listed in part **Safety** of this Operator Manual.

Explanation of "proper use" consists of illustrated descriptions of attaching, using and removing the accessory component.

⚠ CAUTION

Accessories not firmly attached or not used. Accessories overloaded.

Patient injury

- Make sure that accessory parts are attached firmly by pulling and pushing them.
- Follow the instructions in the Operator Manual.
- Observe the information regarding storage and maximum weight when using the accessories.

6.1.2 Safety of accessories

For reasons of safety for patients, operators and the equipment, only original accessories manufactured and sold by Siemens Healthineers or Siemens Healthineers-approved accessories from another manufacturer may be used.

Siemens Healthineers cannot accept any responsibility for accessories not approved by us.

⚠ CAUTION

Use of accessories not intended for the system.

Risk of injury to patient or user; risk of damage to the system

- Use only accessories approved by Siemens Healthineers with this system.
- All accessories are labeled and described in the Operator Manual.

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CAUTION

Riding on wheeled accessories

Accessory overbalances and tips over causing damage to system or injury of persons.

◆ Do not ride or allow anyone else to stand, sit or ride on wheeled accessories while they are being rolled.

CAUTION

Trolley or wheeled accessory tips over or suddenly stops while being rolled.

Injury of persons

◆ Please be careful when pushing trolleys or wheeled accessories. They are not designed to be pushed over a step, cable or other hindrance because they can tip over or suddenly stop.

WARNING

Wear or material fatigue in the system or accessories

Risk of injury or system damage

◆ Follow the maintenance guidelines to maintain safety and proper function of the system.

◆ Check accessories for wear before use.

6.1.3 Orientation of accessories

When using the accessories and to simplify the description, the following directional indications shall apply in this Operator Manual with regard to the working position of the user.

- **In front** means in front of the patient table looking in the direction of the X-ray tube
- **Left** is the head end of the patient table
- **Right** is the foot end of the patient table
- **Frontal** means the longitudinal side of the patient table close to the operating panel (not with Multitom Rax)
- **In back** means the longitudinal side of the patient table away from the operating panel (not with Multitom Rax)
- **Turn toward the right** means tighten accessory with knurled screws or levers (clockwise)
- **Turn toward the left** means remove accessory with knurled screws or levers or other fastening devices (counterclockwise)

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Example:

1 Turn both knurled screws toward the left.

2 Push the grip protection bar on the head end into both accessory rails.

3 Turn both knurled screws toward the right.

4 Check that the grip protection bar is seated securely.

6.1.4 Use of several accessory components

This document describes the use of a single accessory component in each chapter text.

If you need to work with several components during an examination, for example compression belt and shoulder supports, these components must be attached and removed in the correct sequence because most accessory components are attached to the patient table using the accessory rails.

Example:

- Hand grip strip and foot rest are already attached to the patient table.
- You need the shoulder supports in addition because they will be helpful for the examination.

Using of several accessory components

1 First, remove the hand grip strip as described in these instructions.

2 Then position the patient on the table.

3 Next attach the shoulder supports.

4 Next push the hand grip strip back into the accessory rails and secure it.

5 Check that all accessory components are securely attached.

6.1.5 Cleaning and disinfection of accessory parts

For cleaning and disinfection, please refer to Register **System Safety**, Chapter "Cleaning and disinfection".

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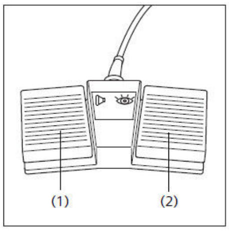
6.2 Description

6.2.1 Foot switch for fluoroscopy and exposure

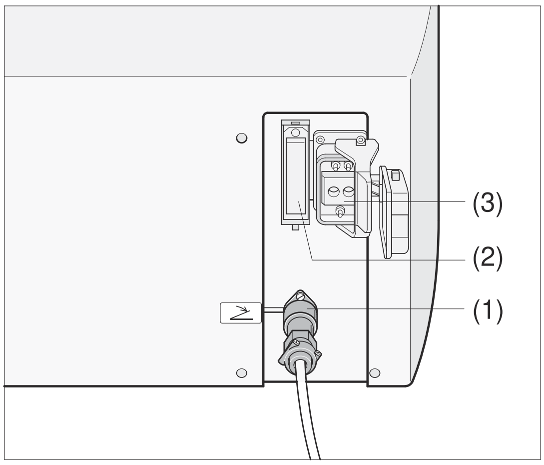


(1) Pedal for exposure (with precontact)
(2) Pedal for fluoroscopy

Application

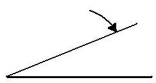
For releasing radiographic or fluoroscopic exposures.

Attachment



Foot switch plug in receptacle and receptacles for injector plug

(1) Foot switch plug in receptacle
(2) Receptacle for injector plug (Siemens Healthineers interface)
(3) Receptacle for injector plug (ODU interface)



Label of foot switch receptacle

Label of injector receptacle

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⚠ CAUTION

Operator or patient trips over cables or foot switches on the floor

Risk of injury

- Position the cables and foot switches on the floor so that nobody trips.

⚠ CAUTION

Unintentional activation of the foot switch

Unnecessary radiation exposure

- Place the foot switch outside the movement range of the unit.
- Make sure there is no danger of collision with the foot switch when tilting or lowering the unit.

1. Insert the foot switch plug into the corresponding receptacle (1) in the electronics cabinet.
2. Turn the locking ring of the plug to the right until the connector is seated properly.
3. Route the cable along the floor appropriately.
4. Position the patient.

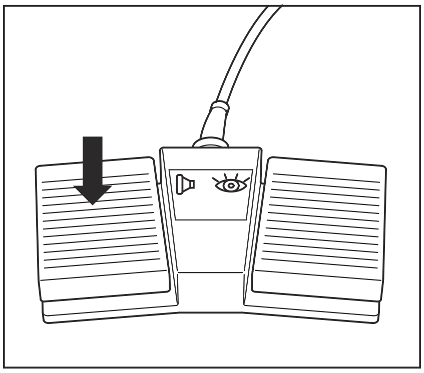
Route the cable carefully to avoid safety hazards (for example tripping).

Do **not** permit the wheels of transport devices such as carts and wheelchairs to run over the pedals of the foot switch assembly.

Such an action can cause inadvertent radiation release.

Ensure proper radiation protection.

Releasing radiography



1. Step on the left pedal (foot switch for radiography).
2. Continue pressing down on the pedal until the exposure has ended.

   The exposure has ended when the radiation ON indicator on the remote control console goes out.

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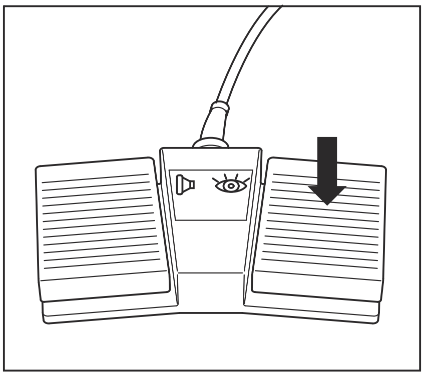
3 Stop pressing down on the pedal.

Releasing fluoroscopy



1 Step on the right pedal (foot switch for fluoroscopy).

2 Continue pressing down on the pedal as long as fluoroscopy should be released.

3 Stop pressing down on the pedal.

Fluoroscopy release ends.

Removal

Depending on the frequency of use, you can leave the foot switch attached, or disconnect it after the examination.

1 Turn the locking ring on the cable plug to the left.

2 Remove the plug from the receptacle.

3 Roll up the foot switch cable.

4 Store the foot switch and cable in an appropriate location.

6.2.2 Multifunctional wireless foot switch

CAUTION

Unintentional activation of the foot switch

Unnecessary radiation exposure

- Place the foot switch outside the movement range of the unit.
- Make sure there is no danger of collision with the foot switch when tilting or lowering the unit.

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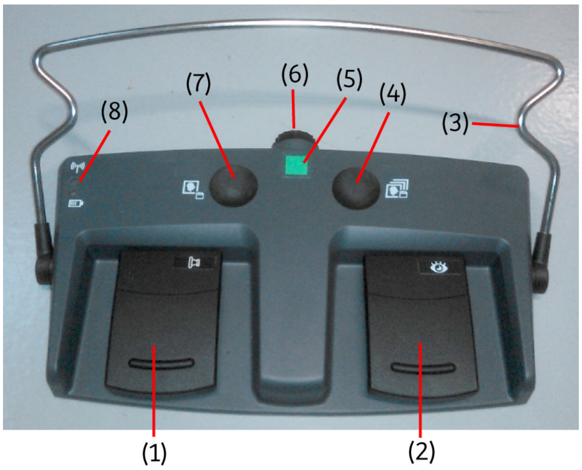


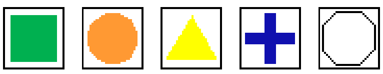
(1) Pedal for exposure (with precontact)
(2) Pedal for fluoroscopy
(3) Clamp for positioning
(4) Store fluoro loop button
(5) Colored symbol for identification
(6) Battery compartment
(7) Snapshot / store image / store LIH button
(8) Status LEDs

The status LEDs on the wireless foot switch indicate the following information:

LED	Operation status	Error condition
	Battery capacity: • Green: Battery full or nearly full • Orange: Battery almost full • Orange (flashing): Battery low	Fatal error or foot switch update: • Orange (both LEDs flashing): Fatal error (for example, battery exhausted or memory error) • Green and orange (both LEDs alternately flashing): Firmware update process running
	Connection status or pairing mode: • Green: Connection active • Orange: No connection active • Orange (flashing): Foot switch not paired • Green and orange (alternately flashing): Pairing mode activated	

Using the wireless foot switch

For unique identification, the following colored symbols are provided on the wireless foot switch:



Make sure that the colored symbols on the foot switch match to your system!

CAUTION

Wrong wireless foot switch used

Undesired radiation exposure

- Take care that the wireless foot switch and the system have the same colored label.
- Use wireless foot switch only when there is visual and acoustic contact with patient.

WARNING

As with many wireless components, the portable detector and the wireless footswitch may have electromagnetic fields that could disturb a life-supporting device such as a pacemaker or ventilator.

Interference with and possible malfunction of a life-supporting device

- Keep a safety separation distance of more than 30 cm between the wireless component and the life-supporting device, except for pacemakers. For pacemakers, a safety distance of minimum 3 cm is required.
- In case of interference with other equipment, increase the distance between the interfering devices.

CAUTION

Weak batteries in the foot switch

Interruption of examination

- Check the LED indication on the foot switch before starting the examination.

1. Position the wireless foot switch on the floor and swivel the clamp into the upright position.

   The wireless footswich will switch on automatically.

2. You can reposition the wireless foot switch with a foot using the clamp.

The wireless foot switch automatically switches to standby mode if it is not used for a certain period of time. It is activated again by actuating any button or pedal.

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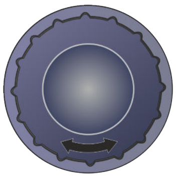
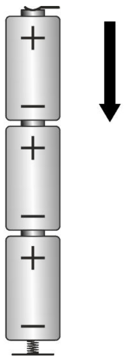
Replacing the batteries

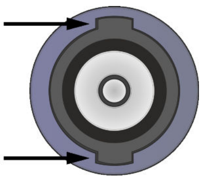


1 To open the battery compartment on the bottom, lightly press in the knob and turn it.

2 Remove batteries.



3 Insert new batteries (LR14) into the wireless foot switch as shown in the diagram (see arrow for insertion direction).

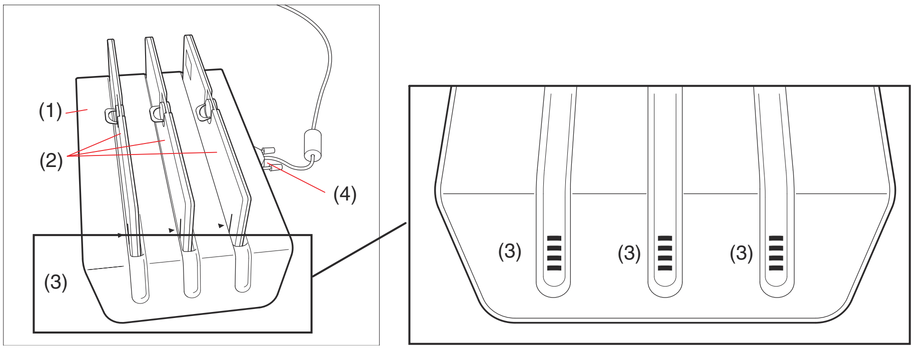


4 To close the battery compartment, insert the knob into the depressions. Press lightly and turn it until it clicks into place.

Technical data

Footswitch / Receiver	
Data transfer	Bluetooth BTA.1LT (EU: 2402 - 2480 MHz without restriction
Radiated power	Class 2: 2.5 mW/ +4dBm transmission power
Supported wireless standards	IEEE 802.15.1
Encryption standard	128 bit AES-CCM

6.2.3 Battery charger for MAX detector batteries

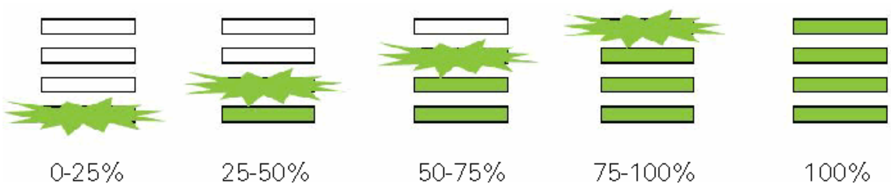


(1) Charger
(2) Batteries
(3) Status LEDs
(4) Power input with green power indicator
Green LED is illuminated when power is on.

Status LEDs

For each charging slot, there are 4 LEDs. When there is no battery to be charged, all LEDs should be off.

There is 1 bottom LED (orange or green) and 3 green LEDs.



Charging levels

Fault signal

Charging a fully discharged battery will take a maximum of 3 hours.

Up to three batteries can be charged at a time in the charger.

Inserting battery into charger

The battery charger is designed to insert a battery in only one direction.

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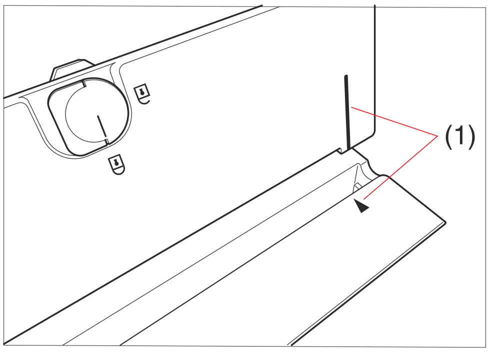

1 Insert the battery into the charger as shown above.

2 Pay attention to the alignment identification (1) on the battery and the charging slot (see above).

3 Gently push the battery onto the connector.

Once the battery is in place correctly, the bottom LED starts blinking.

Charging troubleshooting

Orange fault LED on

The battery is not inserted correctly / battery temperature too high.

1 Remove battery and insert it again correctly.

If the problem remains, the battery is damaged.

2 Remove battery and insert it again with a normal temperature.

The charger slot will remain in a fault condition until the battery is removed and inserted again with a normal temperature.

No green power LED active

◆ Check that the power supply is working.

No charging LEDs active

No battery inserted

Battery not inserted correctly.

1 Remove battery and insert it again correctly.

2 If problem remains, check that connectors are not damaged or battery is not faulty.

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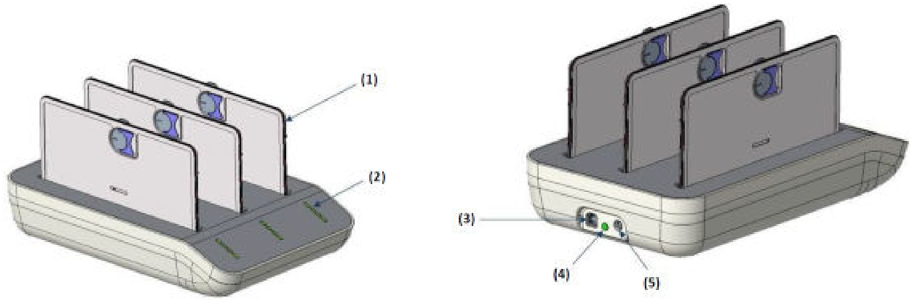
6.2.4 Battery charger for X.wi-D detector batteries



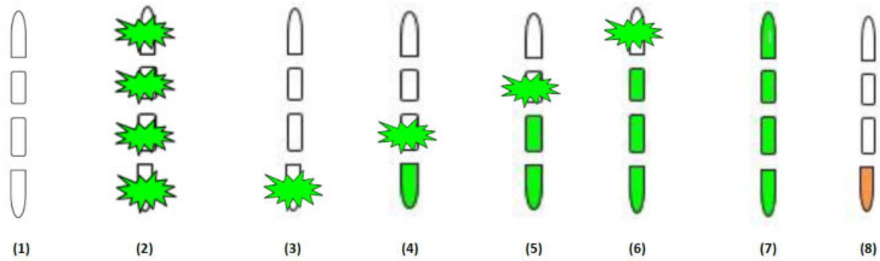
(1) Battery
(2) Status LEDs
(3) USB port
(4) Green power indicator
    Green LED is illuminated when power is on.
(5) Power input

The batteries are inserted vertically onto the charging slots.

Status LEDs

For each charging slot, there are 4 LEDs. When there is no battery to be charged, all LEDs should be off.

There is 1 bottom LED (orange or green) and 3 green LEDs.



(1) No battery inserted
(2) All LEDs blinking one after another when battery is being inserted
(3) Battery charged < 25%
(4) Battery charged 25% - 50%
(5) Battery charged 50% - 75%
(6) Battery charged 75% - 100%
(7) Battery fully charged
(8) Faulty battery

Charging a fully discharged battery will take a maximum of 4 hours.

Up to three batteries can be charged at a time in the charger.

Charging troubleshooting

Orange fault LED on The battery is not inserted correctly.

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- Remove battery and insert it again correctly

  – or –

  or disconnect the charger from power for 10 seconds.

No charging LEDs active when a battery is inserted

1. Check if the main power is applied to the power input.
2. If problem remains, check that connectors are not damaged or battery is not faulty.

6.2.5 Wall charger

The wall charger safely stores one MAX wi-D portable detector, one MAX mini portable detector, one portable grid for the MAX wi-D, and a spare detector battery.

The wall charger can store one MAX wi-D detector and charge it at the same time.

Charging is indicated by a blinking battery LED on the detector.

The imaging system automatically detects which MAX wi-D detector is placed in the wall holder.

The MAX mini and the spare battery cannot be charged.

The wall charger can be mounted to the wall.

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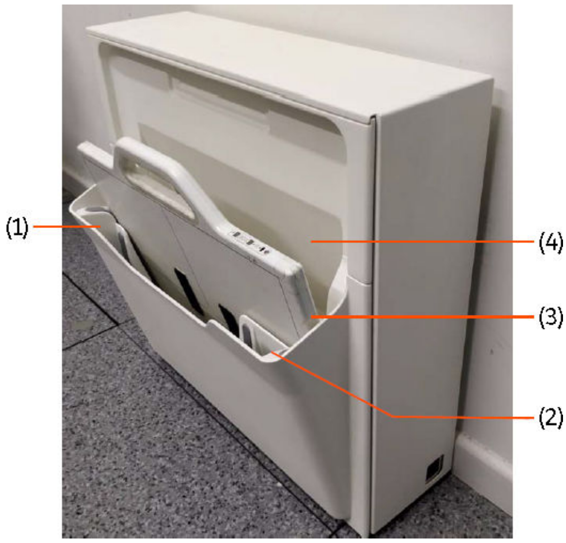

(1) Compartment for one MAX mini (without charging)
(2) Compartment for one spare detector battery (without charging)
(3) Compartment for one MAX wi-D
(4) Compartment for one portable grid for the MAX wi-D

Inserting the MAX wi-D detector correctly for charging

To charge the MAX wi-D detector correctly, proceed as follows:

1. Insert the MAX wi-D detector into the corresponding compartment (charging station) of the wall charger.
2. Make sure that the handle of the detector points up and the charging contacts of the detector face the wall to permit the charging procedure to start.

Removing the MAX wi-D detector

- Grasp the MAX wi-D detector by its handle and pull it out of the wall charger carefully.

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6.2.6 Wall charger for X.wi-D and MAX detectors (option)

(1) Charging compartment for X.wi-D and MAX detectors with cover
(2) Label showing the correct insertion of the detector

The wall charger safely stores one X.wi.D 43, X.wi-D 35, or MAX wi-D wireless detector and charges it at the same time.

Charging is indicated by a blinking battery LED on the detector.

The imaging system automatically detects which wireless detector is placed in the wall charger.

X.wi-D 24 and MAX mini detectors cannot be charged, please use the corresponding battery chargers for that.

The wall charger can be mounted to the wall.

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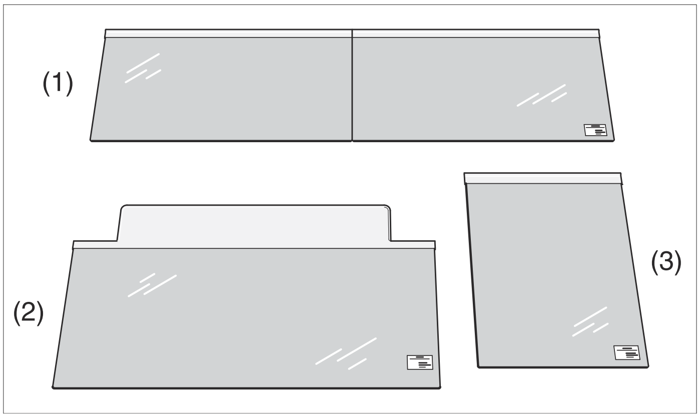
Inserting the wireless detector

◆ Insert the wireless detector as shown above.

6.2.7 Interventional kit

The interventional kit consists of three protective covers as protection against ingress of liquids.



(1) Front cover
(2) Foot end cover
(3) Rear cover

The protective cover consists of plastic sheets for the front, rear, and foot ends of the table. The protective covers are positioned between the tabletop and the table frame to prevent the penetration of liquids.

The front and rear ends can be fixed to the accessory rails of the tabletop. The cover for the foot end can be slid between the tabletop and the table frame.

Attaching the front (1) and rear (3) covers

✓ The tabletop is in the horizontal position

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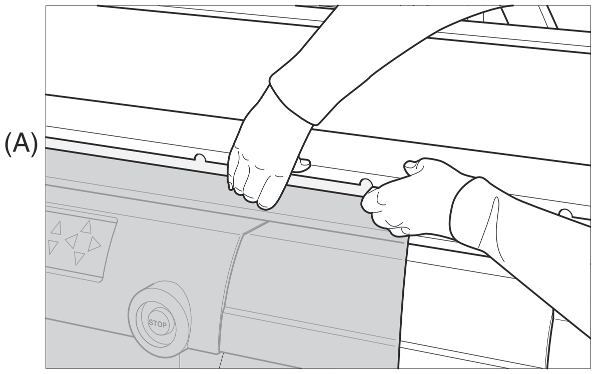
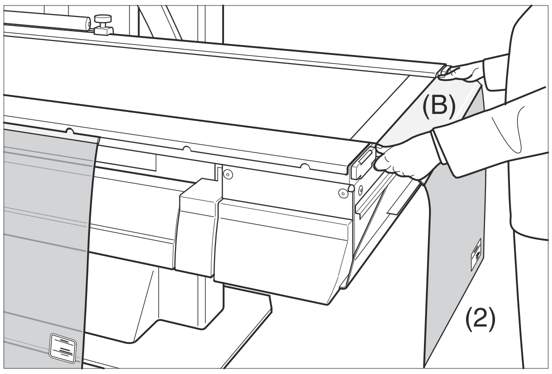


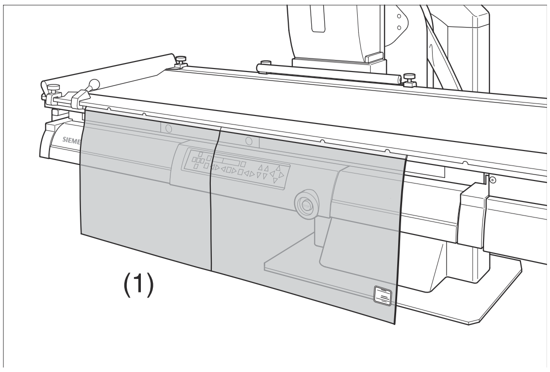
◆ Push the gray strip (A) of the covers from underneath into the accessory rails.

Attaching the cover at the foot end (2)

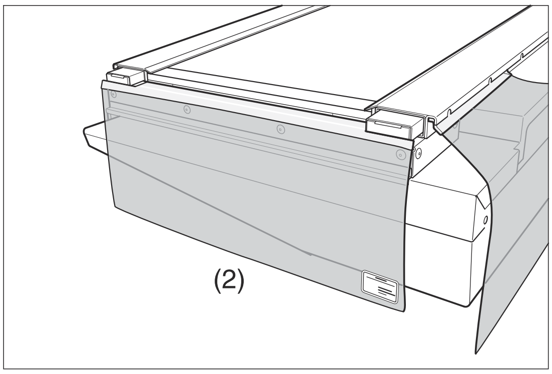
✓ The table is in the horizontal position.



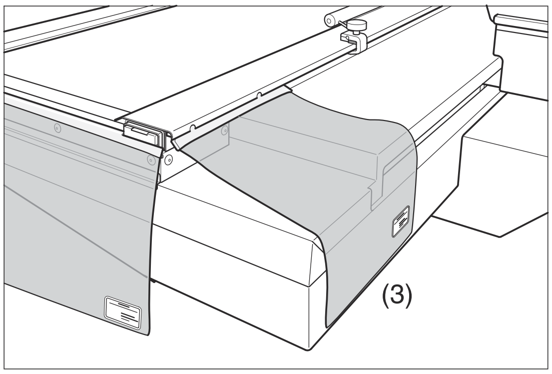
◆ Slide the flat gray end (B) of the cover (2) underneath the tabletop.



Front cover attached (1)



Foot end cover attached (2)



Rear cover attached (3)

Do **not** move tabletop with covers attached!

Removing the covers

✓ The tabletop is in the horizontal position.

◆ Pull the covers out of the accessory rails and from under the tabletop with a smooth motion.

Storing the covers

◆ Store the covers in an appropriate place.

6.2.8 Patient positioning mattress

Application

The mattress is used for comfortable patient positioning and is attached to the protection strip of the tabletop.

Radiation protection value

The attenuation equivalent is < 0.5 mm Al.

Attachment

1 Attach the protection strip at the head end.

2 Place the mattress on the tabletop and pull the loops over the hand screws of the protection strip.

6.2.9 Hand grip, oblique

Application

The hand grips serve for stable and safe patient positioning.

They are provided for the safety of the patient.

To avoid injuries to the patient's hands when moving the tabletop, please ensure that the patient uses the grips and does **not** grasp the ends or sides of the tabletop.

WARNING

Hand grips loosen or are not used

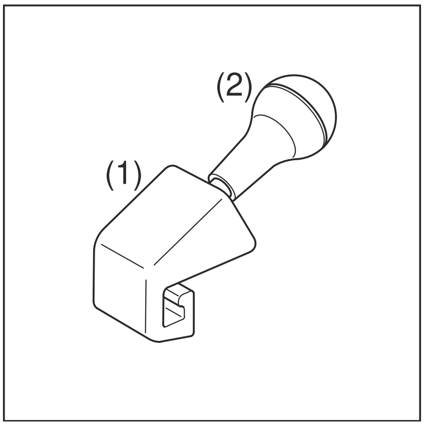
Injury of patient due to falling

◆ Check that the patient hand grips on the table are in the right position.

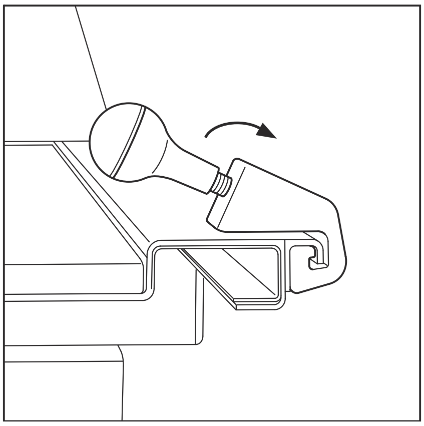
◆ Ensure that hand grips are tightened prior to use.

Attaching the hand grip

The hand grip can be attached to either of the two lateral accessory rails of the patient table.



1 Hold the base (1) of the hand grip and loosen it by turning the hand grip (2) counterclockwise.



2 Insert the hand grip into the groove of the accessory rail and slide it into the desired position.

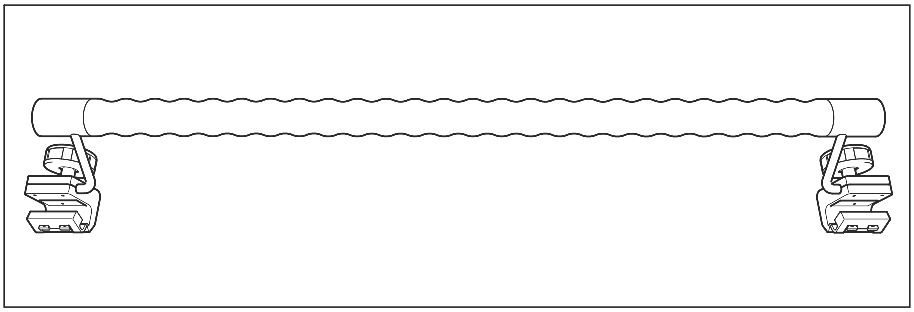
3 Tighten the hand grip by turning it clockwise.

Please make sure that the hand grips are securely fastened.

Removing the hand grip

- Proceed in reverse order.

6.2.10 Hand grip strip



Application

The hand grip strip can be attached to front or back accessory rail in order to assist in positioning the patient.

The maximum permissible load of the hand grip strip is 20 kg.

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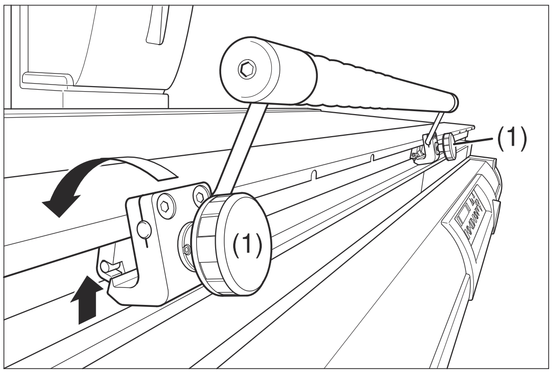
Attachment

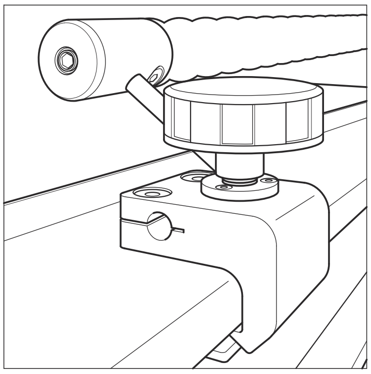


1 Loosen the right and left cap screws (1) on the hand grip strip.

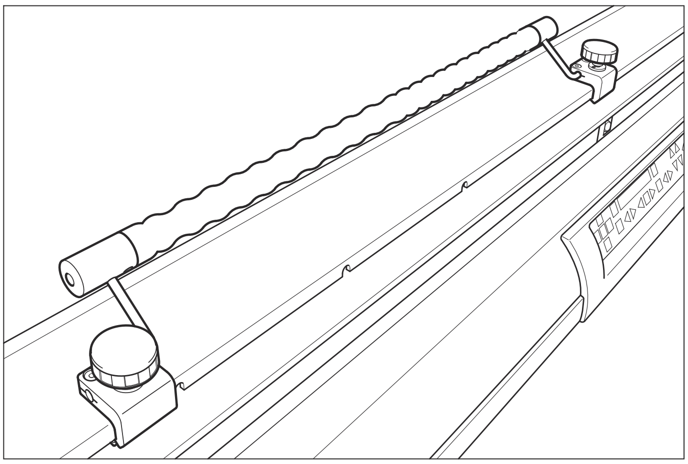
2 Position the hand grip strip from below at the accessory rail as shown in above figure.

3 Lift it up and turn it down (see following figure)

We recommend placing the hand grip strip on the far side of the user.



4 Tighten both cap screws.

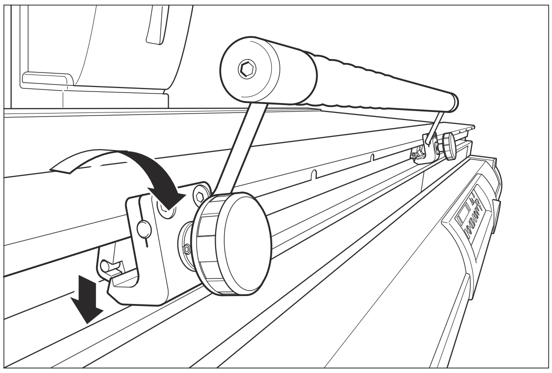


5 Make sure that the hand grip strip is secure by pulling and pressing on it.

The hand grip strip provides additional safety for the patient. To prevent injury to the patient's hands when moving the tabletop, make sure that the patient uses the hand grip strip (not the cap screw) and does not hold onto the headend or the side edge of the patient table.

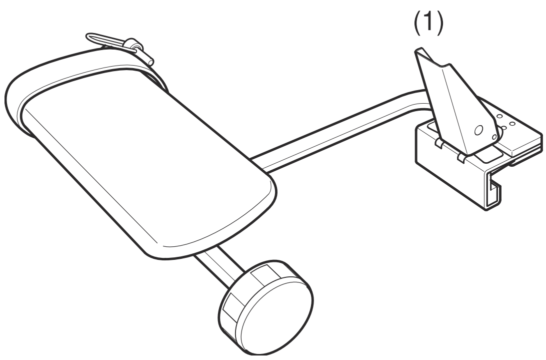
Removal

1 Loosen both cap screws on the hand grip strip.



2 Turn up the hand grip strip and remove it carefully from the accessory rail.

6.2.11 Arm support



Application

For securing the arm of the patient for injections, IVs, or insertion of a catheter.

Attachment

✓ Patient table in horizontal position

1 Position the patient.

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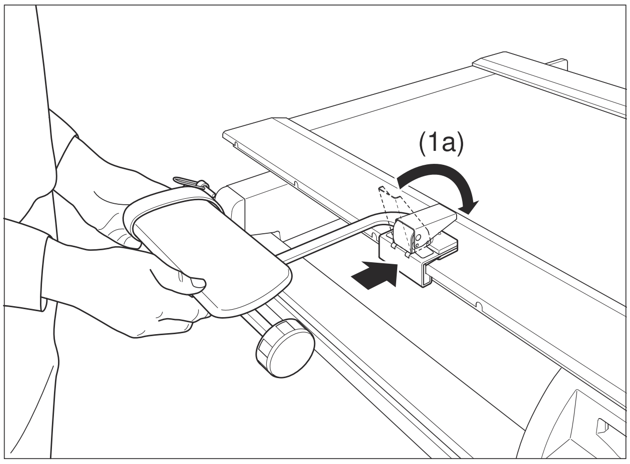
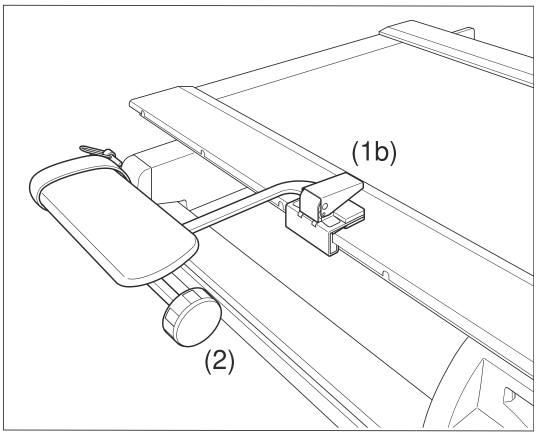


2 Open the clamping lever (1a).

3 Place the arm support onto the front or rear accessory rail and slide it into position.



4 Close the clamping lever (1b).

5 Push and pull on the arm support to ensure it is seated properly.

6 Loosen the cap screw (2).

7 Move the padded arm support to a comfortable position.

8 Tighten the cap screw (2).

9 Ensure that the arm support is seated properly.

10 Place the arm of the patient on the arm support.

11 Place the strap over the arm of the patient and under the pin.

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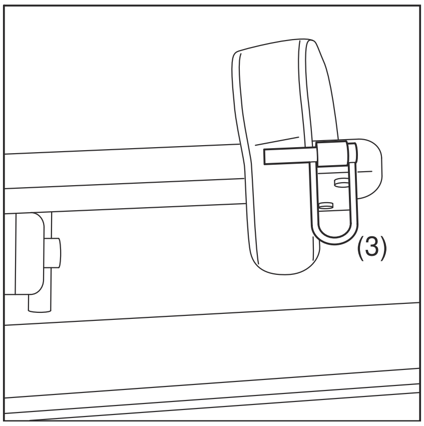


12 Press down on the clamp (3).

13 Ensure that the arm of the patient is stabilized and comfortable.

Removal

✓ Patient table in horizontal position

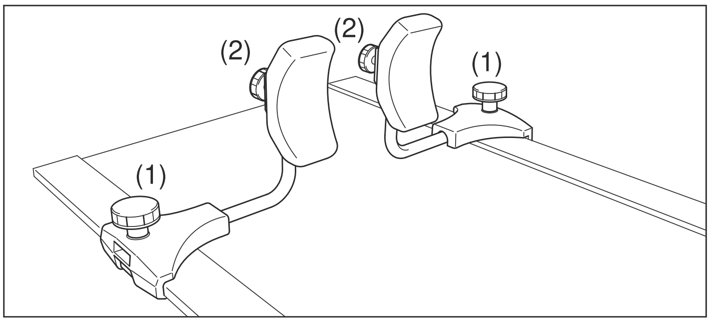
1 Pull the clamp upward and loosen the strap.

2 Raise the arm of the patient from the arm support.

3 Open the clamping lever (1a).

4 Remove the arm support from the accessory rail.

6.2.12 Shoulder supports, 1 pair



Application

To support the patient during examinations in Trendelenburg position, for example Myelography

For >15° Trendelenburg position, the shoulder supports must be used.

Up to 45°, the patient weight must not exceed 180 kg.

In Trendelenburg position >45°, the maximum patient weight of 150 kg must not be exceeded.

The maximum permissible load of the shoulder supports is 75 kg.

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Attachment

✓ Patient table in horizontal position

1 Position the patient.

2 Loosen the cap screws (1) on both supports.

3 Move the supports into position on both accessory rails.

4 Pull the shoulder pads out until they are aligned with the center of the shoulders of the patient.

5 Tighten (1) the cap screws.

6 Check the shoulder supports.

7 Loosen the cap screws (2) for the shoulder pads.

8 Adjust the height of the shoulder pads for secure and comfortable positioning.

9 Tighten both (2) cap screws.

When in Trendelenburg position, the patient is protected against slipping only by the shoulder supports. Pull on the shoulder supports to be sure that they are securely attached before moving the patient into the Trendelenburg position.

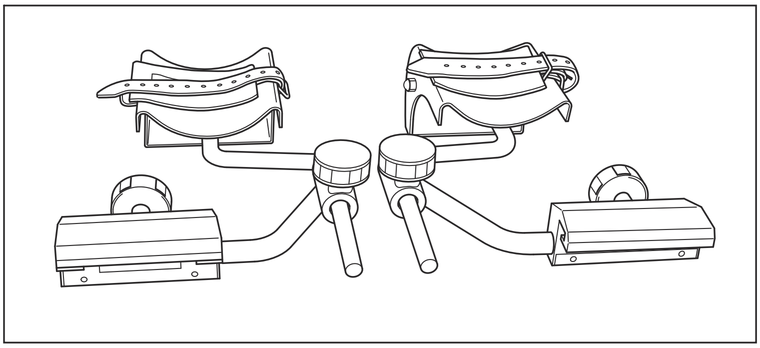
Removal

✓ Patient table in horizontal position

1 Loosen the cap screws (1) on both sides.

2 Pull the shoulder supports out of the accessory rails.

6.2.13 Leg support



Application

For facilitating positioning for gynecological and urological examinations

CAUTION

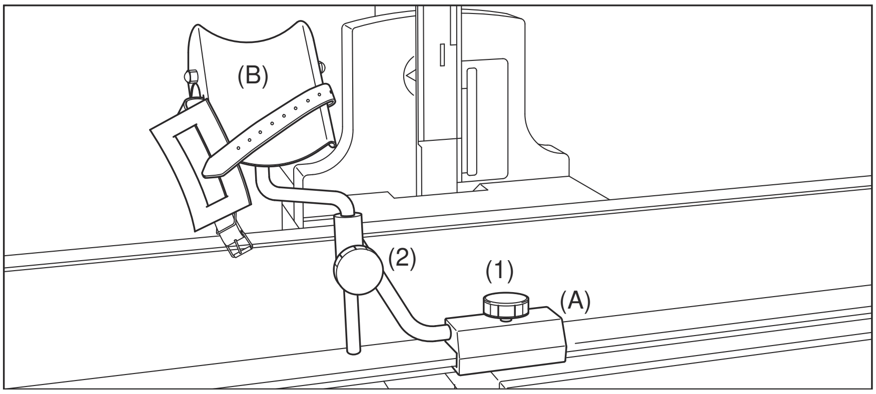
Leg support not properly used

Risk of patient injury

- Do not load the leg support with more than 20 kg each.
- Tighten the cap screws and make sure that the leg support is securely fastened.
- Tighten the straps appropriately.

Attachment

✓ Patient table in horizontal position



1 Loosen both cap screws (1).
2 Slide flanges (A) onto the accessory rails.
3 Insert both molded leg supports (B) into the holders at the cap screws (2).
4 Position the patient on the patient table.
5 Slide the leg supports along the accessory rails to the examination position.
6 Tighten the cap screws (1).
7 Open the securing straps on the molded leg supports.
8 Position the lower legs of the patient on the molded leg supports.

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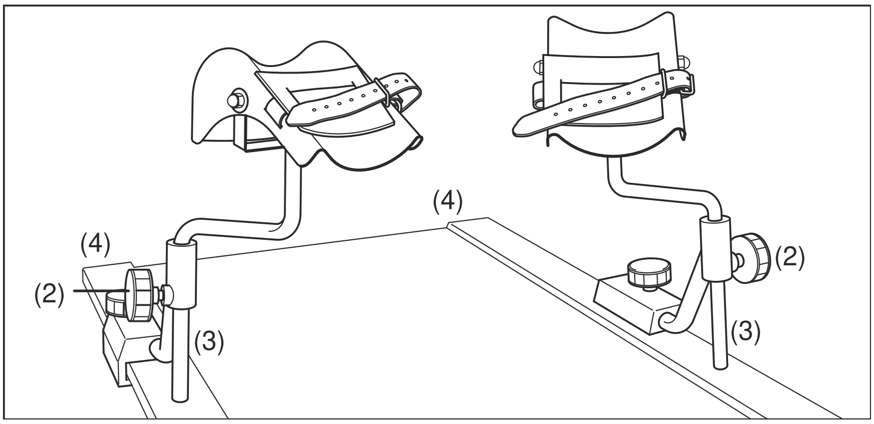

9 Raise the leg supports to the examination position and adjust the position of the legs inward or outward as needed.

When adjusting the height of the leg supports ensure that the distance between the accessory rail (4) and the holder for the leg supports (3) is at least 1 cm.

10 Tighten the cap screws (2).

11 Pull and push on the leg supports and the flanges to ensure they are seated properly.

12 Place the leather pads over the center of the lower legs of the patient.

13 Position the straps around the lower legs of the patient.

14 Tighten the straps appropriately.

Removal

✓ Patient table in horizontal position

1 Release the straps.

2 Lift the legs of the patient from the leg supports.

3 Wrap the straps around the leg supports.

4 If necessary, remove the leg supports by loosening the cap screws (2).

5 Loosen the cap screws (1).

6 Remove the flanges from the accessory rails.

Please place towels between the legs of the patient and the leg supports to avoid direct skin contact with the leg supports.

6.2.14 Foot board

Application

Positioning aid for examinations of patients when standing or sitting and for additional use with the foot holder for recumbent patients in Trendelenburg position.

The maximum permissible patient weight on the foot rest is 230 kg (500 lbs)

WARNING

Foot rest not firmly attached

Patient slides off table with risk of injuries

- Make sure that the foot rest is attached firmly by pulling and pushing it and lifting it up.
- Check that the two green bars on each side of the foot rest are visible.
- Attach and remove only when table is horizontal.

WARNING

Foot board overloaded

Foot board disconnects or breaks resulting in patient injury

- Do not load the foot board with more than 230 kg.
- If the patient weighs more than 230 kg, do not tilt the table more than 45°.
- Take care that the patient can stand securely on the foot board.
- The maximum permissible load of the patient table is 330 kg, including any attachments. Note that the foot board weighs ~15kg. See the list for the weights of other accessories and attachments.

WARNING

Foot rest overloaded

Foot rest disconnects or breaks resulting in patient injury

- Do not load the foot rest with more than 230 kg.
- If the patient weighs more than 230 kg, do not tilt the table more than 45°.
- Take care that the patient can stand securely on the foot rest.
- The maximum permissible weight for the patient table is 300 kg.

CAUTION

Attaching or detaching the foot board

Risk of injury

- Consider the heavy weight when handling the foot board.

  Weight of the foot board: ~ 15 kg.

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The foot board can also be attached to the head end of the patient table, if the patient has to be turned for positioning (left-hand side head end).

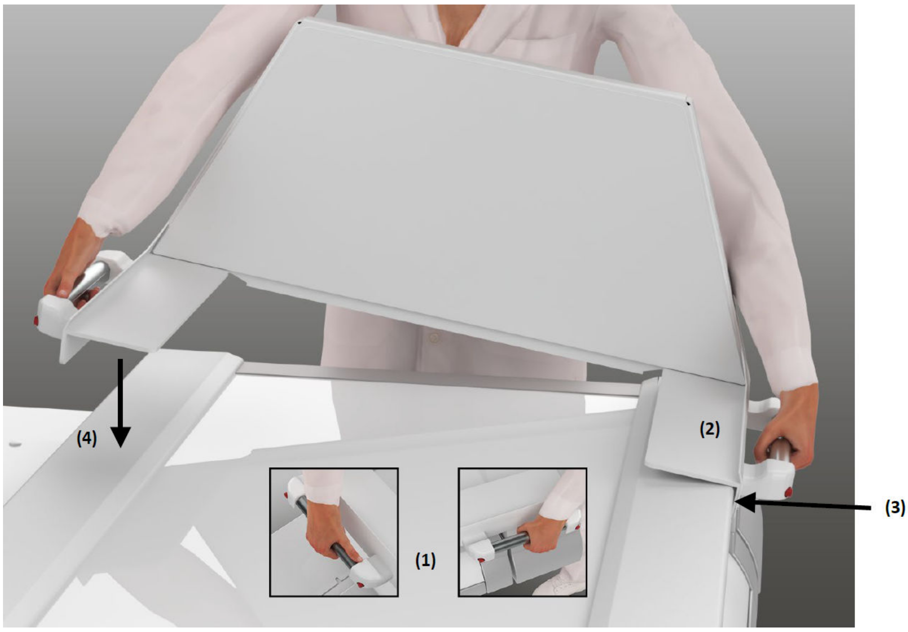
Three lock-in positions every 20 cm at the head end and foot end are available.

The room height calculator does not consider the foot board attached at the head end.

Be extremely careful when raising or tilting the unit with the foot board attached at the head end and extended tabletop.

Attachment

✓ Patient table in horizontal position.



1 Hold foot board laterally on the hand grips and press the spring-loaded parts of the hand grips.

   The two red buttons (1) on each side must be visible.

2 First, tilt the foot board to the left and hook it in the left-hand side accessory rail (2). Press the left hand grip firmly to the accessory rail and release the spring-loaded part of the left hand grip (3).

3 Then lower the right side onto the accessory rail and release the spring-loaded part of the right hand grip (4).

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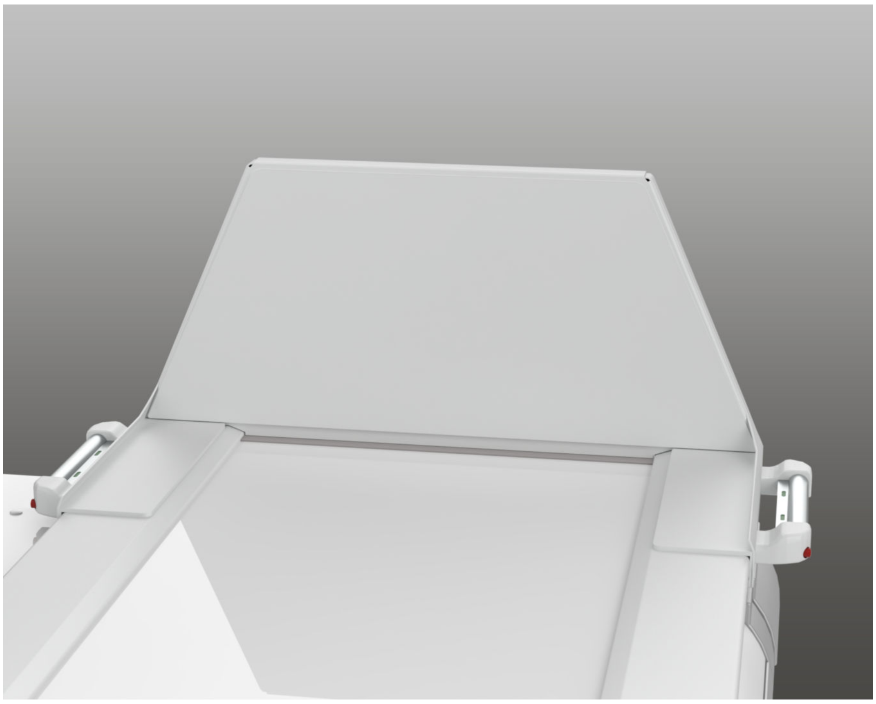
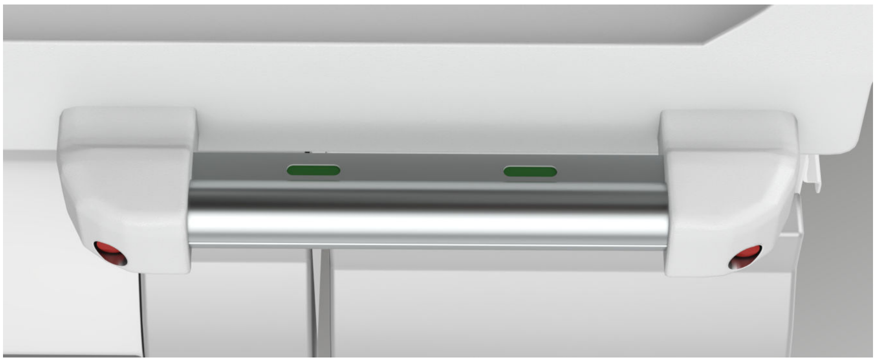


4 Move the foot board firmly back and forth until you hear it click into place.

The two red buttons must be flush with the hand grip and the two green bars must be visible on both hand grips.



5 You can shake the foot board vigorously by grasping the foot board itself (rather than the hand grips). Make sure that both sides of the foot board are securely attached to the table.

6 Position the patient.

Be sure to check that the foot board has locked into position every time before allowing a patient to sit or stand on it.

Make sure that patients standing on the foot board hold on to the hand grip or the hand grip strip.

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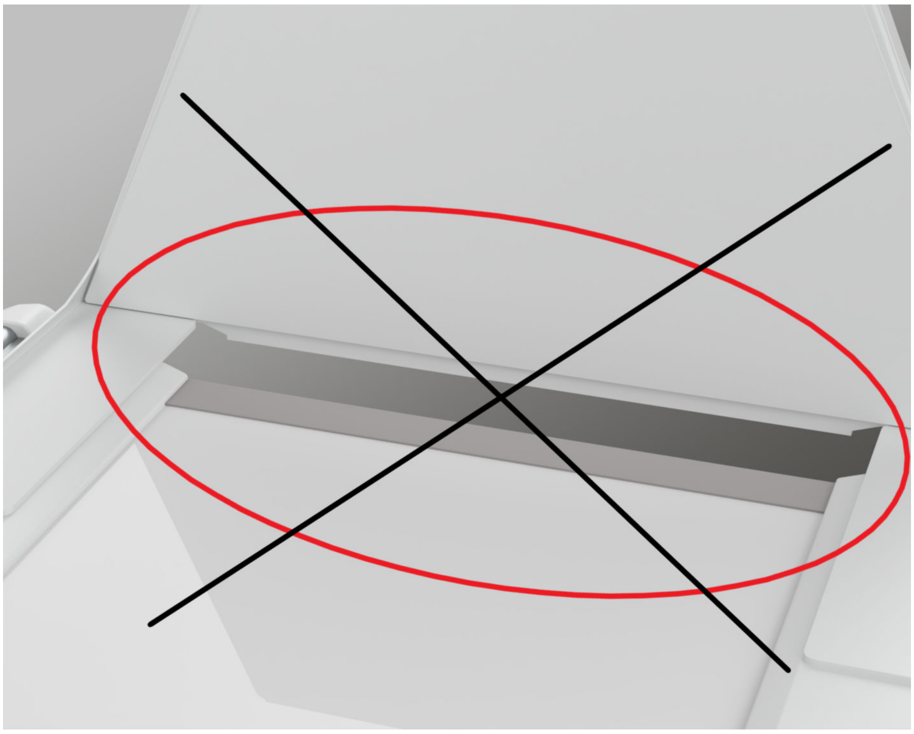
WARNING

Table tilted in Trendelenburg (head-down) position

Patient slides off table head-first

- Use shoulder supports to prevent the patient from sliding off the table.
- If the table must be tilted more than 15° Trendelenburg (head-down), use the foot holder in addition to the shoulder supports.

Samples of incorrect attachment



There should be no visible gap between the end of the tabletop and the foot rest.

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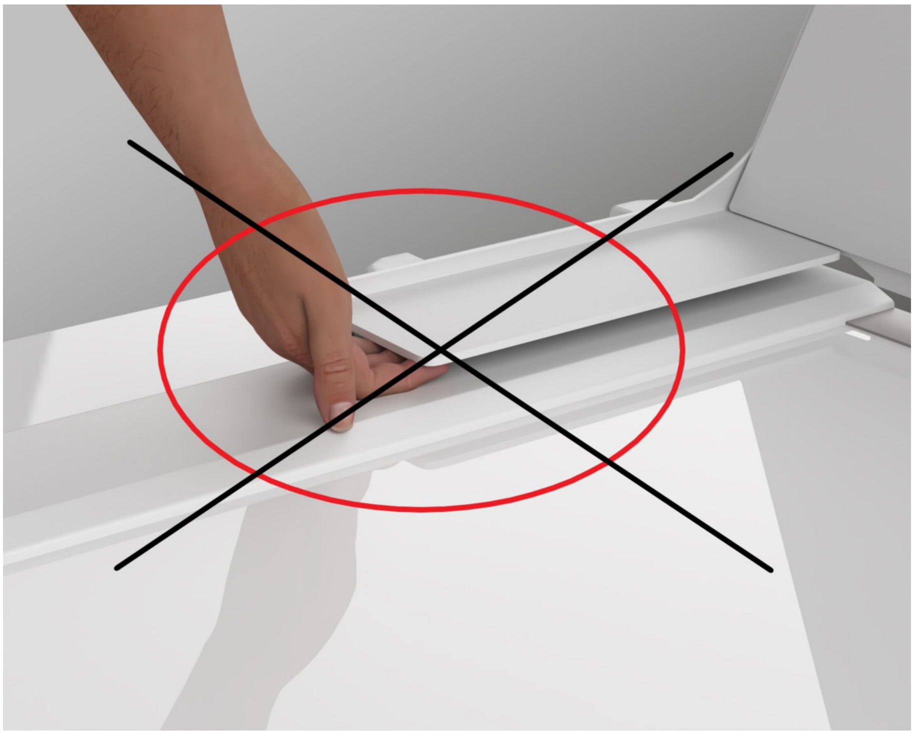

The foot rest must be flush with the tabletop, if there is too much play (like in the Figure above) the attachment is not secure. You should not be able to get your fingers between the foot rest and the table.

Adjustment

1 Press the spring-loaded hand grip strips in the hand grips.

The two red buttons on each side must be visible.

2 Move the foot rest to the desired position (lock-in position).

3 Release the hand grip strips and shift the foot rest slightly back and forth until it locks in.

The two red buttons on each side must be flush with the hand grip.

The foot rest may only be unlocked, adjusted or removed when the table is in the horizontal position. In all other system positions, the foot rest could fall off and present a high risk of accidents.

Removal

✓ Patient table in horizontal position.

1 Press the spring-loaded hand grip strips in the hand grips.

The two red buttons on each side must be visible;

2 Lift the foot rest first on the right-hand side from the accessory rail.

3 Then remove the foot rest on the left-hand side from the accessory rail.

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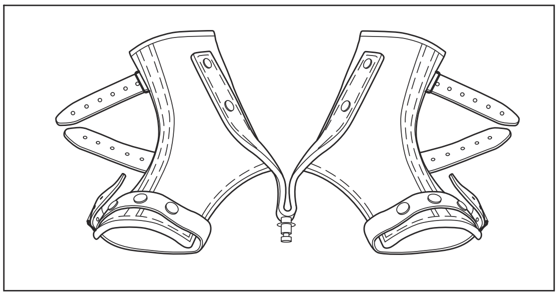
Maintenance note

- To prevent contrast medium from penetrating into the sockets, cover the foot rest surface with an absorbant fleece pad.

6.2.15 Foot holder



Application

For securing the patient during examinations in the Trendelenburg position with a relaxed spinal column.

At a Trendelenburg position >40°, the foot holder must be used in addition to the shoulder supports.

Up to 45° the maximum patient weight must not exceed 180 kg.

The maximum patient weight must not exceed 150 kg at a Trendelenburg position >45°.

Inspection/maintenance note

Check that the foot holder is in a proper condition at regular intervals of about two months.

Check especially that the fastening rivets, the supporting straps and closing straps are firmly attached. Check that their seams are complete and that the leather is free of cracks and porous areas.

If the foot holder shows any signs of defects, take it out of service and obtain a replacement through Siemens Healthineers Customer Service.

We recommend using a commercially available transparent impregnating pump spray for leather for cleaning and care.

Attachment

✓ Patient table in horizontal position

1 Attach the foot rest.

2 Make sure that the foot rest is attached firmly by pulling and pushing it.

3 Position the patient in the prone or supine position with the feet resting against the foot rest.

4 Place the foot holder closely around the ankles of the patient.

5 Tighten the straps.

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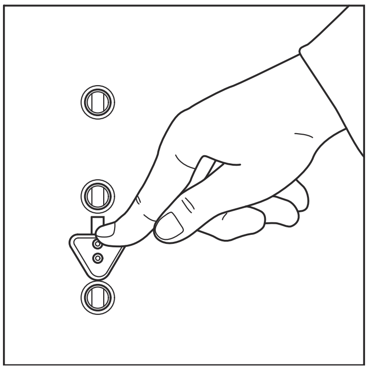
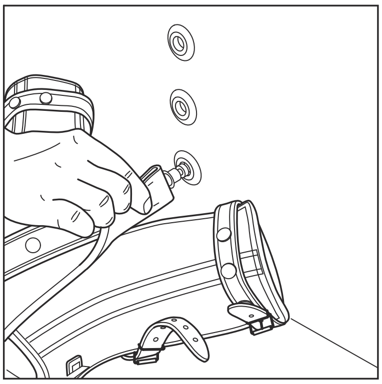


6 Push the triangular locking tab on the back of the foot rest down and hold it firmly.



7 Insert the metal spigot of the foot holder in the hole of the appropriate foot rest that will enable the patient to lie quietly and relaxed.

The heels or toes of the patient should lie lightly on the tabletop.

8 Release the locking tab.

9 Check the locking of the foot holder by pulling on it.

Ensure without fail that the metal spigot of the foot holder is locked in the foot rest.

Instruct the patient to hold the hand grips firmly during this examination!

Removal

✓ Patient table in horizontal position

1 Push the triangular locking tab on the back of the foot rest down and hold it firmly.

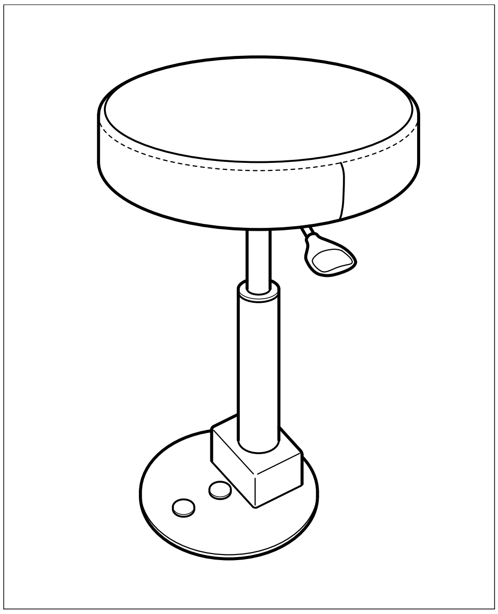
2 Pull the metal spigot of the foot holder out of the hole of the foot rest.

3 Loosen the straps.

4 Lift the feet of the patient out of the foot holder.

Print No. XPD3-530G.620.01.04.02

6.2.16 Swivel stool



Application

The swivel stool is attached to the foot rest and is used for examinations in sitting position.

The rotation range of 360° allows front and side views without standing.

WARNING

Movement of the tube and the table with a patient sitting on the swivel stool

Danger of crushing the patient (knee, hand) with the compression unit

- Pay special attention to the danger zone between the compression unit and the sitting patient.
- Move the table as little as possible when the patient is positioned on the swivel stool.

CAUTION

Patient falls from the swivel stool

Risk of patient injury

- Make sure that the patient is seated securely on the swivel stool.
- Position a hand grip so that the patient can hold on.
- Do not move the table remotely when a patient is positioned on the swivel stool; perform all movements from the table-side control.

The maximum permissible patient weight on the swivel stool is 150 kg.

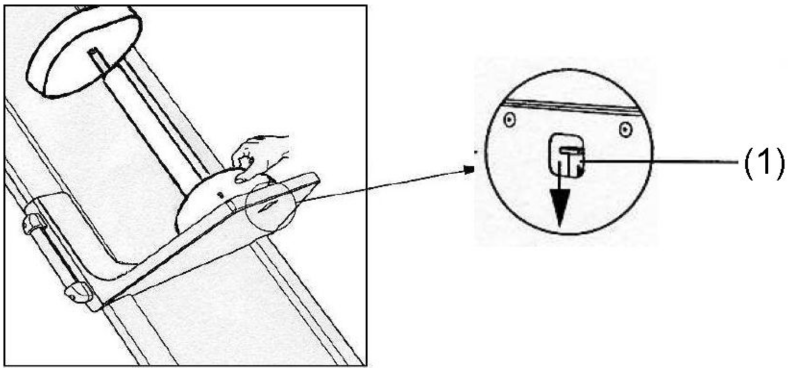
Inspection/maintenance note

Check that foot rest and swivel stool are in working condition at regular intervals of about two months.

Attachment

✓ Foot rest attached. (➔ Page 239 *Foot board*)

✓ Table tilted to 20°.



1 Place the swivel stool at the front edge of the foot rest as shown above.

2 Push the locking tab (1) on the back of the foot rest down and hold it firmly.

3 Insert the metal pins at the base of the swivel stool in the corresponding holes of the foot rest.

4 Release the locking tab.

5 Check the locking of the swivel stool by pulling on it.

⚠ CAUTION

The swivel stool detaches

Risk of patient injury

- Make sure that the swivel stool is attached firmly.
- Push and pull the stool to make sure that it is secure before positioning the patient.

6 Tilt the table to 90°.

Removal

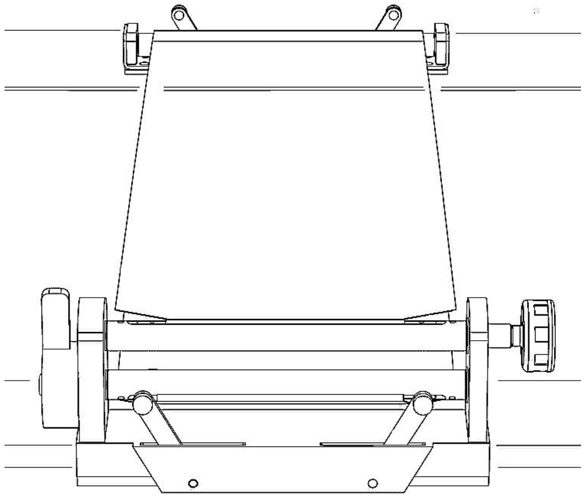
✓ Table tilted to nearly 80°.

1 Push the locking tab on the back of the foot rest down and hold it firmly.

2 Pull the metal pins of the rotating chair out of the holes of the foot rest.

Print No. XPD3-530G.620.01.04.02

6.2.17 Compression belt



WARNING

Compression belt cannot hold patient if table is tilted.

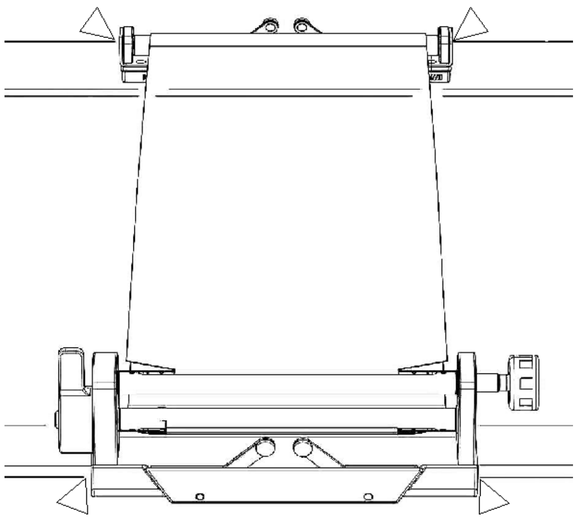
Patient can fall from the table!

- Do not tilt the patient table more than ± 15˚ when using the compression belt.

Risk of crushing on compression belt

It is the responsibility of the operator to attach and remove the equipment only if it is ensured that neither the patient nor third parties can be endangered.

The parts of the unit marked in the drawing show possible danger zones where fingers can be crushed.



▽ Risk of injury for the personnel when attaching compression belt.

Print No. XPD3-530G.620.01.04.02

Functional and safety check of compression belt

Daily checks

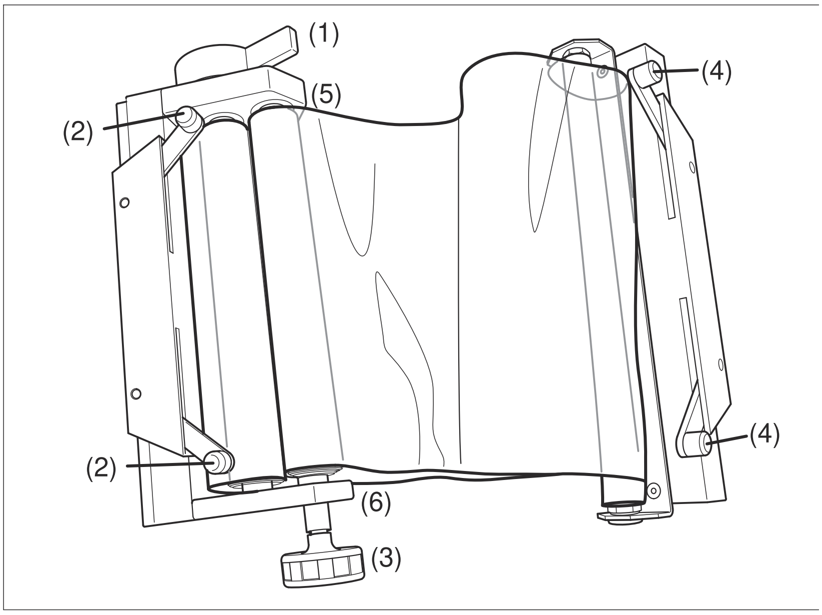
1 Check tension lever.

2 Check cap screw.

3 When the cap screw and tension levers are in locked status, check if the compression belt is locked.

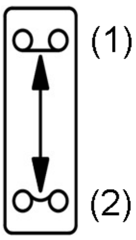
Remove all traces of contrast agent from the compression belt. Otherwise, the X-ray image shows artifacts.

Description of compression belt



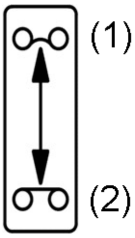
(1) Ratchet handle
(2) Tension lever
(3) Cap screw
(4) Tension lever
(5) Tag for ratchet handle
(6) Tag for cap screw

Ratchet handle



Tag for ratchet handle

(1) upward: tightened
(2) downward: released



Tag for cap screw

(1) upward: released
(2) downward: tightened

Intended use

The compression belt is a mechanical component for X-ray exposures.

It is used for patient compression on the table or BWS, reducing the thickness of the patient which leads to better image quality for some examinations, especially abdominal.

Product features

- High reliability
- Easy attachment
- Increase image quality

Requirements

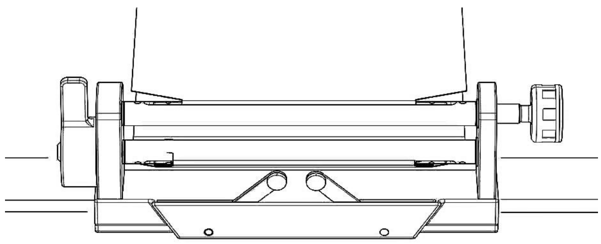
The X-ray imaging system must be operated by trained and qualified personnel. In addition, make sure to follow the requirements of relevant laws and regulations during the operating process.

Using the compression belt on the patient table

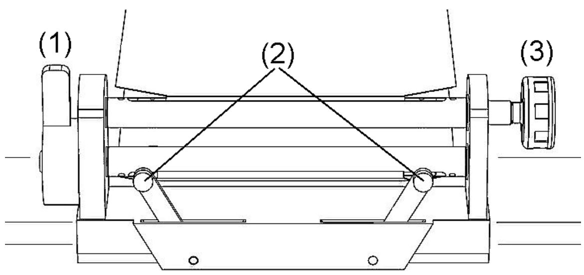
Attachment

1 Move the patient table to the horizontal position.

2 Position the patient on the table.



3 Push the tensioning part of the compression belt into the required position over the front accessory rail.



4 Press both tension levers (2) outward.

5 Check that the tensioning roller is firmly seated by pulling on it.

6 Press the ratchet handle (1) downward.

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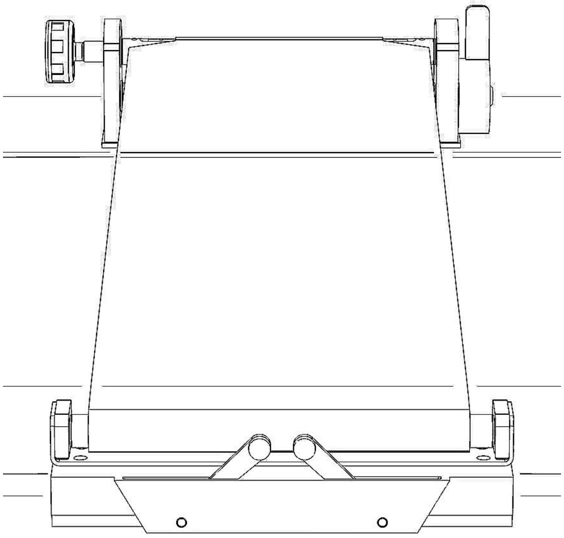
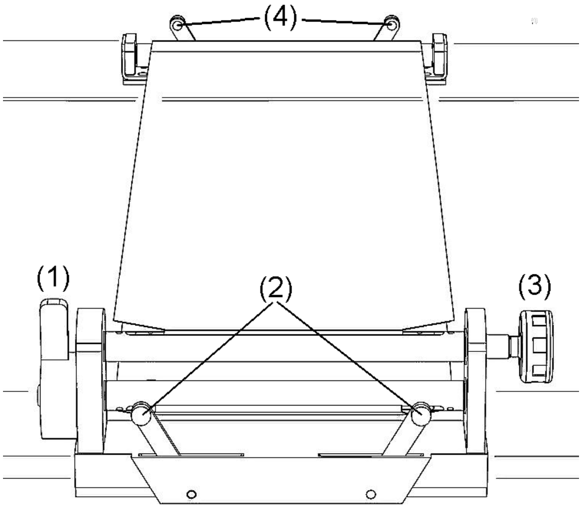
The belt is released.

For obese patients, the belt holder is easier to secure if you are standing at the back of the patient table. In this case, and for restless or frail patients, we recommend that someone assist you in attaching the compression belt

7 Lift the holder with the belt across the patient.



8 Push the holder over the back accessory rail. Make sure that the compression belt attaches straight across the tabletop and not at an angle.



9 Press the tension levers (4) outward.

10 Check that the holder is seated firmly by pulling on it.

11 Turn the cap screws (3) on the tensioning roller toward the left to tighten the belt.

12 Push and pull back on the ratchet handle (1), until the belt tension is optimal.

13 Check the belt tension on the patient once again.

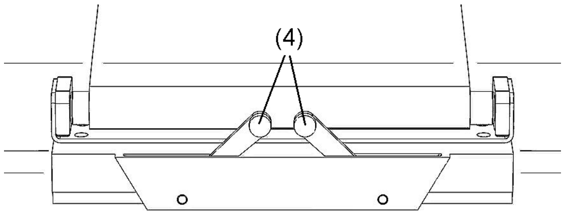
Print No. XPD3-530G.620.01.04.02

Loosening the compression belt

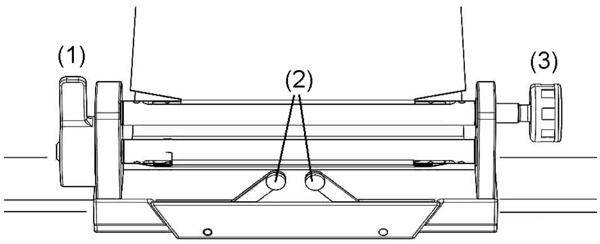
1 Move the patient table to the horizontal position.
2 Press the ratchet handle (1) back.
3 Turn the cap screw (3) on the tensioning roller toward the right.
  The compression belt unclenches.
4 Use the ratchet handle to set a lower belt tension.

Removal

1 Move the patient table to the horizontal position.
2 Press the ratchet handle (1) back.
3 Turn the cap screw (3) on the tensioning roller toward the right.
  The compression belt unclenches.



4 Press both tension levers (4) on the belt holder inward.
5 Pull the belt holder backward off the accessory rail.
6 Lift the belt holder over the patient.
7 Roll up the compression belt halfway using the cap screw.
8 Place the belt holder next to the tensioning roller.



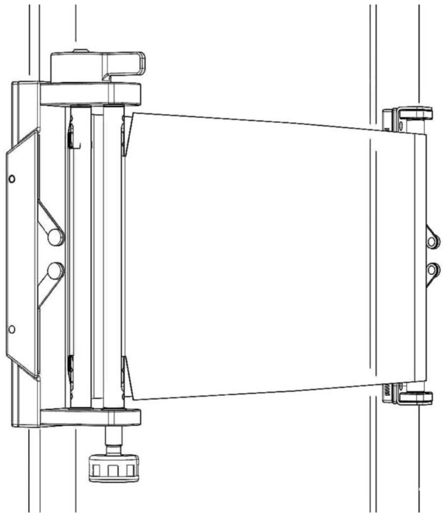
9 Press both levers (2) inward on the tensioning roller.
10 Pull the tensioning roller from the front accessory rail.
11 Help the patient to get off the patient table.
12 Use the cap screw (3) to roll up the belt on the tensioning roller.

Using the compression belt on the wall stand

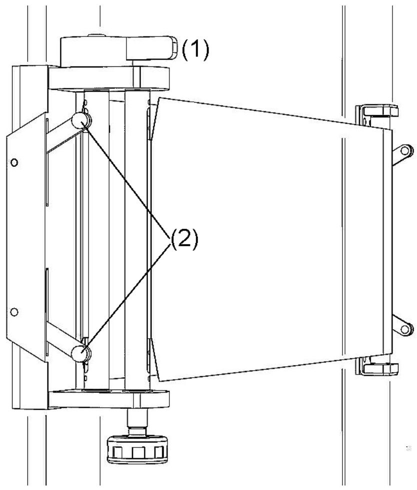
Attachment

1 Position the patient.

Print No. XPD3-530G.620.01.04.02



2 Attach the clamping part of the compression belt to the left accessory rail at the required position.



3 Push both tension levers (2) outwards.

4 Make sure that the clamping part is attached firmly by pulling and pushing it.

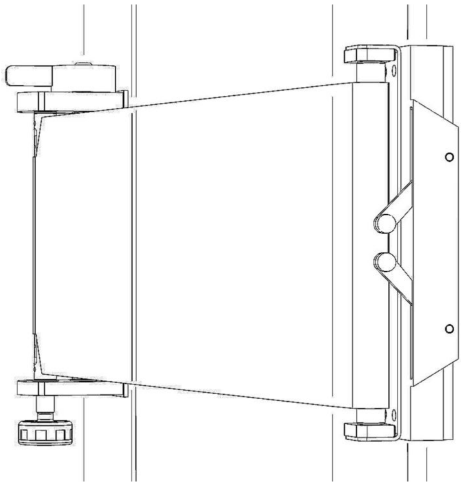
5 Push the ratchet handle (1) to the back.

The belt unclenches.

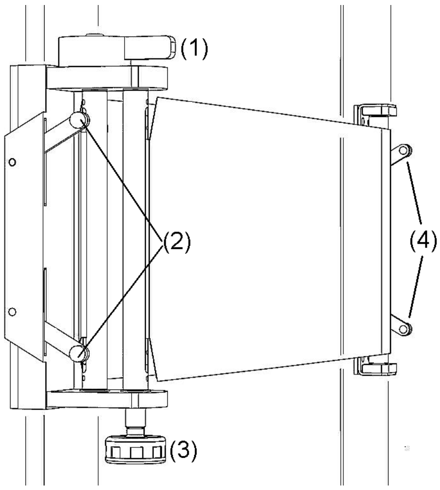
For the attachment to the wall stand, we recommend that two persons fasten the compression belt.

6 Lift the holding part with belt over the patient.

Print No. XPD3-530G.620.01.04.02



7 Push the holding part over the right accessory rail. Take care that the belt runs at right angles to the top and is not applied obliquely.



8 Push both tension levers (4) outwards.

9 Make sure that the holding part is attached firmly by pulling and pushing it.

10 Turn the cap screw (3) on the clamping part to the left.

The belt is tensioned.

11 Push the ratchet handle (1) backwards and forwards until the belt is tensioned optimally.

12 Check the belt tension once again on the patient.

Loosening the compression belt

1 Push the ratchet lever (1) to the back.

2 Turn the cap screw (3) on the clamping part to the right.

The belt unclenches.

3 Set a reduced belt tension once again with the ratchet lever.

Removal

1 Push the ratchet lever (1) to the back.

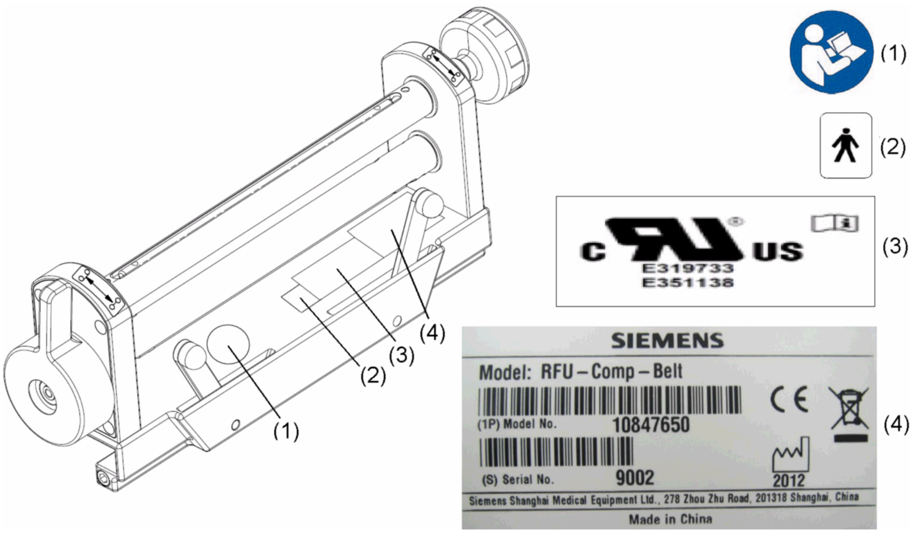
2 Turn the cap screw (3) on the clamping part to the right.

The belt unclenches.

Technical data of compression belt

Dimensions	Width	≤ 100 mm
	Length	≤ 400 mm
	Height	≤ 150 mm
Ambient conditions (Operation)	Temperature:	+10°C to +35°C
	Relative humidity:	20% to 75% rel. air humidity without condensation
	Barometric pressure:	700 hPa to 1060 hPa
Ambient conditions (Transport and storage)	Temperature:	-20°C to +70°C
	Relative humidity:	10% to 95% rel. air humidity without condensation
	Barometric pressure:	500 hPa to 1060 hPa
Aluminum equivalent	≤ 0.8 mm Al	

Location of labels on compression belt



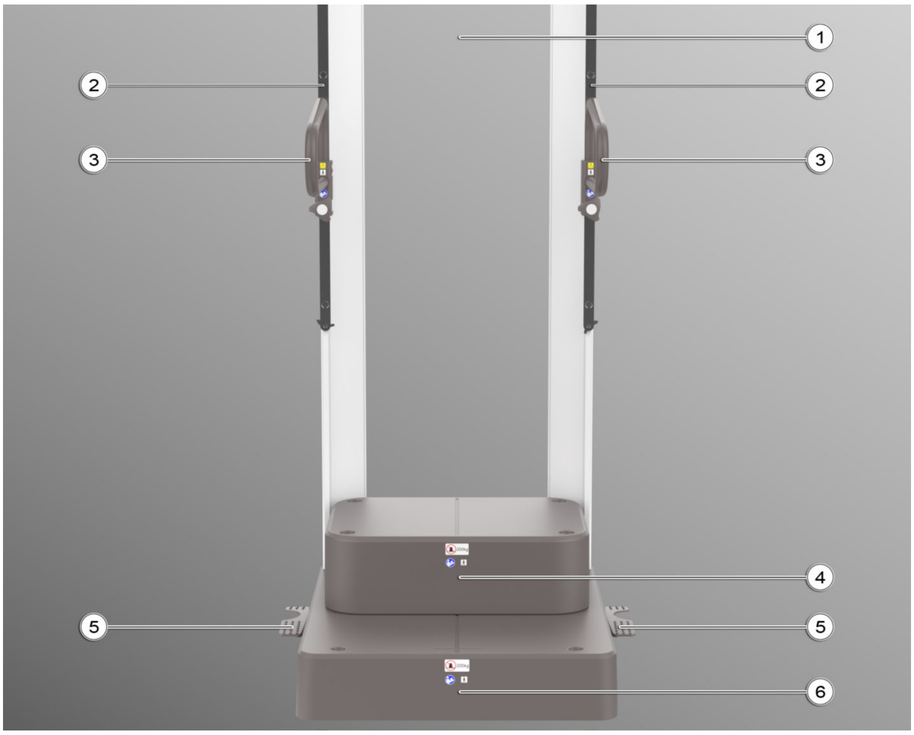
(1) Warning label
(2) Type B equipment label
(3) UR label/ UR
(4) Identification label for compression belt

Print No. XPD3-530G.620.01.04.02

6.2.18 Multipurpose / SmartOrtho stand (11105090)

The SmartOrtho stand is necessary to position the patient correctly for Ortho imaging.

Overview



(1) Radiotransparent rear plate
(2) Accessory rails (on both sides)
for attachment and height adjustment of accessories
(3) Hand grips (on both sides)
(4) Additional removable platform
for smaller patients (for example children) and for examinations of the lower extremities
(5) Brake pedals (on both sides)
(6) Patient platform (with fixing holes for additional platform)

Print No. XPD3-530G.620.01.04.02

Safety

WARNING

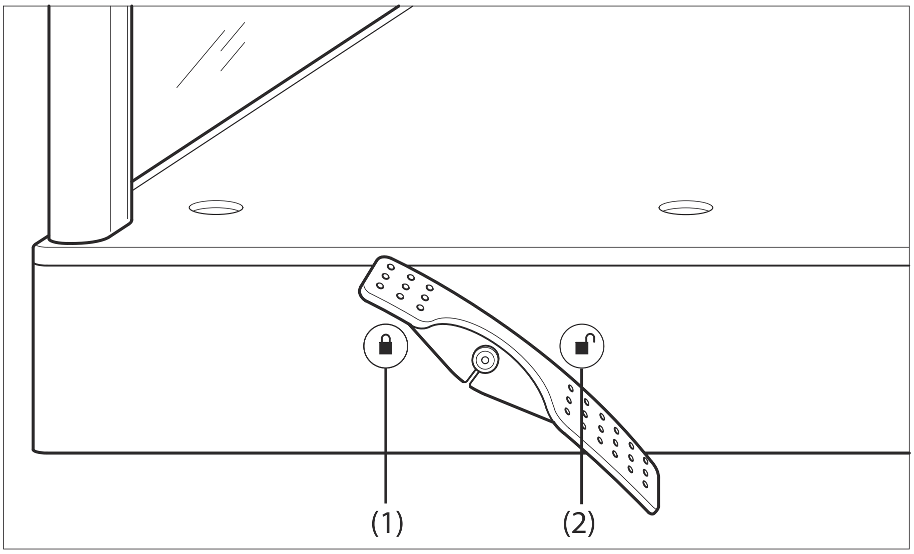
Patient cannot maintain position for ortho

Injury to patient, repetition of examination

- Always use the ortho support for ortho examinations.
- Check that the hand grips are at the right position and are tightened prior to use.
- Make sure that the patient is stable enough to maintain position and is using the hand grips.

Using the brakes

The support is equipped with brakes to enable the secure it. For transport of the support you have to release the brakes. The brakes are operated with the brake pedals on both sides of the patient standing platform.

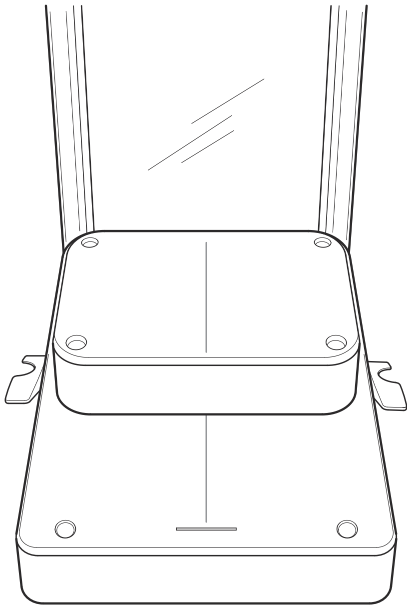


1 To lock the brakes push the pedal down towards the symbol of the closed padlock (1).

2 To release the brakes push the pedal down towards the symbol of the open padlock (2).

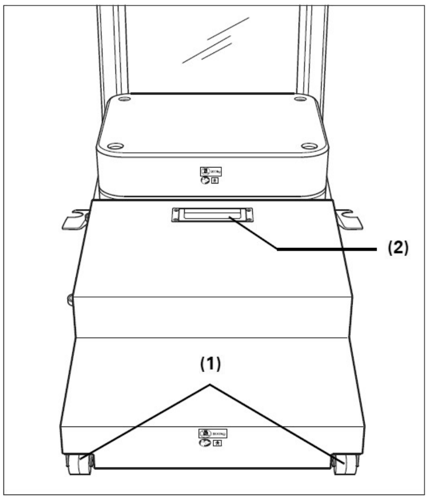
Print No. XPD3-530G.620.01.04.02

Additional platforms



Example: One additional platform

For examinations of the feet with both additional platforms, it may be necessary to help the patient to get easier on the upper platform. In this case, use the additional steps.



(1) Wheels (there are more wheels on the left side)
(2) Handle (another handle is on the right side)

Hand grips

- The hand grips (3) are intended to support the patient while standing during the examination.
- The hand grips can be moved along the guide rails to the required height.
- The measuring scale on the hand grips can be used to estimate the TOD (Table -Object-Distance) in cm or inch.

The maximum load of the hand grips is 30 kg.

Attaching the hand grips

The hand grips can be attached to the guide rails from below for use as patient support or from above for use as patient stretch grip.

WARNING

Hand grips loosen or are not used

Injury of patient due to falling

◆ Check that the patient hand grips on the table are in the right position.
◆ Ensure that hand grips are tightened prior to use.

Print No. XPD3-530G.620.01.04.02

⚠ CAUTION

Patient uses hand grips for support when mounting the ortho support.

Injury of patient

- The hand grips can only support 30 kg*. If necessary, have the patient grasp the vertical plate when mounting the stand.
- Then position the hand grips to the correct height for the examination.
- Make sure that the patient's hands are not on the vertical plate during the examination.

*50 kg for the RAX stand

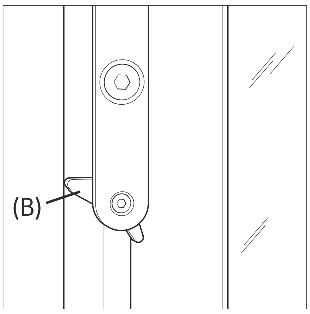
1 Turn the locking screws (A) on the hand grips counterclockwise.

2 Slide the hand grips from below or from above on the guide rail to the required height.

3 Fix the hand grips (A) by turning the locking screws clockwise.

Check that the locking screws on both hand grips are properly fixed to prevent the hand grips from disengaging while the patient is leaning on them.

Disengaging hand grips might cause injuries to the patient and/or damage to the support!

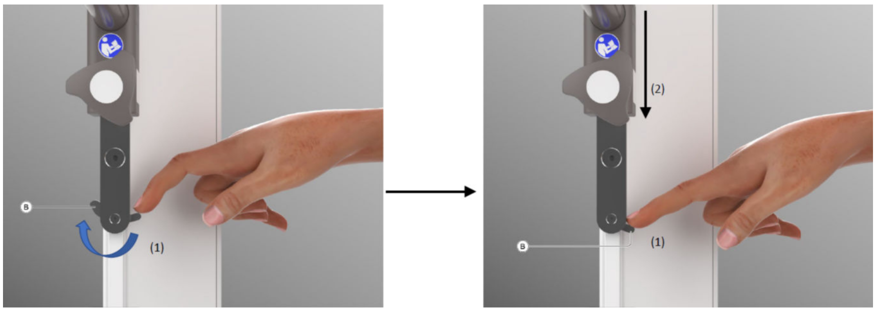


In addition to the locking screws, there is a security lever (B) at the bottom of each guide rail to prevent the hand grips from falling to the ground if the locking screws are not properly fixed.

Removing the hand grips

1 To remove the hand grips when used as stretch grip, just turn the locking screws counterclockwise and remove the hand grips upwards from the guide rails.

2 To remove the hand grips when used as support for the patient, turn the locking screws counterclockwise and slide the hand grips downward to the lower end of the guide rails where they are stopped by the security lever.



3 Press the security lever to the side as shown in above figure and remove the hand grips downwards from the guide rails.

Technical data

Height	196.5 cm (when brake is on) 198 cm (when brake is off)
Depth	85 cm
Width (incl. brake pedals)	82 cm
Weight (incl. additional platform and two hand grips)	112 kg
Max. patient weight	200 kg
Max. patient size	190 cm
Attenuation equivalent of the transparent protection plate acc. to DIN EN 60601-1-3	< 1 mm Al equal - 100 kV 3.2 HVL

Print No. XPD3-530G.620.01.04.02

6.2.19 Multipurpose / Ortho stand (11584200)

The Ortho stand is necessary to position the patient correctly for Ortho imaging at the wall stand.

WARNING

Patient cannot maintain position for ortho

Injury to patient, repetition of examination

- Always use the ortho support for ortho examinations.
- Check that the hand grips are at the right position and are tightened prior to use.
- Make sure that the patient is stable enough to maintain position and is using the hand grips.

CAUTION

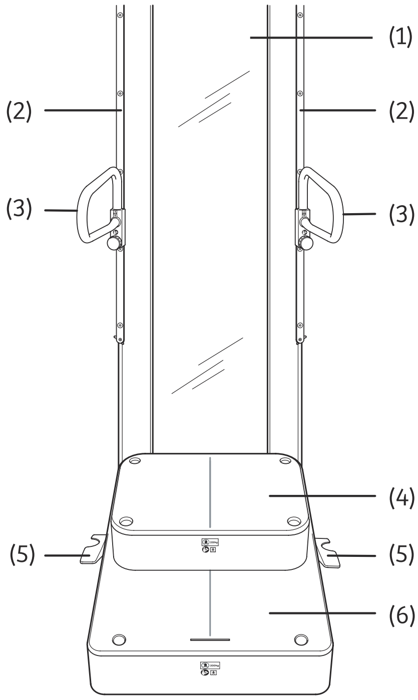
Riding on wheeled accessories

Accessory overbalances and tips over causing damage to system or injury of persons.

- Do not ride or allow anyone else to stand, sit or ride on wheeled accessories while they are being rolled.

Print No. XPD3-530G.620.01.04.02

Description of the Ortho stand



(1) Radiotransparent rear plate
(2) Accessory rails (on both sides)
for attachment and height adjustment of accessories
(3) Hand grips (on both sides) with scales in cm and inch
(4) Additional removable platform for smaller patients (for example children) and for examinations of the lower extremities
(5) Brake pedals (on both sides)
(6) Patient platform (with positioning indentations for additional platform and swivel stool)

Handgrips on the Ortho stand

- The hand grips (3) are intended to support the patient while standing during the examination.
- The hand grips can be moved along the guide rails to the required height.
- The measuring scale on the hand grips can be used to estimate the TOD (Table -Object-Distance) in cm or inch.
- The maximum load of the hand grips is 30 kg.

WARNING

Hand grips loosen or are not used

Injury of patient due to falling

◆ Check that the patient hand grips on the table are in the right position.

◆ Ensure that hand grips are tightened prior to use.

Attaching the handgrips

The hand grips can be attached to the guide rails from below for use as patient support or from above for use as patient stretch grip.

Print No. XPD3-530G.620.01.04.02

CAUTION

Patient uses hand grips for support when mounting the ortho support.

Injury of patient

- The hand grips can only support 30 kg*. If necessary, have the patient grasp the vertical plate when mounting the stand.
- Then position the hand grips to the correct height for the examination.
- Make sure that the patient's hands are not on the vertical plate during the examination.

*50 kg for the RAX stand

1 Turn the locking screws (A) on the handgrips counterclockwise.

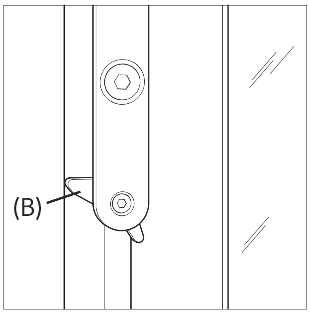
2 Slide the handgrips from below or from above on the guide rail to the required height.

3 Fix the handgrips (A) by turning the locking screws clockwise until you hear a clicking noise.

Check that the locking screws on both handgrips are properly fixed. Thus, you prevent the handgrips from disengaging while the patient is leaning on them.

Disengaging handgrips might cause injuries to the patient and/or damage to the support!

Print No. XPD3-530G.620.01.04.02

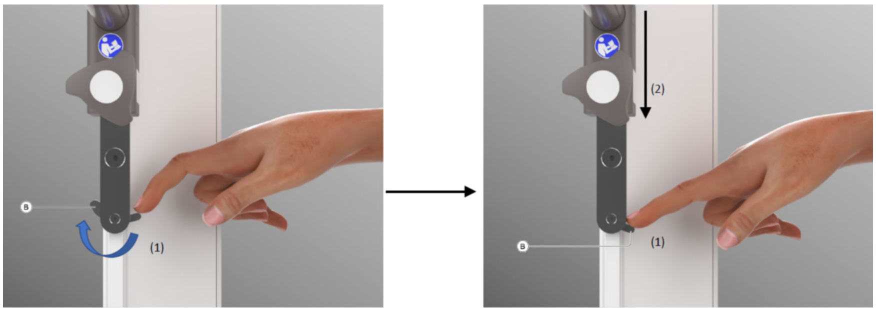


In addition to the locking screws, there is a security lever (B) at the bottom of each guide rail to prevent the hand grips from falling to the ground if the locking screws are not properly fixed.

Removing the handgrips

1 To remove the hand grips when used as stretch grip, just turn the locking screws counterclockwise and remove the handgrips upwards from the guide rails.

2 To remove the handgrips when used as support for the patient, turn the locking screws counterclockwise and slide the handgrips downward to the lower end of the guide rails where they are stopped by the security lever.

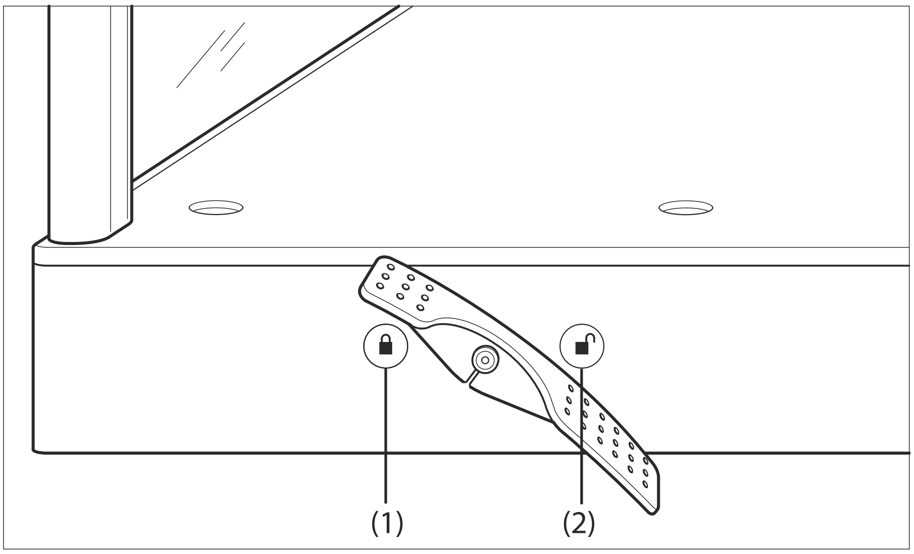


3 Press the security lever (B) to the side as shown in above figure (1) and remove the hand grips downwards (2) from the guide rails.

Brake pedals

The Multipurpose / Ortho stand is equipped with brakes to enable the secure stand of the support. For transport of the support you have to release the brakes. The brakes are operated with the brake pedals on both sides of the patient platform.

Print No. XPD3-530G.620.01.04.02



(1) Padlock closed
(2) Padlock open

Push the brake pedal down towards this symbol to lock the brakes, support cannot be moved.

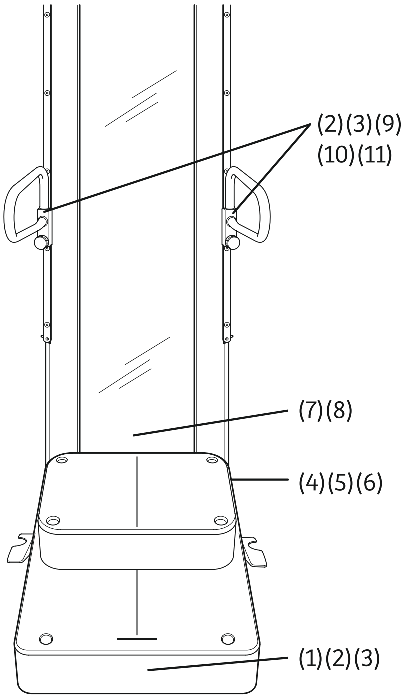
Push the brake pedal down towards this symbol to release the brakes, support can be moved.

Technical data of the Ortho stand

Height	196,5cm when brake is on 198 cm when brake is off
Depth	85 cm
Width	82 cm (incl. brake pedals)
Weight	112 kg (incl. additional platform and two handgrips)
Max. patient weight	200 kg
Max. patient size	190 cm
Attenuation equivalent of the transparent protection plate acc. to DIN EN 60601-1-3	< 1 mm Al - 100 kV 3.6 HVL

Location of labels

All labels displayed in the operator manuals are examples only and may differ from the labels attached to the system and components.



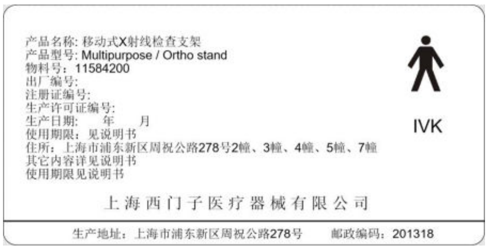
(1) Weight label (200 kg)

(2) Type B equipment label

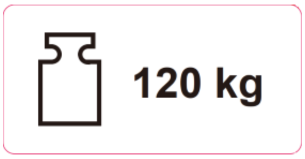
(3) Refer to manual label

Print No. XPD3-530G.620.01.04.02

(4) Ortho stand ID label

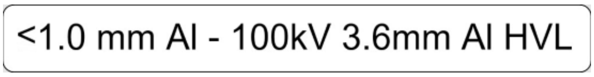


(5) China nameplate label



(6) Weight label (120 kg)

(7) DHHS label



(8) Al equivalent label

(9) Handgrip ID label

Print No. XPD3-530G.620.01.04.02

(10) Weight label (30 kg)

(11) Caution label

6.2.20 Accessories for Multipurpose / Ortho Stand

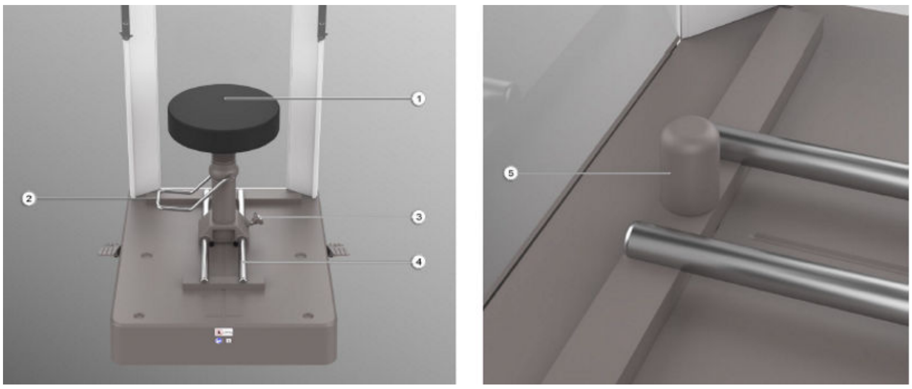
Swivel stool

CAUTION

The swivel stool detaches

Risk of patient injury

- Make sure that the swivel stool is attached firmly.
- Push and pull the stool to make sure that it is secure before positioning the patient.

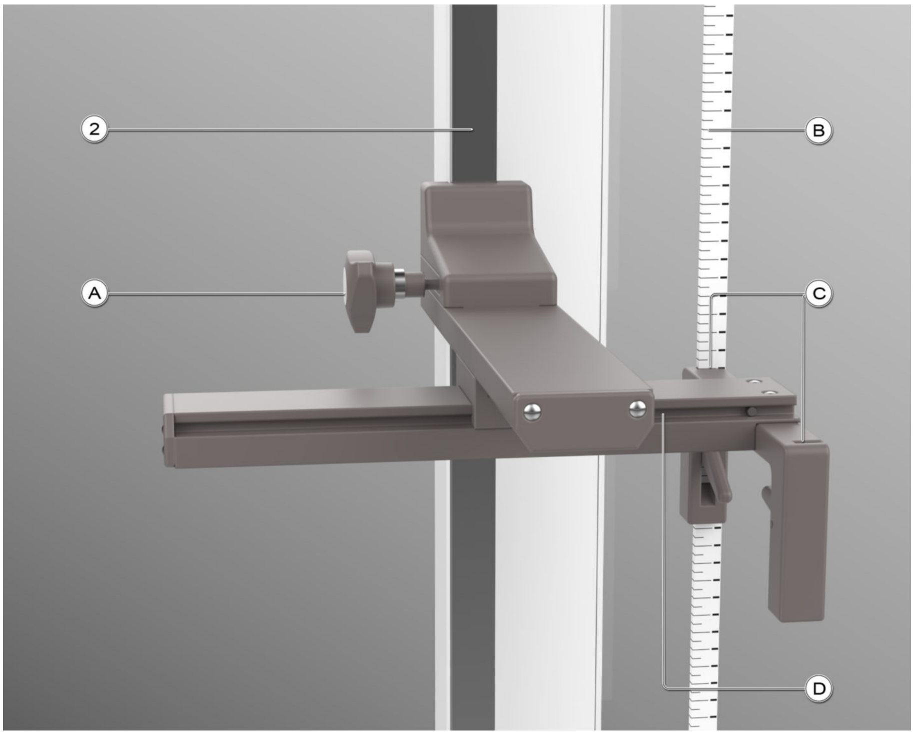


(1) Swiveling seating
(2) Lever for height adjustment / fixing the swivel movement
(3) Screw to fix swivel stool in a certain position on the rails
(4) Rails for longitudinal movement
(5) Fixing bolt for the attachment of the stool on the patient platform of the stand.

Ruler with holding device

The holding device is used to position the ruler on the image.

Print No. XPD3-530G.620.01.04.02



(A) Locking screw

(B) Ruler

(C) Positioning slots

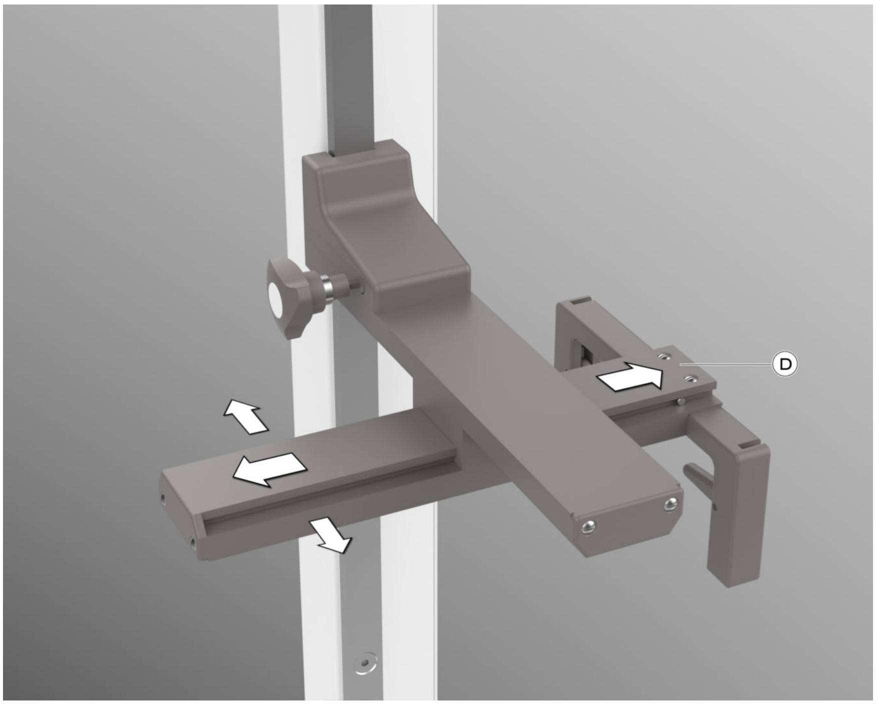
(D) Movable holding arm

(2) Guide rail

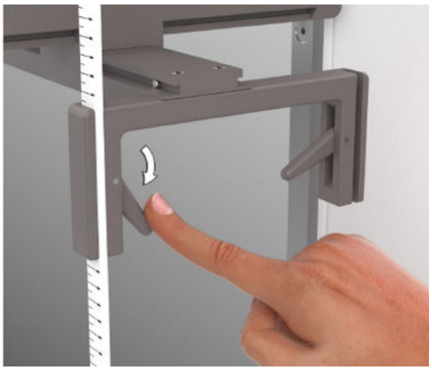
How to use the holder for the ruler

1 Attach the holding device to one of the guide rails (2) and fix it by turning the locking screw (A) clockwise until you hear a clicking noise.

2 To move the holder vertically, release the locking screw (A).



3 The holding arm (D) can be moved horizontally as shown above.



4 To position the ruler in one of the two holding slots first of all open the fixing clamp as shown above left.

5 Position the ruler in the slot and fix it with the fixing clamp as shown above right.

Baby carrier holder for multipurpose/Ortho stand

The baby carrier holder is used to fasten the baby carrier during pediatric examinations of small children.

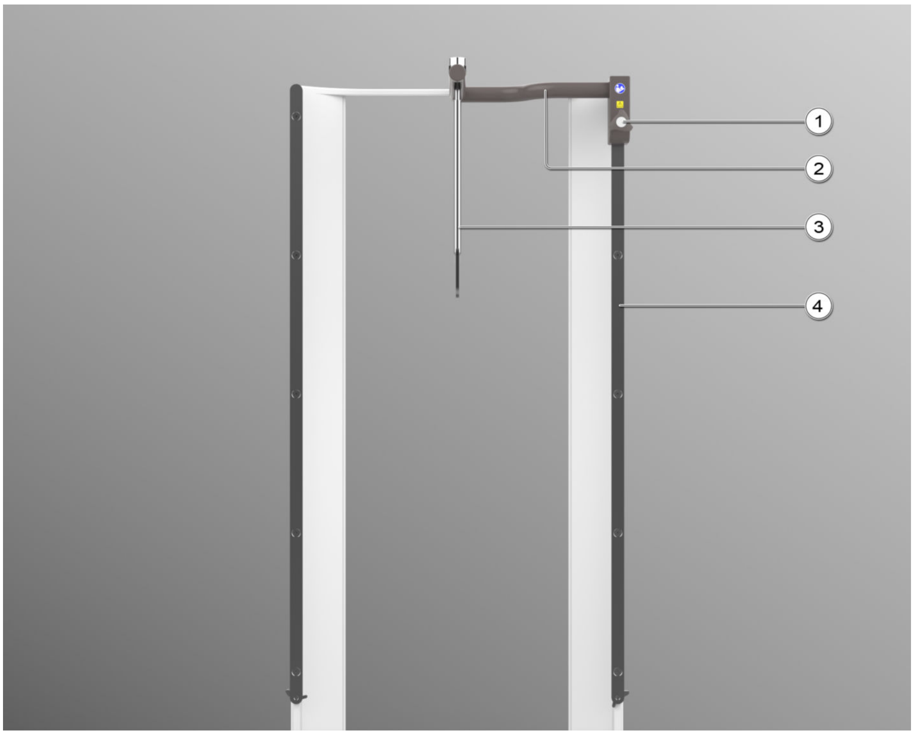
WARNING

Baby carrier holder loaded with more than 10 kg

Injury of persons

- Do **not** load the baby carrier holder with more than 10 kg.

Print No. XPD3-530G.620.01.04.02



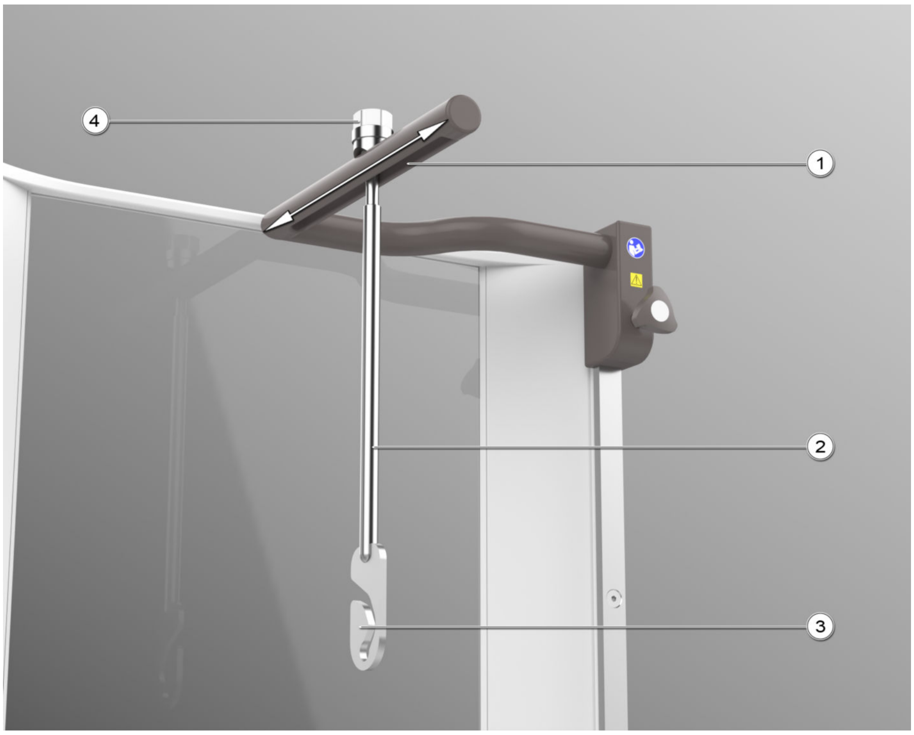
(1) Locking screw
(2) Support arm
(3) Baby carrier holder
(4) Guide rail

Attachment

The baby carrier holder can only be mounted on the right guide rail at the top.

- Slide the support arm (2) with the baby carrier holder (3) from above on the guide rail (4) and fix it by turning the locking screw (1) clockwise until you hear a clicking noise.

Positioning the baby carrier holder



(1) Guide rail
(2) Baby carrier holder
(3) Hook for attaching the BABIX
(4) Holding knob

Horizontal positioning

The baby carrier holder can be positioned moving it back and forth in the guide rail (2).

Turning the hook

- The hook of the baby carrier holder can be turned in steps of 90° around the vertical axis by lifting the baby carrier holder with the holding knob and turning it.

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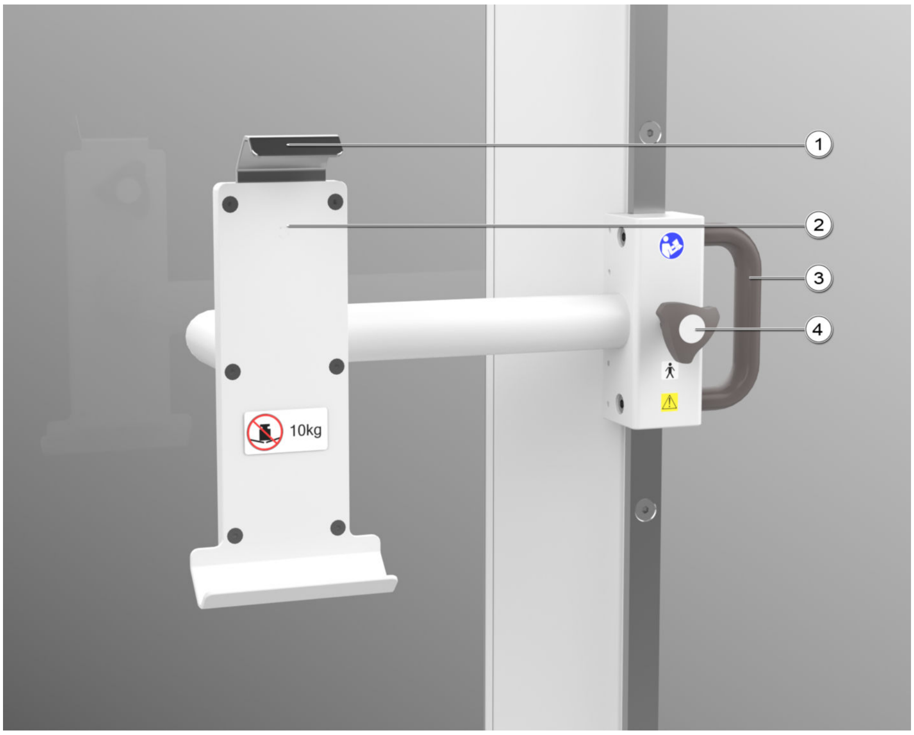
Detector holder



(1) Fixing device for detector
(2) Detector holder
(3) Handle
(4) Locking screw

Attaching the detector holder

The detector holder can be attached only to the right guide rail from below (recommended) or from above.

1 Turn the locking screw (4) counterclockwise.

2 Slide the detector holder from below or from above on the guide rail to the required height using the handle (3).

3 Fix the detector holder by turning the locking screw clockwise until you hear a clicking noise.

Check that the locking screw is properly fixed.

In addition to the locking screws, there is a security lever at the bottom of each guide rail to prevent the detector holder from falling to the ground if the locking screw is not properly fixed. (For more deteails to the security lever, refer to “Attaching the handgrips”.

Placing the detector

1 Lift the fixing device (1).

2 Place the detector on the lower holder.

3 Slide the fixing device downward.

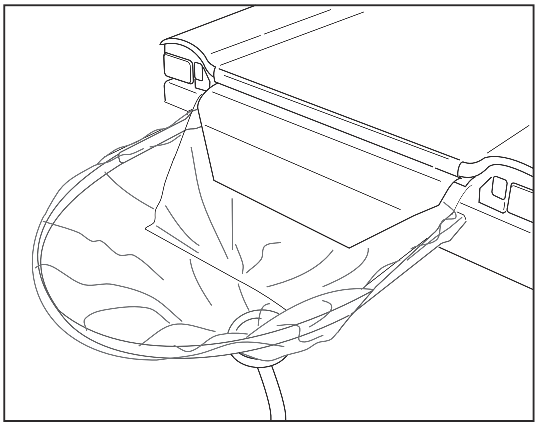
4 Ensure the detector is seated properly in the holder.

Print No. XPD3-530G.620.01.04.02

Removing the detector holder

- Refer to “Removing the hand grips”.

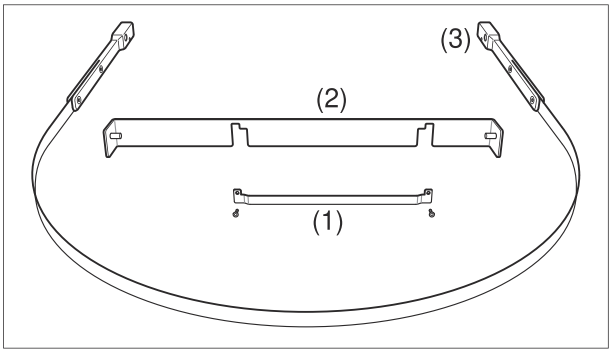
6.2.21 Plastic drain bag assembly



The suspension of the plastic drain bag is spring-loaded and thus allows the object to be approached closely even under unfavorable conditions.

A sieve is welded in the plastic bag. The drain hose which is permanently welded to the bag allows you to connect it quickly to the connection piece of the drain outlet on the unit.

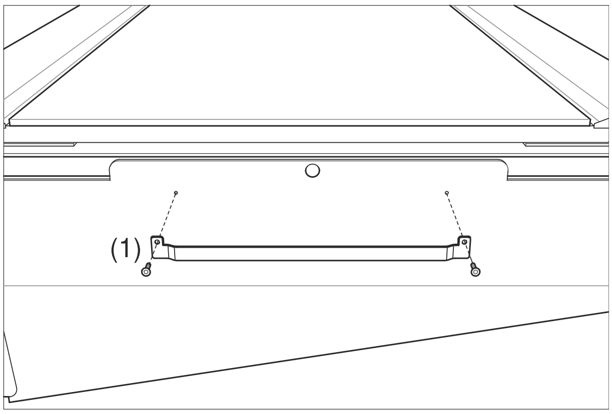
Do **not** use the drain bag several times because of the risk of infection.



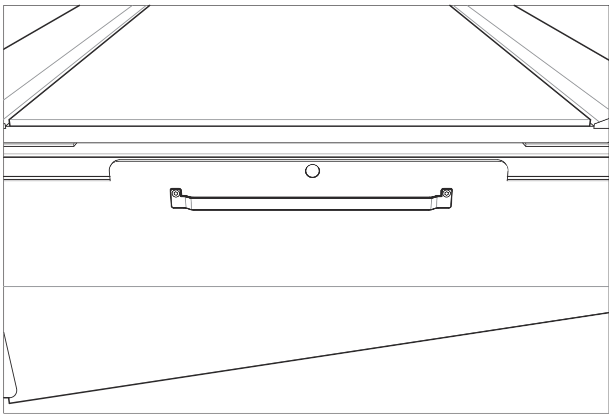
(1) Supporting rod
(2) Holding device
(3) Spring frame

Print No. XPD3-530G.620.01.04.02

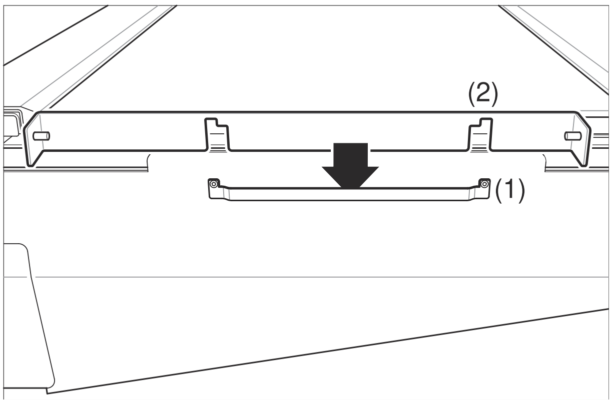
Attachment



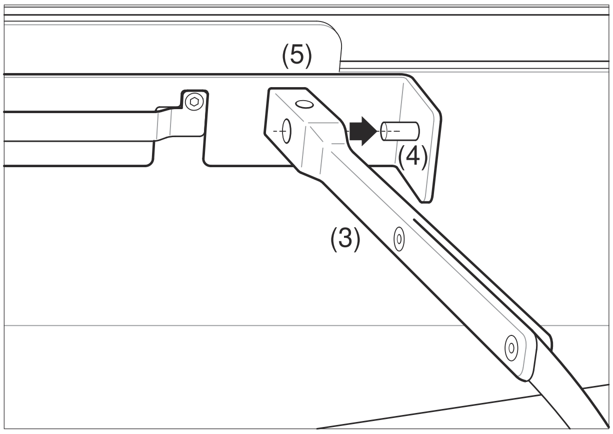
1 Mount the supporting rod (1) with the two provided screws to the foot side of the table.



Mounted supporting rod

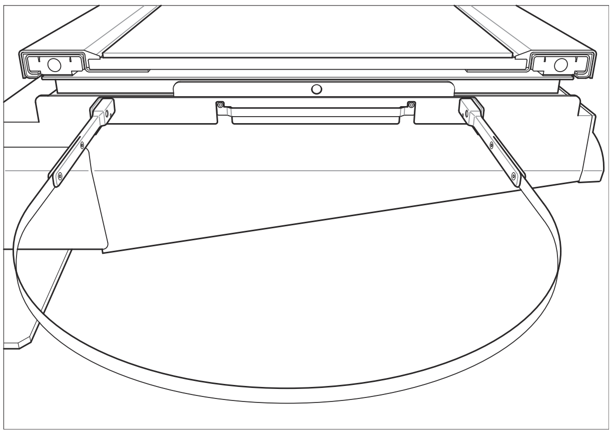


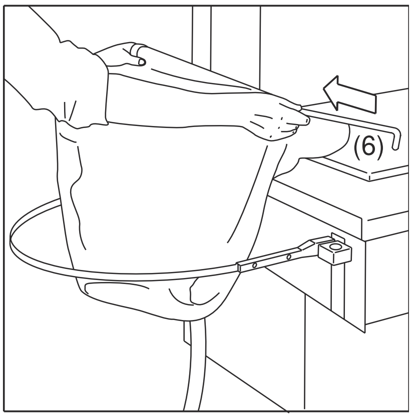
2 Plug the holding device (2) in the supporting rod (1).



3 Attach the end pieces of the spring frame (3) to the bolts (4).

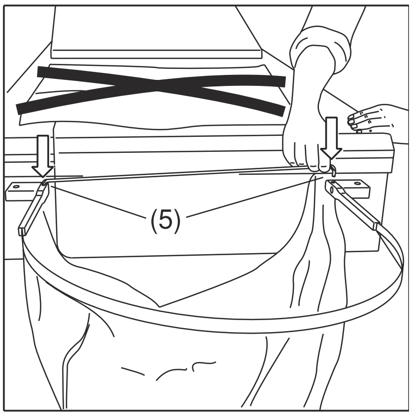
The holes (5) must face upward.



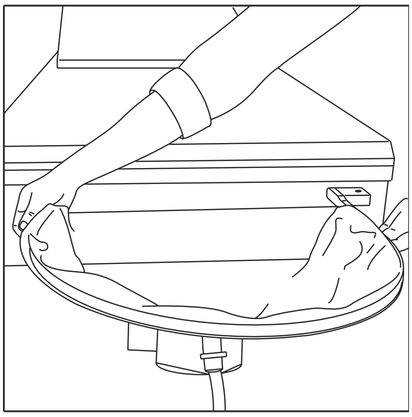


4 Push the locking bar (6) through the hollow seam of the drain bag.

Print No. XPD3-530G.620.01.04.02



5 Insert the locking bar in the two holes (5).



6 Fold the edges of the bag around the spring frame and smooth the edges.

7 Insert the drain hose of the bag into the hose connection piece of the drain outlet on the unit.

Do not place the extension lug too tightly on the table. Form a loop with the extension lug for protection against fluids and insert the end about 1 cm between the tabletop and pad.

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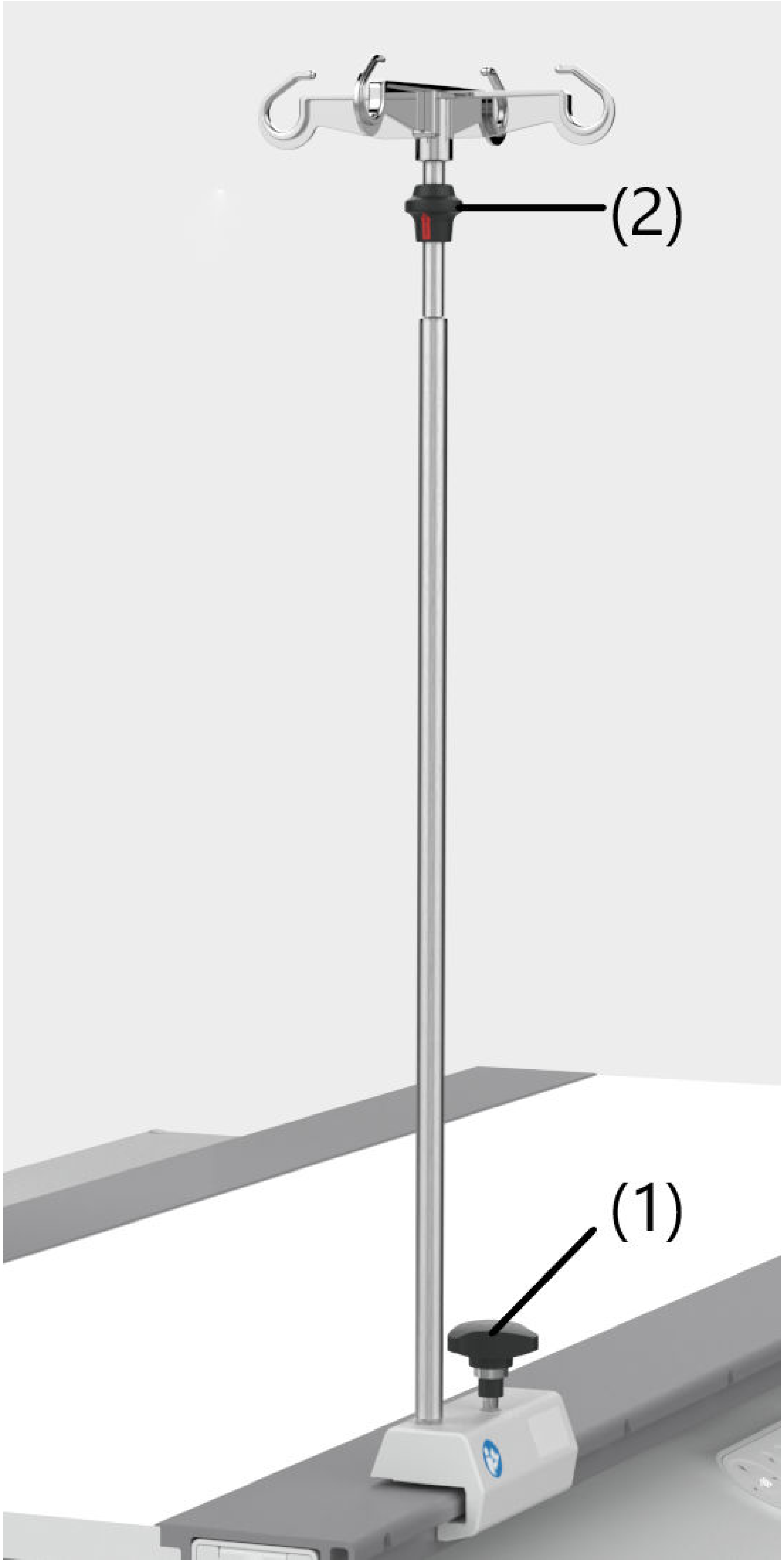
8 After use, dispose of the drain bag in an environmentally compatible manner according to relevant regulations.

Print No. XPD3-530G.620.01.04.02

6.2.22 Infusion bottle holder



Print No. XPD3-530G.620.01.04.02

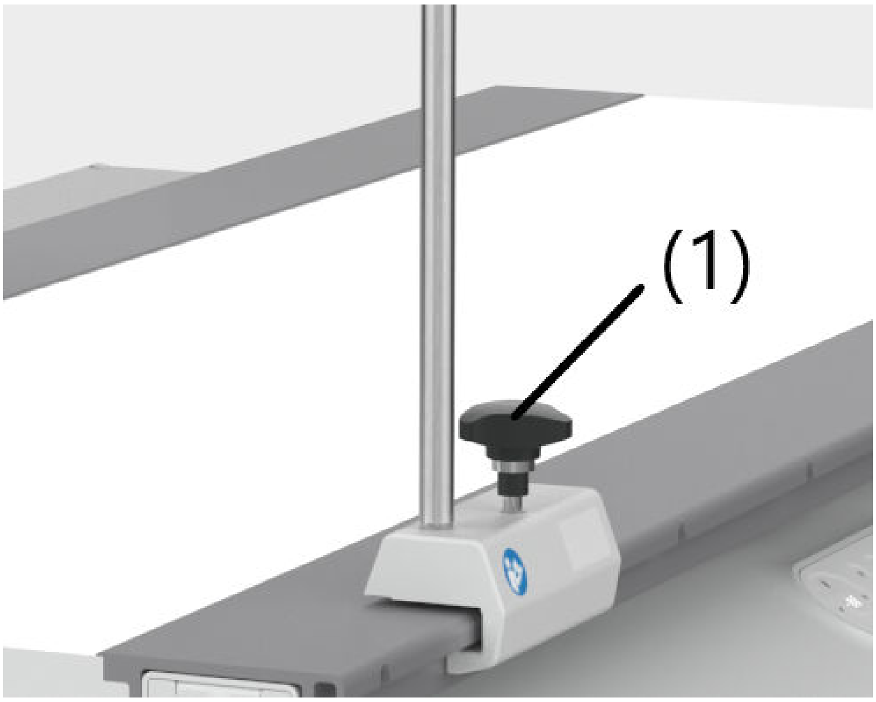
(1) Locking screw
(2) Fixing ring

Application

For intravenous therapy or administration of a contrast agent during the radiological examinations

Attachment

✓ Patient table in horizontal position.



1 Loosen the locking screw (1).

2 Position the infusion bottle holder (IV holder) at the treatment position and slide it onto the accessory rail.

3 Turn the locking screw (1) clockwise until you hear a clicking noise.

4 Push and pull on the IV holder to ensure it is seated properly.

5 Hang one or more IVs on the loops.

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6 To set the height of the IV holder push the fixing ring (2) at the removable upper section of the IV holder upward (as shown above) and move the upper section to the desired height.

7 To fix the height just release the fixing ring (2).

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8 Ensure that the height-adjustable rod is seated properly and secure.

The IV holder can be used in system positions up to ± 15°.

Removal

✓ Patient table in horizontal position.

1 Remove the IVs.

2 Push the fixing ring (2) at the removable upper section of the IV holder upward and move the upper section downward.

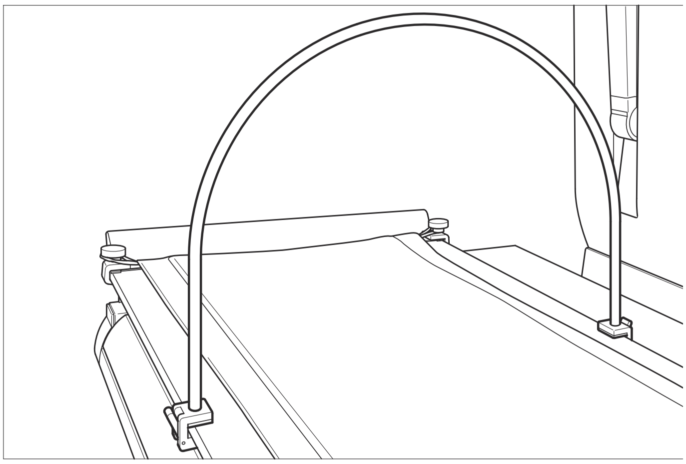
3 Release the fixing ring (2).

4 Ensure that the upper section and extension rod are secured.

5 Loosen the locking screw (1) turning it counter-clockwise.

6 Slide the base of the IV holder from the accessory rail.

6.2.23 Anesthesia screen holder

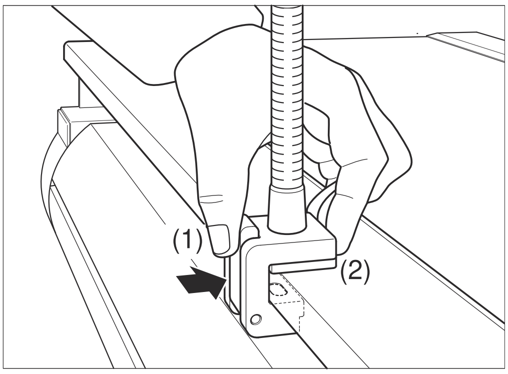


The anesthesia screen holder is used for attaching sterile cloths as an anesthesia screen between head and abdominal area of the patient.

Attachment

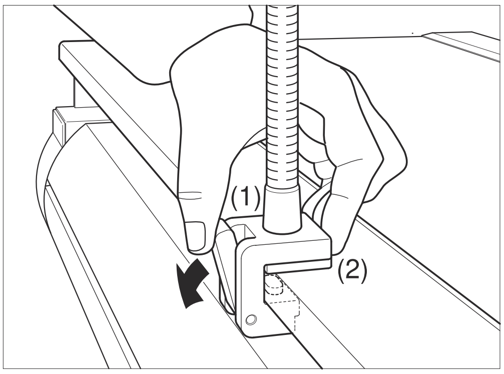
The anesthesia screen holder is attached to the accessory rails of the tabletop.

Print No. XPD3-530G.620.01.04.02



1 Push the lever (1) of the clamping device (2).

2 Hold the lever and put the clamping device straight on the accessory rail.



3 Let go of the lever (1) to fasten the clamping device (2).

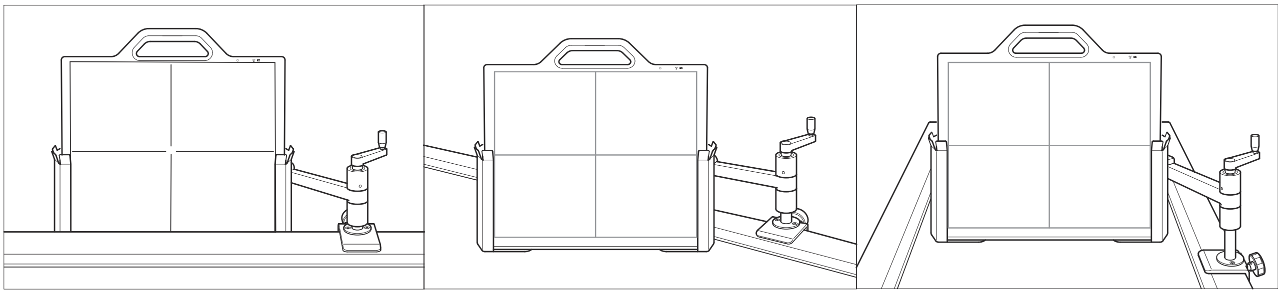
Be sure not to tilt the clamping device when attaching it to the accessory rail.

Be sure that the clamping device engages safely with a “click”.

6.2.24 Detector holder, lateral

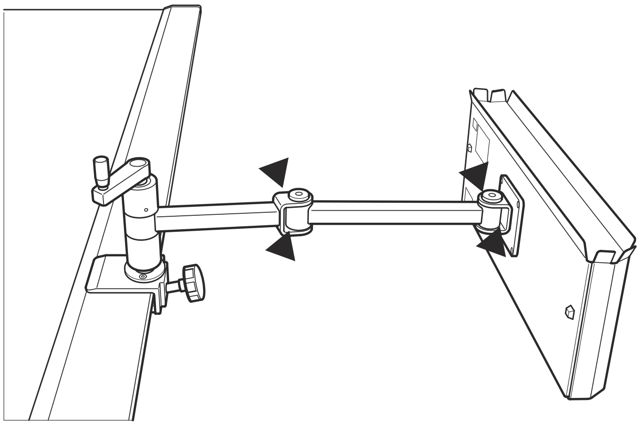
Application

Lateral exposures with the X-ray tube onto the upright detector

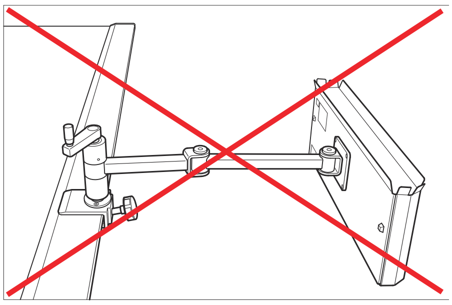


Examples of application

The detector holder is to be used only on or above the tabletop or directly at the side of the tabletop (see examples above).



Danger points



Misuse

Do **not** use the detector holder as shown above.

Do **not** lean on it, you might damage the tabletop.

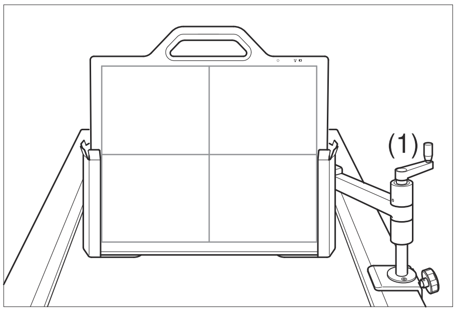
Attaching detector holder

The detector holder can be attached to either of the two lateral accessory rails of the patient table.

1 Take the detector holder with both hands (high weight!) and insert it into the groove of the accessory rail.

2 Slide it into the desired position.

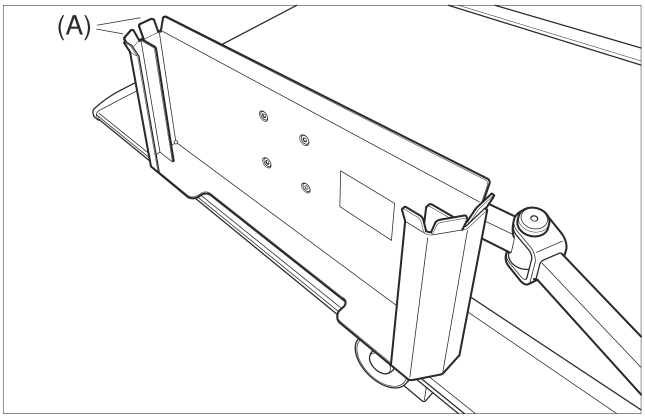
3 Tighten the detector holder.



4 To move the detector holder vertically use the crank (1).

Inserting the detector

1 Insert the detector into the detector holder from above.



2 There are **two** grooves (A) that can be used to insert the detector:

In the front groove **only** the detector can be inserted.

In the back groove the detector with the optional grid attached can be inserted.

Removing detector and detector holder

1 Take the detector and grid out of the detector holder from above.

2 Place the detector and grid in a safe location.

3 Dismantle the detector holder with both hands from the lateral accessory rail of the patient table.

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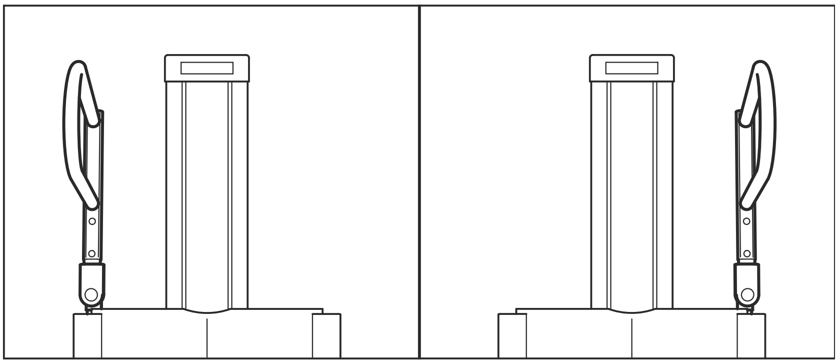
6.2.25 Overhead patient hand grip



Application

The overhead patient hand grip serves to stretch the patient during chest examinations.

Insert the overhead patient hand grip on the left or right side of the Bucky unit/ detector tray. Once in position, you can swivel it 0° and 180° parallel to the detector or 90° vertically in front of the detector. The overhead patient hand grip engages in the 0°, 90°, and 180° positions.

For easy storage of the overhead patient hand grip, a wall holder is provided.

The overhead patient hand grip has no locking mechanism to prevent it from being pulled out from the top.

Please pay special attention when patients use the grip!

The maximum load on the overhead patient hand grip must not exceed 25kg.

Inserting

1 Move the detector tray until its upper edge is approx. at eye level.

2 Push the shaft of the overhead hand grip straight down into the holder on the left or right at the top of the detector tray until the shaft is completely inside the duct.

 In this position, you cannot swivel it about its vertical axis.

Moving and removing

- When moving or removing the overhead patient hand grip, pull it vertically upwards out of its holder.

Adjustment

You can swivel the overhead patient hand grip about its lower center of rotation through 360°. The exception is the position parallel in front of the detector. In this position, you can swivel the overhead hand grip only by 100°.

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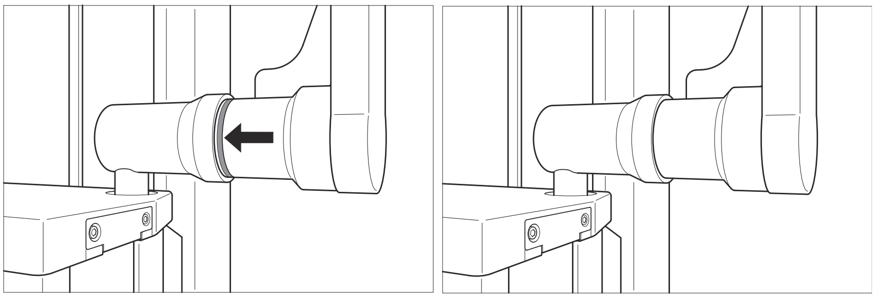
CAUTION

Wall stand hand grip loosens

Injury of patient or user, damage to wall stand

- Check that the overhead hand grip (stretch grip) is firmly attached and that no red mark is visible.
- Check that it cannot swivel before allowing the patient to use it.
- Do not let the patient hang or push down on the hand grip.
- The hand grip is labelled with its maximum allowed load.



1 Pull the overhead patient hand grip axially approximately 1 cm against the spring force to the front.

A red mark is visible to indicate that the overhead patient hand grip is not locked.

2 Swivel it by the required angle.

3 Then let it snap back again into its axial rest position.

If the red mark is still visible, the overhead patient hand grip is not securely locked and should not be used.

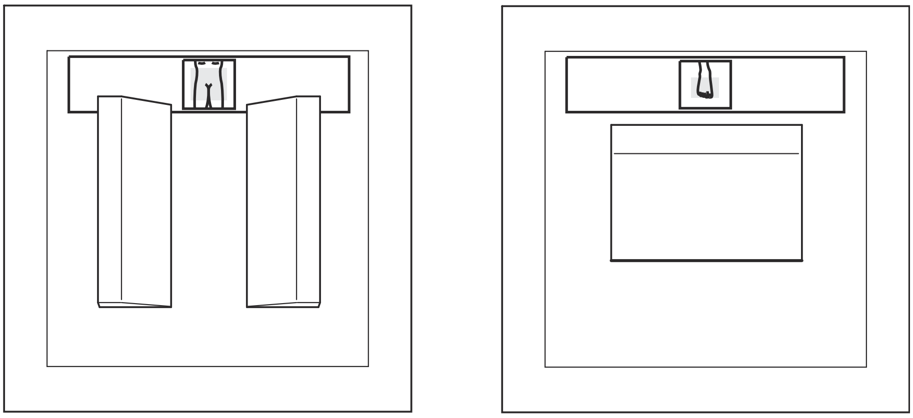
6.2.26 Compensation filters

Application

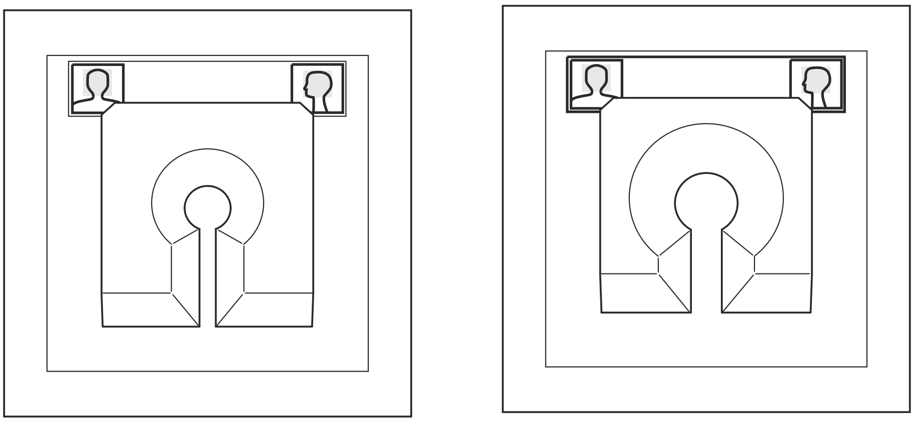
Absorption compensation for acquisitions of:

- Pelvis
- Foot
- Infant skull
- Adult skull
- T-spine, L-spine
- Shoulder

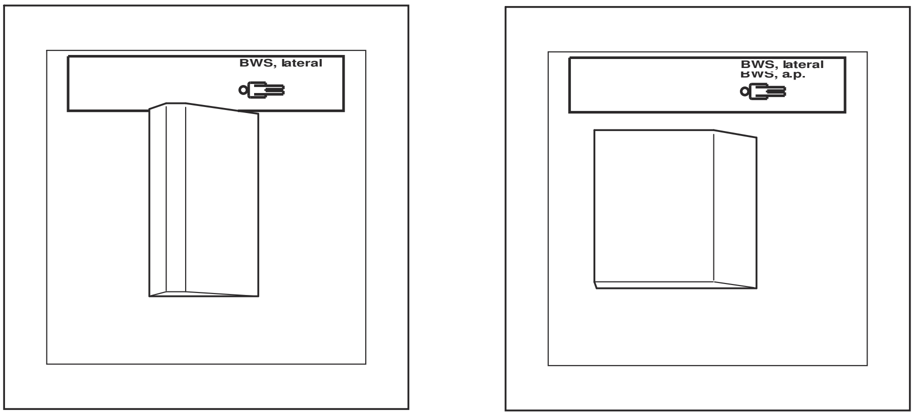
Print No. XPD3-530G.620.01.04.02



Compensation filter for pelvis and compensation filter for foot

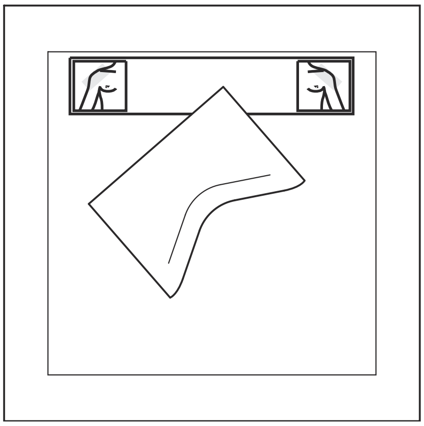


Compensation filter for infant skull and compensation filter for adult skull



Compensation filter for thoracic spine, lat. and compensation filter for thoracic or lumbar spine, lat.

Print No. XPD3-530G.620.01.04.02



Compensation filter for shoulder

Attachment

The operating personnel has to take care that the filters are appropriately attached.

1 Push the filter with the aluminum part **downwards** into the two accessory rails of the multileaf collimator.

The safety spring in the left rail is pushed beside in this case.

2 Push the filter up to the stop.

WARNING

Falling objects because too much weight is hung on the collimator

Injury of persons, damage to system, misalignment of tube

- Do not hang more than 7 kg on the collimator accessory rails.
- Do not exert any force on the multileaf collimator or its rails when inserting the accessory or auxiliary device.
- Check that the accessory is firmly anchored in the collimator rails.

If you exert too strong a pressure on the accessory rails, you might have to change the settings of the RHA because of a change in the position of the tube assembly.

Removal

1 Pull filter out of the accessory rails of the multileaf collimator

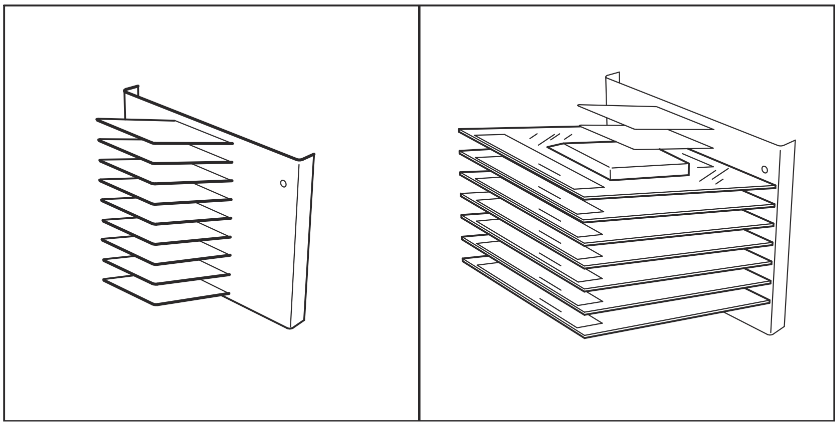
The safety spring in the left rail has to be pushed beside in this case.

2 Keep filter secured in the holder.

Print No. XPD3-530G.620.01.04.02

Please be careful when handling the compensation filters. They are thin, scratch-sensitive, and may become unusable due to rough handling.

6.2.27 Filter holder for eight filters



Application

For storing a maximum of eight compensation filters.

Fastening the wall holder

- Have the holder fastened to the wall at working height at a suitable place with the enclosed wall plugs and screws.

Equipping the wall holder

1. Remove compensation filters from the packaging.
2. Turn the filter so that you can read the designation on the filters reading normally from above.
3. Place the filters in the upwards angled compartments.

Storing the compensation filter

- Place the filter in a free compartment of the wall holder, turn the filter so that you can read the designation on the filter reading normally from above.

6.2.28 Three-field template, set

Application

For exposing the measuring fields on the object to be exposed.

Print No. XPD3-530G.620.01.04.02

The three-field templates are available for the following SIDs:

- 90 cm - 110 cm
- 110 cm - 130 cm
- 130 cm - 175 cm
- 175 cm - 220 cm

Usage

1 Push the locking lever on the left accessory rail to the left.

2 Slide the template into the collimator accessory rail in the correct direction for exposing the measuring field.

The locking lever on the accessory rail springs to the right.

3 Check that the template sits firmly in the collimator.

4 Expose the measuring fields.

To do so, switch on the light of the collimator.

Storage

1 Push the locking lever on the left accessory rail to the left.

2 Pull the template out of the accessory rails.

The locking lever on the accessory rail springs to the right.

3 Store the three-field templates in a suitable location.

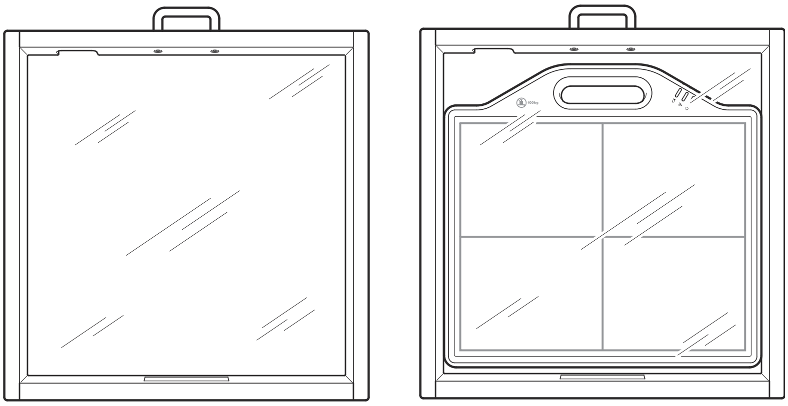
Please be very careful with the three-field templates. They are thin, sensitive to scratching and can become useless if handled carelessly.

6.2.29 Detector protector (patient support)

Application

The detector protector is a protective cover for the detector. It is used for examinations where the patient has to stand on the detector if the patient body weight exceeds 100 kg.

Print No. XPD3-530G.620.01.04.02



The detector protector may only be used without clip-on grids.

The detector protector can be used for all mobile detectors.

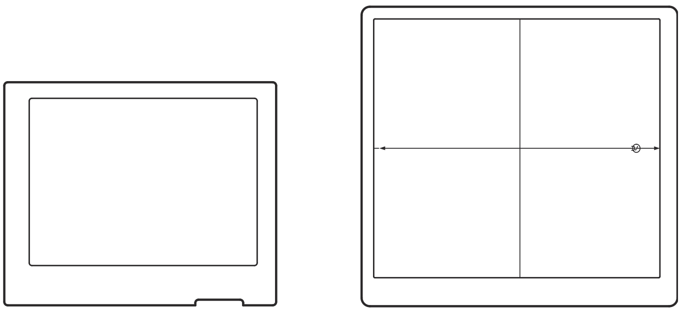
The detector protector is made of acrylic glass.

The maximum load on the detector protector must not exceed the specified value on the label.

Al equivalent

The Al equivalent value is 1.39 (for 3.7 mm Al).

6.2.30 Clip-on grids for MAX detectors



The following clip-on grids are available:

- 15/80 F115 for MAX wi-D
- 5/85 F115 for MAX wi-D
- 5/85 F115 for MAX mini

Print No. XPD3-530G.620.01.04.02

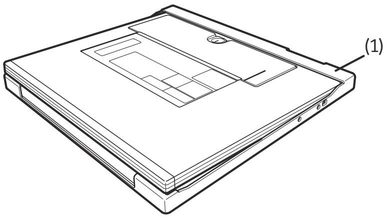
Application

The clip-on grids are used for free exposures on the mobile MAX detectors.

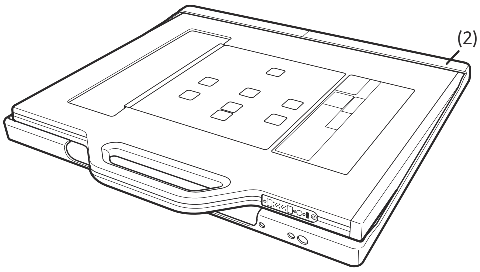
For exposures on the patient table or on the wallstand, use the corresponding internal grid.

When the clip-on grid is not in use, remove it from the detector and store it in a safe place where it cannot fall, for example in the grid holder.

Even if it is not extremely damaged, maybe its characteristics are changed, which may cause a problem in image quality.



MAX mini

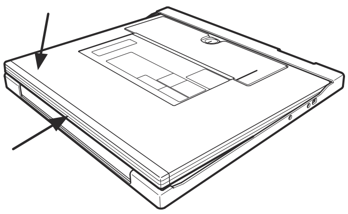


MAX wi-D

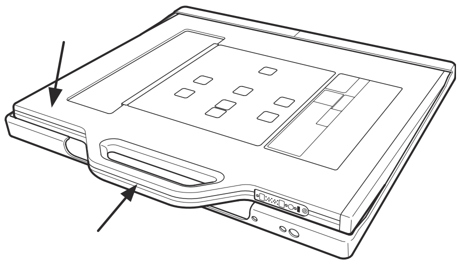
(1) Clip-on grid for MAX mini
(2) Clip-on grid for MAX wi-D

Attaching the clip-on grids

✓ Detector front must face the backside of the clip-on grid.



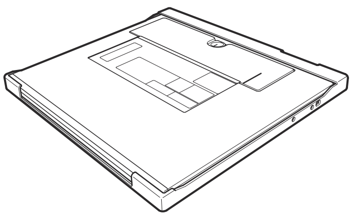
MAX mini



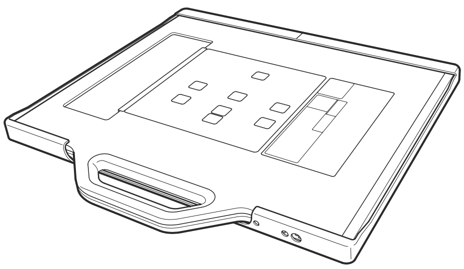
MAX wi-D

1 Put the detector with the bottom in the clip-on grid as show in above figure.

2 Push the detector gently down against the loading spring of the clip-on grid and let it slide smoothly into the clip-on grid and let go.



MAX mini

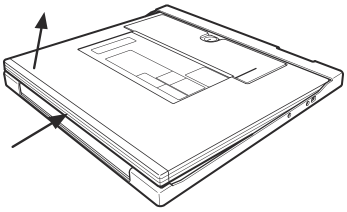


MAX wi-D

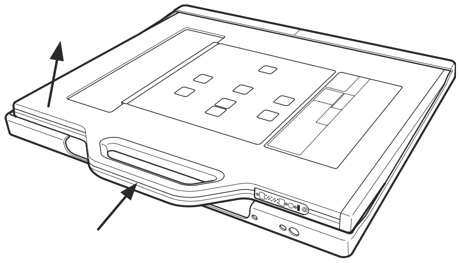
Print No. XPD3-530G.620.01.04.02

Ensure that the grid unit is locked before lifting the detector unit. Otherwise, the grid unit can fall and be damaged.

Removing the clip-on grid



MAX mini



MAX wi-D

- Push the detector gently down against the loading spring of the clip-on grid and then pull the detector out of the clip-on grid and let go.

6.2.31 Clip-on grids for X wi-D detectors

The following clip-on grids are available:

- 5/31 F115 for X wi-D 43
- 5/31 F115 for X wi-D 35

Application

The clip-on grids are used for free exposures on the X wi-D detectors.

For free exposures with the X.wi-D 43 detector, use the clip-on grid only for exposures with large structures creating scatter radiation.

For exposures on the patient table, use the corresponding internal grid.

When the clip-on grid is not in use, remove it from the detector and store it in a safe place where it cannot fall, for example in the grid holder.

Even if it is not extremely damaged, maybe its characteristics are changed, which may cause a problem in image quality.

For details concerning attaching and removing the clip-on grids, refer to

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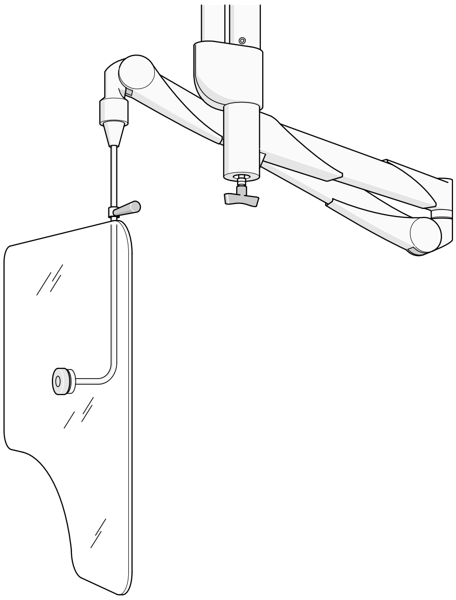
6.2.32 Wall holder for grids

This holder is used for storage of exchangeable grids.

It can be mounted to the wall.

6.2.33 Radiation protection for the upper body



Application

The radiation protection window is used to reduce exposure of the upper body of the examiner to scattered radiation. It is especially designed to decrease the exposure of the eyes and thyroid of the examiner to radiation.

The radiation protection window can be moved in any direction.

The upper body radiation protection is **not** suitable for supporting loads! This means, it is not suitable for carrying any additional weight.

This could pull down the upper body radiation protection.

Do not hang any objects on it.

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During unit movements, especially swiveling movements, there is a risk of collisions and damage (could break off and fall down).

Watch out for possible collisions while performing unit movements.

Lead equivalent

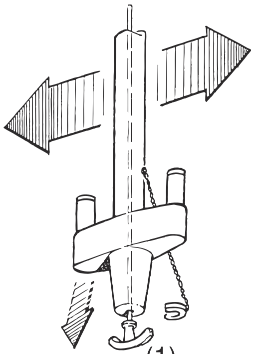
The lead equivalent value is 0.5 mm (according to EN 61331-3:1999).

Sterile cover

- Both window designs can be provided with sterile covers; contact your Siemens Healthineers sales representative.

Positioning

The radiation protection device can be moved along the ceiling rail:



1 Pull down and turn the brake handle (1).

  You can move the carriage freely in the ceiling rail.

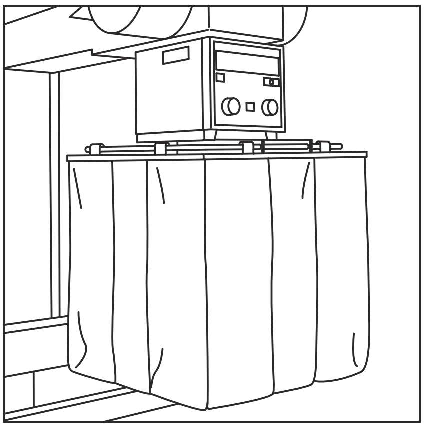
2 Move it to the required position.

3 Release the brake handle.

4 Position the radiation protection window so that it protects the eyes and thyroid of the examiner.

For patient positioning and when not in use, the upper body radiation protection should be positioned outside the swivel range of the system.

6.2.34 Radiation protection, removable



Application

Additional radiation protection in tableside examinations.

CAUTION

Lead radiation protection drops on the patient

Injury of patient

- Make sure that the radiation protection is securely fastened.
- Avoid any collisions with the radiation protection when moving the system.

Use this radiation protection only with the tabletop practically horizontal up to maximum +/- 10°. At a greater tilt there is a danger that the lead-rubber flaps will slip off and drop on the sterile covered patient.

Radiation protection value

The lead equivalent of the double-layer lead-rubber flap is 0.5 mm Pb at 80 kV.

Attachment

1 Recommendation: Set the examination unit horizontal and lower the table to minimum height.

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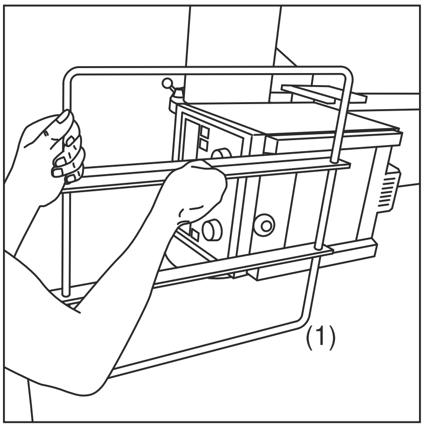
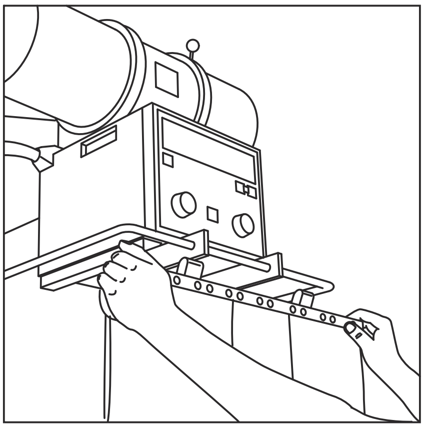


2 Push the bar (1) into the upper groove of the accessory rails up to the stop.

When in position in the stop, a spring lever in the left profile rail engages in a notch of the bar.



3 Check for the firm seating of the bar on the collimator.

4 Be sure to set the examination unit horizontal (± 10°)!

5 As required hook a lead-rubber flap on the front bar of the holder.

6 Hook the second lead-rubber flap according to the position of the examiner on the left or right bar.

Do **not** exceed the maximum permissible additional weight of 70N or 7 kg. This means that at the most two lead-rubber flaps may be hooked onto the bar.

Removal

1 Unhook the lead-rubber flaps from the bars after one another and carefully place them on the side (do not kink them).

2 Pull the bars out from the accessory rails of the collimator.

3 Store the accessory in a suitable place.

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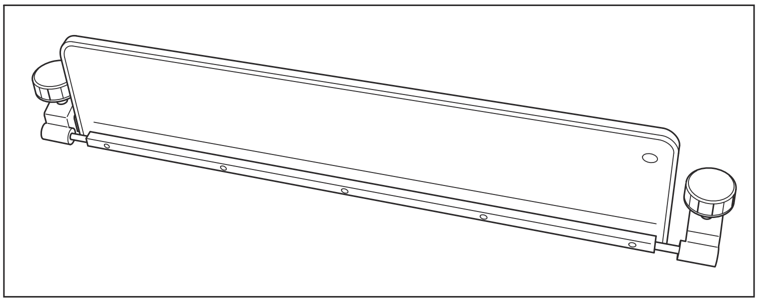
The lead-rubber elements of this accessory are inserted in welded plastic pockets. These pockets are riveted with the rails.

The plastic pockets can be cleaned easily with mild dish-washing liquid solutions.

6.2.35 Radiation protection, lateral



Application

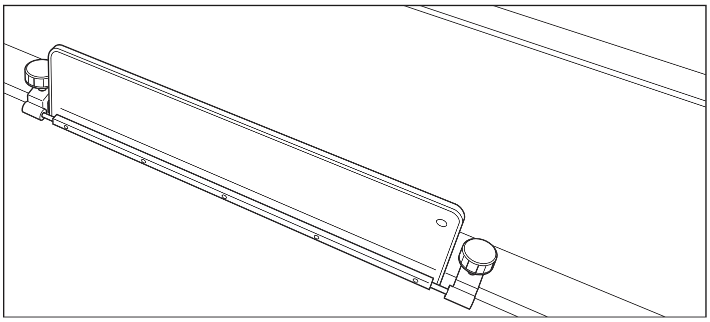
For additional radiation protection during tableside operation of exposures

Attachment

✓ Patient table in horizontal position.

1 Position the patient.

2 Loosen the right and left cap screws on the lateral radiation protection.



3 Insert the radiation shield in the front accessory rail and slide it to the appropriate work position.

4 Tighten both cap screws.

5 Push and pull on the radiation protection to ensure it is seated properly.

6 Tilt the radiation protection into a protective position.

The radiation shield can be tilted inward on the accessory rail as well as outward and completely down.

Removal

✓ Patient table in horizontal position.

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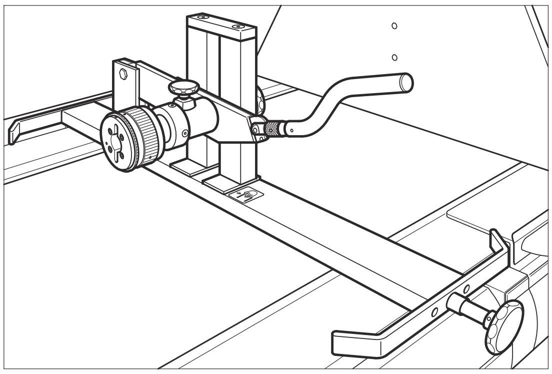
1 Tilt the radiation protection to the vertical position.

2 Loosen both cap screws.

3 Remove the lateral radiation protection from the front accessory rail.

6.2.36 Manual holder for pediatric cradle



CAUTION

Activation of the compression device while attaching the pediatric cradle causes a collision

Injury of child

- Do not use the compression device while using the pediatric cradle.
- Take care not to activate compression unintentionally.

Application

Pediatric exposures in sitting cradles

Attachment of cradle holder

1 Set the tabletop horizontal and if possible move it into its working position.

2 If a mattress is on the tabletop, remove it.

Because of its weight, we recommend that two persons carry the cradle holder to the tabletop and set it down there. It is also mounted more easily by two persons.

3 Position the cradle holder on the table.

The two stop pins (1) on the front and back of the mounting rail must project parallel to the edges of the front and back accessory rail.

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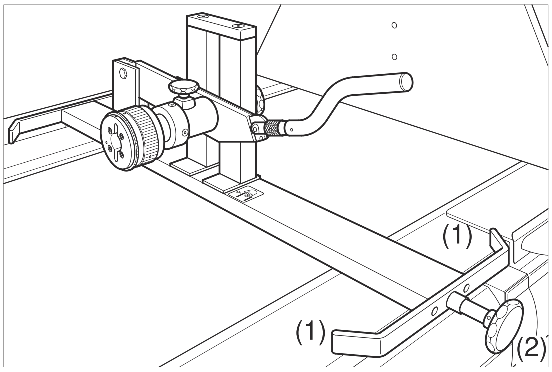
Ensure without fail that the two stop pins are located directly next to the outer edges of the front and back accessory rail. Under no circumstances, may they stand on the accessory rails. The cradle holder does not stand securely in this case and cannot be mounted. It could tilt over or drop down. In this way, you avoid possible injury of persons and material damage.



4 Tighten the cap screw (2).

5 Please tighten the cap screws firmly.

Apart from the clamp, there is no further fastening or securing for the cradle holder.

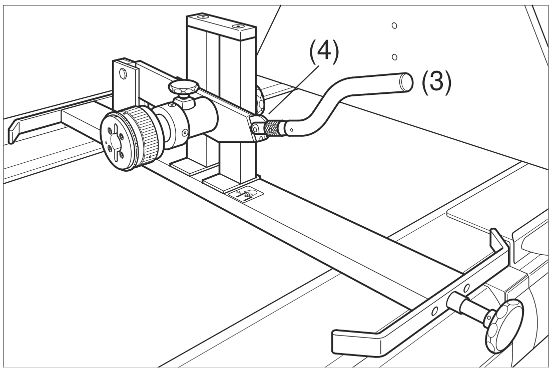
For certain examinations, the tabletop together with the cradle holder is tilted up into the +90 ° position.

With the correct attachment of the cradle holder and the cradle, you avoid the cradle holder dropping down. Thus you avoid possible injury of persons and material damage.

6 Check the firm location of the cradle holder by pulling and pushing.

Setting the handle

It is possible to set the handle (3) of the cradle holder at three working heights.



1 Set the handle to the working height which is suitable for you before fastening the sitting cradle.

That simplifies raising and lowering the cradle.

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2 Pull the knurled ring (4) to the front against the spring tension and hold it tight.

3 Raise or lower the handle to its working height.

4 Let the knurled ring snap into one of the three grooves.

5 Check that the handle is locked by pulling and pushing.

Attachment of the cradle

1 Move the tabletop if possible into its working position upwards or downwards.

2 Prepare the sitting cradle with the infant outside the examination room. Fasten the infant securely in the cradle.

To make your work easier, we recommend that two persons fasten the cradle in the cradle holder and remove it again.

Reason: The sitting cradle together with the infant can weigh up to 18 kg.

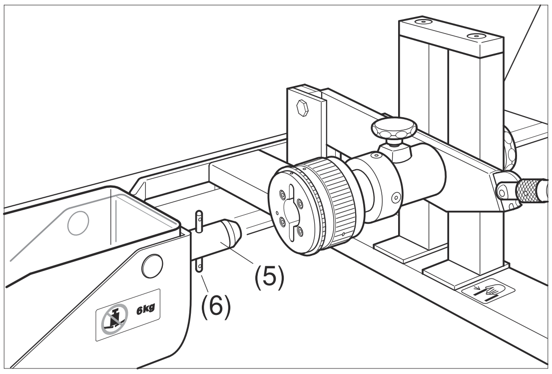
CAUTION

Pediatric cradle is not inserted correctly in the cradle holder.

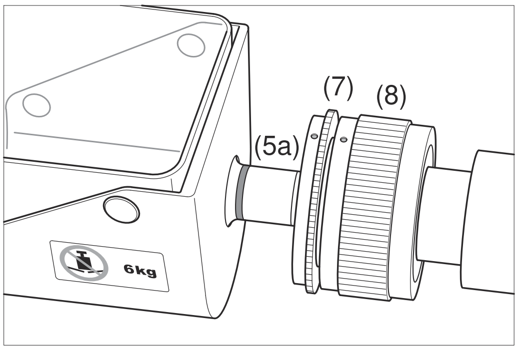
Cradle dropping down onto tabletop

Injury of infant

- Make sure that the pediatric cradle engages correctly in the bayonet closure.
- Check that the cradle is locked in the holder by pulling and pushing on it.
- Be careful when raising, lowering and turning the cradle.
- Follow the instructions in the operator manual.



3 Push the shaft (5) of the cradle horizontally in the round opening of the bayonet closure.



4 Fit the two pins (6) in the corresponding openings of the bayonet closure.

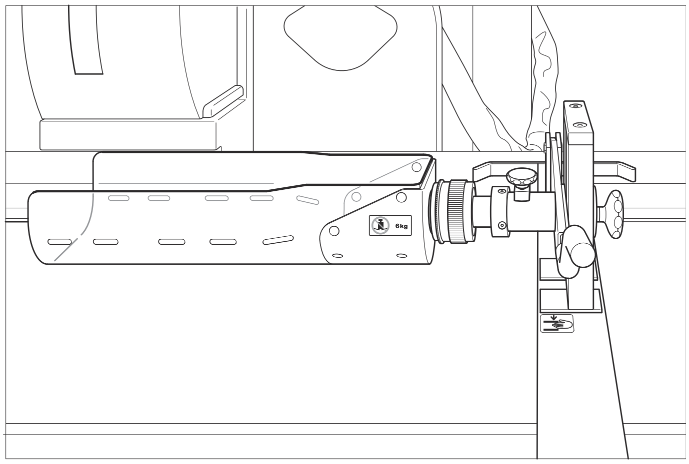
The red dot on the narrow ring (7) must be located opposite the red dot on the wide knurled ring (8) of the cradle holder.

5 Push the sitting cradle up to the stop into the bayonet closure.

The red ring (5a) on the shaft of the cradle must be completely vanished.

The wide knurled ring turns to the left in this case and springs back into its basic position.

The red dot on the narrow ring (7) must be located opposite the red dot on the wide knurled ring (8) of the cradle holder.



Raising and lowering the cradle

You can adjust the distance of the sitting cradle from the tabletop continuously.

1 Hold the handle firmly.

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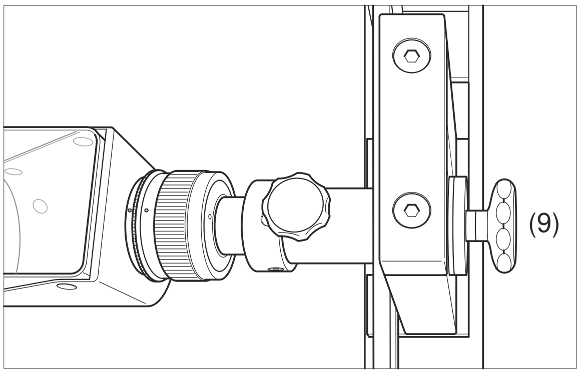


2 Loosen the knurled ring (9) on the back of the cradle holder.

3 Raise or lower the sitting cradle with the handle into the examination height.

4 Tighten the knurled ring firmly.

Please tighten this knurled ring firmly. Except for this clamp, there is no further fastening or securing for the cradle.

By correctly tightening the knurled ring you avoid the cradle dropping down onto the tabletop and thus possible injuries of persons and material damage.

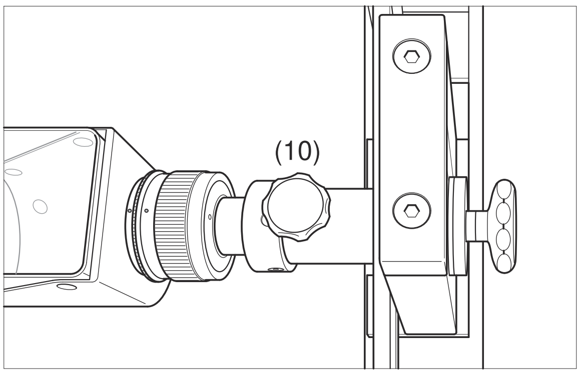
---

5 Check the secure location of the cradle in the holder by pulling on it.

Turning the cradle

You can turn the cradle manually by 360° to the right (back) or to the left (front).

Please take care that the cradle in the holder is in a height above the tabletop which enables the cradle to be turned without touching the tabletop. You thus avoid possible injury of persons and material damage.



1 Loosen the knurled screw (10).

2 Turn the cradle to the right (back) or to the left (front) into the examination position.

Please take care that the infant is strapped safely in the cradle.

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3 Tighten the knurled screw.

4 Check the firm location of the cradle by turning it.

Removal of the cradle

CAUTION

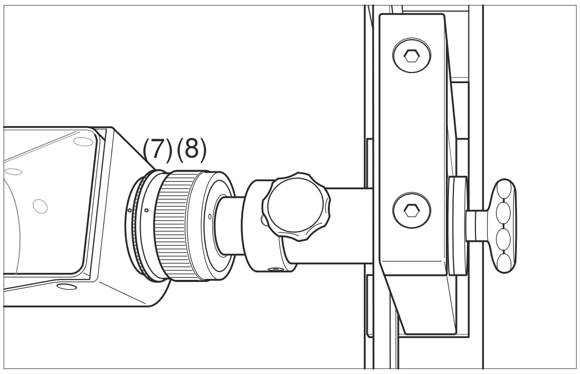
Incorrect removal of pediatric cradle

Injury of infant

- To remove the cradle follow the instructions in the operator manual.
- After unlocking the bayonet closure, hold the cradle firmly with both hands before withdrawing it from the cradle holder.

We recommend that two persons both fasten and also remove the sitting cradles.

1 Hold the cradle firmly.



2 Turn the wide knurled ring (8) together with the narrow knurled ring (7) on the bayonet closure to the left (front) and hold it.

3 Pull the cradle about 4 cm to the left up to the stop and release the wide knurled ring.

The wide knurled ring then turns to the left.

The red dot on it springs back into the basic position.

The red dot on the wide knurled ring (8) now stands again opposite the red dot on the narrow ring (7) on the cradle.

4 Turn the narrow knurled ring (7) on the bayonet closure to the right (back) and hold it.

The cradle is now inserted unsecured in the bayonet closure. Hold the cradle firmly with both hands before you withdraw it completely from the cradle holder. In this way, you prevent the cradle tilting down or dropping onto the tabletop and thus possible injury of persons and material damage.

5 Withdraw the sitting cradle completely from the cradle holder.

6 Remove the sitting cradle with the infant from the X-ray room.

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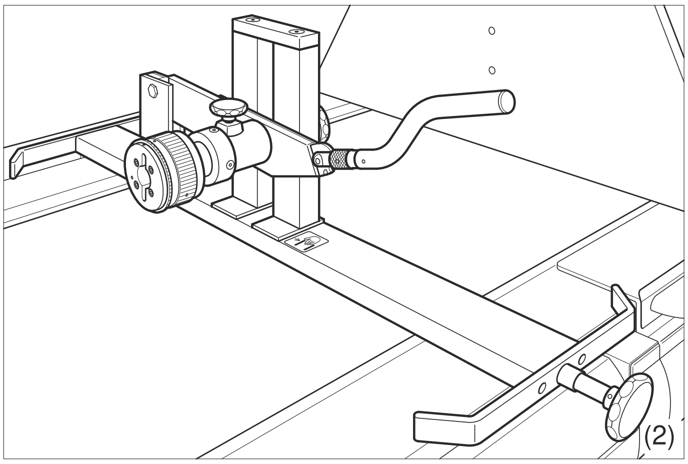
7 Loosen the retaining straps, lift the infant out from the cradle and care for it.

For safety reasons, may you under no circumstances dismantle the complete holder with infant and cradle!

Removal of cradle holder

1 Set the tabletop horizontal.

2 If possible move the tabletop into a height suitable for lifting the cradle holder down.



3 Loosen the knurled screw (2) completely.

The mounting rail is released from the accessory rails.

4 If possible, two persons should lift the cradle holder down off from the accessory rails.

Storing the BABIX cradle

1 Install the holder or the holders for the BABIX cradle at a good working height with the wire hook downwards, for example, on a shelf.

The wire hook must be open to the front.

2 Take care that the distance between the holders is large enough so that the cradles can hang without touching and can be easily removed or suspended.

Refer to the suggestion on the following Figure.

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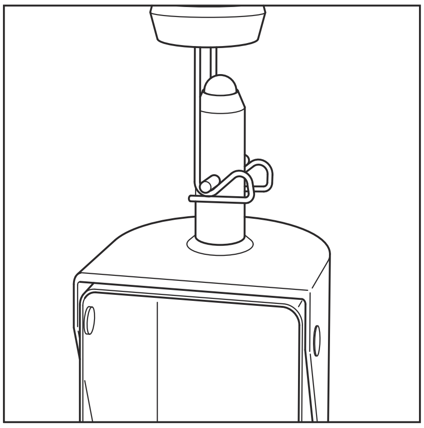
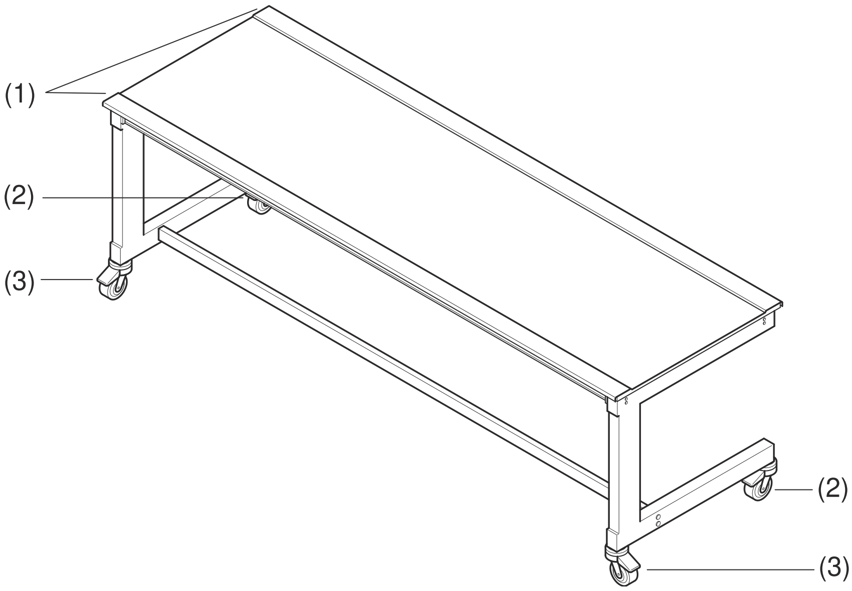


6.2.37 Patient positioning table, mobile



(1) Accessory rails
(2) Castors without brakes
(3) Castors with brake pedal

Castor brake pedal

To enable the mobile patient positioning table to be fixed at any location, the two castors at the front of the table are equipped with brake pedals (as shown in the picture).

When the patient is moving onto or leaving the patient table, this device can be locked to fix the table position.

Intended use

As a patient supporting device of the X-ray imaging system, the mobile patient positioning table can facilitate X-ray imaging of the recumbent patient, for instance the examination of stomach and lower limbs.

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Product features

- Wide application scope
- High reliability
- Easy mobility of the patient table
- Tabletop has high level of X-ray permeability and stiffness
- Large tabletop facilitates emplacement of patients
- The thin tabletop surface can narrow the distance between the detector and the object to be exposed.
- By using the brake control device, it is easy to position the patient table.

The mobile patient positioning table must not be used to carry patients, it is only to be used as auxiliary supporting device when taking exposures.

When the patient is moving onto or leaving the patient table, the brake pedal must stay locked. Besides, give assistance to the patient, otherwise the table might slide to create hazard.

Requirements

As a part of the X-ray imaging system, the mobile patient positioning table is used for the examination of body parts such as abdomen and lower limbs.

The X-ray imaging system must be operated by trained and qualified personnel. In addition, make sure to follow the requirements of relevant laws and regulations during the operating process.

Danger points

When the equipment is handled properly and when patients are positioned, the staff may only grasp the handles provided with their hands.

It is the responsibility of the operator to initiate movements of the equipment only if it is ensured that neither the patient nor third parties can be endangered by these movements.

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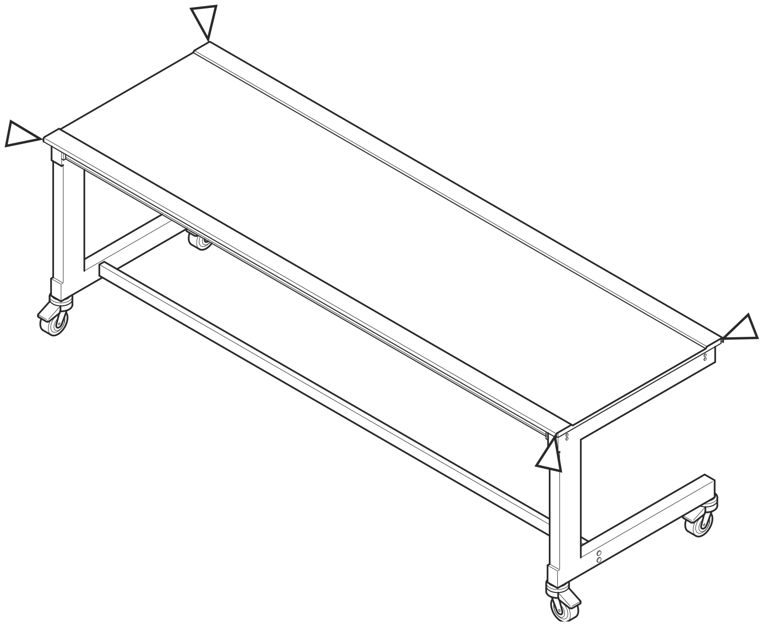
The parts of the unit marked in the drawing below indicate possible danger zones where the patient or staff coned crush their fingers or be struck.



Risk of injury for the operating staff when placing the tabletop in position.

Functional and safety check

Daily checks

1. Check the unit movement.
2. Check the brakes.
3. Check the mobility of the castors.
4. Check that the necessary devices for patient holding and immobilization are attached properly to the unit.
5. When the castor brake pedal is in locked status, check if the patient table position is locked.

Operating the mobile patient positioning table

Before using the table, make sure to remove all nearby objects such as chairs which might collide with the table.

When moving the mobile patient positioning table:

- Make sure to avoid collision with X-ray the detector or other devices.
- Make sure to notice and avoid squeezing of hands and similar incidents.
- Make sure that both the patient and the operating personnel stay inside controllable safety zone.
- Make sure to take care and avoid bringing any unnecessary harm to patients, operators or other personnel, and also tube and detector.

1 Before exposure, place the detector and the tube at the required exposure position.

2 Adjust the height of the detector moving it at pre-set height matching the patient table height.

3 Move the table to a position above the detector, make sure the table cannot collide with the detector and the post.

4 Move the table to an appropriate position.

5 Lock both castors and ensure the table is fixed.

6 Help the patient onto the bed and adjust the position as required.

7 Release the brakes of both castors, move the table to obtain the required exposure position, and then lock the brakes again.

8 Switch on the light localizer at the collimator, and prepare the final exposure position.

9 Adjust the light field of the collimator and release an exposure.

10 Release the brakes on both castors and move the table to a position convenient for the patient to leave the table;

11 Lock both castors and help the patient from the table.

Technical data

Dimensions	Tabletop	2000 mm (+/- 5 mm) X 660 mm (+/- 5 mm)
	Distance from tabletop to the ground	700 mm (+/-10mm)
	Usable width of tabletop between standing posts	1900 mm
Maximum load of the tabletop	135 kg	
Aluminum equivalent of tabletop	< 0.8 mm Al (according to measurement results based on GB 9706.12(IEC60601-1-3:2008) (100 kV/3.7 mm Al)	
Environmental conditions	Temperature	+10 °C to +35°C

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	Relative humidity	30% to 70%
	Atmospheric pressure	700 hPa to 1060 hPa

6.2.38 Mobile detector holder

Refer to the original Operator Manual of the manufacturer.

CAUTION

Removing the detector from the mobile detector holder. Due to the spring loading, the holder will move upwards when the detector is removed or the brake is released.

Risk of injury

- Be careful when removing the detector from the detector holder.
- Turn the knob to lock the vertical position before removing the detector.
- Hold the horizontal arm while slowly removing the detector to make sure that the holder remains at that position.

CAUTION

Collision with the mobile detector holder positioned above the table

Injury of patient or user

- Be careful lifting the table when the mobile detector holder is positioned near the table.

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6.2.39 Trolley for remote control console

Remote control console on a trolley in the examination room.

- For interventional examinations
- Complete unit and system control with joysticks and TUI
- Identical to remote control console in the control room

Do **not** lean onto the trolley!

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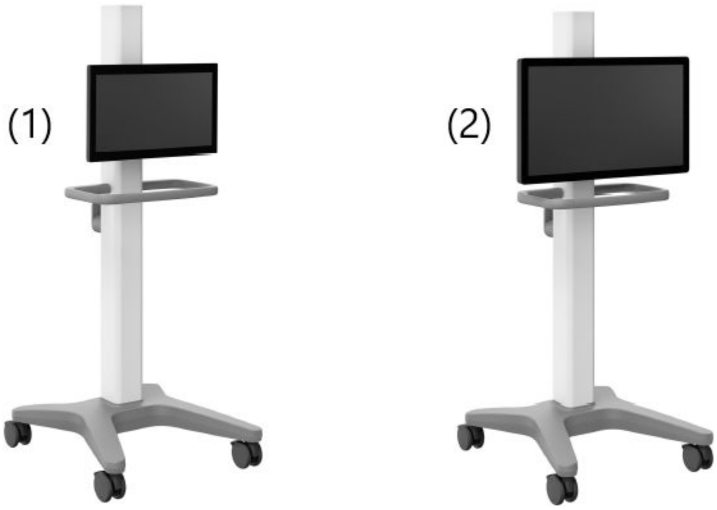
6.2.40 Display trolley



Example of trolley with a 24" display and a 32" display

(1) Trolley with 24" display
(2) Trolley with 32" display

You can move the trolley with the handle into any desired observation position.

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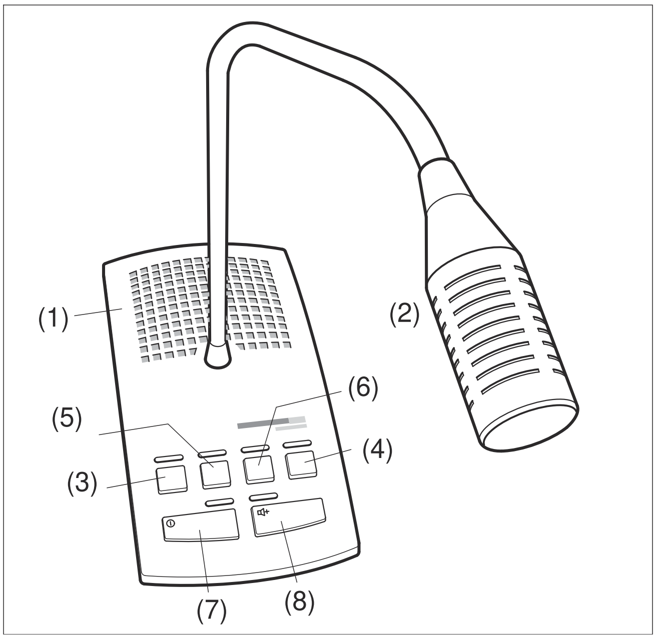
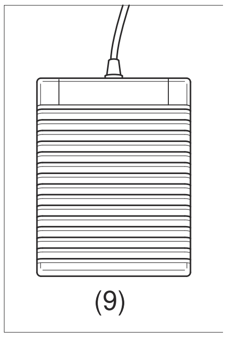
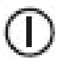
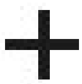
6.2.41 Intercom system in control room





(1) Loudspeaker
(2) Microphone
(3) ON or OFF button
(4) Audio IN ON/OFF
(5) Reduce volume
(6) Increase volume
(7) Loudspeaker ON/OFF
(8) Microphone ON/OFF
(9) Foot switch (same function as the ON/OFF button)

Operating Elements

Key	Function	LED State	System Operating State
	Intercom System ON/OFF	on off flashes	ready for use switched off set-up mode
	Loudspeaker ON/OFF	on off	loudspeaker ON loudspeaker OFF
	Microphone ON/OFF	on off	microphone ON microphone OFF
	Audio IN ON/OFF	on off	Audio IN ON Audio IN OFF
	Increase volume	brightens flashes	maximum volume

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Key	Function	LED State	System Operating State
	Reduce volume	darkens flashes	minimum volume

"Push to talk" (PTT) mode

Talking permanently possible from the examination room to the control room;

Talking from the control room to the examination room only possible by pushing a button.

Talking

- Push and hold the "Microphone" key to switch the microphone on.

Switching ON PTT

1. Switch the intercom system off.
2. Push and hold this key (approximately 5 s) until the LED above the "Microphone" key flashes.

Switching OFF PTT

1. Switch the intercom system off.
2. Push and hold this key (approximately 5 s) until the LED above the "Microphone" key flashes.

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7 Technical Description

7.1 Technical data

All technical data represent typical values unless specific tolerances are stated.

Siemens Healthineers reserves the right to change the design and specifications without notice.

7.1.1 System

Installation data	The entire system is connected via a single line voltage	
	Line voltage	3/N/PE, 380V/400V/440V/480V (50/60Hz) (± 10%)
	Frequency	≥ 340 kHz
	Internal fuses	3 x 50 A slow-blow
	External fuses	3 x 63 A
	Power consumption Standby Maximum power for long term fluoro Maximum power for short term radiography	max. 145 kVA < 2.5 kVA < 40 kVA < 127 kVA

Radio interference suppression/EMC	IEC 60601-1-2	
Protection class 	The system belongs to protection class I with a type B applied part according to IEC 60601-1	
	Protection against ingress of water	• IPx8: Foot switches • IPx3: Tableside control panel • IP22: Control Room Module • IPx0: Rest of system
Explosion protection (AP, APG after IEC 60601-1-3:2008; IEC 60601-1:2005)	The system is not intended for operation in explosion-endangered areas (for example highly flammable mixtures of anaesthesia gases with air or oxygen or nitrous oxide).	
Operating mode	Pulsed fluoroscopy	
Weight	Patient table, complete	Approx. 1640 kg
	Bucky wall stand	Approx. 290 kg (incl. detector)
	X-ray tube support	From 264 kg to 384 kg
	Imaging system hardware	Approx. 20 kg
	Generator cabinet	Approx. 280 kg

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	System remote control console	Approx. 4.5 kg
	Mobile tableside control console	Approx. 40 kg
Degree of contamination	Class 2, according to IEC 60601-1-3:2008; IEC 60601-1:2005	
Total inherentfiltration	2.5 mm Al at 70 kv/2.5 mm Al HVL (DIN EN 61267, RQR5) (combined with collimator Al equivalent 1.0 mm Al and tube Al equivalent 1.5 mm Al) + optional DAP Al equivalent 0.5 mm Al	
DAP chamber	Measurement of DAP	The air kerma values are measured in a distance of 30 cm from the detector with a 20 cm PMMA phantom according to IEC 60601-1-3:2008; IEC 60601-1:2005.
	Tolerance of DAP values	± 35%
AEC chambers	AEC chambers attenuation equivalent	0.7 mm Al
	Tolerance of AEC values	± 3%
Applied parts	• Tabletop • Compression device • Wall stand • Portable detector	
Floor space required	Maximum 3.80 m x 2.32 m without restrictions to movement, traffic area and safety distances Minimum 2.80 m x 2.32 m with restrictions to movement (+90°/-40°, maximum SID 1.15 m at + 90°), without traffic area and safety distances.	
Room height	• System without ceiling-suspended X-ray tube: min. 250 cm (with restrictions) • System with ceiling-suspended X-ray tube: min. 262 cm (with restrictions) • System without restrictions: min 320 cm	

7.1.2 Unit

Table and tabletop

Table tilt	• Motorized, +90° / -50°, or +90° / -90° • Two-level tilt speed 3°/s and 6°/s • Selectable automatic stop in horizontal position (0°) • Digital angle indicator on remote control console and tableside control panel
Table height	• 50 cm to 100 cm, continuously adjustable at 4 cm/s • 48 cm to 98 cm for in-floor installation
Tabletop dimensions	• Outside 210 cm x 80 cm • Radiolucent 201 cm x 53.5 cm

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Table and tabletop

Tabletop Al equivalent	0.7 mm Al at 100 kV / 3.6 mm Al HVL (IEC 60601-1-3:2008)
Patient weight	• Up to 180 kg: Full speed • 180 - 210 kg: – All movements possible – All speeds reduced up to 20% Restriction of movement range: – Table longitudinal +/- 40 cm – Table transverse: Between center and +/- 7.5 cm (operator side) Restriction for patient position: – Patient prone or supine position, centered on the table • 210 - 260 kg – Table tilt -15°/+90°; reduced speed – Lift range not limited; reduced speed – Tabletop: Centered in both directions; no movement allowed Restriction for patient position: – Patient prone or supine position, centered on the table • 260 - 330 kg: – Table tilt -15°/+45°; reduced speed – Tabletop: Centered in both directions; no movement allowed – Table height adjustments are possible with reduced speed. Restriction for patient position: – Patient prone or supine position, centered on the table • 330 kg: – Maximum patient weight • Because of CPR requirements, interventional examinations are limited to patients up to 280 kg (with additional 50 kg CPR equipment on centered table). The permitted nominal patient weights were established in accordance with IEC 60 601-1 with fourfold test load.
Longitudinal travel	• 160 cm motorized, 80 cm in each direction (head and foot end) • Max. speed 9 cm/s
Transverse travel	• 35 cm motorized, 17.5 cm to the left and right • Max. speed 4.5 cm/s
Foot rest	• Attachable at foot or head end, height-adjustable (three positions) • Can be lowered to 4 cm above floor with table in vertical position • Maximum weight capacity up to 230 kg with table in vertical position

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X-ray tube stand

Source-image distance (SID)	115 cm, 150 cm, and 180 cm; motorized adjustment at approx. 5 cm/s
Oblique projection	• ± 45° • Motorized adjustment of pivot point height 1 cm to 30 cm above the tabletop (= parallax correction of the object) • Digital angle display for remote and tableside control
Tube rotation	Motorized from +90° to -180°
Scattered radiation grid	Stationary 15:1, 80 lines/cm, f_0= 140 cm
Travel range	Maximum 113 cm travel at up to 10 cm/s
Central beam height above the floor	• 75 cm to 188 cm with table at + 90° • 55 cm to 168 cm with table at -90° (only with ±90° option)
Table-detector distance	7.0 cm (minimum)
Central beam distance to end of table	minimum 38 cm (head end) with orthogonal beam path

X-ray tube	**OPTITOP 150/40/80HC-100**
Maximum exposure voltage	150 kV (IEC 60613)
Focal spot nominal value	0.6 (IEC 60336) 1.0 (IEC 60336)
Nominal anode input power	• 40 kW / 80 kW (IEC 60613) - thermal anode reference output 300 W • 52 kW / 103 kW (IEC 60613) - thermal anode reference output 0 W
Radiographic anode input power	47 kW/85 kW (IEC 60613)
Continuous anode input power	350 W
Anode heat dissipation rate	120,000 J/min. (170,000 HU/min.)
Anode heat storage capacity	580,000 J (820,000 HU)
Maximum heat storage capacity of tube housing	1,800,000 J (2,530,000 HU)
Anode drive	150/180 Hz (9,000 to 10,800 rpm)
Leakage radiation (at 150 kV, 3mA at 1 m distance)	≤ 0.8 mGy/h (IEC 60601-1-3:2021; IEC 60601-1:2020)
Total filtration	≥ 2.5 mm Al/80 kV (IEC 60601-1-3:2008; IEC 60601-1:2005)

Collimator

Maximum field size	• 35 cm x 35 cm at 0.7 m SID • 43 cm x 43 cm at 1.0 m SID • 42.5 cm x 42.5 cm at 1.0 m SID (for collimator with DSA module)
Minimum field size	< 3.0 cm x 3.0 cm at 1.0 m SID

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Collimator

Angle of rotation	± 45° about the central beam axis
LED light	• Only original Siemens Healthineers replacement lamps may be used! • 24 V / 25 W / part no.: 10398386 LED module mounted
Inherent filtration	1.0 mm Al at 75 kV/1.5 mm Al at 75 kV
Source to collimator flange distance	8 cm
Cu pre-filters (Al equivalent)	None, 0.1 mm (3.5 mm), 0.2 mm (7.1 mm), 0.3 mm (10.8 mm)
Radiation protection	depending on regulations, up to max. tube voltage of 150 kV
Full-field light localizer	Very efficient 4 W high power LED technology; high energy efficiency enabling low-noise design without external cooling system, lifetime approx. 100,000 h, timer functionality, laser line light localizer (coverable)
Dimensions (height x width x depth)	• 183 mm x 232 mm x 352 mm (without DSA module) • 262 mm x 233 mm x 380 mm (with DSA module)
Maximum weight (without accessories)	10.5 kg to 15.5 kg (with DSA module)
Maximum weight of the accessories	≤ 7kg
Connected load	30 - 40 V AC ; 50/60 Hz; 8A
Aperture angle	28° / 28°
Attenuation of the useful radiation	
Attenuation factor without cone	m = 2.48
Attenuation factor with cone	m = 2.50

Compression device

	• Remote-controlled • Radiolucent cone • Retractable • Park position outside the radiation field • Separate soft cover for cone, cone can be detached • Automatic safety switch-off for system movements beginning at 50 N
Compression force	5 N to maximum 155 N or 80 N*, continuously adjustable
Speed	Compression or decompression about 7.5 cm/s
Force display	Digital display (LCD) in increments from 3 to 15 (1 increment approx. 10 N)

* configuration for Japan only

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7.1.3 Ceiling-suspended X-ray tube

X-ray tube support

Al equivalent	< 0.45 mm Al (100 kV, 3.6 mm Al HVL)	
Horizontal travel range	Longitudinal	346 cm
	Transverse	• With 300 cm trolley: 220 cm • With 400 cm trolley: 355 cm
Travel range and speed	Vertical travel range	180 cm (70.9 "), manual or motorized
	Vertical travel speed	Up to max. 0.3 m/s (11.8"/s)
	Rotation range around the vertical axis	• -154° to +182° manual • Mechanical stops at -90°; 0°; +90°
	Rotation range around horizontal axis	• ±140° manual; max. 25 °/s • Mechanical stops at -90°; 0°; +90°
	Rotation speed around the horizontal axis	Max. 40°/s
Minimum focus-ceiling distance	83 cm (32.7")	

X-ray tube	**OPTITOP 150/40/80HC-100**
Maximum exposure voltage	150 kV (IEC 60613)
Focal spot nominal value	0.6 (IEC 60336) 1.0 (IEC 60336)
Nominal anode input power	• 40 kW / 80 kW (IEC 60613) - thermal anode reference output 300 W • 52 kW / 103 kW (IEC 60613) - thermal anode reference output 0 W
Radiographic anode input power	47 kW/85 kW (IEC 60613)
Optical anode angle	12° (IEC 60788)
Anode heat dissipation rate	120,000 J/min. (170,000 HU/min.)
Anode heat storage capacity	580,000 J (820,000 HU)
Maximum heat storage capacity of tube housing	1,800,000 J (2,530,000 HU)
Anode drive	150/180 Hz (9,000 to 10,800 rpm)
Leakage radiation (at 150 kV, 3mA at 1 m distance)	≤ 0.8 mGy/h (IEC 60601-1-3:2021; IEC 60601-1:2020)
Total filtration	≥ 2.5 mm Al/80 kV (IEC 60601-1-3:2008; IEC 60601-1:2005)

Multileaf collimator

Inherent filtration	1mm Al at 70 kV

Multileaf collimator

Full field light localizer	Very efficient 4 W high power LED technology; High energy efficiency enabling low-noise design without external cooling system, lifetime approx. 100,000 h, timer functionality, laser line light localizer (coverable)
Cu prefilter	Without filter; 0.1 mm; 0.2 mm; 0.3 mm; motorized and positionable via clinical protocol steps
Rotation	±45° manually
Collimator control	Manual and motorized, preset via clinical protocol steps

7.1.4 Bucky wall stand

Bucky wall stand for use with ceiling-suspended X-ray tube

Travel range (central beam to floor)	From 31.5 cm (12.4") to 175 cm (68.9") (for all detector configurations)
Detector unit	Tiltable from - 20° to + 90° with 0° detent Non tiltable for small rooms
Central beam alignment / Top alignment	Tube alignment: Central X-ray beam is centered to center of wall detector.
	Top alignment: Light field is offset to upper border of wall detector.
Tracking	tube automatically tracks wall Bucky height adjustments with detector tray at 0°, 90° or intermediate tilt angles
Bucky wall stand front - detector distance	≤ 4.2 cm (1.7")
X-ray absorption	≤ 0.6 mm Al (at 100 kV/3.7 mm Al HVL; DIN IEC 60601-1-3:2008)

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7.1.5 Anti-scatter grids

Bucky wall stand for use with ceiling-suspended X-ray tube

Type	Transparent grid, stationary, focused
Universal grid	Pb 13/40, f0 = 115 to f0 = 180 cm; Pb with paper interspacing
Grid	Pb 13/40, f0 = 180 cm; Pb with paper interspacing Standard delivery with BWS
Grid	Pb 13/40, f0 = 300 cm; Pb with paper interspacing
Clip-on grids for MAX wi-D(option)	Grid, Pb 5/85, f0 = 115 cm; Pb with aluminum interspacing
	Grid, Pb 15/80, f0 = 115 cm; Pb with paper interspacing
Clip-on grid for MAX mini (option)	Grid, Pb 5/85, f0 = 115 cm; Pb with aluminum interspacing
Clip-on grids for X.wi-D 43	Pb 5/31, f0 = 115 cm; Pb with paper interspacing
Clip-on grids for X.wi-D 35	Pb 5/31, f0 = 115 cm, Pb with paper interspacing

7.1.6 Detectors

X.fluoro detector (integrated flat detector in table)

Detector technology	Amorphous silicon flat detector with Cesium iodide scintillator High-resolution 2840 x 2874 matrix with 148 µm pixel size and 16-bit digitization depth High-performance fiber-optic connection to digital imaging system	
Input fields (active field)	Zoom F43 Zoom F30 Zoom F23 Zoom F15	42 cm x 42.5 cm (17.0” x 17.0”) 30 cm x 30 cm (11.8” x 11.8”) 23 cm x 23 cm (9.0” x 9.0”) 15 cm x 15 cm (5.9” x 5.9”)
Material	aSi with CsI scintillator	
Pixel size	148 µm	
Spatial resolution (Nyquist frequency)	3.4 lp/mm	
Maximum acquisition speed	Up to 8f/s (Fluoro 30 p/s)	
Matrix	Up to 2840 x 2874 pixels	
Digitization depth	16 bits	
DQE in %; 2 µGy (RQA5) (IEC 62220)	73 % at 0.05 lp/mm 51 % at 1.0 lp/mm 42 % at 2.0 lp/mm 28 % at 3.0 lp/mm 19 % at 3.4 lp/mm Nyquist	

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DQE in %; 200 nGy (RQA5) (IEC 62220)	73 % at 0.05 lp/mm 51 % at 1.0 lp/mm 42 % at 2.0 lp/mm 27 % at 3.0 lp/mm 19 % at 3.4 lp/mm Nyquist
DQE in %; 20 nGy (RQA5) (IEC 62220)	70 at 0.05 lp/mm 47% at 1.0 lp/mm 31% at 2.0 lp/mm 16% at 3.0 lp/mm 11% at 3.4 lp/mm Nyquist
MTF in % (RQA5) (IEC 62220)	66 % at 1.0 lp/mm 35 % at 2.0 lp/mm 19 % at 3.0 lp/mm 15 % at 3.4 lp/mm Nyquist
Modulation depth at the Nyquist frequency	Approx. 13%
Weight	25 kg
Anti-scatter grid	Stationary, Pb 15:1, 80 lines/cm, f_0 = 125 cm, removable
Tabletop-detector distance	7.3 cm
X-ray absorption	≤ 0.5 mm Al (at 100 kV/3.7 mm Al HVL; IEC 60601-1-3 and IEC 60601-2-54)

MAX detectors

MAXcharge (option)	Charging MAX wi-D in the detector tray and/or wall holder (option)
MAXswap	MAXswap is the right way to share, allowing you to swap the MAX wi-D and MAX mini between multiple systems so you always have the right detector when and where you need it (depends on system configuration)

MAX static (integrated flat detector)

Detector technology	Cesium iodide scintillator coupled to TFT matrix with amorphous silicon technology
Dimensions (active area)	42.3 cm x 42.5 cm
Active detector matrix	2860 x 2874
Pixel size	148 µm
Spatial resolution	3.4 lp/mm
Semiconductor material	Amorphous silicon (a-Si)

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Scintillator	Cesium iodide (CsI)
Acquisition depth	16 bits
DQE in %; 2µGy (RQA5) (IEC 62220)	• 67% at 0.05 lp/mm • 58 % at 0.5 lp/mm • 51 % at 1 lp/mm • 46 % at 1.5 lp/mm • 42 % at 2 lp/mm • 36 % at 2.5 lp/mm • 27 % at 3 lp/mm • 18 % at Nyquist
MFT in % (RQA5) (IEC 62220)	• 80 % at 0.5 lp/mm • 62% at 1 lp/mm • 47 % at 1.5 lp/mm • 35% at 2 lp/mm • 26% at 2.5 lp/mm • 19% at 3 lp/mm • 15% at Nyquist
Data transmission	<3.5 s preview, <6 s full image
X-ray absorption	≤ 0.5 mm Al (at 100 kV/3.7 mm Al HVL; IEC 60601-1-3 and IEC 60601-2-54)

MAX wi-D (wireless detector)

Detector technology	Cesium iodide scintillator coupled to TFT matrix with amorphous silicon techno-logy
Dimensions (active area)	34.8 cm x 42.4 cm Can be inserted in the detector tray in landscape and portrait format.
Dimensions with detector housing	44.1 cm x 46.1 cm x 1.9 cm
Active detector matrix	2350 x2866
Pixel size	148 µm
Semiconductor material	Amorphous silicon (a-Si)
Scintillator	Cesium iodide (CsI)
Acquisition depth	16 bits

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DQE in %; 2µGy (RQA5) (IEC 62220)	• 70 % at 0.05 lp/mm • 60% at 0.5 lp/mm • 51% at 1 lp/mm • 47% at 1.5 lp/mm • 42% at 2 lp/mm • 35% at 2.5 lp/mm • 29% at 3 lp/mm • 19% at Nyquist
MFT in % (RQA5) (IEC 62220)	• 81% at 0.5 lp/mm • 63% at 1 lp/mm • 47% at 1.5 lp/mm • 35% at 2 lp/mm • 26% at 2.5 lp/mm • 19% at 3 lp/mm • 15% at Nyquist
Data transmission	WLAN* < 2 s preview; < 5 s full image
WLAN Standard	IEEE 802.11n, 2 x 2 mimo
Thickness	19 mm
Weight	3.3 kg
Max. load capacity	• 300 kg with patient recumbent • 100 kg with patient standing
Battery	Lithium-ion, rechargeable, exchangeable
Charging time	3 h in battery charger
Battery operation time	• Up to 1050 images • Up to 6.5 hours during regular utilization • Up to 11.7 hours in standby mode
Charging location	Table detector tray, Bucky wall stand detector tray, battery charger (option) and wall charger (option)
IEC regulations	Electromagnetic compatibility: compliance with IEC 60601-1-2 ed. 4.1 This detector does not affect pacemakers that fulfill DIN EN 45502-2-1, Section 27.
X-ray absorption	≤0.5 mm Al

* The preview/full image transmission time depends on the quality of the Wi-Fi link and the selected processing parameters.

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If there is a WLAN or other wireless equipment in your working environment, please consult your Siemens Healthineers representative for optimal set-up of the wireless connection.

MAX mini (wireless detector)

Detector technology	Cesium iodide scintillator coupled to TFT matrix with amorphous silicon technology
Dimensions (active area)	22.5 cm x 28.4 cm
Dimensions with detector housing	26.9 cm x 32.9 cm x 1.6 cm
Active detector matrix	1520 x 1920
Pixel size	148 µm
Semiconductor material	Amorphous silicon (a-Si)
Scintillator	Cesium iodide (CsI)
Acquisition depth	16 bits
DQE in %; 2µGy (RQA5) (IEC 62220)	• 70% at 0.05 lp/mm • 60% at 0.5 lp/mm • 51% at 1 lp/mm • 47% at 1.5 lp/mm • 42% at 2 lp/mm • 35% at 2.5 lp/mm • 29% at 3 lp/mm • 19% at Nyquist
MFT in % (RQA5) (IEC 62220)	• 81% at 0.5 lp/mm • 63% at 1 lp/mm • 47% at 1.5 lp/mm • 35% at 2 lp/mm • 26% at 2.5 lp/mm • 19% at 3 lp/mm • 15% at Nyquist
Data transmission	WLAN* < 1.5 s preview; < 3.5 s full image
WLAN Standard	IEEE 802.11n, 2 x 2 mimo
Thickness	16 mm
Weight	1.6 kg
Max. load capacity	• 300 kg with patient recumbent • 100 kg with patient standing
Battery	Lithium-ion, rechargeable, exchangeable

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Charging time	3 h in battery charger
Battery operation time	• Up to 1050 images • Up to 6.5 hours during regular utilization • Up to 11.7 hours in standby mode
Charging location	Battery charger
IEC regulations	Electromagnetic compatibility: compliance with IEC 60601-1-2 ed. 4.1 This detector does not affect pacemakers that fulfill DIN EN 45502-2-1, Section 27.
X-ray absorption	≤0.5 mm Al

* The preview/full image transmission time depends on the quality of the Wi-Fi link and the selected processing parameters.

If there is a WLAN or other wireless equipment in your working environment, please consult your Siemens Healthineers representative for optimal set-up of the wireless connection.

X.wi-D detectors

MAXcharge	Charging X.wi-D 43 or X.wi-D 35 in the table detector tray

Detector technology	Cesium iodide scintillator coupled to TFT matrix with amorphous silicon technology	
Dimensions (active area)	X.wi-D 24	22.9 cm x 28.4 cm
	X.wi-D 35	34.5 cm x 42.4 cm
	X.wi-D 43	42.5 cm x 42.5 cm
Dimensions with detector housing	X.wi-D 24	26.9 cm x 32.9 cm x 1.6 cm
	X.wi-D 35	38.5 cm x 46.1 cm x 1.6 cm
	X.wi-D 43	46.1 cm x 46.1 cm x 1.6 cm
Active detector matrix	X.wi-D 24	2319 mm x 2879 mm
	X.wi-D 35	3495 mm x 4298 mm
	X.wi-D 43	4302 mm x 4301 mm
Pixel size	99 µm	
Semiconductor material	Amorphous silicon (a-Si)	
Scintillator	Cesium iodide (CsI)	
Acquisition depth	16 bits	

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DQE in %; 2 µGy (RQA5) (IEC 62220)	• 70% at 0.05 lp/mm • 60% at 0.5 lp/mm • 53% at 1.0 lp/mm • 49% at 1.5 lp/mm • 45% at 2.0 lp/mm • 42% at 2.5 lp/mm • 38% at 3.0 lp/mm • 33% at 3.5 lp/mm • 27% at 4.0 lp/mm • 21% at 4.5 lp/mm • 14% at Nyquist	
MTF in % (RQA5) (IEC 62220)	• 84% at 0.5 lp/mm • 68% at 1.0 lp/mm • 54% at 1.5 lp/mm • 43% at 2.0 lp/mm • 34% at 2.5 lp/mm • 27% at 3.0 lp/mm • 21% at 3.5 lp/mm • 16% at 4.0 lp/mm • 13% at 4.5 lp/mm • 10% at Nyquist	
Data transmission	X.wi-D 24	< 3.0 s preview; < 7.5 s final
	X.wi-D 35	< 3.0 s preview; < 7.5 s final
	X.wi-D 43	< 3.5 s preview; < 10.0 s final
WLAN Standard	IEEE 802.11ax, 2 x 2 mimo	
Thickness	16 mm	
Weight (including battery)	X.wi-D 24	1.5 kg
	X.wi-D 35	2.6 kg
	X.wi-D 43	3.0 kg
Max. load capacity	• 300 kg with patient recumbent • 100 kg with patient standing with 1 foot on detector surface	
Drop height	120 cm	
Battery	Lithium-ion, rechargeable, exchangeable with Battery Hotswap	
Charging time	< 3.5 h in battery charger	

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Battery operation time	• Up to 810 images • Up to 7 hours during regular utilization • Up to 48 hours in standby mode
Charging location	Table detector tray (X.wi-D 43 and X.wi-D35); Battery charger
IEC regulations	Electromagnetic compatibility: compliance with IEC 60601-1-2 ed. 4.1
Protection class	IP 67
X-ray absorption	≤0.3 mm Al

* The preview/full image transmission time depends on the quality of the Wi-Fi link and the selected processing parameters.

If there is a WLAN or other wireless equipment in your working environment, please consult your Siemens Healthineers representative for optimal set-up of the wireless connection.

7.1.7 Generator POLYDOROS RFX

Technical data	X-ray generator POLYDOROS RFX 65 (150 kV/ 65 kW) or POLYDOROS RFX 80 (150 kV/80 kW) All technical data are typical values unless specific tolerances are stated.	
Nominal voltage	380 V/ 400 V/ 440 V/ 480 V +/- 10%, 50/60 Hz 3-phases	
Internal line resistance (max.)	POLYDOROS RFX 65	• 0,16 Ohm (380 V) • 0,17 Ohm (400 V) • 0,20 Ohm (440 V) • 0,24 Ohm (480 V)
	POLYDOROS RFX 80	• 0,10 Ohm (380 V) • 0,11 Ohm (400 V) • 0,14 Ohm (440 V) • 0,16 Ohm (480 V)
Capacity	Generator	• Long-term 2.5 kVA (380 V, 400V, 440 V, 480 V) • Short-term 126 kVA (380 V, 400 V, 440 V, 480 V)
	External connection	• 380 V/ 400 V, 440 V, 480 V 25 A 3-phases • 230 V max 8 A 1-phase
Radio interference suppression/EMC	IEC 60601-1-2 limit-value-class B	
Type of protection	Protective class I, in accordance with IEC 60601-1	
Grade of protection (according to IEC 60529)	IPx0 = no special protection	

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Application part	none	
Degree of protection against inflammable gas mixtures	No use in combination with inflammable gas mixtures (not AP, APG examined)	
Operation mode	Continuous operation	
High-voltage waveform	Multipulse	
Nominal shortest irradiation time	For LUMINOS Q.namix R unit and ceiling suspended X-ray tube: 2.8ms at 80kV (with AEC chamber 10664277, anti-scatter grid, large focus, 80% power)	
	POLYDOROS RFX 65	POLYDOROS RFX 80
Nominal Power	65 kW (100 kV, 650 mA, 0.1 s)	80 kW (100 kV, 800 mA, 0.1 s)
Exposure	430 mA at 150 kV 650 mA at 100 kV 800 mA at 79 kV	520 mA at 150 kV 800 mA at 100 kV 1000 mA at 79 kV
	0.5 - 800mAs / 0.5 - 1000 mAs	
Fluoroscopy	40 kV to 150 kV, 4 mA to 84 mA, 2 ms to 10 ms, pulsed fluoroscopy	
Radiography (AEC)	40 kV to 150 kV	
Automatic System	• 0- and 1- point technique with continuosly falling load • 2- and 3- point technique with constant load	
Switching time (Minimum)	• 1- point technique: 1 ms at 100 kV • 2- point technique: 2 ms at 100 kV • 3- point technique: 20 ms at 100 kV	
Tolerances	• Exposure time tolerance ± (10 % +1 ms) • kV tolerance ± 8 % • mAs tolerance ≤ ± (10 % + 0.2 mAs) • mA tolerance ≤ ± 20 % • Variation coefficient of radiation output ≤ 0.05	
Ambient conditions (Operation)	Temperature	+10°C to +35°C
	Relative humidity	20% to 75%, non-condensing
	Barometric pressure	700 hPa to 1060 hPa
Ambient conditions (Transport and storage)	Temperature	-20°C to +60°C
	Relative humidity	10% to 95%, non-condensing
	Barometric pressure	500 hPa to 1060 hPa

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7.1.8 Imaging system

Computer system	Equipment	Intel Core i5 microprocessor, min. 3.7 GHz, 32 GB RAM, S-ATA drive, USB 3.0 and interface cards for the detector/X-ray system
	Operating system	Windows 10 Enterprise LTSB 2021 (64 Bit)
	Noise level	≤ 50 dBA
Archiving and documentation	Acquisition memory on hard disk	SSD: 2x2TB – up to 200,000 DFR images (1 RAD image equals 8 DFR images)
	Background functions	Preprocessing and data transfer is done in the background.
	Archiving	USB in DICOM 3.0 format
	Export	USB in DICOM 3.0 format
	Documentation	• DICOM Print • Virtual Film Sheet
	DICOM Worklist, MPPS	Data exchange via HIS/RIS
Electrical data	Power supply	Powered by system power supply
	Nominal frequency	50; 60 Hz
	Power	< 700 VA (incl. monitor)
Ambient conditions	Operation	Climate class S
	Temperature range	+10 °C to +35°C
	Relative humidity	20% to 75% not condensing
	Air pressure	800 hPa to 1060 hPa
	Height	≤2000 m

7.1.9 Displays

24" LCD touch monitor

24” TFT high-contrast color display for flicker-free, distortion-free live image and reference image display for X-ray diagnostics

Light weight, high luminance and contrast values

Screen size	24" (60.5 cm)
Display area (W x H)	52.7 cm x 27.5 cm
Pixel number	1920 x 1080

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Typical brightness	600 cd/m²
Surface	Anti-glare and anti-fingerprint coating
Typical contrast ratio	1000 : 1
Power consumption	< 46 W; in power save mode: < 2 W (Touch on)
Weight	6.2 kg (13.6 lbs)
Horizontal viewing area	178° (H and V)
Dimensions (W x H x D)	55.7 cm x 33.9 cm x 5.4 cm

32" LCD touch monitor

32” TFT high-contrast color display for flicker-free, distortion-free live image and reference image display for X-ray diagnostics

Light weight, high luminance and contrast values

Screen size	31.5" (80 cm)
Display area (W x H)	69.7 cm x 39.2 cm
Pixel number	Full/single image: 2048 x 2048 Splitscreen: 1884 x 1884
Typical brightness	1000 cd/m²
Surface	Anti-glare and anti-fingerprint coating
Typical contrast ratio	1300 : 1
Power consumption	< 72 W; in power save mode: < 7 W
Weight	14.7 kg (32.4 lbs)
Horizontal viewing area	178° (H and V)
Dimensions (W x H x D)	75.8 cm x 45.5 cm x 7.5 cm

7.1.10 Display trolley

Suitable for one 24" or one 32" touch monitor

One radiation-ON indicator

Tilt range	+ 20 ± 2° backwards and - 12 ± 2° forwards
Max weight of the Trolley (24”and 32”)	55 kg (121 Ibs)
Vertical travel range for monitor	13cm

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7.1.11 Display ceiling suspension (DCS)

Ceiling suspension system with extension and spring arm for one 24" or one 32" touch monitor Two radiation-ON indicators and mechanical brakes	
Length of longitudinal rails	251 cm (98.8")
Travel range of ceiling-mounted carriage	202 cm (79.5”)
Vertical lift (height adjustment)	73 cm (28.7") (for room height between 300 cm (118.1") and 356 cm (140.2"))
Rotation range of the ceiling-mounted support about the rail axis	Maximum 360°
Rotation range of displays	Maximum 360°
Max weight (24" and 32")	Approx. 111 kg (244.7 lbs) including ceiling rail assembly

7.1.12 WLAN access point SCALANCE W7x8

Data transfer	Ethernet transfer rate	BASE-T 10/100/10000Mbps
	Wireless transmission rate	1 ... 450 Mbps
	Wireless standards supported	• IEEE 802.11a • IEEE 802.11b • IEEE 802.11g • IEEE 802.11h • IEEE 802.11n
	Power supply standards supported	IEEE 802.3at type 1 (802.3af) and 802.3at type 2 (Power over Ethernet)
Interfaces	Power	Power over Ethernet IEEE 802.3at min. 36 VDC, max. 57 VDC via RJ-45 jack The following applies to PoE: Current supply lines must be electrically isolated according to IEEE 802.3at, dielectric resistance > 2 Mohms.
	Data W7x8-x RJ-45	RJ-45 jack for Ethernet 3 antenna sockets of type R-SMA female
Electrical data	Maximum power consumption	15.6 W
Construction	Dimensions without antennae (H x W x D)	W7x8-x RJ-45 (IP30): 158 mm x 200 mm x 79 mm
	Weight	W7x8-x RJ45 (IP 30): 1.7 kg
Permitted ambient conditions	Operating temperature	-20°C to +60°C

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	Transportation/storage temperature	-40°C to +70°C
	Degree of protection	Tested to IP30

7.1.13 Environment

CAUTION

Operation in unsuitable environment

Malfunction of system

- The system may only be run in an environment approved or authorized by the manufacturer.
- The specified environmental conditions must be complied with.

Ambient conditions

Operation	Temperature range	+10°C to +35°C
	Relative humidity	20% to 75%, non-condensing
	Barometric pressure	700 hPa to 1060 hPa
Transport and storage	Temperature range	-20°C to +70°C
	Relative humidity	10% to 95%, non-condensing
	Barometric pressure	500 hPa to 1060 hPa

Please note that the maximum permitted ambient temperature is 28°C when combined with certain components.

7.2 Location of system labels

This chapter shows the location of the system labels. The numbers and parameters on the labels are examples only. Different numbers and parameters (for example serial number) may appear on the actual label.

The following symbols are contained in the labels, if applicable:

MD	Medical device
CONFIG	Configurable device A configurable device is a device that consists of several components which can be assembled by the manufacturer in multiple configurations. Those individual components may be devices in themselves.

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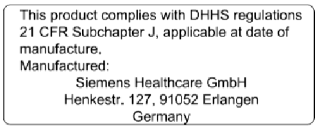
The labels shown below are only showing the position of the real labels on the system. Material numbers or serial numbers shown are partially only place holders. Please refer to the real labels on the system for material numbers etc.

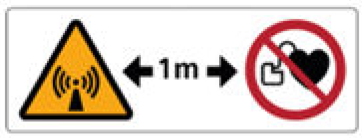
7.2.1 On the table

7.2.2 On the collimator of the ceiling-suspended tube assembly



7.2.3 On the generator

on both sides and at the front



at the front

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7.2.4 Label plate with second set of labels on the PC container

on the left side of the PC container

7.3 Electromagnetic compatibility (EMC)

7.3.1 Essential Performance

According to the IEC 60601-2-54 Standard the product provides following Essential Performance:

- Accuracy of Loading Factors
- Reproducibility of the Radiation output
- Automatic Control System
- Imaging performance

According to the IEC 60601-2-43:2010 + AMD1:2017 + AMD2:2019 Standard, the product provides following Essential Performance:

- Recovery management
- Radiation dose documentation

7.3.2 Precautions for EMC - 4.1 Edition

Degradation of the essential performance caused by electromagnetic interferences

The operator may experience degradations caused by electromagnetic interference or a defect in the system.

Possible consequences: Deterioration of the image quality or accuracy of the radiation parameters

Remedy: Remove the source of interference (for instance, a mobile phone) from the area. Shut down the system and turn it on again. Pay attention to error messages during restart.

Declaration of immunity and emission compliance

The product complies with following test levels of IEC 60601-1-2:2014+A1:2020:

Emission test:	**Basic EMC standard or test method:**	**Compliance level:**
Conducted and radiated RF emissions	CISPR11	Group 1, Class A

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Immunity test:	Basic EMC standard or test method:	Compliance level:
Electrostatic discharge	IEC 61000-4-2	• ± 8 kV contact • ± 2, 4, 8, 15 kV air
Electrical fast transient / burst	IEC 61000-4-4	• ± 2 kV for power supply lines; 100 kHz repetition frequency • ± 1 kV for input / output lines; 100 kHz repetition frequency
Surge	IEC 61000-4-5	• ± 0.5, 1, 2 kV line to ground • ± 0.5, 1 kV line to line
Voltage dips on power supply lines	IEC 61000-4-11	N/A
Voltage interruptions on power supply lines	IEC 61000-4-11	0% U_T; 250/300 cycles
Power frequency (50 or 60 Hz) magnetic field	IEC 61000-4-8	N/A
Conducted disturbances induced by RF fields	IEC 61000-4-6	• 150 kHz to 80 MHz: 3 Vrms • 6 Vrms in ISM bands between 0.15 MHz and 80 MHz
Radiated RF EM fields	IEC 61000-4-3	80 MHz - 2.7 GHz: 3 V/m
Proximity fields from RF wireless communications equipment	IEC 61000-4-3	• 385 MHz: 27 V/m; Pulse modulation 18 Hz • 450 MHz: 28 V/m; Frequency modulation ± 5 kHz deviation 1 kHz sine • 710, 745, 780 MHz: 9 V/m; Pulse modulation 217 Hz • 810, 870, 930 MHz: 28 V/m; Pulse modulation 18 Hz • 1720, 1845, 1970 MHz: 28 V/m; Pulse modulation 217 Hz • 2450 MHz: 28 V/m; Pulse modulation 217 Hz • 5240, 5500, 5785 MHz: 9 V/m; Pulse modulation 217 Hz
Immunity to proximity magnetic fields	IEC 61000-4-39	• 134.2 kHz; 65 A/m • 13.56 MHz; 7.5 A/m

The XP product should not be used adjacent to or stacked with other equipment and that if adjacent or stacked use is necessary, the equipment or product should be observed to verify normal operation in the configuration in which it will be used.

Declaration of EMC environment

XP product is suitable for professional healthcare facility environment according IEC60601-1-2 ed. 4.1.

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Warning for adjacent or stacked equipment

Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.

List of cables, transducers and accessories

Fixed product cabling, which cannot be removed by the user, is not listed. This cabling is part of the product and was present at all EMC-measurements. Without this cabling there is no complete functionality of the product. Deviations or additions to this document are added to the product-specific documents.

Warning for usage of other accessories, cables and transducers

CAUTION

Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.

Dysfunctional system, Repetition of examination

- Only use accessories, converters and cables that are sold by the manufacturer of the unit or system as replacement parts for internal components.

Warning for portable RF communication equipment

Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the LUMINOS Q.namix R, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.

Note for CISPR11 class A equipment

The emission characteristics of this equipment make it suitable for use in industrial areas and hospitals (CISPR 11 class A). If it is used in a residential environment (for which CISPR 11 class B is normally required) this equipment might not offer adequate protection to radio-frequency communication services. The user might need to take mitigation measures, such as relocating or re-orienting the equipment.

Instructions for maintaining basic safety and essential performance with regard to electromagnetic disturbances for the expected service life

Perform an annual visual inspection of accessible connectors, screws and conducting cabinet door to ensure that its electrical connection has not deteriorated. In case of any degradation call Siemens Healthineers Service to replace or repair the corresponding part. With this it is ensured that the design measures regarding to the electromagnetic compatibility of the equipment remain effective during the whole service life time.

Warning against use of equipment nearby of emitters

This equipment has been tested for radiated RF immunity only at selected frequencies; use of emitters in close proximity at other frequencies could result in improper operation.

8 Glossary

ACSS	Automatic Collimation Size Selection
Auto Composing	If set in the clinical protocol step, images will be composed automatically. The algorithm for image composition (spine or leg) is automatically selected.
Blend mode	The single images are blended at the transitions between each pair of images.
CARE	The CARE package (Combined Applications to Reduce Exposure) comprises features for dose reduction. A complete set of CARE features can be found on your system, consisting of: • CAREFILTER: Automatic copper filters to reduce skin dose • CAREMATIC: Automatic X-ray control system for fully automatic calculation and optimization of the exposure data based on fluoroscopic values • CAREMAX: Dose area product measurement and monitoring • CAREVISION: Pulsed fluoroscopy with selectable pulse frequencies • CAREPROFILE: Radiation-free collimation using Last Image Hold (LIH) for orientation • CAREPOSITION: Radiation-free positioning via graphic display of the field-of-view in the Last-Image-Hold (LIH) (only together with CAREPROFILE)
CAREPOSITION	With CAREPOSITION, any move of table or stand is indicated in the last acquired image or LIH.
CAREPROFILE	With the CAREPROFILE function, the position of the collimator blades and the semi-transparent filter positions are displayed graphically in the last image. In this way you can change the collimation without additional fluoroscopy.
Clinical EXI	The **Clinical EXI** (**cEXI**) analyzes the mean gray value of the diagnostic relevant part in the image via histogram, without the peaks for direct radiation and collimation.
Clinical protocol	A clinical protocol (CP) defines the steps for an examination (acquisitions). It comprises one or more clinical protocol steps.
Clinical Protocol Editor	The **Clinical Protocol Editor** allows to define and manage clinical protocols that consist of one or more clinical protocol steps, create additional clinical protocols or clinical protocol steps (“organ programs”) and define the acquisition parameters for the clinical protocol steps.
Clinical protocol step	A clinical protocol step (“organ program”) comprises the parameters for acquisitions.
Composed image	A large image created out of several single images by sewing them together with a suitable algorithm.
Composing	**Composing** allows you to compose digital, **overlapping single images** that were acquired for this purpose.
CP	Clinical protocol
CR	Computer Radiography
Cut mode	The single images are placed side by side without blended transitions.
DAP	Dose Area Product

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Deviation Index	The **Deviation Index** (**DI**) detects the tolerance of the exposure in correlation of the **Clinical EXI** value. The Deviation Index tells something about under- and overexposure of the diagnostic area.
DFR	Digital Fluoro Radiography (DFR mode)
DSA	Digital Subtraction Angiography
DX	Digital X-ray
EP	Exposure Points, for correcting exposure data.
FluoroLoop	Retrospective storage and review of fluoroscopy sequences
Harmonization	Harmonization is a method of image processing that emphasizes the detail contrast of small structures by contrast compensation of coarse structures.
HIS	Hospital Information System
Image	DICOM term: An image can be a single image (frame), e.g. an X-ray acquisition, or a multiframe object, e.g. a fluoroscopic scene, an Ortho series, or a 3D volume.
kV	kilo-Volts (1 kV = 1,000 V), unit for tube voltage of an exposure
mAs	milli-Ampere seconds, unit for the product of tube current (1 mA = 0.001 A) and time (s) of an exposure Equivalent to the applied radiation dose
MAX	MAX is a Siemens Healthineers innovation that is available with the Luminos Agile Max, Luminos dRF Max, Luminos Fusion Max, LUMINOS Lotus Max fluoroscopy systems, as well as the Ysio Max, YSIO X.pree, Mobilett Mira Max, and MOBILETT Elara Max radiography systems.
MAX dynamic	Flat detector for radiography and fluoroscopy imaging
MAX mini	Small wireless mobile detector for more difficult-to-position exams such as orthopedic, pediatric, and trauma imaging
MAX static	Flat detector which is integrated in the Bucky wall stand or the patient table
MAXswap	Swap a pool of MAX wi-D, MAX mini, and X.wi-D detectors between multiple systems
MAX wi-D	Wireless mobile detector for use with patient table, Bucky wall stand, and free examinations
ms	milli-seconds (1 ms = 0.001 s), unit for the duration of an X-ray pulse of an exposure
myExam Cockpit	System configuration interface
Patient	DICOM term: All examination results, i.e. studies, are uniquely assigned to a patient. Unique assignment is usually achieved with the patient’s personal data, i.e. last name, first name, date of birth, sex and patient identification number.
Physical EXI	The **Physical EXI** (**pEXI**) is a measure of the mean gray value in the middle area of the image.
Procedure step	DICOM term: A modality performed procedure step (MPPS) divides an examination into several procedure steps.
RAD	Digital Radiography (RAD mode)
RF	Radiography, Fluorography
RIS	Radiology Information System

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RIS Matching	Matching of the names of the clinical protocols of your system to the names used by the connected RIS (Radiology Information System).
ROI	Region of Interest
Series	DICOM term: A series consists of one or more images.
SID	Source-Image Distance
SmartOrtho	Automated tilting technique for long leg and full spine imaging.
SmartView	SmartView allows you to display and switch between images from different modalities.
SOD	Source-Object Distance
STD	Source-Table Distance
Study	DICOM term: A study contains the data of an examination, series and images.
TOD	Table-Object Distance
Tube UI	Color touchscreen for showing and setting examination and system data
X.wi-D	Wireless mobile detector for use with patient table and free examinations
XA	X-ray, Angiography

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